6. A girl standing 6·0 m from a convex mirror sees her virtual image 1·0 m behind the mirror. The focal length of the mirror is : *1*

(A) 6·0 m (B) 1·2 m

(C) 2·4 m (D) 0·8 m

CBSE 26C

6. A girl standing 6·0 m from a convex mirror sees her virtual image 1·0 m behind the mirror. The focal length of the mirror is : *1*

(A) 6·0 m (B) 1·2 m

(C) 2·4 m (D) 0·8 m

CBSE 26C

(B) 1.2 m

20. What is meant by linear magnification of an image formed by a spherical mirror ? Write an expression for the magnification produced by a concave mirror in terms of the object distance and the image distance. CBSE 26C 2

20. What is meant by linear magnification of an image formed by a spherical mirror ? Write an expression for the magnification produced by a concave mirror in terms of the object distance and the image distance. CBSE 26C 2

Linear magnification is the ratio of the size of image to the size of object. Alternatively: $$m = \frac{h_I}{h_O} = \frac{\text{Size of image}}{\text{Size of object}}$$	1
$$m = -\frac{v}{u} = -\frac{\text{Image distance}}{\text{Object distance}}$$	1

25. A thin convex lens of focal length 25 cm is cut into two pieces 0·5 cm parallel and above the principal axis. An object is placed at a distance of 50 cm in front of the smaller part. Find : (i) the position of the image, and (ii) the magnification produced by this part. CBSE 26C 3

25. A thin convex lens of focal length 25 cm is cut into two pieces 0·5 cm parallel and above the principal axis. An object is placed at a distance of 50 cm in front of the smaller part. Find : (i) the position of the image, and (ii) the magnification produced by this part. CBSE 26C 3

(i)	$f = +25\text{ cm},\ u = -50\text{ cm},\ v = ?$	½
	$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$	½
	$\frac{1}{25} = \frac{1}{v} - \frac{1}{-50} = \frac{1}{v} + \frac{1}{50}$	
	$\frac{1}{v} = \frac{1}{25} - \frac{1}{50} = \frac{2-1}{50} = \frac{1}{50}$	½
	$v = +50cm$	½
(ii)	$m = \frac{v}{u}$	½
	$m = \frac{50}{-50} = -1$	½

33. (a) (i) Explain the phenomenon of total internal reflection. State two conditions that must be satisfied for total internal reflection to occur. Derive the relation between the critical angle and the refractive index of a medium.

(ii) A ray of light is travelling from a denser medium of refractive index 1·5 to a rarer medium of refractive index 1·0. Calculate critical angle for total internal reflection. CBSE 26C 5

(i) When a ray of light enters from a denser medium to a rarer medium, it bends away from the normal. As the angle of incidence increases at the interface, angle of refraction also increases till the angle of incidence is $\frac{\pi}{2}$ (i.e the refracted ray grazes the surface of the interface). If the angle of incidence is further increased, the incident ray totally gets reflected back. This is called total internal reflection. | 2

Conditions

1. Light must travel from a denser medium to a rarer medium. | ½

2. At the interface, the angle of incidence must be greater than the critical angle. | ½

Relation

When light goes from a denser medium (refractive Index n_2) towards a rarer medium (refractive Index n_2), then for the value of critical angle, we have relation-

$n_2 \sin i_c = n_1 \sin 90^\circ$ | ½

$\sin i_c = \frac{n_1}{n_2}$

or, $\frac{n_2}{n_1} = \frac{1}{\sin i_c}$

or,

$n_{21} = \frac{1}{\sin i_c}$ | ½

(ii) Calculating critical angle

$n_2 = 1.5 = \frac{3}{2}$, $n_1 = 1$

$\sin i_c = \frac{1}{n_{21}} = \frac{2}{3}$ | ½

Therefore,

$i_c = \sin^{-1}\left(\frac{2}{3}\right)$ | ½

9. A ray of light is incident at an angle of 60° on face AB of prism ABC with $\angle A = 30°$. The ray emerging out of the prism makes an angle of 30° with the incident ray. The angle of emergence ($\angle e$) is equal to :

(A) 0°

(B) 30°

(C) 60°

(D) 90°

CBSE 26C

9. A ray of light is incident at an angle of 60° on face AB of prism ABC with $\angle A = 30°$. The ray emerging out of the prism makes an angle of 30° with the incident ray. The angle of emergence ($\angle e$) is equal to :

(A) 0°
(B) 30°
(C) 60°
(D) 90°

CBSE 26C

(A) 0^0

10. A point source in air is placed in front of convex spherical glass surface (n = 1·5 and R = 10 cm). Its real image is formed at the same distance as the surface from the source. The distance of the source from the surface is :

(A) 25 cm

(B) 50 cm

(C) 75 cm

(D) 100 cm

CBSE 26C

10. A point source in air is placed in front of convex spherical glass surface (n = 1·5 and R = 10 cm). Its real image is formed at the same distance as the surface from the source. The distance of the source from the surface is :

(A) 25 cm

(B) 50 cm

(C) 75 cm

(D) 100 cm

CBSE 26C

(B) 50 cm

(b) (i) The lower half of a concave mirror's reflecting surface is covered with an opaque non-reflecting material. What effect will this have on the image of an object placed in front of the mirror ? *1*

(ii) A lens made of glass of refractive index 1·5 disappears, when immersed in a trough of liquid. What is the refractive index of the liquid ?

Give reasons for your answer in each case. CBSE 26C *1*

(b) (i) The lower half of a concave mirror's reflecting surface is covered with an opaque non-reflecting material. What effect will this have on the image of an object placed in front of the mirror ? *1*

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Give reasons for your answer in each case. CBSE 26C *1*

(i) The image will be formed that of the whole object. However, the intensity of the image will be low (or half)	1
(ii) The refractive index of the liquid = 1.5	½
Since lens disappears inside the liquid which is possible only when both have same refractive indices.	½

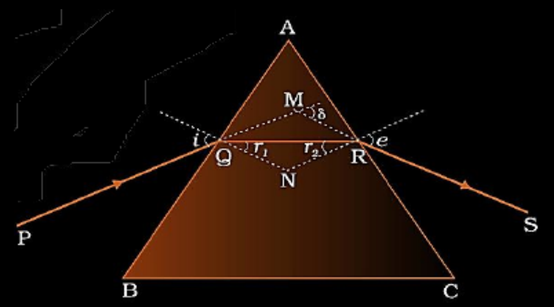
24. Derive an expression for refractive index of an equilateral prism in terms of the angle of minimum deviation and the angle of the prism. Plot a graph to show the variation of the angle of deviation as a function of angle of incidence. *3*

CBSE 26C

24. Derive an expression for refractive index of an equilateral prism in terms of the angle of minimum deviation and the angle of the prism. Plot a graph to show the variation of the angle of deviation as a function of angle of incidence. 3

CBSE 26C

In quadrilateral AQNR,



$\angle A + \angle QNR = 180°$

From triangle QNR, ½

$r_1 + r_2 + \angle QNR = 180°$

Comparing the above equations,

$r_1 + r_2 = A$

The total deviation δ is the sum of deviations at the two faces:

$\delta = (i - r_1) + (e - r_2)$

$\delta = i + e - A$ ½

At minimum deviation,

$\delta = D_m$, $i = e$ *which implies* $r_1 = r_2$

$r = \frac{A}{2}$

$D_m = 2i - A$ or $i = \frac{A+D_m}{2}$ ½

The refractive index of the prism is

$n = \frac{\sin i}{\sin r}$ 1

$n = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$ ½

27. (a) Draw a labelled ray diagram of a refracting telescope in normal adjustment. Briefly explain its working. *2*

(b) Why is the eyepiece of a telescope of short focal length, while the objective is of large focal length ? Explain. *1*

CBSE 26C

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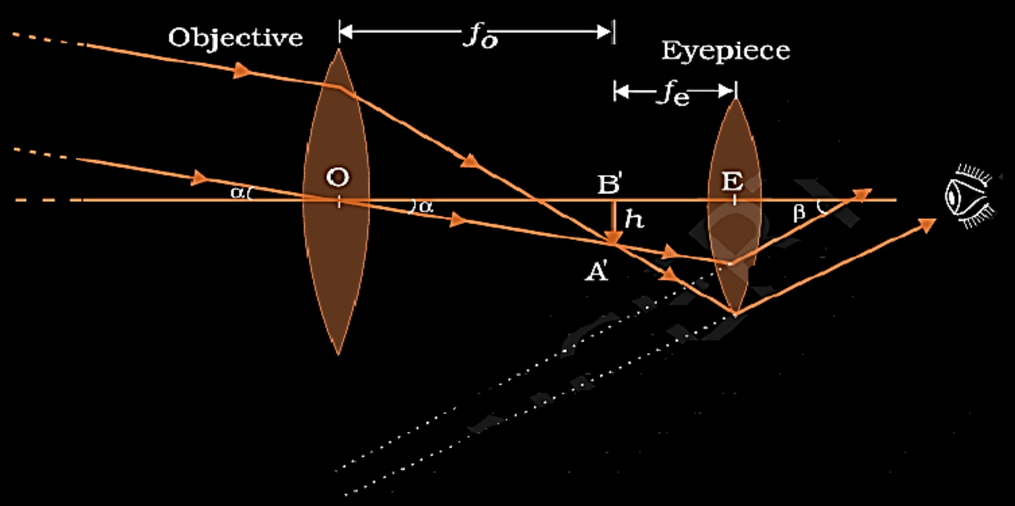
CBSE 26C

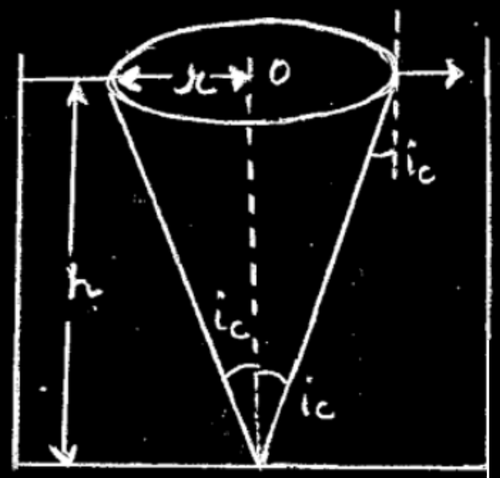


	1
Working: When light from a distant object enters the objective lens, a real image is formed in the tube at its second focal point. The eyepiece magnifies this image, producing a final inverted image.	1
(b) To increase the magnifying power or angular magnification.	1

33. (a) A small bulb is placed at the bottom of a tank containing water to a depth of 1·00 m. Find the area of the surface of water through which light from the bulb can emerge from. The refractive index of water is (4/3). *3*

(b) An object is placed at a distance of 15 cm from a convex lens of focal length 10 cm. On the other side of the lens, a convex mirror is placed coaxially at a distance of 10 cm from the lens. If the image formed by this combination coincides with the object itself, draw the ray diagram and find the focal length of the convex mirror. *2*

CBSE 26C



1

$\sin i_c = \frac{1}{n}$ ½

$\tan i_c = \frac{r}{h}$

$r = h \tan i_c$ & $\tan i_c = \frac{1}{\sqrt{n^2-1}}$

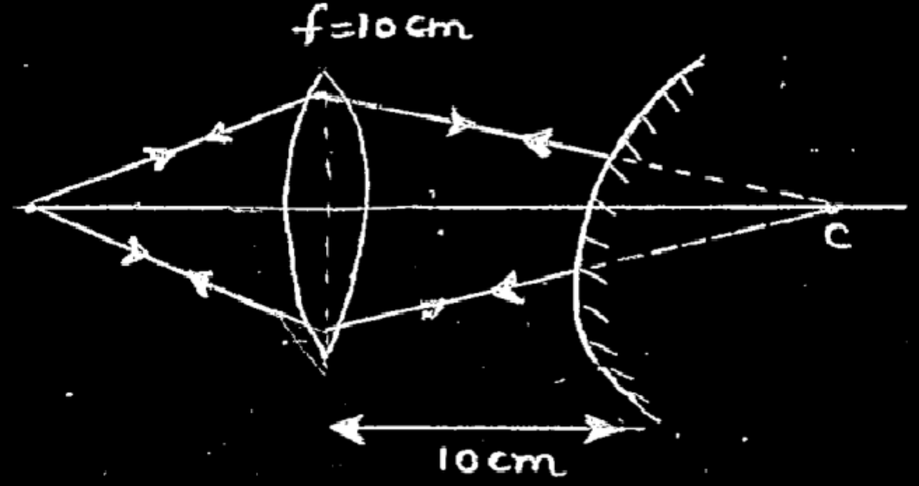
$A = \pi r^2 = \pi h^2 \tan^2 i_c$ ½

$A = \frac{\pi h^2}{n^2-1}$ ½

$A = \frac{\pi \times 1}{\frac{16}{9}-1}$ ½

$= \frac{9\pi}{7}$ m^2 $= \frac{9\times 3.14}{7}$ =4.04 m^2

(b)



1

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{10} - \frac{1}{15}$

$= \frac{1}{30}$

$v = 30$ cm ½

Radius of curvature of the mirror,

$R = 30 - 10 = 20$ cm ½

$f = \frac{R}{2}$

$f = 10$ cm

9. The focal lengths of the objective and the eyepiece of a compound microscope are 2 cm and 3 cm respectively. If they are kept 25 cm apart, the tube length of the microscope is :

(A) 23 cm (B) 22 cm

(C) 20 cm (D) 30 cm

CBSE 26C

9. The focal lengths of the objective and the eyepiece of a compound microscope are 2 cm and 3 cm respectively. If they are kept 25 cm apart, the tube length of the microscope is :

(A) 23 cm (B) 22 cm

(C) 20 cm (D) 30 cm

CBSE 26C

(C) 20 cm

24. (a) The distance of an object from a concave mirror is increased from 12 cm to 15 cm. It is observed that the new image becomes $\left(\frac{2}{5}\right)^{\text{th}}$ the size of the initial image. Calculate the focal length of the mirror.

(b) A light ray is incident on the horizontal upper face of a glass cube making an angle of 45° with the face. While emerging, the ray just grazes the vertical face of the cube. Find the refractive index of the glass. CBSE 26C 3

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(a)

For $u = -12\ cm$

$$m_1 = \frac{h_1}{h_o} = \frac{f}{f+12} \quad \ldots(1)$$ ½

For $u = -15\ cm$

$$m_2 = \frac{h_2}{h_o} = \frac{f}{f+15} \quad \ldots(2)$$ ½

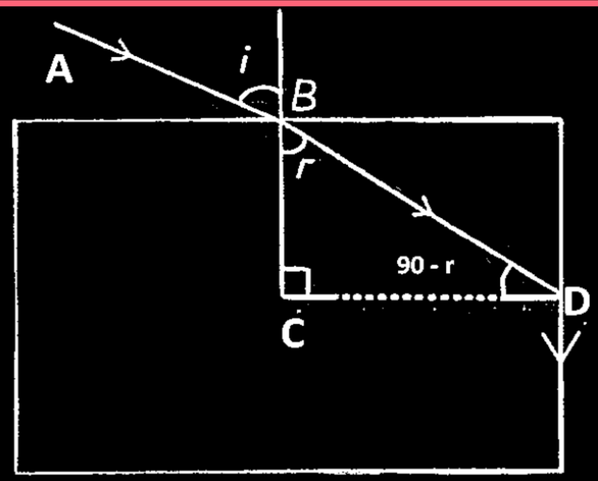
Dividing (1) by (2)

$$\frac{m_1}{m_2} = \frac{h_1}{h_2} = \frac{f+15}{f+12}$$

$$h_2 = h_1\left(\frac{2}{3}\right)$$

On solving,

$$f = -10\ cm$$ ½



At air-glass interface,

$$n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin 45^\circ}{\sin r}$$

$$\sin r = \frac{\sin 45^\circ}{n} = \frac{1}{\sqrt{2}\,n} \quad \ldots(1)$$ ½

At glass-air interface, light suffers Total Internal Reflection (TIR) ($i_C = 90°\text{-}r$).

$$n = \frac{1}{\sin i_C}$$

$$n = \frac{1}{\sin(90^\circ - r)} = \frac{1}{\cos r}$$

$$n = \frac{1}{cosr}$$

$$\cos r = \frac{1}{n} \quad \ldots(2)$$ ½

Squaring and adding (1) and (2)

$$1 = \frac{1}{2n^2} + \frac{1}{n^2}$$

On solving,

$$n = \sqrt{\frac{3}{2}}$$ ½

27. A thin convex lens with its two surfaces of radii of curvature R_1 and R_2 is made of a material of refractive μ_2. A medium of refractive index μ_1 ($< \mu_2$) surrounds it. Using a ray diagram, derive the Lens Maker's formula when a point object placed on the principal axis in front of the surface of radius of curvature R_1 produces an image I on the other side of the lens. *3*

CBSE 26C

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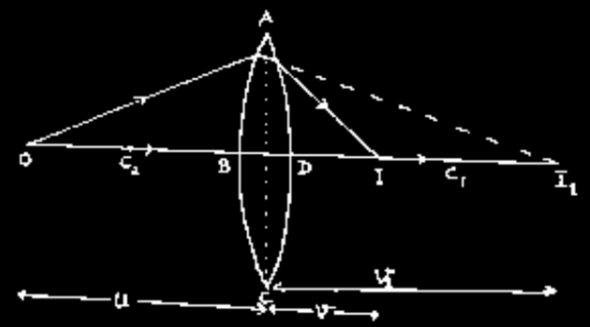
CBSE 26C



1

The first refracting surface forms the image of O at I_1.
For refraction through the first refracting surface (ABC),

$\frac{\mu_2}{v_1} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1}$ .....(1) ½

Similarly for refraction through second refracting surface ADC, the image I_1 acts as a virtual object and forms the final image at I.

$\frac{\mu_1}{v} - \frac{\mu_2}{v_1} = \frac{\mu_1 - \mu_2}{R_2}$ .....(2) ½

Adding equations (1) and (2),

$\frac{\mu_1}{v} - \frac{\mu_1}{u} = (\mu_2 - \mu_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ½

Suppose the object is at infinity i.e. $u = \infty$, and image at *f*.

$\frac{\mu_1}{f} = (\mu_2 - \mu_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ½

3. A ray of light passes from a medium of refractive index $\sqrt{2}$ into the medium of refractive index 1. For the ray to just graze along the surface of separation, the angle of incidence should be :

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$

(D) $\frac{2\pi}{3}$

CBSE 26C

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(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{2\pi}{3}$

CBSE 26C

(B) $\pi/4$

24. Two thin converging lenses of focal lengths f_1 and f_2 are placed coaxially in contact. Draw a ray diagram to show the image formation by the combination. Hence, derive the expression for the focal length of the combination. CBSE 26C 3

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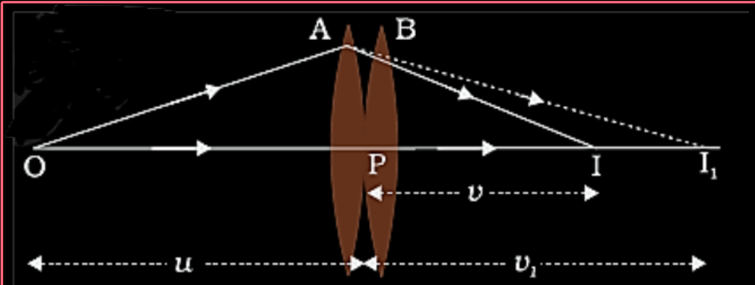
CBSE 26C 3



1

For the image formed by the first lens,

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \quad \text{---------- (1)}$$ ½

For the second lens,

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \quad \text{---------- (2)}$$ ½

Adding equations (1) and (2),

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} \quad \text{---------- (3)}$$ ½

For the two-lens system being equivalent to a single lens of focal length f, the equation (3) becomes

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{------ (4)}$$

On comparing equations (3) and (4),

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$ ½

27. Two converging lenses L_1 and L_2, of focal lengths 144 cm and 6 cm respectively, are arranged coaxially in a tube 147 cm apart. The arrangement is used to view a far off tree, keeping the eye close to L_2.

(i) Draw the ray diagram depicting image formation by the combination.

(ii) How far from L_2 will the final image be formed ?

(iii) Calculate the magnification produced by the combination. *3*

CBSE 26C

(ii) $L = 147 \text{ cm} > f_1 + f_2 = (150 \text{ cm})$

The first image is formed between the focus and lens L_2.

i.e. $u_2 < f_2$

$L = f_1 + | u_2 |$

$| u_2 | = L - f_1 = 147 - 144 = 3 \text{ cm}$ ½

For L_2

$u_2 = -3 \text{ cm}, \ f_2 = 6 \text{ cm}$

$v_2 = \frac{u_2 f_2}{u_2 + f_2}$

$v_2 = \frac{(-3)\times 6}{-3+6}$ ½

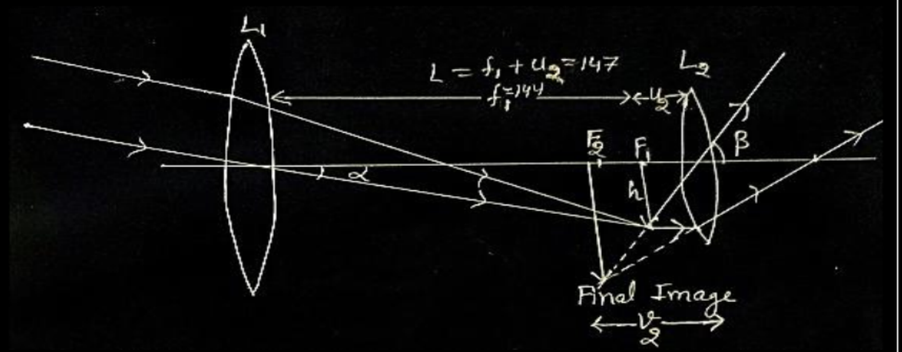
$= -6 \text{ cm}$

Final image is formed at 6 cm from L_2 at left side.

(iii) Magnification, $m = \frac{tan\beta}{tan\alpha}$

$= \frac{h/u_2}{h/f_1}$ ½

(i)



$m = \frac{f_1}{u_2}$

$m = f_1 \left(\frac{1}{v_2} - \frac{1}{f_2}\right)$ 1

$m = 144 \left(\frac{1}{-6} - \frac{1}{6}\right)$

$m = -48$ ½

19. A tank is filled with a liquid to a height of 12·5 m. The apparent depth of a needle lying at the bottom of the tank is measured to be 9·0 m. Calculate the speed of light in the liquid. CBSE 2026 *2*

19. A tank is filled with a liquid to a height of 12·5 m. The apparent depth of a needle lying at the bottom of the tank is measured to be 9·0 m. Calculate the speed of light in the liquid. CBSE 2026 *2*

$$n = \frac{\text{speed of light in vacuum(c)}}{\text{speed of light in medium(v)}} \quad \ldots\ldots\ (i)$$ ½

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$ ½

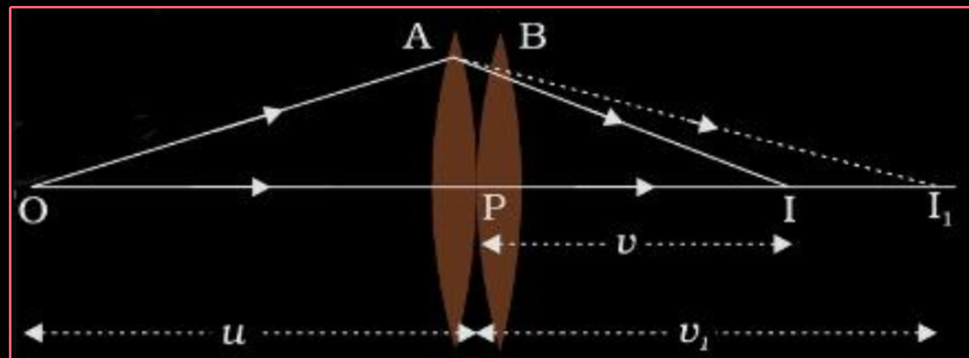
$$n = \frac{12.5}{9} \quad \ldots\ldots\ldots\

20. Two thin lenses of focal length f_1 and f_2 are placed in contact with each other coaxially. Prove that the focal length f of the combination is given by $f = \dfrac{f_1 f_2}{f_1 + f_2}$. 2

CBSE 2026

20. Two thin lenses of focal length f_1 and f_2 are placed in contact with each other coaxially. Prove that the focal length f of the combination is given by $f = \dfrac{f_1 f_2}{f_1 + f_2}$. 2

CBSE 2026



½

For the image formed by lens A

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \quad \text{.....(i)}$$

½

For the image formed by second lens B

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \quad \text{.....(ii)}$$

½

Adding (i)and(ii)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_2} + \frac{1}{f_1}$$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$f = \frac{f_1 f_2}{f_1 + f_2}$$

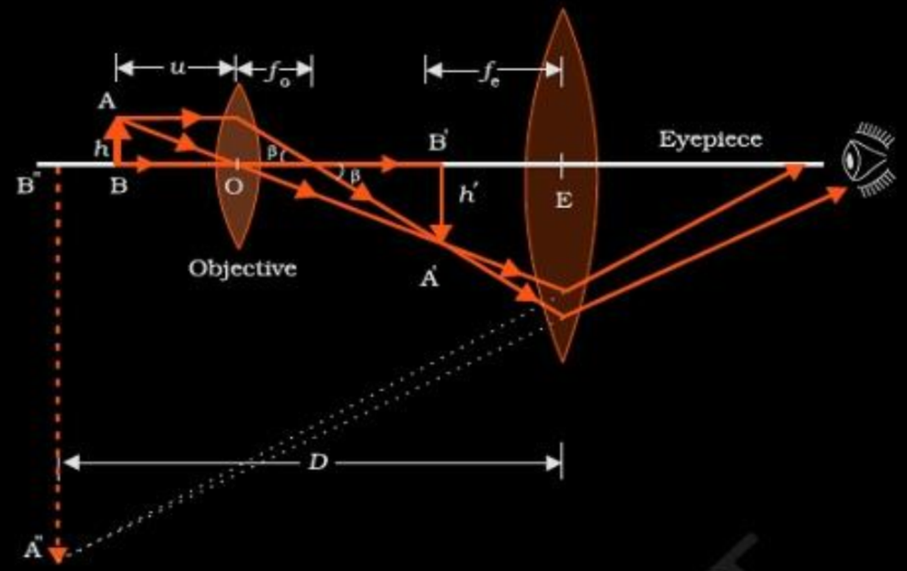
½

(b) (i) With the help of a ray diagram, describe the construction and working of a compound microscope.

(ii) (I) The real image of an object placed between f and 2f from a convex lens can be seen on a screen placed at the image location. If the screen is removed, is the image still there ? Explain.

(II) Plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ? Explain. CBSE 2026 5

(i)



1

Construction: It consists of two convex lenses, objective lens with small aperture placed near the object and the eye piece with large aperture placed near the eye. Both lenses have small focal length. These lenses are placed coaxially in a narrow tube with rack and pinion arrangement.

1

Working:
The lens near to the object called objective forms a real, inverted & magnified image of the object. This serves as the object for eye piece which produces the final image, enlarged & virtual.

1

(ii) I Yes ½

Real image is formed by actual intersection of rays. ½

II Yes ½

If the object is virtual. ½

Note: 1. If students do not write construction award full marks for well-labelled diagram representing objective lens, eye piece, along with image position at near point or normal adjustment.

2. Deduct ½ mark if arrows are not drawn.

21. A point source, in air, is placed at a distance of 6 cm in front of a convex spherical surface (n = 1·5 and radius of curvature = 24 cm). Find the position and nature of the image formed. CBSE 2026 *2*

21. A point source, in air, is placed at a distance of 6 cm in front of a convex spherical surface (n = 1·5 and radius of curvature = 24 cm). Find the position and nature of the image formed. CBSE 2026 *2*

$$\frac{n_2 - n_1}{R} = \frac{n_2}{v} - \frac{n_1}{u} \quad \text{½}$$

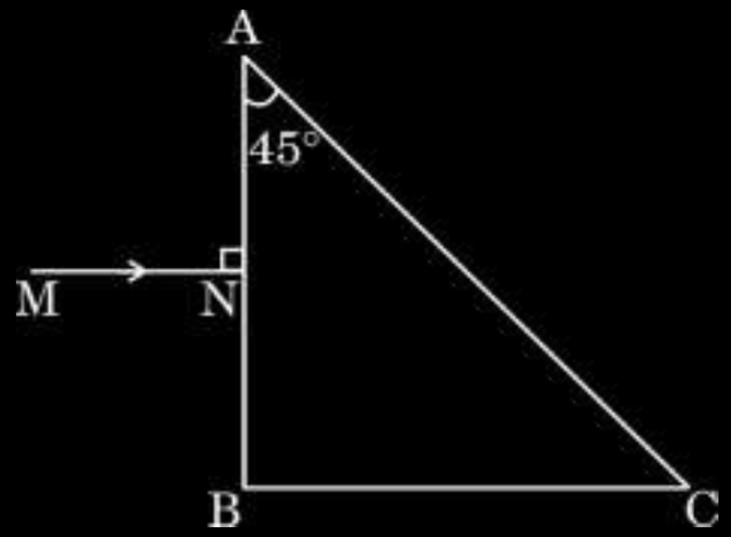
$$\frac{1.5 - 1.0}{24} = \frac{1.5}{v} - \frac{1.0}{-6} \quad \text{½}$$

After solving

$$v = \frac{-72}{7} \approx -10.28cm \quad \text{½}$$

Nature: Virtual ½

18. A ray of light MN is incident normally on the face corresponding with side AB of a prism with an isosceles right-angled triangular base ABC. Trace the path of the ray as it passes through the prism when the refractive index of the prism material is (i) $\sqrt{2}$, and (ii) $\sqrt{3}$. *2*



CBSE 2026

(i)

$$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}}$$

$$i_c = 45^0$$

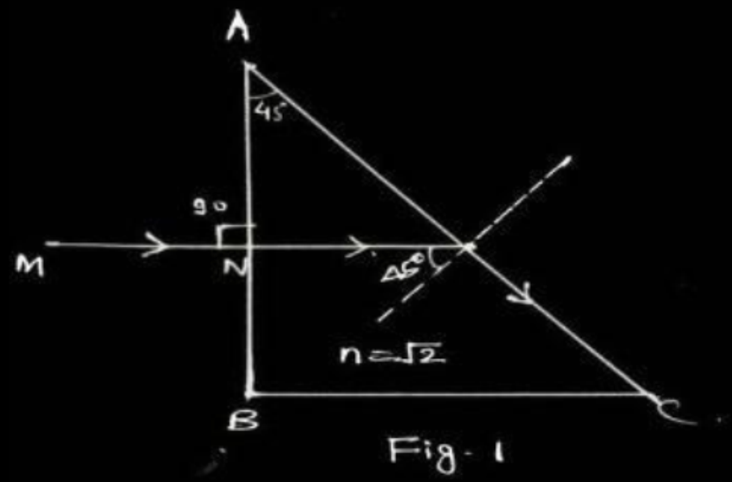


Fig - 1

1

Since $i = i_c$ so the emergent ray grazes alongside AC.

(ii)

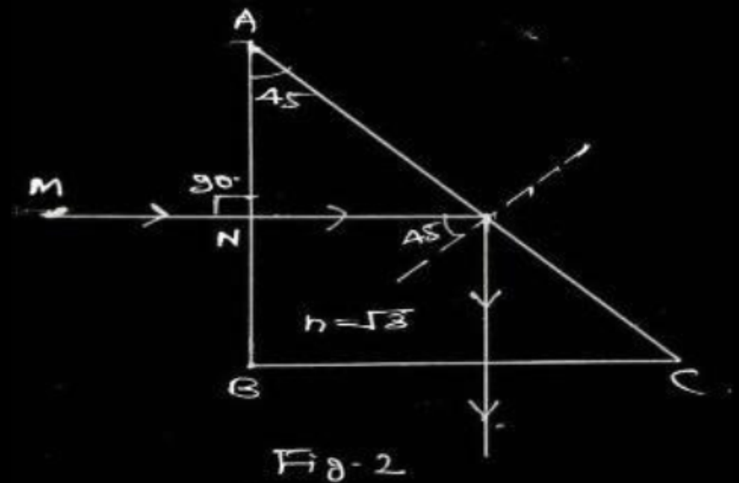


Fig - 2

1

$$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{3}}$$

$$i_c < 45^0$$

Since $i > i_c$ so, the ray suffers from Total Internal Reflection (TIR)

15. *Assertion (A) :* When a convex lens made of glass is immersed in water, its converging power increases.

Reason (R) : The focal length of a lens depends only on the radii of curvature of its two faces.

CBSE 2026

15. *Assertion (A) :* When a convex lens made of glass is immersed in water, its converging power increases.

Reason (R) : The focal length of a lens depends only on the radii of curvature of its two faces.

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(D) Both assertion(A) and Reason (R) are false.

19. A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges. *2*

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19. A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges. *2*

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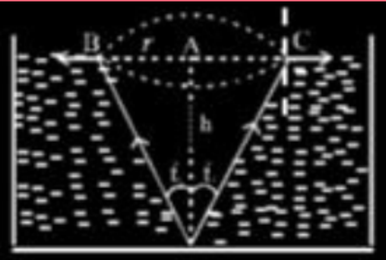

$r = h \times \tan i \quad [\text{as } i = i_c]$ ½

$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}} \Rightarrow i_c = 45°$ ½

$r = h$

$\text{Area} = \pi r^2$ ½

$= 3.14 \times (1)^2$

$\text{Area} = 3.14\ m^2$ ½

30. An astronomical telescope consists of two converging lenses. One of them of large aperture and large focal length is called objective lens and the other one, of smaller focal length and smaller aperture is called the eyepiece. It is used to see distant objects which are not seen clearly with naked eyes. The image formed by the objective lens acts as an object for the eyepiece and the final image produced by the eyepiece is magnified.

(i) The images formed by the objective lens and the eyepiece are respectively : *1*

(A) virtual, real

(B) real, virtual

(C) virtual, virtual

(D) real, real

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30. An astronomical telescope consists of two converging lenses. One of them of large aperture and large focal length is called objective lens and the other one, of smaller focal length and smaller aperture is called the eyepiece. It is used to see distant objects which are not seen clearly with naked eyes. The image formed by the objective lens acts as an object for the eyepiece and the final image produced by the eyepiece is magnified.

(i) The images formed by the objective lens and the eyepiece are respectively : *1*

(A) virtual, real

(B) real, virtual

(C) virtual, virtual

(D) real, real

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(i) (B) real, virtual

(ii) The magnification produced by the telescope does ***not*** depend upon the : *1*

(A) colour of light

(B) focal length of objective lens

(C) focal length of eyepiece

(D) apertures of objective lens and eyepiece

(iii) Which of the following statements is ***not*** correct for this telescope ? *1*

(A) The focal length of objective lens (f_o) is larger than the focal length of eyepiece (f_e).

(B) Its magnifying power can be increased by increasing the focal length of objective lens (f_o).

(C) The distance between two lenses is more than ($f_o + f_e$).

(D) The magnifying power can be decreased by increasing the focal length of eyepiece.

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(ii) The magnification produced by the telescope does ***not*** depend upon the : *1*

(A) colour of light

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(C) focal length of eyepiece

(D) apertures of objective lens and eyepiece

(iii) Which of the following statements is ***not*** correct for this telescope ? *1*

(A) The focal length of objective lens (f_o) is larger than the focal length of eyepiece (f_e).

(B) Its magnifying power can be increased by increasing the focal length of objective lens (f_o).

(C) The distance between two lenses is more than ($f_o + f_e$).

(D) The magnifying power can be decreased by increasing the focal length of eyepiece.

(ii) (D) apertures of objective lens and eyepiece

(iii) (C) The distance between two lenses is more than $(f_0 + f_e)$

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(iv) (a) An astronomical telescope has objective lens and eyepiece of focal lengths 80 cm and 4 cm respectively. To view the image in normal adjustment, the lenses must be separated by a distance of : *1*

(A) 84 cm (B) 76 cm

(C) 20 cm (D) 320 cm

OR

(b) Consider the telescope described in question (iv) (a). Its magnifying power in normal adjustment will be : *1*

(A) 320 (B) 84

(C) 76 (D) 20

(iv) (a) An astronomical telescope has objective lens and eyepiece of focal lengths 80 cm and 4 cm respectively. To view the image in normal adjustment, the lenses must be separated by a distance of : *1*

(A) 84 cm (B) 76 cm

(C) 20 cm (D) 320 cm

OR

(b) Consider the telescope described in question (iv) (a). Its magnifying power in normal adjustment will be : *1*

(A) 320 (B) 84

(C) 76 (D) 20

(iv) (A) 84 cm

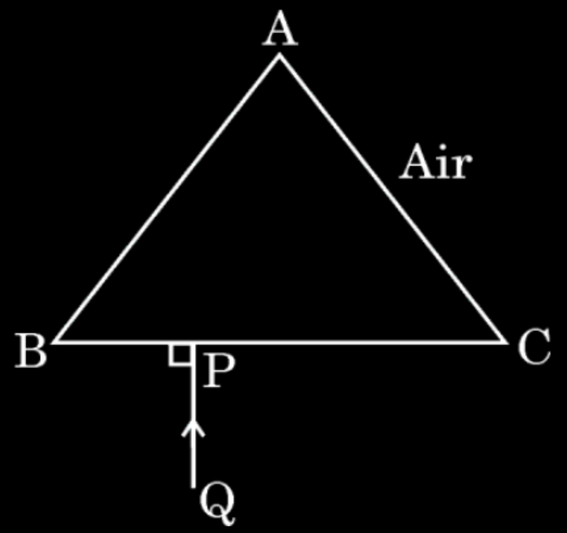
OR

(D) 20

31. (a) (i) Define refractive index of a medium in terms of speed of light.

(ii) Derive the relation for the refractive index (μ) of a prism in terms of angle of minimum derivation (δ_m) and angle of prism (A).

(iii) A ray of light QP is incident normally on the face BC of a triangular prism ABC of refractive index 1·5 kept in air, as shown in the figure. Trace the path of the ray as it passes through the prism and give relevant explanation. *5*



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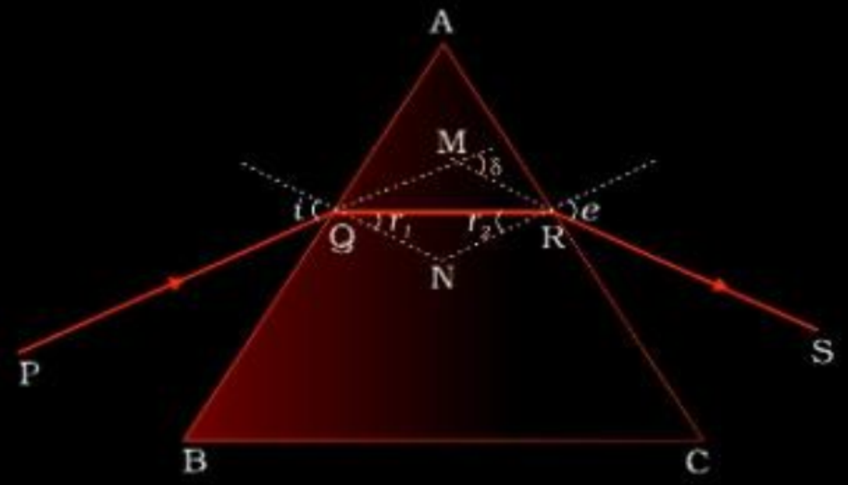
(i) Refractive index of a medium is the ratio of the speed of light in vacuum to the speed of light in the medium. — 1

Alternatively

$$n_{21} = \frac{c}{v}$$

(ii)



In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral is 180°.

$\angle A + \angle QNR = 180°$

From the triangle QNR,

$r_1 + r_2 + \angle QNR = 180°$

Comparing these two equations, we get

$r_1 + r_2 = A$ .....(i) — ½

The total deviation δ is the sum of the deviations at the two faces.

$\delta = i + e - A$ .....(ii) — ½

For $\delta = \delta_m$; i = e which implies $r_1 = r_2$

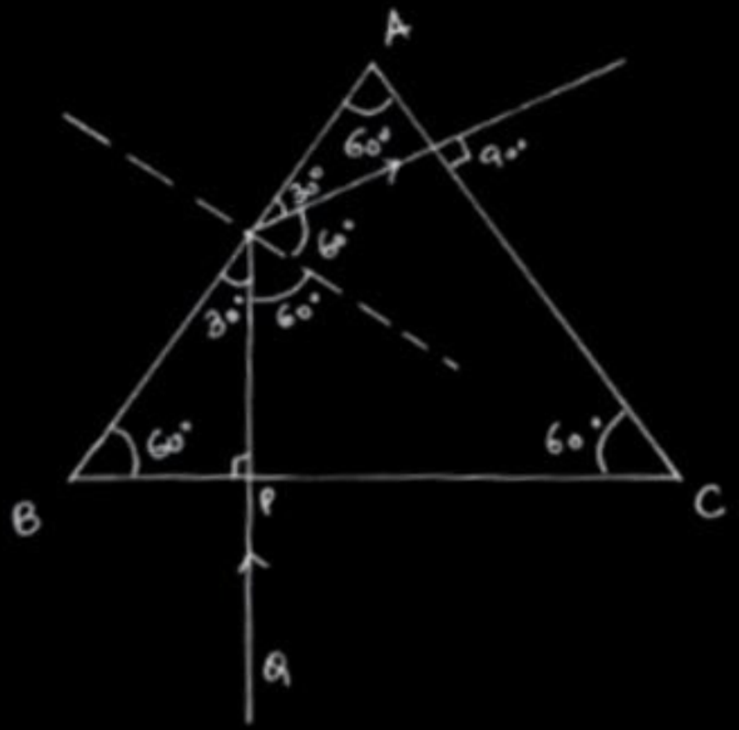
From equation (i) and (ii) — ½

$$r = \frac{A}{2}; i = \frac{A + \delta_m}{2}$$

Refractive index of the prism

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin\frac{(A + \delta_m)}{2}}{\sin\frac{A}{2}}$$ — ½

(iii)

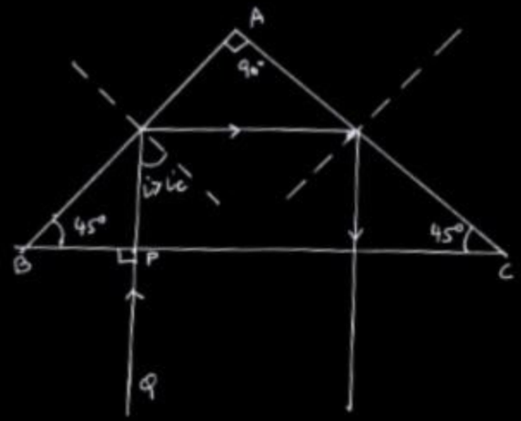


1

The ray QP strikes the face BC normally and hence goes undeviated. As $i > i_c$, it undergoes TIR. It strikes the face AC normally and comes out undeviated.

1

Alternatively



The ray QP strikes the surface BC normally and hence goes undeviated. As $i > i_c$, it undergoes TIR. As at face AC, $i > i_c$, it undergoes TIR. The ray comes out of the face BC undeviated.

(Award full credit of this part if the student takes the value of A other than 60° or 90° and traces the path correctly and according to the angle taken)

OR

19. A concave mirror of radius of curvature 16 cm produces two-times enlarged real image of an object kept in front of it. Find the position of the object. *2*

CBSE 2026

19. A concave mirror of radius of curvature 16 cm produces two-times enlarged real image of an object kept in front of it. Find the position of the object. *2*

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Given: $R = -16$ cm (concave mirror)

$m = -\frac{v}{u} = -2$

$v = 2u$ ½

Now for concave mirror,

$f = -8\text{cm}, v = 2u, u = ?$

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ ½

$\frac{1}{2u} + \frac{1}{u} = \frac{1}{-8}$ ½

$\frac{1+2}{2u} = \frac{1}{-8}$

$\frac{2u}{3} = -8$

$u = -\frac{24}{2} = -12\text{cm}$

$\therefore u = -12\text{cm}$ ½

20. A convex lens of refractive index 1·5 has a focal length of 20 cm in air. Find its nature and focal length when it is immersed in a transparent liquid of refractive index 1·25. *2*

CBSE 2026

20. A convex lens of refractive index 1·5 has a focal length of 20 cm in air. Find its nature and focal length when it is immersed in a transparent liquid of refractive index 1·25. *2*

CBSE 2026

$$\frac{1}{f} = ({}^{a}\mu_{g} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad ½$$

$$\frac{1}{f'} = ({}^{\ell}\mu_{g} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

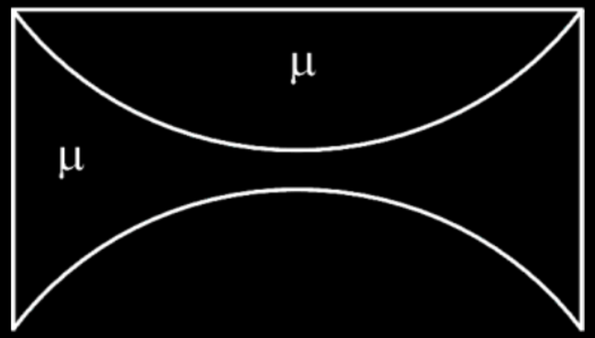
$$\frac{f'}{f} = \frac{({}^{a}\mu_{g} - 1)}{({}^{\ell}\mu_{g} - 1)} = \frac{1.5 - 1}{\left(\frac{1.5}{1.25} - 1\right)} = \frac{0.5}{0.25} \times 1.25 \qquad ½$$

$$\frac{f'}{f} = 2.5$$

or $f' = 2.5 \times 20 = 50\text{cm}$ ½

Nature ⟶ converging. ½

4. A thin plano-convex lens and a thin equi-concave lens are kept coaxially in contact as shown in the figure. Assuming both the lenses are made of glass of refractive index μ, and R is the radius of curvature of each curved surface, the focal length of the combination is :



(A) $\frac{R}{\mu - 1}$

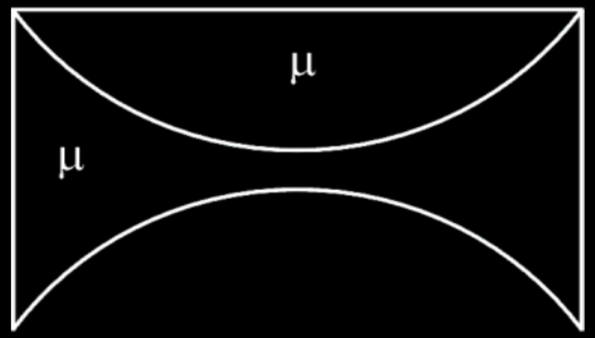
(B) $-\frac{R}{\mu - 1}$

(C) $\frac{2R}{\mu - 1}$

(D) $-\frac{2R}{\mu - 1}$

CBSE 2026

4. A thin plano-convex lens and a thin equi-concave lens are kept coaxially in contact as shown in the figure. Assuming both the lenses are made of glass of refractive index μ, and R is the radius of curvature of each curved surface, the focal length of the combination is :



(A) $\frac{R}{\mu - 1}$

(B) $-\frac{R}{\mu - 1}$

(C) $\frac{2R}{\mu - 1}$

(D) $-\frac{2R}{\mu - 1}$

CBSE 2026

(B) $\frac{-R}{\mu-1}$

6. A telescope has an objective lens of focal length 144 cm and an eyepiece of focal length 6·0 cm. The magnifying power and the length of telescope tube will be respectively :

(A) 24, 150 cm

(B) 42, 138 cm

(C) 24, 138 cm

(D) 42, 150 cm

CBSE 2026

6. A telescope has an objective lens of focal length 144 cm and an eyepiece of focal length 6·0 cm. The magnifying power and the length of telescope tube will be respectively :

(A) 24, 150 cm

(B) 42, 138 cm

(C) 24, 138 cm

(D) 42, 150 cm

CBSE 2026

(A) 24, 150 cm

7. In which of the following, total internal reflection does ***not*** occur ?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

CBSE 2026

7. In which of the following, total internal reflection does ***not*** occur ?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

CBSE 2026

(A) Twinkling of stars

26. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface. *3*

CBSE 2026

26. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface. *3*

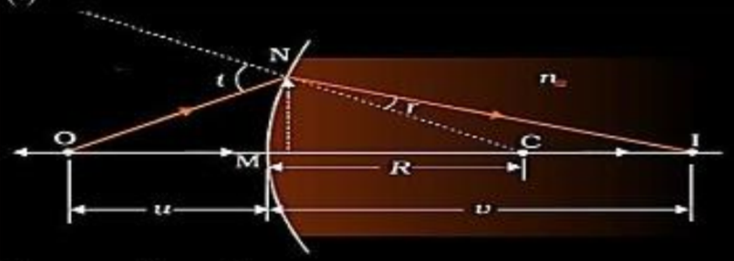
CBSE 2026

$\sin i_C = \frac{1}{n}$ $= \frac{2}{3}$	1
For glass – water surface $\frac{3}{2}\sin i_C = \frac{4}{3}\sin r$	½
$\frac{3}{2} \times \frac{2}{3} = \frac{4}{3}\sin r$ $\sin r = \frac{3}{4}$	
For water – air surface. $\frac{4}{3}\sin r = \sin e$	½
$\frac{4}{3} \times \frac{3}{4} = \sin e$	½
$\sin e = 1$ $e = 90^\circ$	½

33. (a) (i) A point object is kept in front of a convex spherical surface of radius of curvature R. Draw the ray diagram to show the formation of image and derive the relation between the object and image distance (u and v) in terms of refractive index n of the medium and R.

(ii) A convex lens of focal length of 20 cm is used to form the image of an object placed 30 cm away from the lens. Find the position and nature of the image formed. CBSE 2026 *5*

(i)



1

For small angles

$\tan\angle NOM = \frac{MN}{MO}$

$\tan\angle NCM = \frac{MN}{MC}$

$\tan\angle NIM = \frac{MN}{MI}$

$i = \angle NOM + \angle NCM$

$= \frac{MN}{MO} + \frac{MN}{OC}$ ----------(1) ½

Similarly

$r = \angle NCM - \angle NIM$

$= \frac{MN}{MC} - \frac{MN}{MI}$ ----------(2) ½

By Snell's law for small angles

$1 \times i = n \times r$

substituting i and r from (1) and (2)

$\frac{1}{MO} + \frac{n}{MI} = \frac{n-1}{MC}$ ----------(3) ½

Using cartesian sign conventions

MO = -u, MI = +v, MC = +R

Substituting in (3)

$\frac{n}{v} - \frac{1}{u} = \frac{n-1}{R}$ ½

(Note : Give full credit of this part if student derives the relation by forming virtual image.)

(ii) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ ½

$\frac{1}{20} = \frac{1}{v} - \frac{1}{-30}$ ½

v = 60 cm ½

Nature of image: Real and inverted ½

(b) (i) Two thin converging lenses of focal length f_1 and f_2 are placed coaxially in contact. Derive expression for the focal length of the combination.

CBSE 2026

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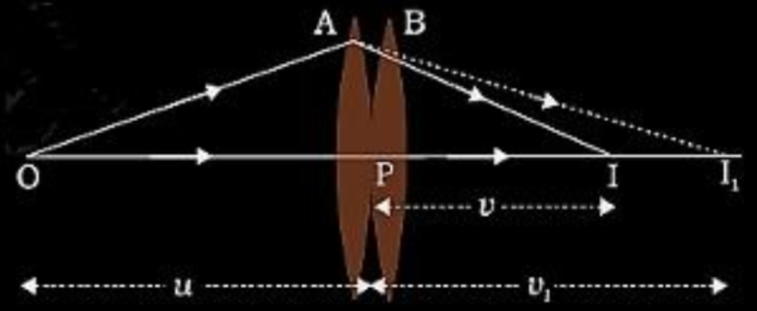
CBSE 2026

(i)



½

For image formed by first lens A

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$$

½

For image formed by second lens B

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$$

½

Adding and comparing with lens formula

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

½

3. An equiconvex lens of focal length 15 cm is cut into two equal plano-convex halves. The focal length of each half will be :

(A) 30 cm (B) 20 cm

(C) 15 cm (D) 7·5 cm

CBSE 2026

3. An equiconvex lens of focal length 15 cm is cut into two equal plano-convex halves. The focal length of each half will be :

(A) 30 cm (B) 20 cm

(C) 15 cm (D) 7·5 cm

CBSE 2026

(A) 30 cm

5. A plano-convex lens, made of glass (n = 1·5) has a curved surface whose radius is 40 cm. The image formed of an object is of the same size as that of the object. The distance of the object from the lens is :

(A) 40 cm (B) 80 cm

(C) 160 cm (D) 200 cm

CBSE 2026

5. A plano-convex lens, made of glass (n = 1·5) has a curved surface whose radius is 40 cm. The image formed of an object is of the same size as that of the object. The distance of the object from the lens is :

(A) 40 cm (B) 80 cm

(C) 160 cm (D) 200 cm

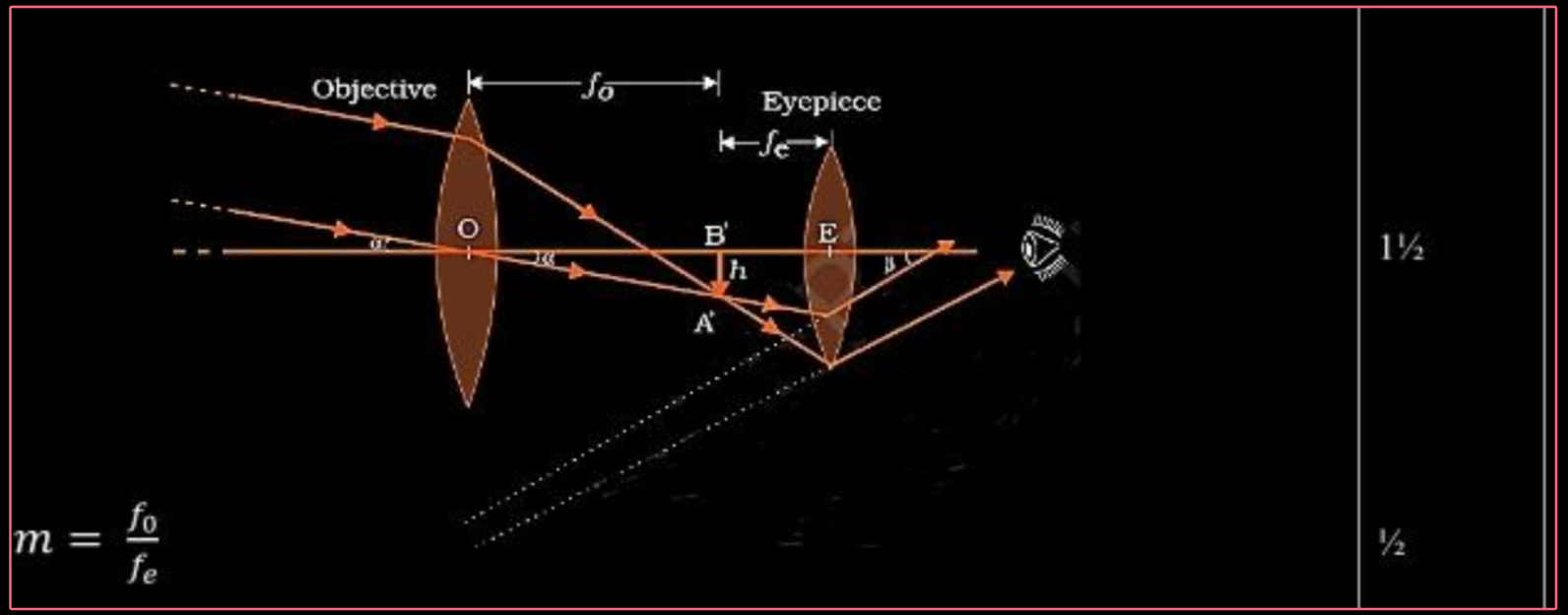
CBSE 2026

(C) 160 cm

20. Draw a labelled ray diagram of an astronomical telescope when final image is formed at infinity. Write the expression for its magnifying power. CBSE 2026 *2*

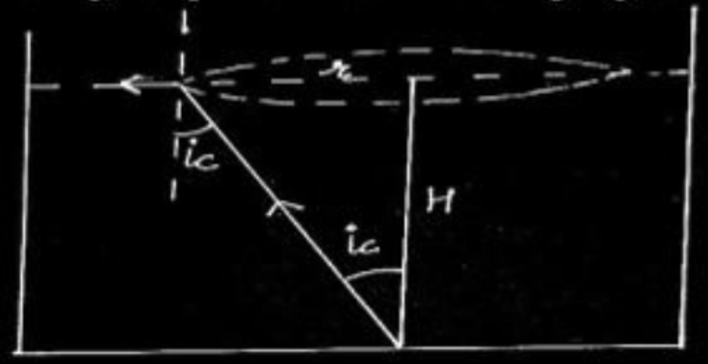
20. Draw a labelled ray diagram of an astronomical telescope when final image is formed at infinity. Write the expression for its magnifying power. 2

CBSE 2026



23. Define the term 'critical angle'. A small bulb is placed at the bottom of a tank containing a liquid of refractive index μ at a depth H. It is observed that the light emerges from a circular area of radius r of the surface. Obtain the expression for r in terms of H and μ. CBSE 2026 *3*

23. Define the term 'critical angle'. A small bulb is placed at the bottom of a tank containing a liquid of refractive index μ at a depth H. It is observed that the light emerges from a circular area of radius r of the surface. Obtain the expression for r in terms of H and μ. CBSE 2026 *3*

Critical angle: The angle of incidence corresponding to the angle of refraction 90° for a given pair of media when light goes from denser to rarer medium.	1
	½
$\text{Sin } i_c = \frac{1}{\mu} = \frac{r}{\sqrt{r^2+H^2}}$	½
$\mu^2 r^2 = r^2 + H^2$	½
$(\mu^2 - 1)r^2 = H^2$	
$r = \frac{H}{\sqrt{\mu^2-1}}$	½

9. A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases ?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) Total internal reflection occurs for both green and blue lights.

(D) None of them will undergo total internal reflection.

CBSE 2026

9. A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases ?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) Total internal reflection occurs for both green and blue lights.

(D) None of them will undergo total internal reflection.

CBSE 2026

(C) Total internal reflection occurs for both green and blue lights.

23. A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the :

(a) angle of minimum deviation for the prism, and

(b) refractive index of the material of the prism. *3*

CBSE 2026

23. A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the :

(a) angle of minimum deviation for the prism, and

(b) refractive index of the material of the prism. *3*

CBSE 2026

(a) As ray of light passes symmetrically through the prism $\angle r_1 = \angle r_2 = r \ , \ i = e$ $\delta_m = 2i - A$ $= 2 \times 45° - 60°$ $= 30°$	½ ½ ½
(b) $r = \frac{A}{2} = 30°$ $\mu = \sin \frac{(\frac{A+\delta_m}{2})}{sin\frac{A}{2}}$ $= \sin \frac{(\frac{60° + 30°}{2})}{sin30°}$ $= \frac{sin45°}{sin30°}$ $= \sqrt{2}$	½ ½ ½
(Alternatively) $\mu_a sini = \mu_g sinr$ $1 \times sin45° = \mu_g \times sin30°$ $\mu_g = \sqrt{2}$	

4. The process that allows light to travel through a tube which is not straight is – **1**

(A) reflection of light
(B) dispersion of light
(C) scattering of light
(D) total internal reflection

CBSE 2026

4. The process that allows light to travel through a tube which is not straight is – **1**

(A) reflection of light
(B) dispersion of light
(C) scattering of light
(D) total internal reflection

CBSE 2026

(D) total internal reflection of light

18. (a) A ray of light travelling along the principal axis is incident on a spherical mirror and reflected back along the same path. Explain, why ? **2**

OR

(b) Write all the sign convention followed for a convex lens. **2**

CBSE 2026

18. (a) A ray of light travelling along the principal axis is incident on a spherical mirror and reflected back along the same path. Explain, why ? **2**

OR

(b) Write all the sign convention followed for a convex lens. **2**

CBSE 2026

When a ray of light travelling along the principal axis is incident on a spherical mirror, then incident light is normal to the surface of spherical mirror, so it reflects back along the same path. Alternatively, The angle of incidence along the principal axis is zero therefore from the laws of reflection, the angle of reflection will be zero. Thus, it reflects back along the same path.	**2**
1. All the distances are measured from the optical center of the lens. 2. Distances measured along the direction of the incident ray are taken as positive. 3. Distance measured opposite to the direction of incident ray are taken as negative. 4. Height above the principal axis is taken as positive and height below the principal axis is taken as negative.	**½ ×4**

19. A small object is placed in air in front of a convex glass surface (n = 1.5 and radius of curvature 20 cm) at a distance of 50 cm from it. Find the position and nature of the image. CBSE 2026 **2**

19. A small object is placed in air in front of a convex glass surface (n = 1.5 and radius of curvature 20 cm) at a distance of 50 cm from it. Find the position and nature of the image. CBSE 2026 **2**

$\frac{-n_1}{u}+\frac{n_2}{v}=\frac{n_2-n_1}{R}$	½
$\frac{-1}{(-50)}+\frac{1.5}{v}=\frac{1.5-1}{20}$	½
$\frac{1}{50}+\frac{1.5}{v}=\frac{0.5}{20}$	
$\frac{1.5}{v}=\frac{1}{40}-\frac{1}{50}$	
v = 300 cm or 3m	½
Nature of the image is real and inverted	½

(a) Briefly explain the working of a reflecting type telescope. Write its two advantages over a refracting telescope.

(b) A telescope consists of two lenses of focal length 125 cm and 5 cm. Find its magnifying power when final image is at (i) infinity (ii) at a distance of 25 cm from eye. CBSE 2026 **5**

(i) Parallel rays from a distant object (star) enter in the telescopes parallel to principal axis of the mirror tend to collect at the focus of the concave mirror. These reflected rays encounter a secondary convex mirror before meeting at focus, the convex mirror reflected them to the eye piece. Final image is seen through the eye piece. Final image formed is inverted w.r.t the object.	**2**
Advantages: (1) Free from chromatic aberration (2) Free from Spherical aberration (3) Require Less mechanical support (4) low Cost (Any Two)	**½ + ½**
(ii) Magnifying power, (I) $m = \frac{-f_0}{f_e} = \frac{-125}{5} = -25$	**½ + ½**
(II) $m = -\frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right)$	**½**
$m = \frac{-125}{5}\left(1+\frac{5}{25}\right) = -30$	**½**
NOTE:- Award full marks if student does not use negative sign in the formula of magnification.	

19. A ray of light in air is incident at angle ∠i on a face of an equilateral glass prism and is refracted through the prism. As ∠i is varied, it is observed that the ray undergoes minimum deviation, when the ∠i is three-fourth of the angle of the prism. Calculate the speed of light in the prism. CBSE 2026 **2**

19. A ray of light in air is incident at angle ∠i on a face of an equilateral glass prism and is refracted through the prism. As ∠i is varied, it is observed that the ray undergoes minimum deviation, when the ∠i is three-fourth of the angle of the prism. Calculate the speed of light in the prism. CBSE 2026 **2**

Angle of incidence $i = \frac{3}{4}A = 45^o$	1/2
At minimum deviation $r = \frac{A}{2} = 30^o$	1/2
$n = \frac{\sin i}{\sin r} = \frac{\sin 45^o}{\sin 30^o}$	
$n = \sqrt{2}$	1/2
$n = \frac{c}{v}$	
$v = \frac{3}{\sqrt{2}} \times 10^8$ m/s = 2.12×10^8m/s	1/2

21. A 5 cm long pencil is placed along the principal axis of a concave mirror of focal length 20 cm such that its nearest end is at a distance of 25 cm from the mirror. Calculate the length of the image of the pencil. CBSE 2026 **2**

21. A 5 cm long pencil is placed along the principal axis of a concave mirror of focal length 20 cm such that its nearest end is at a distance of 25 cm from the mirror. Calculate the length of the image of the pencil. CBSE 2026 **2**

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	$^1/_2$
$\frac{1}{v_1} = \frac{1}{f} - \frac{1}{u_1}$	
$= \frac{1}{(-20)} - \frac{1}{(-25)}$ (For near end of the pencil)	
$v_1 = -100\text{cm}$	$^1/_2$
$\frac{1}{v_2} = \frac{1}{(-20)} - \frac{1}{(-30)}$ (For far end of the pencil)	
$v_2 = -60$ cm	$^1/_2$
Length of the image L = $\lvert v_1 \rvert - \lvert v_2 \rvert = 40\text{cm}$	$^1/_2$

30. In an experiment with convex lens of focal length f, the screen is fixed at a distance D from the object. A student slowly moves the lens away from the object towards the screen and finds that she is able to form sharp image of the object for two positions of the lens. The distance between these two positions of the lens is d.

(i) The value of d is 1

(A) $\sqrt{D(D-4f)}$ (B) $\sqrt{D(D-2f)}$

(C) $2\sqrt{Df}$ (D) $\sqrt{D(D-f)}$

(ii) Compared to the size of the object, the images formed in the two positions of the lens are respectively 1

(A) reduced, enlarged (B) reduced, reduced

(C) enlarged, enlarged (D) enlarged, reduced

CBSE 2026

30. In an experiment with convex lens of focal length f, the screen is fixed at a distance D from the object. A student slowly moves the lens away from the object towards the screen and finds that she is able to form sharp image of the object for two positions of the lens. The distance between these two positions of the lens is d.

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(C) $2\sqrt{Df}$ (D) $\sqrt{D(D-f)}$

(ii) Compared to the size of the object, the images formed in the two positions of the lens are respectively 1

(A) reduced, enlarged (B) reduced, reduced

(C) enlarged, enlarged (D) enlarged, reduced

CBSE 2026

(i) (A) $\sqrt{D(D-4f)}$

(ii) (D) Enlarged, reduced

(iii) If the distance between object and screen is 80.00 cm and the lens forms sharp images at two positions separated by 20.00 cm., the focal length of convex lens is **1**

(A) 15.50 cm (B) 18.75 cm

(C) 20.50 cm (D) 22.75 cm

(iv) Consider a convex lens of focal length 15 cms. For which of the following values of object-screen distance, two positions of the object can be found to obtain sharp image on the screen ? **1**

(A) 45 cm (B) 50 cm

(C) 55 cm (D) 65 cm

OR

(iv) A thin convex lens of focal length 10 cm and another thin lens of focal length 'f' are placed coaxially in contact. If the power of their combination is $\frac{10}{3}$ D, the value of 'f' is **1**

(A) –15 cm (B) –10 cm

(C) –20 cm (D) –30 cm

(iii) If the distance between object and screen is 80.00 cm and the lens forms sharp images at two positions separated by 20.00 cm., the focal length of convex lens is 1

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(A) –15 cm (B) –10 cm

(C) –20 cm (D) –30 cm

(iii) (B) 18.75 cm

(iv) (D) 65cm

OR

(A) -15 cm

(a) Draw a labelled ray diagram of a refracting telescope when it forms image of a distant object at infinity. Derive expression for its magnifying power.

(b) (i) In a telescope the objective has much larger aperture than the eye piece. Why ?

(ii) Write two advantages of reflecting telescope over refracting telescope.

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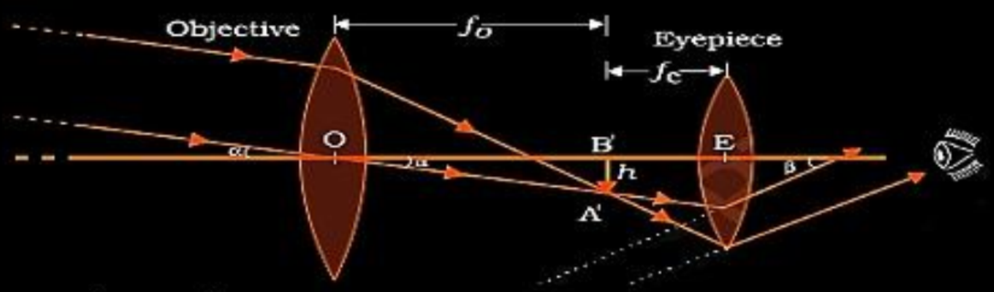
(a)



1

- $m = \frac{\beta}{\alpha} \approx \frac{\tan\beta}{\tan\alpha}$ — 1

$= \frac{h / f_e}{h / f_o}$ — ½

$= \frac{f_o}{f_e}$ — ½

(b) (i) With larger aperture even the fainter objects can be observed clearly. — 1

Alternatively:

With larger aperture, the light gathering power and its resolution / resolving power increases.

Alternatively:

Large aperture helps in correcting more light from distant object which forms a sharper/ brighter images.

(ii)
- There is no chromatic aberration in a mirror.
- Spherical aberration in case of parabolic reflectors is removed.
- Requires less mechanical support
- Cost effective
- Brighter / sharper image is formed

(Any two from above or any other appropriate advantages) — (½+½)

19. A ray of light is incident on the face of a triangular prism of refracting angle 60° and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is $\sqrt{2}$. CBSE 2026 **2**

19. A ray of light is incident on the face of a triangular prism of refracting angle 60° and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is $\sqrt{2}$.

CBSE 2026 **2**

$\sqrt{2}(\sin i_c) = 1$ — ½

$(\sin i_c) = \frac{1}{\sqrt{2}}$

$i_c = 45^0$

$r_1 + r_2 = A$ (Here $r_2 = i_c$)

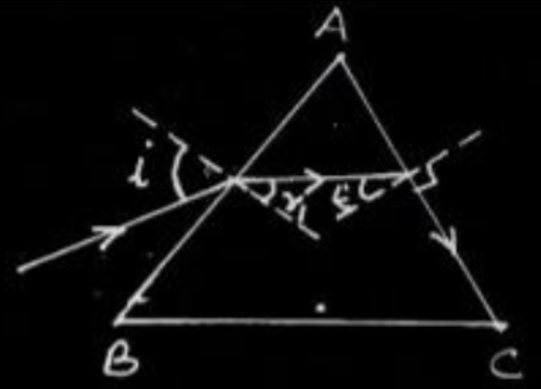
$r_1 = A - i_c$ — ½

$= 60^0 - 45^0$

$= 15^0$ — ½

$\text{Sin } i = \sqrt{2}\sin 15^0$

$i = \sin^{-1}[\sqrt{2}\sin 15^0] = \sin^{-1}(0.366)$ — 1/2



21. A ray of light is incident at an angle 2A on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path.

Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism. **2**

CBSE 2026

21. A ray of light is incident at an angle 2A on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path.

Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism. **2**

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1

From Snell's Law

$n_1 . \sin i = n_2 . sinr$ ½

$A = r_1 + r_2$

$A = r_1 \quad \because r_2 = 0$

$1 \times sin2A = n \times sinA$

$2\ sinA\ cosA = n \times sinA$

$n = 2cosA$ ½

9. A concave lens of focal length 10 cm is cut into two identical plano-concave lenses. The focal length of each lens will be **1**

(A) 20 cm

(B) 30 cm

(C) 40 cm

(D) 5 cm

CBSE 2026

9. A concave lens of focal length 10 cm is cut into two identical plano-concave lenses. The focal length of each lens will be **1**

(A) 20 cm
(B) 30 cm
(C) 40 cm
(D) 5 cm

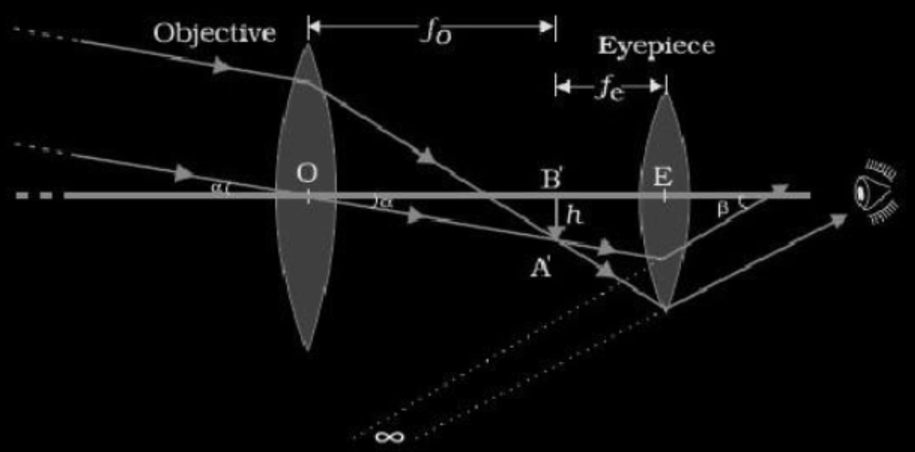
CBSE 2026

(A) 20 cm

27. (a) Draw the ray diagram to show the image formation by a refracting telescope and write the expression for angular magnification for the telescope in normal adjustment.

(b) Give two reasons to explain why a reflecting telescope is preferred over a refracting telescope. CBSE 2026 **3**

(a)



1½

$$|m| = \frac{f_0}{f_e}$$

½

(b)

Reasons:

(i) No Chromatic aberration

(ii) Less mechanical support is required to install it.

½ + ½

(iii) Image formed is brighter and clear.

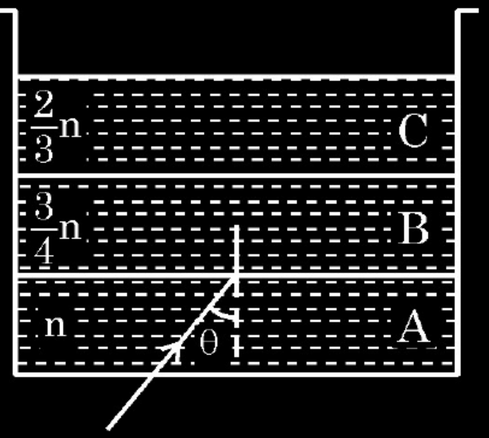
(iv) Cost efficient.

(v) No / less Spherical aberration

[Any two reasons]

28. (a) State the two conditions under which total internal reflection occurs. **3**

(b)



A transparent container contains layers of three immiscible transparent liquids A, B and C of refractive indices n, $\frac{3}{4}$ n and $\frac{2}{3}$ n, respectively. A laser beam is incident at the interface between A and B at an angle θ as shown in figure. Prove that the beam does not enter region C at all for $\sin\theta \geq \frac{2}{3}$.

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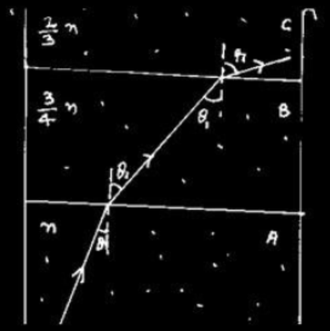
(a) (i) Light must travel from denser medium to rarer medium. ½

(ii) Angle of incidence must be greater than Critical angle. ½

$i > i_c$

(b)



For refraction at AB interface

$$\frac{\sin\theta}{\sin\theta_1} = \frac{3n/4}{n}$$

$$\sin\theta = \frac{3}{4}\sin\theta_1 \quad \text{---------- (1)}$$ ½

For refraction at BC interface

$$\frac{\sin\theta_1}{\sin r} = \frac{2n/3}{3n/4}$$

$$\sin\theta_1 = \frac{8}{9}\sin r \quad \text{---------- (2)}$$ ½

From equations (1) and (2)

$$\frac{3}{2}\sin\theta = \sin r$$ ½

for $\sin\theta = \frac{2}{3}$

$\sin r = 1$ i.e $r = 90^0$ ½

Ray grazes the surface

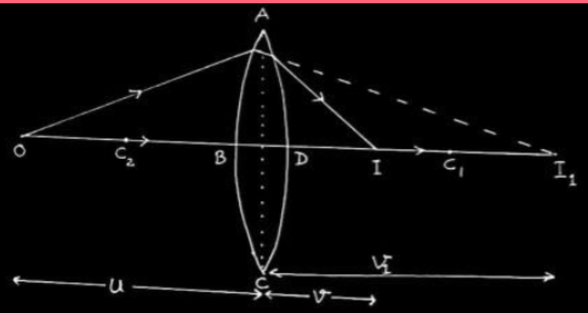
Hence for $\sin\theta \geq \frac{2}{3}$ does not enter the region C at all

32. (a) Using the relation for refraction at a curved spherical surface, derive the expression for lens maker's formula. **5**

(b) Three lenses L_1, L_2 and L_3, each of focal length 40 cm, are placed coaxially. The distance between L_1 and L_2 and between L_2 and L_3 are 120 cm and 20 cm respectively. An object is kept at a distance of 80 cm to the left of lens L_1.

Find the distance of the final image formed from the object. CBSE 2026

(a)



1

The first refracting surface forms the image of the object O at I_1

For refraction from first interface ABC

$$\frac{n_1}{OB}+\frac{n_2}{BI_1}=\frac{n_2-n_1}{BC_1} \qquad \text{---------- (1)}$$ ½

Similarly for refraction from second interface ADC. The image I_1 acts as a virtual object for the second surface ADC

$$-\frac{n_2}{DI_1}+\frac{n_1}{DI}=\frac{n_2-n_1}{DC_2} \qquad \text{---------- (2)}$$ ½

For thin lens $BI_1 = DI_1$

By adding equation (1) and (2)

$$\frac{n_1}{OB}+\frac{n_1}{DI}=(n_2-n_1)\left[\frac{1}{BC_1}+\frac{1}{DC_2}\right]$$

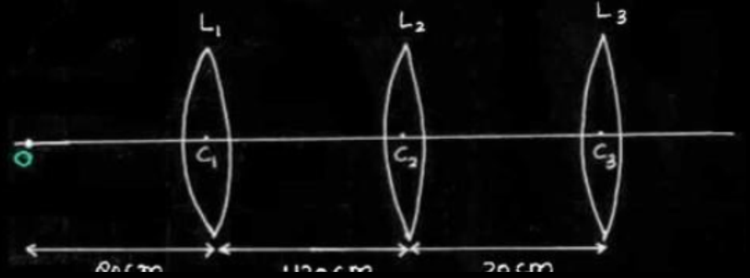
$$-\frac{n_1}{u}+\frac{n_1}{v}=(n_2-n_1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$ ½

$$\frac{1}{v}-\frac{1}{u}=\left(\frac{n_2}{n_1}-1\right)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

If object is kept at ∞, image will form at focus hence

$$\frac{1}{f}=\left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$ ½

(b)



For lens L_1

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$$\frac{1}{40}=\frac{1}{v_1}+\frac{1}{80}$$

$$\frac{1}{40}=\frac{1}{v_1}+\frac{1}{80}$$

$$\frac{1}{v_1}=\frac{1}{80}$$

$v_1 = 80\ \text{cm}$

For lens L_2

$u_2 = 120-80$

$u_2 = 40\ \text{cm}$

$$\frac{1}{40}=\frac{1}{v_2}+\frac{1}{40}$$

$$\frac{1}{v_2}=0$$

$v_2 = \infty$

For lens L_3

$u_3 = \infty$

$$\frac{1}{40}=\frac{1}{v_3}+\frac{1}{\infty}$$

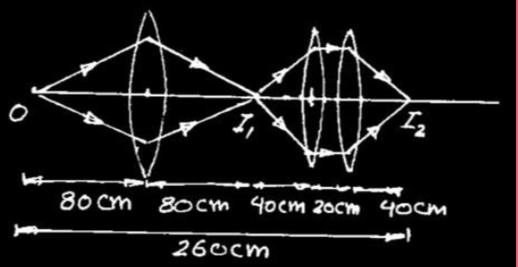
$v_3 = 40\ \text{cm}$

Distance between final image and object

= 80+120+20+40

=260 cm

Alternatively:



(a) Draw a ray diagram to show the image formation by a concave mirror when the object is kept between its focus and the centre of curvature. Using this diagram, derive the mirror formula.

(b) A concave mirror produces a two times magnified virtual image of an object kept 10 cm in front of it. Calculate the focal length of the mirror.

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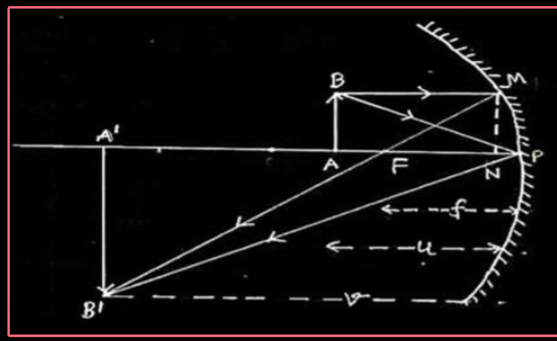


1

ΔBAP and $\Delta B'A'P$ are similar then

$$\frac{BA}{B'A'} = \frac{AP}{A'P} \qquad \text{----------(1)}$$ ½

$\because BA = MN$

ΔMNF and $\Delta B'A'F$ are similar then

$$\frac{MN}{B'A'} = \frac{NF}{A'F}$$

$$\frac{BA}{B'A'} = \frac{NF}{A'F} = \frac{PF}{A'F}$$ (N is very close to P and BA = MN) --------(2) ½

From equation (1) and (2)

$$\frac{AP}{A'P} = \frac{PF}{A'F}$$

$$\frac{-u}{-v} = \frac{-f}{-v+f}$$ ½

$$-uv + uf = -vf$$

$$\frac{-uv}{uvf} + \frac{uf}{uvf} = \frac{-vf}{uvf}$$

$$\frac{-1}{f} + \frac{1}{v} = \frac{1}{-u}$$

$$\boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$$ ½

Note: Award full marks if student derives mirror formula by using any other justified method

(b) $m = \frac{-v}{u}$	½
$m = 2$	
$\because v = -2u \Rightarrow v = -2(-10) = 20\ cm$	½
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	
$\frac{1}{20} - \frac{1}{10} = \frac{1}{f}$	½
$\frac{1-2}{20} = \frac{1}{f}$	
$f = -20\ cm$	½

5. A concave lens of focal length 40 cm is coaxially in contact with two convex lenses, each of focal length 20 cm, on each side. The focal length of the combination is **1**

(A) zero

(B) $\frac{20}{3}$ cm

(C) $-\frac{20}{3}$ cm

(D) $\frac{40}{3}$ cm

CBSE 2026

5. A concave lens of focal length 40 cm is coaxially in contact with two convex lenses, each of focal length 20 cm, on each side. The focal length of the combination is **1**

(A) zero

(B) $\frac{20}{3}$ cm

(C) $-\frac{20}{3}$ cm

(D) $\frac{40}{3}$ cm

CBSE 2026

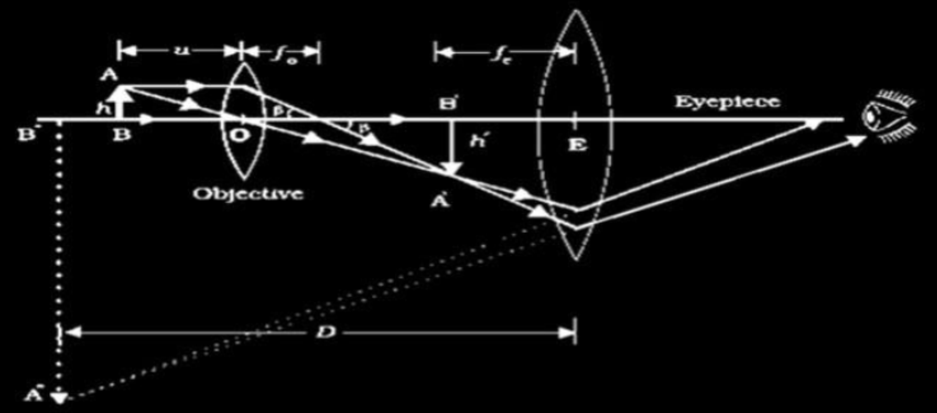
(D) $\frac{40}{3}$ cm

28. Draw a labelled ray diagram showing the formation of image by a compound microscope when final image is formed at least distance of distinct vision. Derive an expression for its magnifying power for this case.

CBSE 2026 **3**

28. Draw a labelled ray diagram showing the formation of image by a compound microscope when final image is formed at least distance of distinct vision. Derive an expression for its magnifying power for this case.

CBSE 2026 **3**



1½

Magnification due to the objective lens $m_0 = \frac{h'}{h} = \frac{L}{f_0}$ ½

using $\tan\beta = \left(\frac{h}{f_0}\right) = \left(\frac{h'}{L}\right)$ where L is the length of the tube

When final image is formed at least distance of distinct vision ,then magnification produced by eye lens $m_e = \left(1 + \frac{D}{f_e}\right)$ ½

Total magnification $m = m_o \times m_e$ ½

$$m = \frac{L}{f_0}\left(1 + \frac{D}{fe}\right)$$

Note: Give full marks for any other alternative method.

7. Light from a small object in air falls on a spherical glass surface (n = 1.5) of radius of curvature R. A real image of the object will be formed if the object distance u is related to R as : **1**

(A) $u < \frac{R}{2}$

(B) $\frac{R}{2} < u < R$

(C) $R < u < 2R$

(D) $u > 2R$

CBSE 2026

7. Light from a small object in air falls on a spherical glass surface (n = 1.5) of radius of curvature R. A real image of the object will be formed if the object distance u is related to R as : **1**

(A) $u < \frac{R}{2}$

(B) $\frac{R}{2} < u < R$

(C) $R < u < 2R$

(D) $u > 2R$

CBSE 2026

(D) u>2R

28. (a) Why does one prefer to view the image formed at infinity than that formed at near point in microscope/telescope ? **3**

(b) Consider lenses L_1, L_2 and L_3 as specified in the following table. Which of them will you select as objective and eyepiece for constructing best possible (i) telescope (ii) compound microscope ? Give reason for your answer.

CBSE 2026

Lens	L_1	L_2	L_3
Power	6 D	3 D	10 D
Aperture	1 cm	8 cm	1 cm

(a) In the case of normal adjustment, the eye muscles remain relaxed while they are strained in the case when image is formed at near point while viewing the object, hence it is preferred.	½
(b) • For Telescope	
Lense L_2 is suitable as objective Lens	½
Lense L_3 is suitable as eyepiece	½
• For Compound microscope	
Lense L_3 is suitable as objective lens	½
Lense L_1 is suitable as eyepiece	½
Reason • For telescope, objective lens should have large aperture and large focal length	
• For microscope, objective and eye lens should have moderate aperture but the objective should have large power.	½

16. *Assertion (A) :* The image of an object placed between f and 2f from a convex lens can be seen on a screen. If the screen is removed, image is not formed.

Reason (R) : Rays from a given point on the object placed between f and 2f, after passing through a convex lens, do not converge on a point in space.

CBSE 2025 C

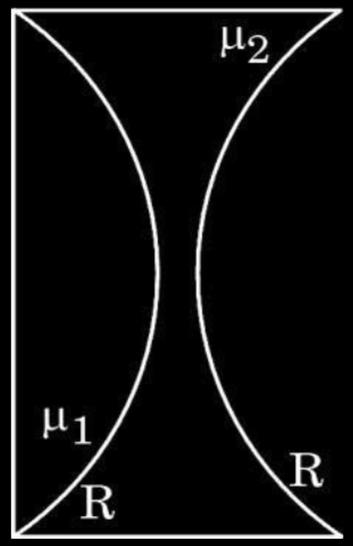
16. *Assertion (A) :* The image of an object placed between f and 2f from a convex lens can be seen on a screen. If the screen is removed, image is not formed.

Reason (R) : Rays from a given point on the object placed between f and 2f, after passing through a convex lens, do not converge on a point in space.

CBSE 2025 C

(D) Both Assertion (A) and Reason (R) are false.

19. (a) A plano-convex lens of refractive index μ_1 is placed coaxially in contact with a biconcave lens of refractive index μ_2 as shown in the figure. All curved faces are of radius of curvature R each. Obtain the expression for the focal length of the combined lens. *2*



CBSE 2025 C

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For plano convex lens

½

$$\frac{1}{f_1} = (\mu_1 - 1)\left(\frac{1}{\infty} - \frac{1}{-R}\right)$$

$$\frac{1}{f_1} = \frac{(\mu_1 - 1)}{R}$$

For concave lens

½

$$\frac{1}{f_2} = (\mu_2 - 1)\left(\frac{1}{-R} - \frac{1}{R}\right)$$

$$\frac{1}{f_2} = -2\frac{(\mu_2 - 1)}{R}$$

½

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

½

$$f = \frac{R}{\mu_1 - 2\mu_2 + 1}$$

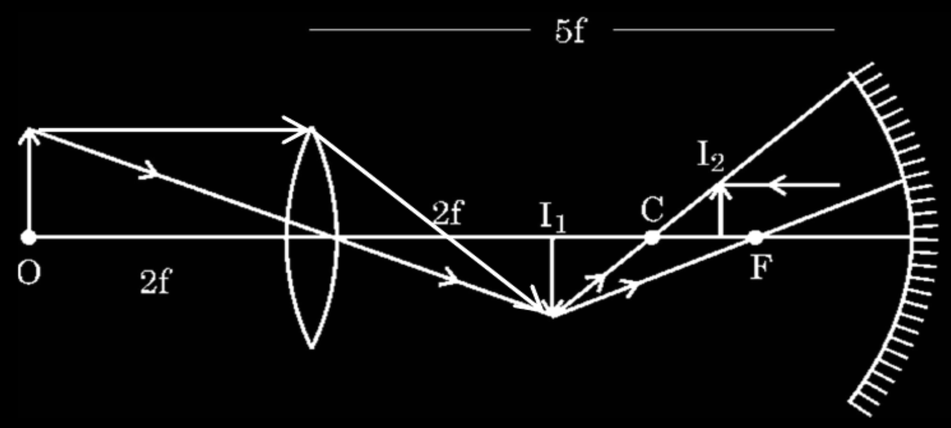
(b) A convex lens is kept coaxially on the left side of a concave mirror at a distance of 5f from it where f is focal length of each of them. An object is kept at a distance of 2f on the left side of the convex lens. Draw the ray diagram showing the formation of the image by the combination. Find the distance of the final image from the mirror. *2*

CBSE 2025 C

CBSE 2025 C

CBSE 2025 C

(b) A convex lens is kept coaxially on the left side of a concave mirror at a distance of 5f from it where f is focal length of each of them. An object is kept at a distance of 2f on the left side of the convex lens. Draw the ray diagram showing the formation of the image by the combination. Find the distance of the final image from the mirror. *2*



1

$u = -3f$

$\frac{1}{\upsilon} + \frac{1}{u} = \frac{1}{f}$ ½

$\frac{-1}{f} = \frac{-1}{3f} + \frac{1}{\upsilon}$

$\upsilon = -\frac{3}{2}f$ ½

30. A compound microscope is an optical instrument used for observing highly magnified images of tiny objects. It consists of two convex lenses. The lens near the object is called the objective and the lens near the eye is called the eyepiece. The magnifying power of a compound microscope is given by $m = m_o m_e$, where m_o is the magnification produced by objective lens and m_e is the magnification produced by eyepiece. The expression for m_e depends on whether the final image is formed at the near point or at infinity (normal adjustment).

CBSE 2025 C

(i) (a) A compound microscope consists of an objective lens of focal length $f_o = 2{\cdot}0$ cm and an eyepiece of focal length $f_e = 6{\cdot}25$ cm separated by a distance of 15 cm. How far should an object be placed from the objective so that the final image is formed at infinity ? *1*

(A) 3·45 cm (B) 5 cm

(C) 1·29 cm (D) 2·59 cm

30. A compound microscope is an optical instrument used for observing highly magnified images of tiny objects. It consists of two convex lenses. The lens near the object is called the objective and the lens near the eye is called the eyepiece. The magnifying power of a compound microscope is given by $m = m_o m_e$, where m_o is the magnification produced by objective lens and m_e is the magnification produced by eyepiece. The expression for m_e depends on whether the final image is formed at the near point or at infinity (normal adjustment).

CBSE 2025 C

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(A) 3·45 cm (B) 5 cm

(C) 1·29 cm (D) 2·59 cm

(D) 2.59 cm

(b) A person with normal near point 25 cm, using a compound microscope with an objective of focal length 0·8 cm and eyepiece of focal length 2·5 cm can bring an object placed at 0·9 cm from the objective in sharp focus. If the image formed by the objective is at a distance of 2·27 cm from the eyepiece, the separation between the lenses is : *1*

(A) 7·47 cm (B) 8·47 cm

(C) 9·47 cm (D) 10·47 cm

(b) A person with normal near point 25 cm, using a compound microscope with an objective of focal length 0·8 cm and eyepiece of focal length 2·5 cm can bring an object placed at 0·9 cm from the objective in sharp focus. If the image formed by the objective is at a distance of 2·27 cm from the eyepiece, the separation between the lenses is : *1*

(A) 7·47 cm (B) 8·47 cm

(C) 9·47 cm (D) 10·47 cm

(C) 9.47 cm

(ii) A compound microscope consists of an objective of focal length 2 cm and an eyepiece of focal length 10 cm. The eyepiece is kept 20 cm away from the objective. Find the magnifying power of the microscope if the final image is formed at infinity. *1*

(A) 7 (B) 10

(C) 14 (D) 16

(ii) A compound microscope consists of an objective of focal length 2 cm and an eyepiece of focal length 10 cm. The eyepiece is kept 20 cm away from the objective. Find the magnifying power of the microscope if the final image is formed at infinity. *1*

(A) 7 (B) 10

(C) 14 (D) 16

(B) 10

(iii) The intermediate image formed in a compound microscope is : *1*

(A) real, inverted and magnified

(B) real, erect and magnified

(C) virtual, erect and magnified

(D) virtual, inverted and magnified

(iii) The intermediate image formed in a compound microscope is : *1*

(A) real, inverted and magnified

(B) real, erect and magnified

(C) virtual, erect and magnified

(D) virtual, inverted and magnified

(A) Real inverted and magnified.

(iv) The focal lengths of the objective and the eyepiece of a compound microscope are f_o and f_e respectively. Which of the following statements is correct, in normal adjustment ? *1*

(A) The object is at a distance less than f_o from the objective.

(B) The object is at a distance more than f_o and less than $2f_o$ from the objective.

(C) The intermediate image is formed at a distance slightly more than f_e from the eyepiece.

(D) The intermediate image is formed at $2f_e$ from the eyepiece.

(iv) The focal lengths of the objective and the eyepiece of a compound microscope are f_o and f_e respectively. Which of the following statements is correct, in normal adjustment ? *1*

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(B) The object is at a distance more than f_o and less than $2f_o$ from the objective.

(C) The intermediate image is formed at a distance slightly more than f_e from the eyepiece.

(D) The intermediate image is formed at $2f_e$ from the eyepiece.

(B) The object is at a distance more than f_0 and less than 2 f_0 from the objective

32. (a) A ray of light is incident on one face of a triangular glass prism of refractive index n and is refracted out from the opposite face. Deduce the expression for refractive index of glass prism in terms of the angle of minimum deviation and angle of the prism. Write the condition on n for refraction to take place on the opposite face. *5*

CBSE 2025 C

$$A + \delta = i + e \qquad \text{½}$$

and $A = r_1 + r_2$ ½

at angle of minimum deviation , $\delta = D_m$

$i = e$, $r_1 = r_2 = r$ ½

$A + D_m = 2i$

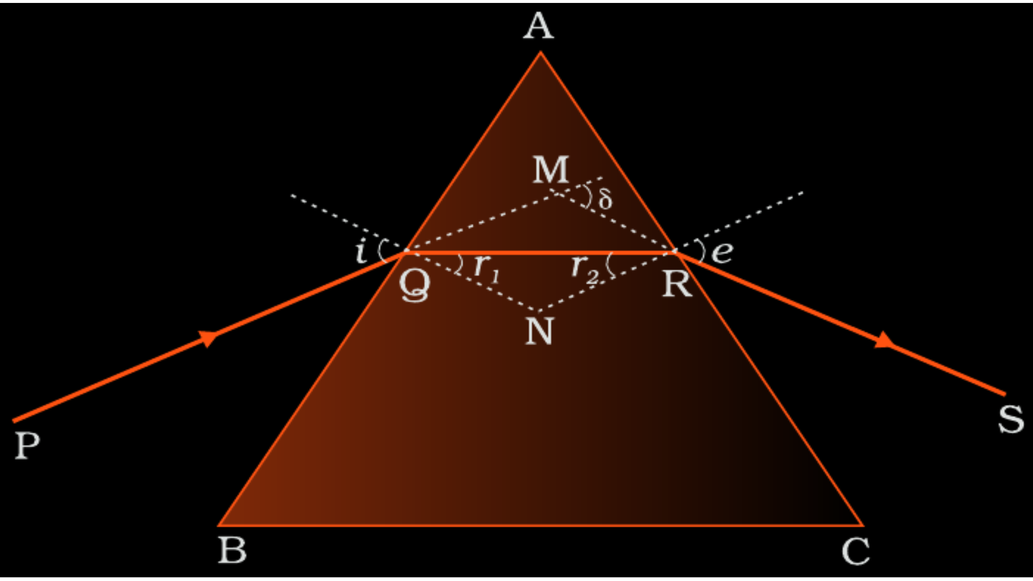
$$i = \frac{A + D_m}{2}$$

$$A = 2r \Rightarrow r = \frac{A}{2} \qquad \text{½}$$

From Snell's law

$$n = \frac{\sin i}{\sin r} \qquad \text{½}$$

$$n = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \qquad \text{½}$$



1

For refraction to take place on opposite face

$$r_2 < i_c$$

$$n = \frac{1}{\sin i_c} \qquad \text{½}$$

$$i_c = \sin^{-1}\left(\frac{1}{n}\right)$$

$$r_2 < \sin^{-1}\left(\frac{1}{n}\right) \qquad \text{½}$$

8. The depth of a swimming pool (refractive index of water = $\frac{4}{3}$) is 1 m. If one looks straight vertically down, the depth of the pool will appear to be :

(A) 0·25 m (B) 0·75 m

(C) 0·95 m (D) 1·0 m

CBSE 2025 C

8. The depth of a swimming pool (refractive index of water = $\frac{4}{3}$) is 1 m. If one looks straight vertically down, the depth of the pool will appear to be :

(A) 0·25 m (B) 0·75 m

(C) 0·95 m (D) 1·0 m

CBSE 2025 C

(B) 0.75 m

33. (a) (i) An equilateral triangular prism is made of material of refractive index $\sqrt{2}$. Find the value of the angle of minimum deviation for the prism. How will this value be affected when the prism is immersed in water ?

(ii) A ray of light is incident normally on one face of this prism. Find the minimum angle at which the ray must be incident on the adjacent face inside the prism so that the ray experiences total internal reflection at this face. *5*

CBSE 2025 C

(i) $n = \sqrt{2}\,,\ A = 60^0$

$$n = \frac{\sin\left(\frac{D_m + A}{2}\right)}{\sin\left(A/2\right)}$$ ½

$$\sqrt{2} = \frac{\sin\left(\frac{D_m + 60^0}{2}\right)}{\sin\left(60^0/2\right)}$$ ½

$$\sin\left(\frac{D_m + 60^0}{2}\right) = \frac{1}{\sqrt{2}}$$

$$\sin\left(\frac{D_m}{2} + 30^0\right) = 45^0$$ ½

$D_m = 30^0$ ½

As $n < \frac{n}{n_w}$, value of D_m decreases in water 1

(ii) $\sin i_c = \frac{1}{n}$ ½

$$i_c = \sin^{-1}\left(\frac{1}{n}\right) = \sin\left(\frac{1}{\sqrt{2}}\right)$$ ½

$i_c = 45^0$ ½

For total internal reflection, angle of incidence > 45º ½

16. *Assertion (A) :* A ray of light is incident normally on the face of a prism. The emergent ray will graze along the opposite face of the prism when the critical angle at glass-air interface is equal to the angle of the prism.

Reason (R) : The refractive index of a prism depends on angle of the prism.

CBSE 2025

16. *Assertion (A) :* A ray of light is incident normally on the face of a prism. The emergent ray will graze along the opposite face of the prism when the critical angle at glass-air interface is equal to the angle of the prism.

Reason (R) : The refractive index of a prism depends on angle of the prism.

CBSE 2025

(C) Assertion(A) is true and Reason(R) is false.

20. (a) Using the mirror equation and the formula of magnification, deduce that "the virtual image produced by a convex mirror is always diminished in size and is located between the pole and the focus." CBSE 2025 *2*

20. (a) Using the mirror equation and the formula of magnification, deduce that "the virtual image produced by a convex mirror is always diminished in size and is located between the pole and the focus." CBSE 2025 *2*

Let, $u = nf$

From the Mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \qquad \text{½}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= \frac{1}{f} + \frac{1}{nf}$$

$$v = \frac{nf}{n+1}$$

$n = +ve \quad \therefore v < f$ ½

$$m = \frac{-v}{u} = \frac{1}{n+1}$$ ½

m is always positive & less than 1. ½

Note: **Award full credit if correctly concluded by any other method.**

(b) A convex lens of focal length 10 cm, a concave lens of focal length 15 cm and a third lens of unknown focal length are placed coaxially in contact. If the focal length of the combination is +12 cm, find the nature and focal length of the third lens, if all lenses are thin. Will the answer change if the lenses were thick ? CBSE 2025 *2*

(b) A convex lens of focal length 10 cm, a concave lens of focal length 15 cm and a third lens of unknown focal length are placed coaxially in contact. If the focal length of the combination is +12 cm, find the nature and focal length of the third lens, if all lenses are thin. Will the answer change if the lenses were thick ? CBSE 2025 *2*

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} \quad \text{½}$$

$$\frac{1}{12} = \frac{1}{10} - \frac{1}{15} + \frac{1}{f_3}$$

$$\frac{1}{f_3} = \frac{5-6+4}{60}$$

$$f_3 = 20\,\text{cm} \quad \text{½}$$

Nature: Convex ½

Yes ½

30. When light travels from an optically denser medium to an optically rarer medium, at the interface it is partly reflected back into the same medium and partly refracted to the second medium. The angle of incidence corresponding to an angle of refraction 90° is called the critical angle (i_c) for the given pair of media. This angle is related to the refractive index of medium 1 with respect to medium 2.

Refraction of light through a prism involves refraction at two plane interfaces. A relation for the refractive index of the material of the prism can be obtained in terms of the refracting angle of the prism and the angle of minimum deviation. For a thin prism, this relation reduces to a simple equation.

Laws of refraction are also valid for refraction of light at a spherical interface. When an object is placed in front of a spherical surface separating two media, its image is formed. A relation between object and image distance, in terms of refractive indices of two media and the radius of curvature of the spherical surface can be obtained. Using this relation for two surfaces of a lens, 'lens maker formula' is obtained. CBSE 2025

(i) An object is placed in front of a convex spherical glass surface (n = 1·5 and radius of curvature R) at a distance of 4R from it. As the object is moved slowly close to the surface, the image formed is : *1*

(A) always real

(B) always virtual

(C) first real and then virtual

(D) first virtual and then real

(i) An object is placed in front of a convex spherical glass surface (n = 1·5 and radius of curvature R) at a distance of 4R from it. As the object is moved slowly close to the surface, the image formed is : *1*

(A) always real

(B) always virtual

(C) first real and then virtual

(D) first virtual and then real

(C) First real and then virtual

(ii) A double-convex lens, made of glass of refractive index 1·5, has focal length 10 cm. The radius of curvature of its each face, is : *1*

(A) 10 cm (B) 15 cm

(C) 20 cm (D) 40 cm

(ii) A double-convex lens, made of glass of refractive index 1·5, has focal length 10 cm. The radius of curvature of its each face, is : *1*

(A) 10 cm (B) 15 cm

(C) 20 cm (D) 40 cm

(A) 10cm

(iii) A small bulb is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H. The radius of the circular area of the surface of liquid, through which light from the bulb can emerge out, is R. Then $\left(\frac{R}{H}\right)$ is : *1*

(A) $\frac{1}{\sqrt{n^2-1}}$

(B) $\sqrt{n^2-1}$

(C) $\frac{1}{\sqrt{n^2+1}}$

(D) $\sqrt{n^2+1}$

(iii) A small bulb is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H. The radius of the circular area of the surface of liquid, through which light from the bulb can emerge out, is R. Then $\left(\frac{R}{H}\right)$ is : *1*

(A) $\frac{1}{\sqrt{n^2-1}}$

(B) $\sqrt{n^2-1}$

(C) $\frac{1}{\sqrt{n^2+1}}$

(D) $\sqrt{n^2+1}$

(A) $1/\sqrt{n^2-1}$

(iv) (a) A parallel beam of light is incident on a face of a prism with refracting angle 60°. The angle of minimum deviation is found to be 30°. The refractive index of the material of the prism is close to : *1*

(A) 1·3 (B) 1·4

(C) 1·5 (D) 1·6

(B) 1.4

(b) The angle of minimum deviation for a ray of light incident on a thin prism, made of crown glass (n = 1·52) is D_m. If the prism was made of dense flint glass (n = 1·62) instead of crown glass, the angle of minimum deviation will : *1*

(A) decrease by 4%

(B) increase by 4%

(C) decrease by 19%

(D) increase by 19%

(b) The angle of minimum deviation for a ray of light incident on a thin prism, made of crown glass (n = 1·52) is D_m. If the prism was made of dense flint glass (n = 1·62) instead of crown glass, the angle of minimum deviation will : *1*

(A) decrease by 4%

(B) increase by 4%

(C) decrease by 19%

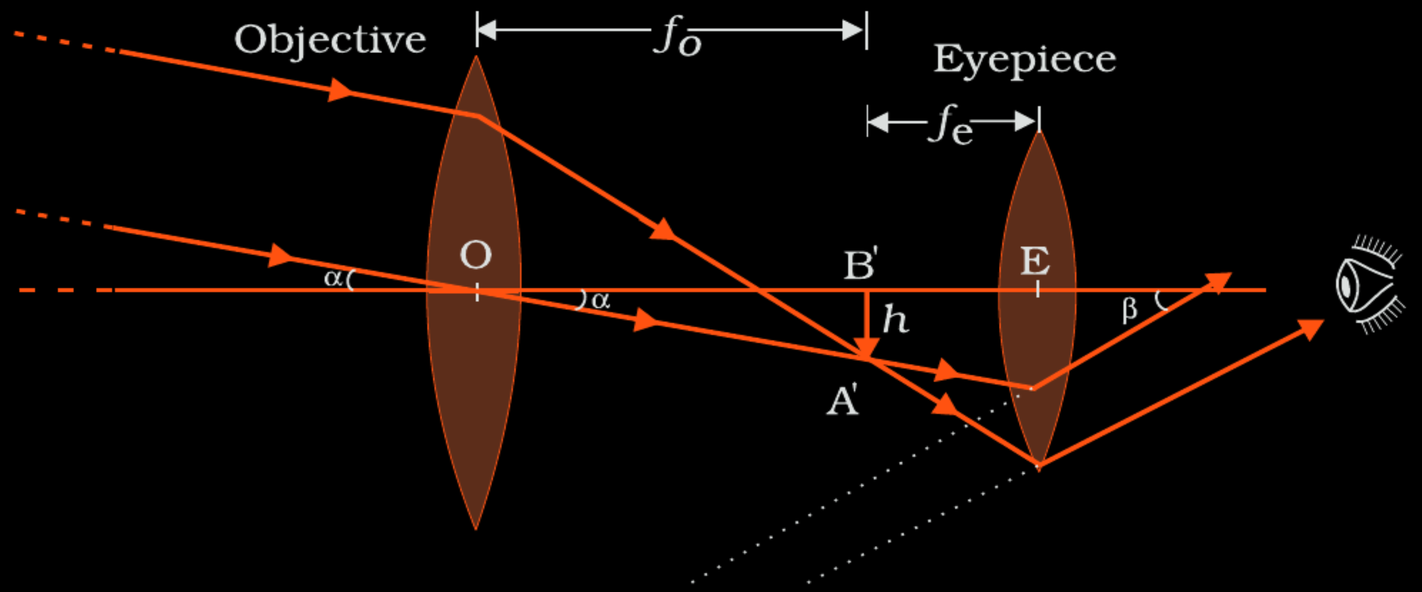
(D) increase by 19%

(D) increase by 19%

32. (a) (i) Explain with the help of a labelled ray diagram the formation of final image by an astronomical telescope at infinity. Write the expression for its magnifying power.

(ii) The total magnification produced by a compound microscope is 20. The magnification produced by the eyepiece is 5. When the microscope is focussed on a certain object, the distance between the objective and eyepiece is observed to be 14 cm. Calculate the focal lengths of the objective and the eyepiece. (Given that the least distance of distinct vision = 25 cm) *5*

CBSE 2025



(Note: Deduct ½ mark, for not showing arrows with the rays) 1½

Light from distant object enters the objective lens & forms a real image A'B' at f_o .

This image A'B' acts as an object for eye piece and eye piece forms a magnified image at infinity. ½

Magnifying Power = f_o / f_e 1

(ii) Image is formed at least distance of distinct vision

$$20 = m_o \times m_e$$

$$m_o = \frac{20}{5} = 4$$

½

$$m_e = 1 + \frac{D}{f_e}$$

$$f_e = \frac{25}{4} cm$$

½

$$\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$$

$$\frac{1}{-25} - \frac{1}{u_e} = \frac{4}{25}$$

$$u_e = -5cm$$

$$L = v_0 + |u_e|$$

$$v_0 = 9cm$$

½

Given, $\frac{v_0}{u_0} = 4$

$$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\frac{1}{f_0} = \frac{1}{9} - \left(-\frac{4}{9}\right)$$

$$f_0 = \frac{9}{5} cm$$

½

15. *Assertion (A) :* In a reflecting telescope, the image does not have chromatic aberration.

Reason (R) : Chromatic aberration occurs only due to refraction of light through an optical medium.

CBSE 2025

15. *Assertion (A) :* In a reflecting telescope, the image does not have chromatic aberration.

Reason (R) : Chromatic aberration occurs only due to refraction of light through an optical medium.

CBSE 2025

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A)

(b) A point source of light in air is kept at a distance of 12 cm in front of a convex spherical surface of glass of refractive index 1·5 and radius of curvature 30 cm. Find the nature and position of the image formed.

CBSE 2025 *2*

(b) A point source of light in air is kept at a distance of 12 cm in front of a convex spherical surface of glass of refractive index 1·5 and radius of curvature 30 cm. Find the nature and position of the image formed. CBSE 2025 *2*

$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$	½
$\frac{1.5}{v} - \frac{1}{(-12)} = \frac{1.5-1}{30}$	½
v = -22.5 cm	½
Image is virtual and erect.	½

21. A tank is filled with a liquid of refractive index $\sqrt{2}$, up to a height of 30 cm. A tiny bulb is glowing at the bottom of the tank. Calculate the diameter of an opaque disc floating symmetrically on the liquid surface that can cut off completely the light from the bulb that comes out of the liquid surface.

CBSE 2025 *2*

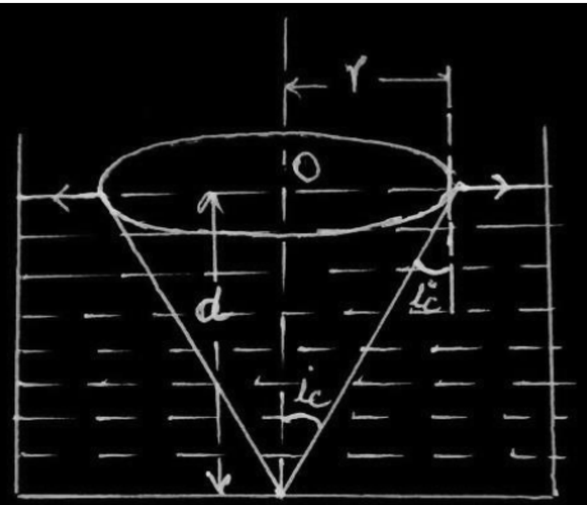
21. A tank is filled with a liquid of refractive index $\sqrt{2}$, up to a height of 30 cm. A tiny bulb is glowing at the bottom of the tank. Calculate the diameter of an opaque disc floating symmetrically on the liquid surface that can cut off completely the light from the bulb that comes out of the liquid surface.

CBSE 2025 | 2

$\sin i_c = \frac{1}{\sqrt{2}}$ ½

$i_c = 45^0$

$\tan i_c = \frac{r}{d}$



$r = d \tan 45^0$ ½

$r = 30 \times 1$ cm

$D = 2r = 60$ cm 1

Alternatively:

$D = 2r = \frac{2d}{\sqrt{n^2 - 1}}$ 1

$= \frac{2 \times 30}{\sqrt{(\sqrt{2})^2 - 1}}$ ½

$D = 60$ cm ½

25. (a) A concave mirror has radius of curvature 20 cm. Calculate the distance of an object from the mirror so as to form an image of magnification – 2. Also find the location of the image.

(b) If the silver coating around the centre of a concave mirror is removed, will the mirror still form the image of an object ? Justify your answer.

CBSE 2025

3

25. (a) A concave mirror has radius of curvature 20 cm. Calculate the distance of an object from the mirror so as to form an image of magnification – 2. Also find the location of the image.

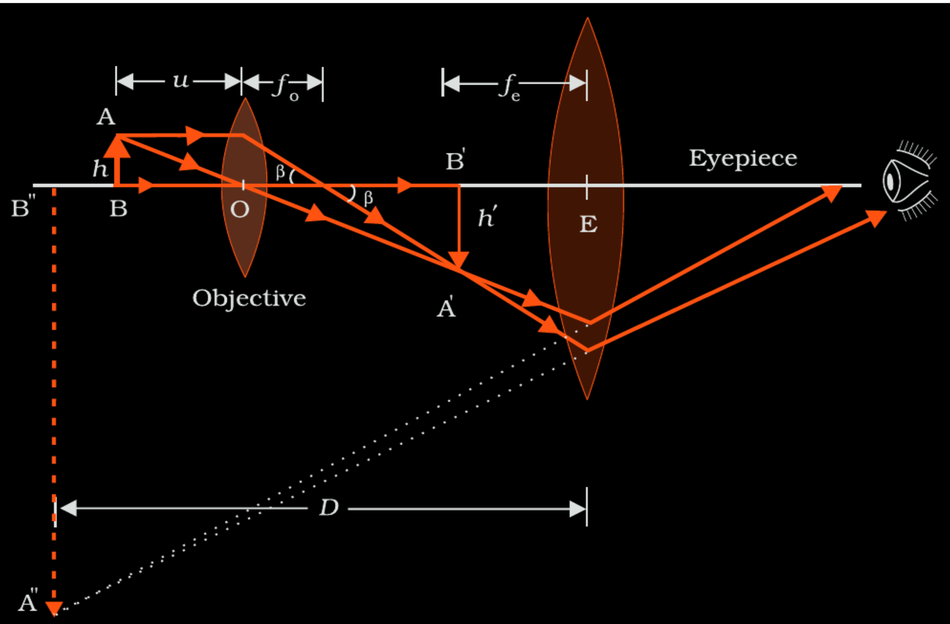
(b) If the silver coating around the centre of a concave mirror is removed, will the mirror still form the image of an object ? Justify your answer.

CBSE 2025 *3*

(a) $m = -\frac{v}{u}$	½
$-2 = -\frac{v}{u}$	
$v = 2u$	
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	
$\frac{1}{-10} = \frac{1}{2u} + \frac{1}{u}$	½
$u = -15\,\text{cm}$	½
$v = -30\,\text{cm}$	½
(b) Yes, same image is formed with reduced intensity, because reflecting area get reduced and laws of reflection still hold good for remaining part of the mirror.	1

33. (a) (i) Draw a ray diagram to show the image formation by a compound microscope. Obtain the expression for the total magnification of the microscope when the final image is formed at infinity.

(ii) In a compound microscope, an object is placed at a distance of 1·5 cm from the objective of focal length 1·25 cm. The eyepiece has a focal length of 5 cm. The final image is formed at infinity. Calculate the distance between the objective and the eyepiece. CBSE 2025 *5*



Note: Deduct ½ mark for not showing arrows with the rays. 1½

Magnification produced by objective

$$m_0 = \frac{h'}{h} = \frac{L}{f_o}$$ ½

Magnification produced by eye-piece

$$m_e = 1 + \frac{D}{f_e}$$

If the final image is formed at infinity

$$m_e = \frac{D}{f_e}$$ ½

Total magnification

$$m = m_0 \times m_e$$

$$= \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$$ ½

(i) $$\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_0}$$ ½

$$\frac{1}{v_o} - \frac{1}{(-1.5)} = \frac{1}{1.25}$$

$v_o = 7.5\,\text{cm}$ ½

$L = |v_o| + |f_e|$ as final image is formed at infinity ($v_e = \infty, u_e = f_e$) ½

L = 7.5 + 5

L= 12.5 cm ½

(ii) Use mirror formula to deduce that a convex mirror always produces a virtual image of an object kept in front of it.

(ii) Use mirror formula to deduce that a convex mirror always produces a virtual image of an object kept in front of it.

(ii)	$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$	½
	$u<0 \ , \ f>0$	
	$\frac{1}{v}+\frac{1}{(-u)}=\frac{1}{f}$	½
	$\frac{1}{v}=\frac{1}{f}+\frac{1}{u}$	½
	$\frac{1}{v}$ is positive	
	$\therefore$ v is positive $\Rightarrow$ virtual image	½

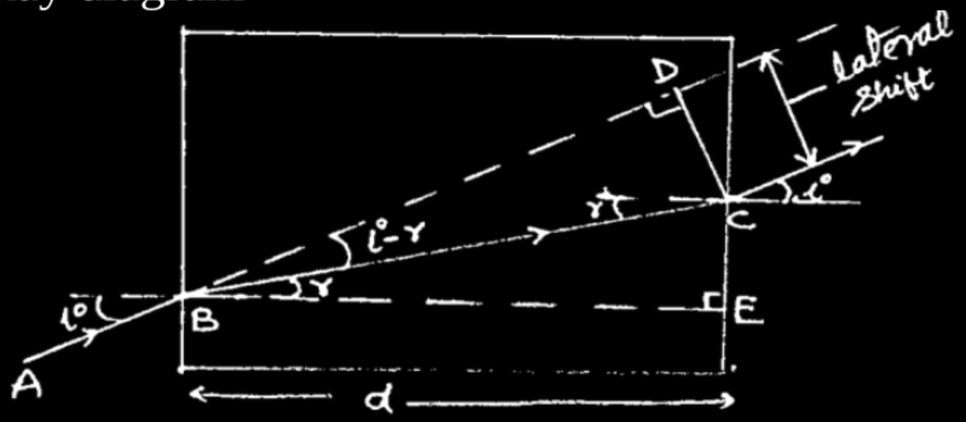
20. A ray of light is incident on face AB of a prism ABC with angle of prism A and emerges out from face AC. The prism is set in the position of minimum deviation with angle of deviation δ. Find (i) the angle of incidence and (ii) the angle of refraction on face AB. CBSE 2025 *2*

20. A ray of light is incident on face AB of a prism ABC with angle of prism A and emerges out from face AC. The prism is set in the position of minimum deviation with angle of deviation δ. Find (i) the angle of incidence and (ii) the angle of refraction on face AB. CBSE 2025 *2*

In a prism we know $A+\delta=i+e \quad and\ A=r_1+r_2$	½
At minimum deviation $i=e \ \ and \ \ r_1=r_2=r$	½
$A+D_m=2i$ $i=\frac{A+D_m}{2}$	½
$A=2r$ $r=A/2$	½

24. A ray of light is incident at an angle i on a parallel sided glass slab of thickness 'd' and gets refracted into the slab at angle r. Draw a ray diagram to show its path as its emerges out of the slab. Hence, obtain an expression for the lateral shift of the ray. Under what condition will the shift be minimum ?

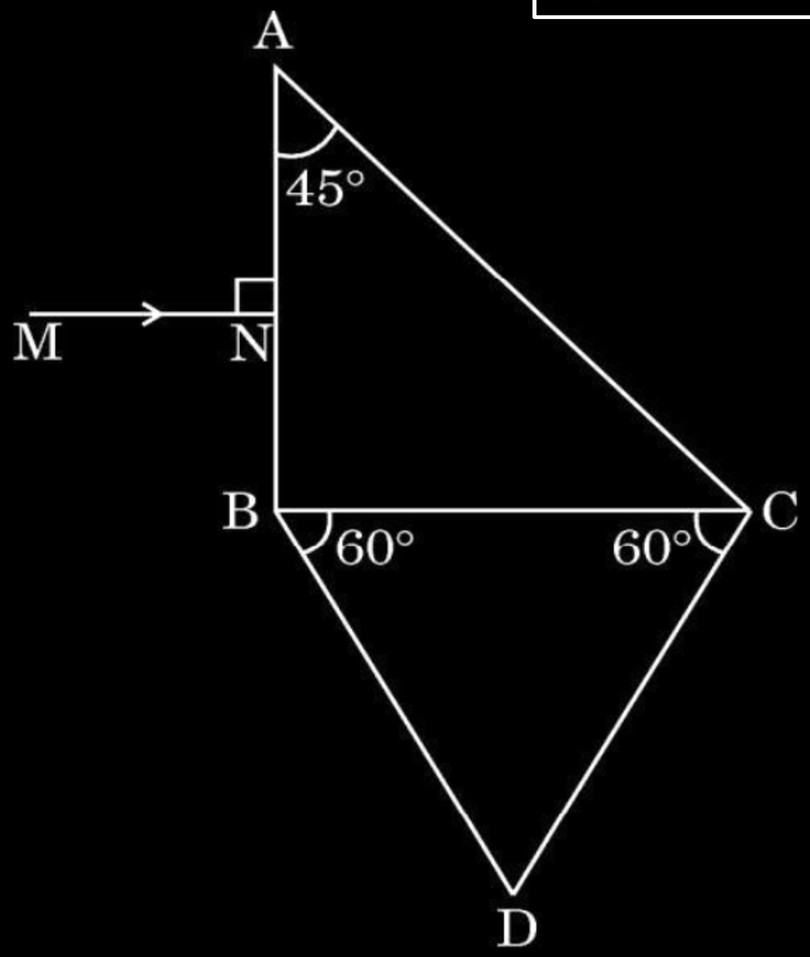
CBSE 2025 *3*

Ray diagram  	1
$\sin(i-r)=\frac{CD}{BC}$ ---------(1)	½
$\cos r=\frac{BE}{BC}=\frac{d}{BC}$	½
Putting in equation (1)	
$CD=\frac{d\sin(i-r)}{\cos r}$	½
Shift will be minimum for minimum angle of incidence.	½

20. A right angled isosceles glass prism ABC is kept in contact with an equilateral triangular prism DBC as shown in the figure. Both prisms are made of the same glass of refractive index 1·6. Trace the path of the ray MN incident normally on face AB as it passes through the combination.

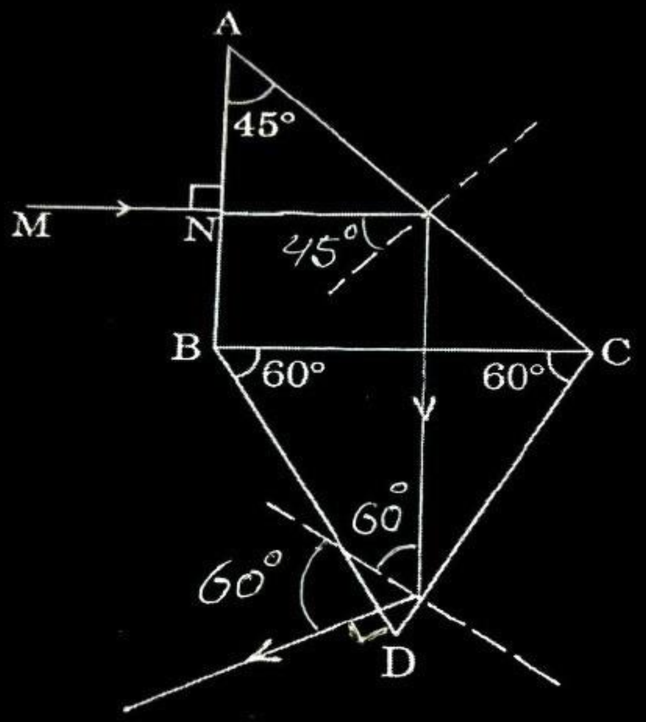
CBSE 2025

2



20. A right angled isosceles glass prism ABC is kept in contact with an equilateral triangular prism DBC as shown in the figure. Both prisms are made of the same glass of refractive index 1·6. Trace the path of the ray MN incident normally on face AB as it passes through the combination.

CBSE 2025 *2*



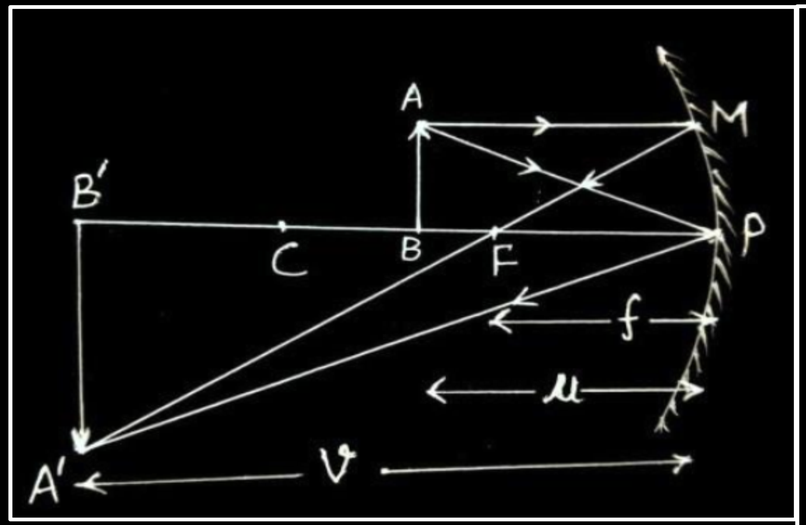
1

1

Note: Please deduct ½ mark for not showing arrows with the rays.

26. Draw a ray diagram showing the image formation when a concave mirror produces a real, inverted and magnified image of an object and hence obtain the mirror formula. CBSE 2025 *3*

26. Draw a ray diagram showing the image formation when a concave mirror produces a real, inverted and magnified image of an object and hence obtain the mirror formula. CBSE 2025 *3*



Note: Please deduct ½ mark of this diagram if not showing arrows with the rays.

In similar triangles

$\Delta A'B'F$ *and* ΔMPF

$$\frac{A'B'}{MP}=\frac{B'F}{FP}$$

or $$\frac{A'B'}{AB}=\frac{B'F}{FP} \quad (\because MP=AB) \quad \text{---------(1)}$$ ½

In similar triangles $\Delta A'B'P$ *and* ΔABP

$$\frac{A'B'}{AB}=\frac{PB'}{PB} \quad \text{-------(2)}$$ ½

from equation (1) and (2)

$$\frac{B'F}{FP}=\frac{PB'}{PB}$$

$$\frac{PB'-PF}{FP}=\frac{PB'}{PB}$$

$$\frac{(-v)-(-f)}{(-f)}=\frac{(-v)}{(-u)}$$ ½

on solving we get

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$ ½

8. Two thin lenses of focal lengths +40 cm and –20 cm and placed coaxially in contact. An object is placed at infinity, in front of the combination. The image formed by the combination will lie – **1**

(A) at a distance of 60 cm, in front of the combination.
(B) at a distance of 40 cm, in front of the combination.
(C) at a distance of 20 cm, behind the combination.
(D) at infinity

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8. Two thin lenses of focal lengths +40 cm and –20 cm and placed coaxially in contact. An object is placed at infinity, in front of the combination. The image formed by the combination will lie – **1**

(A) at a distance of 60 cm, in front of the combination.
(B) at a distance of 40 cm, in front of the combination.
(C) at a distance of 20 cm, behind the combination.
(D) at infinity

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(B) at a distance of 40 cm in front of the combination

18. (a) The real image of an object is formed at a distance twice the distance of the object from a concave mirror of focal length 10 cm. Find the distance between the object and the image. **2**

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18. (a) The real image of an object is formed at a distance twice the distance of the object from a concave mirror of focal length 10 cm. Find the distance between the object and the image. **2**

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$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \qquad ½$$

$u=-x,\ \ v=-2x$ ½

$$\frac{1}{-10}=\frac{1}{-x}+\frac{1}{-2x}=-\frac{3}{2x} \qquad ½$$

$x=15\,\text{cm}$

Distance between object and image,
d = 2x- x = 30- 15 = 15cm ½

(b) An object is placed 1.50 m in front of a lens of power 4.0 D. Find the nature and position of the image formed. **2**

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(b) An object is placed 1.50 m in front of a lens of power 4.0 D. Find the nature and position of the image formed. **2**

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$$f = \frac{1}{P} = \frac{1}{4} = 0.25 \text{ m} = 25 \text{cm}$$ ½

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$ ½

$$\frac{1}{25} = \frac{1}{v} - \left(\frac{1}{-150}\right)$$

On solving, v =30 cm ½

The image formed is real and inverted. ½

24. (a) State the conditions under which total internal reflection occurs. **3**

(b) A glass slab 3.0 cm thick, is placed over a dark ink dot. Find the height through which the image of the dot is raised. The refractive index of the glass is 1.5. CBSE 2025

24. (a) State the conditions under which total internal reflection occurs. **3**

(b) A glass slab 3.0 cm thick, is placed over a dark ink dot. Find the height through which the image of the dot is raised. The refractive index of the glass is 1.5. CBSE 2025

a) (i) The ray of light should pass from a denser to a rarer medium	½
b) (ii)The angle of incidence is greater than the critical angle for the pair of media.	½
c) $n = \frac{\text{Real depth}}{\text{Apparant depth}}$	½
$\text{Apparant depth} = \frac{\text{Real depth}}{n}$	
$= \frac{3\text{ cm}}{1.5} = 2\text{cm}$	½
The height through which the image of the dot raised (i.e. Normal shift)	½
= Real depth - Apparent depth	
= 3 - 2 = 1 cm	½

(b) (A) A ray of light is incident at an angle of incidence, $\frac{A}{2}$ on face AB of a prism ABC of angle of prism A. The ray emerges normally from the opposite face AC. If the refractive index of the material of the prism be n, find the relation between A and n.

(B) The focal length of a convex lens of refractive index 1.5 is 20 cm. Find the focal length of this lens when immersed in a liquid of refractive index 1.25. **5**

CBSE 2025

(A) As refracted ray emerges normally from opposite face AC of the prism, So $r_2 = 0$ ½

$$r_1 + r_2 = A$$ ½

$$r_1 = A$$

$$n = \frac{\sin i_1}{\sin r_1} = \frac{\sin A/2}{\sin A}$$ ½

$$n = \frac{\sin A/2}{2 \sin A/2 \cos A/2}$$ ½

$$n = \frac{1}{2\cos A/2}$$ ½

(B) From lens maker formula

$$\frac{1}{f} = ({}^{1}n_2 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ ½

$$\frac{1}{f_{air}} = (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f_{air}} = 0.5\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ ½

$$\frac{1}{f_{liguid}} = \left(\frac{1.5}{1.25} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f_{liguid}} = 0.2\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ ½

$$\frac{f_l}{f_{air}} = \frac{5}{2}$$ ½

$$f_l = \frac{5}{2} \times f_{air} = \frac{5}{2} \times 20 = 50\,\text{cm}$$ ½

20. (a) A point object is placed in air at a distance R/3 in front of a convex surface of radius of curvature R, separating air from a medium of refractive index n (< 4). Find the nature and position of the image formed.

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2

20. (a) A point object is placed in air at a distance R/3 in front of a convex surface of radius of curvature R, separating air from a medium of refractive index n (< 4). Find the nature and position of the image formed. CBSE 2025 *2*

For refraction at convex surface

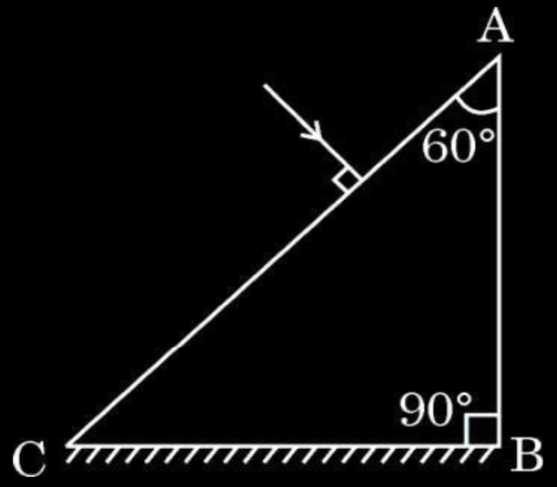
$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$$ ½

$$\frac{n}{v} = \frac{[n-1-3]}{R}$$

$$v = \frac{nR}{n-4}$$ ½

For all values of n < 4, the value of v is negative and greater than R
Therefore the nature of image is virtual and is formed in front of convex surface. **1**

25. A right-angled prism ABC (refractive index $\sqrt{2}$) is kept on a plane mirror as shown in the figure. A ray of light is incident normally on the face AC.

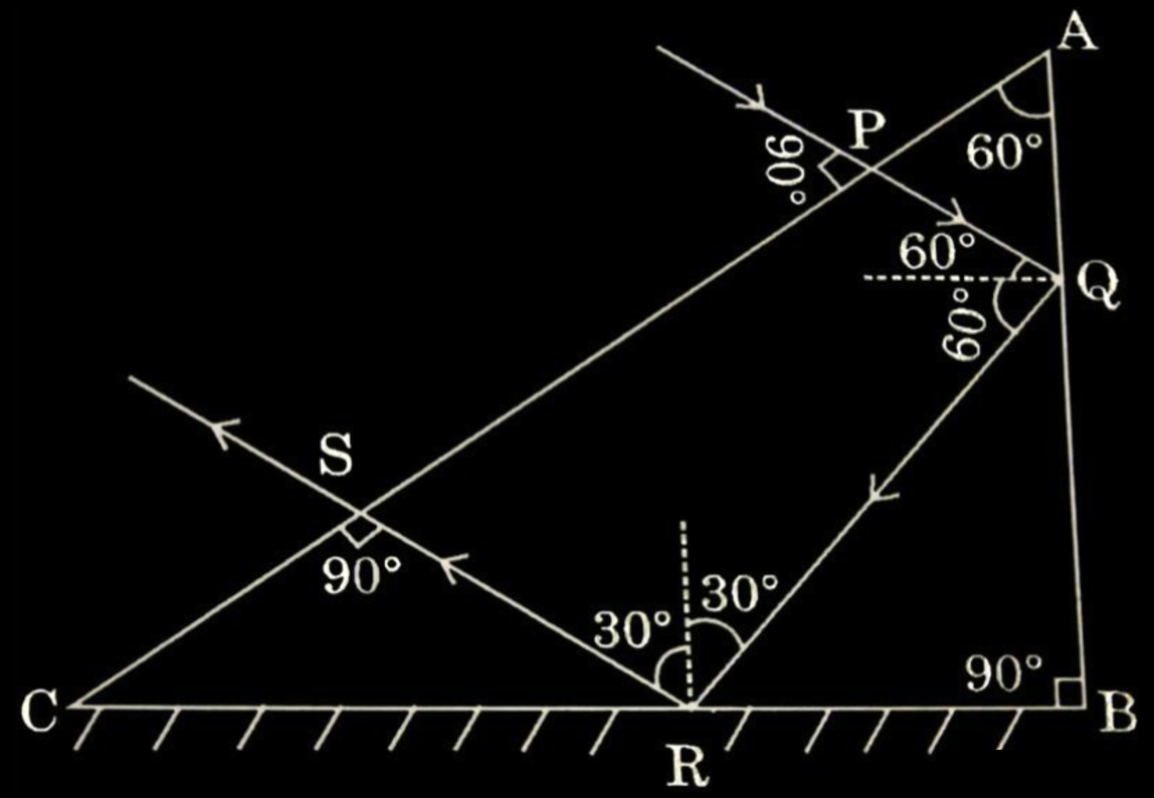


(a) Trace the path of the ray as it passes through the prism.

(b) Find the angle of deviation produced by the prism. CBSE 2025 *3*

(a) $\sin i_c = \frac{1}{\sqrt{2}}$ ½

$i_c = 45^0$



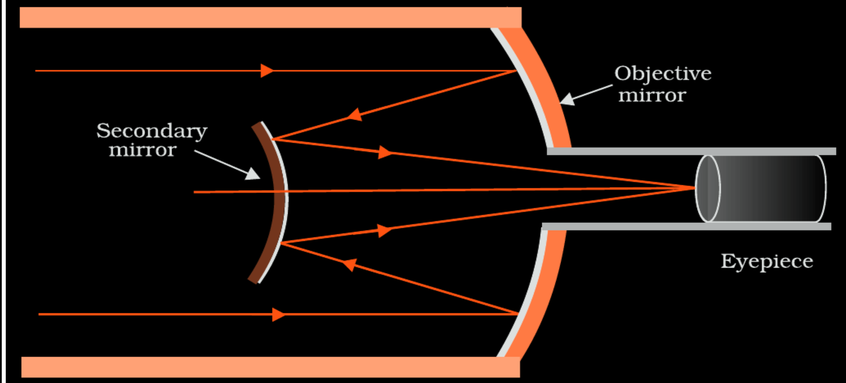
2

(Deduct ½ mark for not showing arrow for the ray)

(b) The angle of deviation is 180° from diagram ½

31. (a) (i) Draw a ray diagram of a reflecting telescope (Cassegrain) and explain the formation of image. State two important advantages that a reflecting telescope has over a refracting telescope.

(ii) In a refracting telescope, the focal length of the objective is 50 times the focal length of the eyepiece. When the final image is formed at infinity, the length of the tube is 102 cm. Find the focal lengths of the two lenses. CBSE 2025 *5*



The parallel rays from a distant object are reflected by a large concave mirror. These rays are then reflected by a convex mirror placed just before the focus of concave mirror and are converged to a point outside the hole. The final image is viewed through eye piece. **1**

1

Advantages (any two)

1) No chromatic aberration.
2) Less spherical aberration
3) Less mechanical support required **½ + ½**
4) Brighter Image
5) High resolving power.
6) High magnifying power

(i) For image at infinity

$|f_0| + |f_e| = L$ **½**

According to question

$f_0 = 50 \times f_e$ **½**

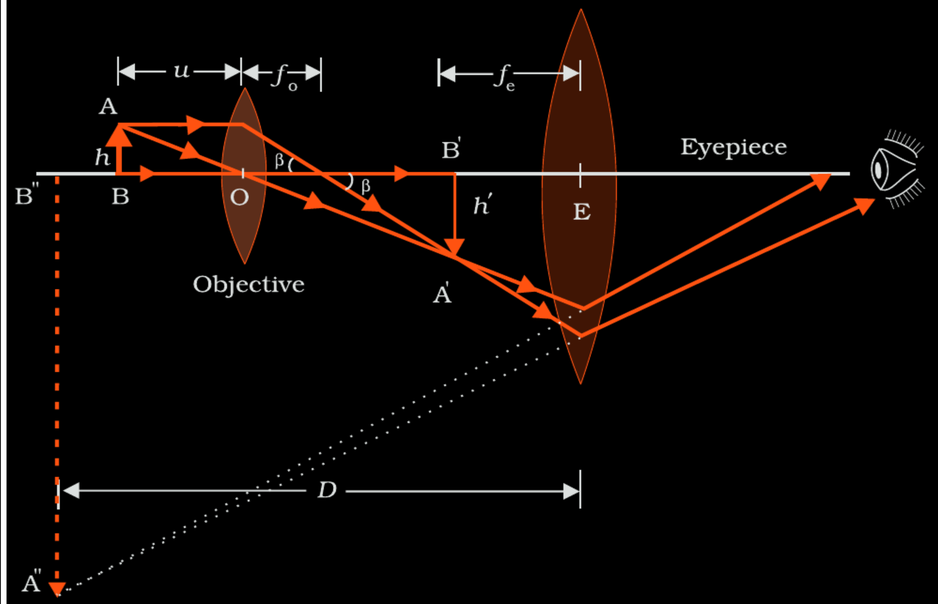
$f_e + 50fe = 102$

$f_e = 2$ cm **½**

$f_0 = 100$ cm **½**

(b) (i) Write any two advantages of a compound microscope over a simple microscope. Draw a ray diagram for the image formation at the near point by a compound microscope and explain it.

(ii) A thin planoconcave lens with its curved face of radius of curvature R is made of glass of refractive index n_1. It is placed coaxially in contact with a thin equiconvex lens of same radius of curvature of refractive index n_2. Obtain the power of the combination lens. CBSE 2025 *5*



(i) Advantages (any two) ½ + ½

1) Larger magnification
2) Brighter image

Any other valid advantage

(deduct ½ mark for not showing arrow for ray diagram)

The lens nearest the object, called the objective, forms a real, inverted, magnified image of the object. This serves as the object for the second lens, the eye piece, functions like a simple microscope and produces final image which is enlarged and virtual. **1**

(ii) Power of plano concave lens $= P_1 = \frac{-(n_1-1)}{R}$ ½

Power of convex lens $= P_2 = (n_2-1)\left(\frac{2}{R}\right)$ ½

$P = P_1 + P_2$ ½

$= \frac{(2n_2 - n_1 - 1)}{R}$ ½

26. An air bubble is trapped at point P (CP = 1·75 cm) in a spherical glass ball (n = 1·5) of radius 7 cm as shown in the figure. Find the nature and position of the image when viewed from side B. Show the image formation by drawing a ray diagram.

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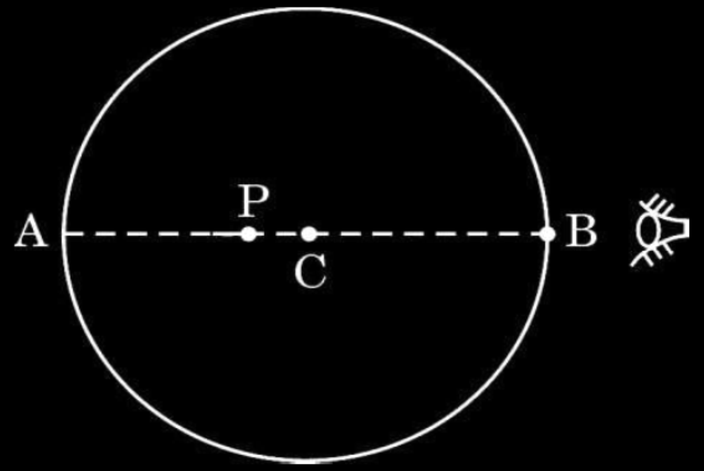
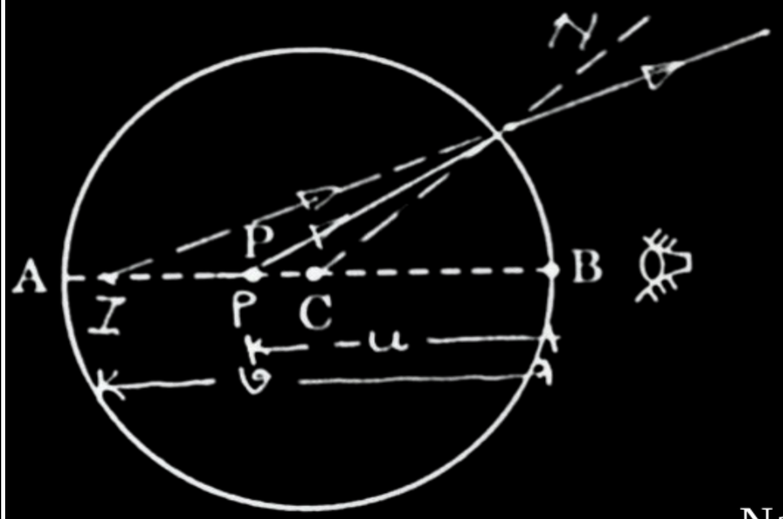
3



26. An air bubble is trapped at point P (CP = 1·75 cm) in a spherical glass ball (n = 1·5) of radius 7 cm as shown in the figure. Find the nature and position of the image when viewed from side B. Show the image formation by drawing a ray diagram. CBSE 2025 *3*



For light going from denser to rarer medium

$$\frac{n_2}{-u} + \frac{n_1}{v} = \frac{n_1 - n_2}{R} \qquad \underline{1}$$

$$\frac{1.5}{8.75} + \frac{1}{v} = \frac{(1-1.5)}{-7} \qquad ½$$

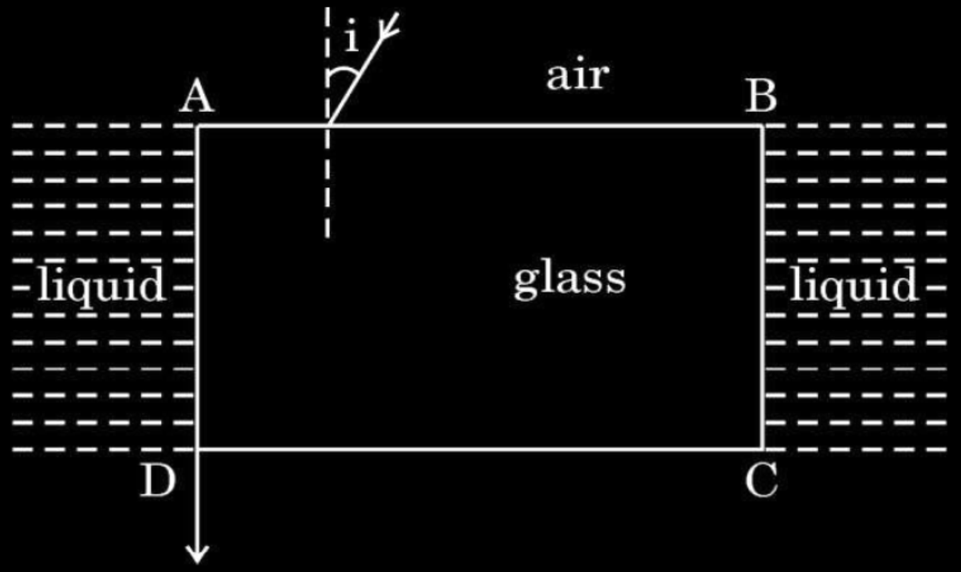
$$\frac{1}{v} = \frac{0.5}{7} - \frac{1.5}{8.75} \qquad ½$$

$$= \frac{-1}{10} \qquad ½$$

$$v = -10 \text{ cm} \qquad ½$$

Nature of image is virtual

27. A rectangular glass slab ABCD (refractive index 1·5) is surrounded by a transparent liquid (refractive index 1·25) as shown in the figure. A ray of light is incident on face AB at an angle i such that it is refracted out grazing the face AD. Find the value of angle i. *3*



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For glass- liquid interface

$$\sin i_c = \frac{1}{n_{21}} \qquad ½$$

$$= \frac{1.25}{1.5} \qquad ½$$

$$= \frac{5}{6} \qquad ½$$

$$i_c + r = 90^0 \qquad ½$$

$$\sin r = \sqrt{1-\cos^2 r} = \frac{\sqrt{11}}{6}$$

Since

$$\frac{Sin\ i}{Sin\ r} = n \qquad ½$$

Therefore, $Sin\ i = \frac{\sqrt{11}}{4}$ or $i = Sin^{-1}\frac{\sqrt{11}}{4}$ ½

9. The magnification produced by a spherical mirror is – 2·0. The mirror used and the nature of the image formed will be

(A) Convex and virtual

(B) Concave and real

(C) Concave and virtual

(D) Convex and real

CBSE 2025

9. The magnification produced by a spherical mirror is – 2·0. The mirror used and the nature of the image formed will be

(A) Convex and virtual

(B) Concave and real

(C) Concave and virtual

(D) Convex and real

CBSE 2025

(B) Concave and real

19. The two surfaces of a biconvex lens are of radius of curvature 'R' each. Obtain the condition under which its focal length 'f' be equal to 'R'. If one of the two surfaces of this lens is made plane, what will be the new focal length of the lens ?

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2

19. The two surfaces of a biconvex lens are of radius of curvature 'R' each. Obtain the condition under which its focal length 'f' be equal to 'R'. If one of the two surfaces of this lens is made plane, what will be the new focal length of the lens ?

CBSE 2025 *2*

$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$	½
For $f=R, R_1=R, R_2=-R$	
$\frac{1}{R}=(n-1)\left(\frac{1}{R}+\frac{1}{R}\right)$	
$n=\frac{3}{2}$	½
For convex lens	
$\frac{1}{f}=(n-1)\left(\frac{2}{R}\right)$	
For plano Convex lens $R_1=R \quad R_2=\infty$	½
$\frac{1}{f_p}=(n-1)\left(\frac{1}{R}\right)$	
$f_p=2f$	½

31. (a) (i) An object is placed 30 cm from a thin convex lens of focal length 10 cm. The lens forms a sharp image on a screen. If a thin concave lens is placed in contact with the convex lens, the sharp image on the screen is formed when the screen is moved by 45 cm from its initial position. Calculate the focal length of the concave lens.

(ii) Calculate the angle of minimum deviation of an equilateral prism. The refractive index of the prism is $\sqrt{3}$. Calculate the angle of incidence for this case of minimum deviation also. *5*

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For real image form by Convex lens	
$\frac{1}{f_1}=\frac{1}{v_1}-\frac{1}{u_1}$	½
$\frac{1}{10}=\frac{1}{v_1}-\frac{1}{(-30)}$	
$v_1=15\,cm$	½
For Combination of lenses, let the focal length of combination of lens is f_3	
$\frac{1}{f_3}=\frac{1}{v_3}-\frac{1}{u_3}$	
$\frac{1}{f_3}=\frac{1}{(15+45)}+\frac{1}{30}$	½
$f_3=20cm$	½
Let the focal length of concave lens is f_2	
$\frac{1}{f_3}=\frac{1}{f_1}+\frac{1}{f_2}$	½
$\frac{1}{f_2}=\frac{1}{20}-\frac{1}{10}$	
$f_2=-20cm$	½

(ii)

Angle of minimum deviation

$$\mu = \frac{\text{Sin}\,\frac{(A+\delta_m)}{2}}{\text{Sin}\,\frac{A}{2}}$$

½

$$\sqrt{3} = \frac{\text{Sin}\,\frac{(60^\circ+\delta_m)}{2}}{\text{Sin}\,30}$$

$$\frac{\sqrt{3}}{2} = Sin\,\frac{(A+\delta_m)}{2}$$

$$60^\circ = \frac{(A+\delta_m)}{2}$$

½

$$\delta_m = 60^o$$

Angle of incidence

½

$$i + e = A + \delta$$

$$2i = A + \delta_m$$

$$i = \frac{A+\delta_m}{2}$$

½

$$i = 60^\circ$$

19. Find the focal length of plano-convex lens of refractive index 1·5 and radius of curvature 10 cm when it is immersed in a liquid of refractive index 1·25. CBSE 2025 *2*

19. Find the focal length of plano-convex lens of refractive index 1·5 and radius of curvature 10 cm when it is immersed in a liquid of refractive index 1·25. CBSE 2025 *2*

$$\frac{1}{f}=\left(\frac{n_2}{n_1}-1\right)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$ ½

For plano convex lens

$R_1 = R$ and $R_2 = \infty$

$$\frac{1}{f}=\left(\frac{n_2}{n_1}-1\right)\frac{1}{R}$$ ½

$$=\left(\frac{1.5}{1.25}-1\right)\times\frac{1}{10}$$ ½

$$\frac{1}{f}=\frac{1}{50}$$

$\therefore f = 50$ cm ½

19. An equiconvex lens is made of glass of refractive index 1·55. If the focal length of the lens is 15·0 cm, calculate the radius of curvature of its surfaces. CBSE 2025 *2*

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$\frac{1}{f}=(n_{21}-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$	½
$R_1=R$ and $R_2=-R$	½
$\frac{1}{15}=(1.55-1)\left[\frac{2}{R}\right]$	½
$R = 16.5$ cm	½

9. The plane face of a planoconvex lens is silvered. The refractive index of material and radius of curvature of the curved surface of the lens are n and R respectively. This lens will behave as a concave mirror of focal length **1**

(A) $\frac{R}{n}$

(B) $\frac{R}{(n-1)}$

(C) nR

(D) $\frac{R}{2(n-1)}$

CBSE 2025

9. The plane face of a planoconvex lens is silvered. The refractive index of material and radius of curvature of the curved surface of the lens are n and R respectively. This lens will behave as a concave mirror of focal length **1**

(A) $\frac{R}{n}$

(B) $\frac{R}{(n-1)}$

(C) nR

(D) $\frac{R}{2(n-1)}$

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(D) $\frac{R}{2(n-1)}$

19. A light beam converges at a point O. In the path of this beam, a concave lens of focal length 15 cm is placed at a distance of 10 cm before point O. The beam now converges at a point O'. Find the magnitude and the direction of shift OO'. CBSE 2025 **2**

19. A light beam converges at a point O. In the path of this beam, a concave lens of focal length 15 cm is placed at a distance of 10 cm before point O. The beam now converges at a point O'. Find the magnitude and the direction of shift OO'. CBSE 2025 **2**

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	½
$\frac{1}{v} = -\frac{1}{15} + \frac{1}{10}$	
On solving: -	
$v = 30\,cm$	½
Magnitude of OO' = 30-10 = 20 cm	½
Direction is away from the lens.	½

28. (a) When a parallel beam of light enters water surface obliquely at some angle, what is the effect on the width of the beam ? 3

(b) With the help of a ray diagram, show that a straw appears bent when it is partly dipped in water and explain it.

(c) Explain the transmission of optical signal through an optical fibre by a diagram.

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28. (a) When a parallel beam of light enters water surface obliquely at some angle, what is the effect on the width of the beam ? 3

(b) With the help of a ray diagram, show that a straw appears bent when it is partly dipped in water and explain it.

(c) Explain the transmission of optical signal through an optical fibre by a diagram.

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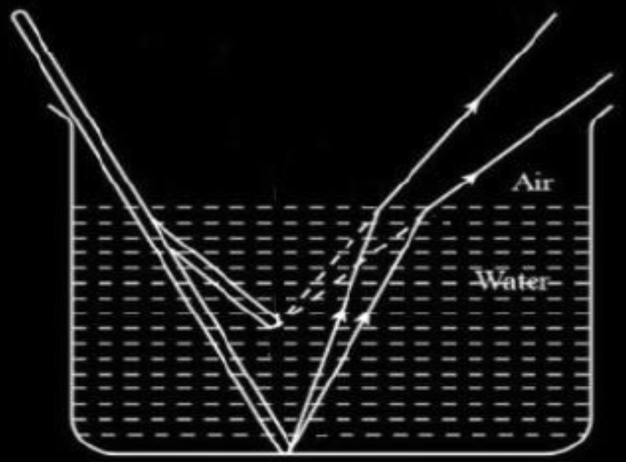
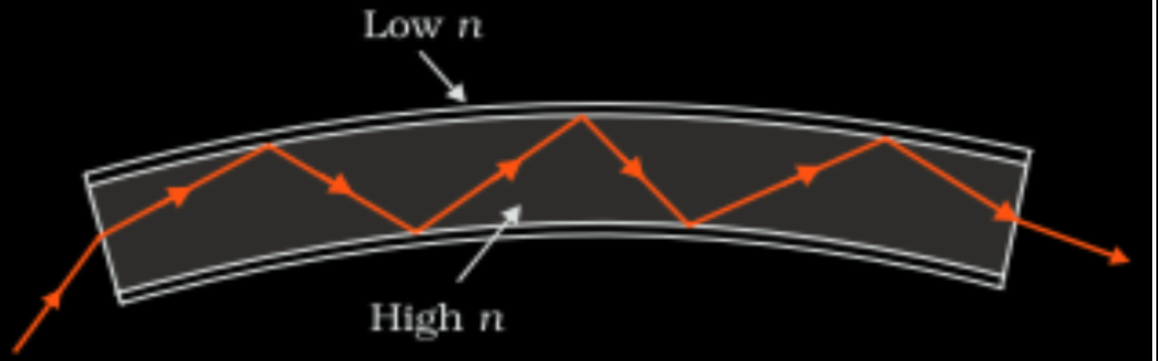
(a) Width of the parallel beam of light increases in water. 1

Alternatively: -
If a student explains using diagram, full credit to be given.

(b) Due to refraction of light, the image of the portion immersed in water appears to be raised.

(c)





32. (a) (i) A thin pencil of length (f/4) is placed coinciding with the principal axis of a mirror of focal length f. The image of the pencil is real and enlarged, just touches the pencil. Calculate the magnification produced by the mirror. **5**

(ii) A ray of light is incident on a refracting face AB of a prism ABC at an angle of 45°. The ray emerges from face AC and the angle of deviation is 15°. The angle of prism is 30°. Show that the emergent ray is normal to the face AC from which it emerges out. Find the refraction index of the material of the prism.

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(i) As the pencil lies between f and 2f such that one end of the pencil coincides with 2f.

Position of the other end (u) $= -\left(2f - \frac{f}{4}\right) = -\frac{7f}{4}$ ½

Magnification (m) $= \frac{f}{f - u}$ ½

$= \frac{-f}{-f - \left(-\frac{7f}{4}\right)}$ ½

$m = -\frac{4}{3}$ 1

Alternatively: -

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} - \frac{4}{7f} = -\frac{1}{f}$

$\frac{1}{v} = -\frac{1}{f} + \frac{4}{7f}$

$v = -\frac{7f}{3}$

$m = -\frac{v}{u} = -\frac{4}{3}$

(ii) For prism; ½

$i + e = A + \delta$ ½

$45^o + e = 30^o + 15^o$

$\therefore e = 0^o$ ½

Hence, $r_2 = 0^o$

$\therefore$ Emergent ray is perpendicular to face AC.

Alternatively: - If the same is shown using diagram full credit to be given.

$r_1 + r_2 = A$ ½

As $r_2 = 0$, hence $r_1 = 30^o$

Refractive index(n) = $\frac{\sin i}{\sin r}$

$= \frac{\sin 45^o}{\sin 30^o}$

½

$n = \sqrt{2}$

9. The focal length of a concave mirror in air is f. When the mirror is immersed in a liquid of refractive index $\frac{5}{3}$, its focal length will become **1**

(A) $\frac{5}{3}$ f

(B) $\frac{3}{5}$ f

(C) $\frac{2}{3}$ f

(D) f

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9. The focal length of a concave mirror in air is f. When the mirror is immersed in a liquid of refractive index $\frac{5}{3}$, its focal length will become **1**

(A) $\frac{5}{3}$ f

(B) $\frac{3}{5}$ f

(C) $\frac{2}{3}$ f

(D) f

CBSE 2025

(D) f

19. Two convex lenses A and B, each of focal length 10.0 cm, are mounted on an optical bench at 50.0 cm and 70.0 cm respectively. An object is mounted at 20.0 cm. Find the nature and position of the final image formed by the combination. CBSE 2025 **2**

19. Two convex lenses A and B, each of focal length 10.0 cm, are mounted on an optical bench at 50.0 cm and 70.0 cm respectively. An object is mounted at 20.0 cm. Find the nature and position of the final image formed by the combination. CBSE 2025 **2**

For the first lens: -

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$ ½

$$\frac{1}{v_1} + \frac{1}{30} = \frac{1}{10}$$ ½

$v_1 = 15$ cm

For the second lens: -

$u_2 = -20 - (-15) = -5$ cm

$$\frac{1}{v_2} - \left(-\frac{1}{5}\right) = \frac{1}{10}$$

On solving: -

$v_2 = -10$ cm ½

Image is virtual. ½

9. A tub is filled with a transparent liquid to a height of 30.0 cm. The apparent depth of a coin lying at the bottom of the tub is found to be 16.0 cm. The speed of light in the liquid will be **1**

(A) 1.6×10^8 m s^{-1} (B) 2.0×10^8 m s^{-1}

(C) 3.0×10^8 m s^{-1} (D) 2.5×10^8 m s^{-1}

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(C) 3.0×10^8 m s^{-1} (D) 2.5×10^8 m s^{-1}

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(A) 1.6 x 10^8 ms^{-1}

19. Two concave lenses A and B, each of focal length 8.0 cm are arranged coaxially 16 cm apart as shown in figure. An object P is placed at a distance of 4.0 cm from A. Find the position and nature of the final image formed.

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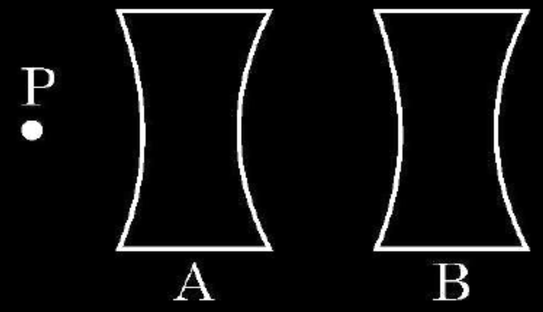
2



For the first lens: -

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$ ½

$$\frac{1}{v_1} + \frac{1}{4} = -\frac{1}{8}$$

$$v_1 = -\frac{8}{3}\text{cm}$$ ½

For the second lens: -

$$u_2 = -16 - \frac{8}{3} = -\frac{56}{3}\text{cm}$$

$$\frac{1}{v_2} - \left(-\frac{3}{56}\right) = -\frac{1}{8}$$

$$v_2 = -5.6\text{cm}$$ ½

Image is virtual. ½

8. A point source is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H. The area of the surface of the liquid through which light from the source can emerge out is **1**

(A) $\frac{\pi H^2}{(n-1)}$

(B) $\frac{\pi H^2}{(n^2-1)}$

(C) $\frac{\pi H^2}{\sqrt{n^2-1}}$

(D) $\frac{\pi H^2}{(n^2+1)}$

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(D) $\frac{\pi H^2}{(n^2+1)}$

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(B) $\frac{\pi H^2}{(n^2-1)}$

19. A double convex lens of glass has both faces of the same radius of curvature 17 cm. Find its focal length if it is immersed in water. The refractive indices of glass and water are 1.5 and 1.33 respectively. **2**

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19. A double convex lens of glass has both faces of the same radius of curvature 17 cm. Find its focal length if it is immersed in water. The refractive indices of glass and water are 1.5 and 1.33 respectively. **2**

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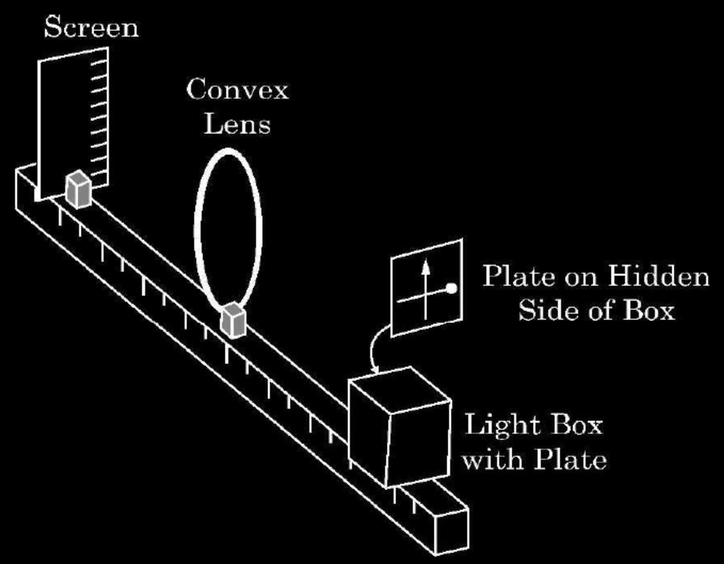
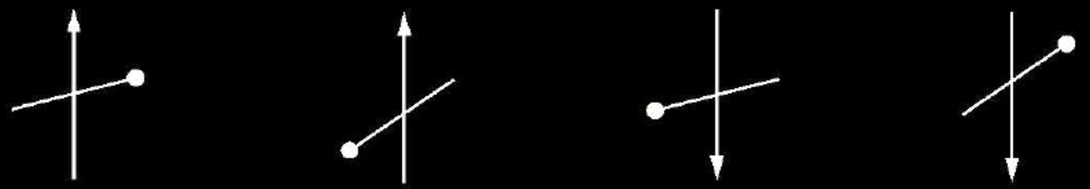
$\frac{1}{f} = \left(\frac{n_g}{n_w} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$	½
For double convex lens R_1=R and R_2= -R	½
$\frac{1}{f} = \left(\frac{1.5}{1.33} - 1\right)\left(\frac{2}{R}\right)$	½
$= \left(\frac{1.5 - 1.33}{1.33}\right)\left(\frac{2}{17}\right)$	
$f = 66.5$ cm	½

29. A thin lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, one can obtain the 'lens maker formula' and then the 'lens formula'. A lens has two foci – called 'first focal point' and 'second focal point' of the lens, one on each side. CBSE 2025 $4 \times 1 = 4$

(i) Consider the arrangement shown in figure. A black vertical arrow and a horizontal thick line with a ball are painted on a glass plate. It serves as the object. When the plate is illuminated, its real image is formed on the screen.

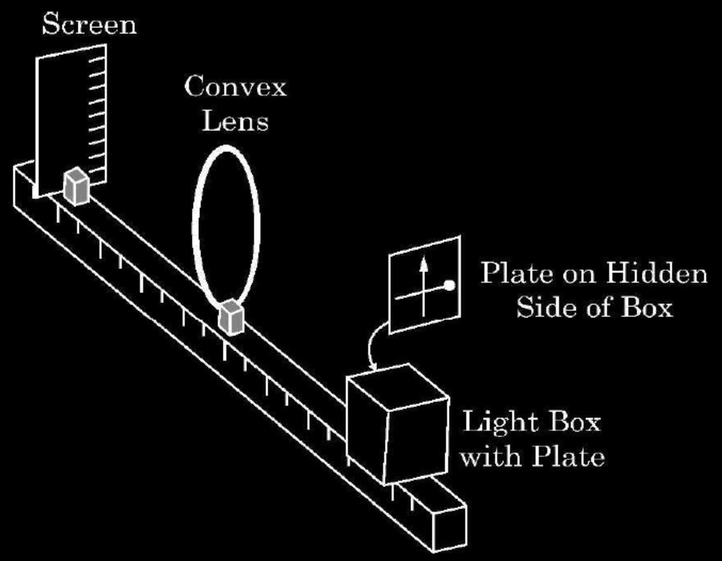


Which of the following correctly represents the image formed on the screen ?



(A) (B) (C) (D)

29. A thin lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, one can obtain the 'lens maker formula' and then the 'lens formula'. A lens has two foci – called 'first focal point' and 'second focal point' of the lens, one on each side. CBSE 2025 $4 \times 1 = 4$

(i) 

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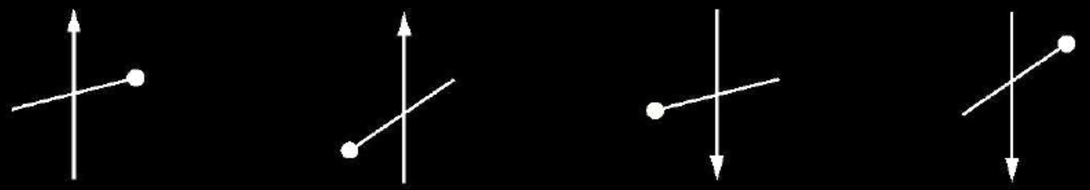
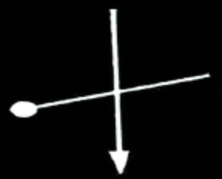


(A) (B) (C) (D)

i) (C)



(ii) Which of the following statements is incorrect ?

(A) For a convex mirror magnification is always negative.

(B) For all virtual images formed by a mirror magnification is positive.

(C) For a concave lens magnification is always positive.

(D) For real and inverted images, magnification is always negative.

(ii) Which of the following statements is incorrect ?

(A) For a convex mirror magnification is always negative.

(B) For all virtual images formed by a mirror magnification is positive.

(C) For a concave lens magnification is always positive.

(D) For real and inverted images, magnification is always negative.

(A) For a convex mirror magnification is always negative

(iii) A convex lens of focal length 'f' is cut into two equal parts perpendicular to the principal axis. The focal length of each part will be :

(A) f

(B) 2 f

(C) $\frac{f}{2}$

(D) $\frac{f}{4}$

(iii) A convex lens of focal length 'f' is cut into two equal parts perpendicular to the principal axis. The focal length of each part will be :

(A) f

(B) 2 f

(C) $\frac{f}{2}$

(D) $\frac{f}{4}$

(B) 2f

(iii) If an object in case (i) above is 20 cm from the lens and the screen is 50 cm away from the object, the focal length of the lens used is

(A) 10 cm

(B) 12 cm

(C) 16 cm

(D) 20 cm

(iii) If an object in case (i) above is 20 cm from the lens and the screen is 50 cm away from the object, the focal length of the lens used is

(A) 10 cm
(B) 12 cm
(C) 16 cm
(D) 20 cm

(B) 12 cm

(iv) The distance of an object from first focal point of a biconvex lens is X_1 and distance of the image from second focal point is X_2. The focal length of the lens is

(A) $X_1 X_2$

(B) $\sqrt{X_1 + X_2}$

(C) $\sqrt{X_1 X_2}$

(D) $\sqrt{\frac{X_2}{X_1}}$

(iv) The distance of an object from first focal point of a biconvex lens is X_1 and distance of the image from second focal point is X_2. The focal length of the lens is

(A) $X_1 X_2$

(B) $\sqrt{X_1 + X_2}$

(C) $\sqrt{X_1 X_2}$

(D) $\sqrt{\frac{X_2}{X_1}}$

(C) $\sqrt{X_1 X_2}$

(ii) A beam of light coming from a distant source is refracted by a spherical glass ball (refractive index 1.5) of radius 15 cm. Draw the ray diagram and obtain the position of the final image formed.

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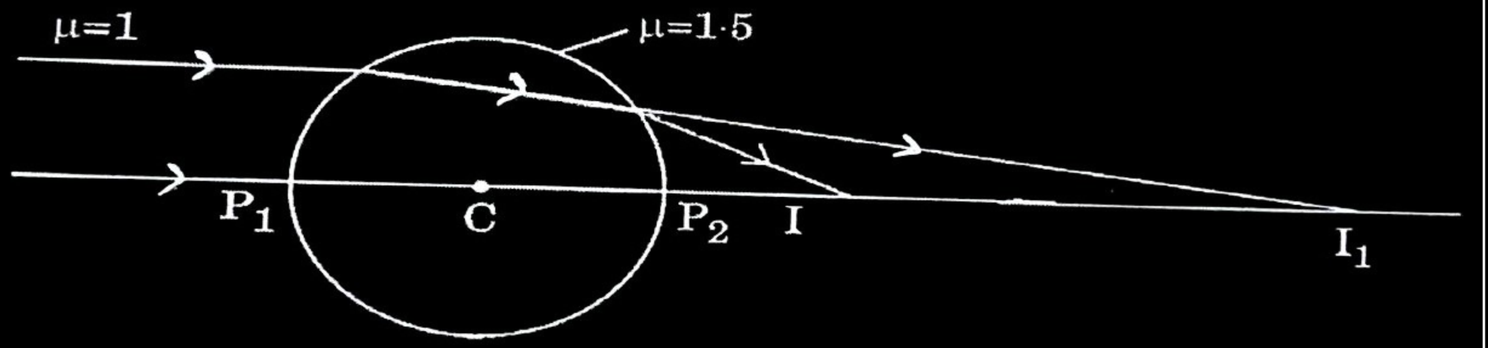
ii)



1

From I[st] surface , Refraction is from rarer to denser medium and object is at ∞

$n_1 = 1$, $n_2 = 1.5$, R = 15 cm , u= ∞

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$ ½

$$\frac{1.5}{v} - \frac{1}{\infty} = \frac{1.5 - 1}{15}$$

v= 45 cm ½

From 2[nd] surface , Refraction is from denser to rarer medium and object is at 15 cm

$n_1 = 1$, $n_2 = 1.5$, R = -15 cm , u= 15cm

$$\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}$$ ½

$$\frac{1}{v} - \frac{1.5}{15} = \frac{1 - 1.5}{-15}$$ ½

v= 7.5 cm

8. The speed of light in two media '1' and '2' are v_1 and v_2 ($> v_1$) respectively. For a ray of light to undergo total internal reflection at the interface of these two media, it must be incident from **1**

(A) medium '1' and at an angle greater than $\sin^{-1}\left(\frac{v_1}{v_2}\right)$

(B) medium '1' and at an angle greater than $\cos^{-1}\left(\frac{v_1}{v_2}\right)$

(C) medium '2' and at an angle greater than $\sin^{-1}\left(\frac{v_1}{v_2}\right)$

(D) medium '2' and at an angle greater than $\cos^{-1}\left(\frac{v_1}{v_2}\right)$

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(A) medium '1' and at an angle greater that $\sin^{-1}\left(\frac{v_1}{v_2}\right)$

19. A spherical convex surface of radius of curvature R separates glass (refractive index 1.5) from air. Light from a point source placed in air at distance R/2 from the surface falls on it. Find the position and nature of the image formed. **2**

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Refraction from rarer to denser medium	
$\frac{n_1}{-u}+\frac{n_2}{v}=\frac{n_2-n_1}{R}$	½
$u=-\frac{R}{2}, n_1=1, n_2=1.5$	
$\frac{2}{R}+\frac{1.5}{v}=\frac{1.5-1}{R}$	½
$\frac{1.5}{v}=\frac{0.5}{R}-\frac{2}{R}$	
$\frac{1.5}{v}=-\frac{1.5}{R}$	
$v = - R$	½
The image is virtual in air at distance R.	½

8. A beaker is filled with water (refractive index $\frac{4}{3}$) upto a height H. A coin is placed at its bottom. The depth of the coin, when viewed along the near normal direction, will be **1**

(A) $\frac{H}{4}$

(B) $\frac{3H}{4}$

(C) H

(D) $\frac{4H}{3}$

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(A) $\frac{H}{4}$

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(C) H

(D) $\frac{4H}{3}$

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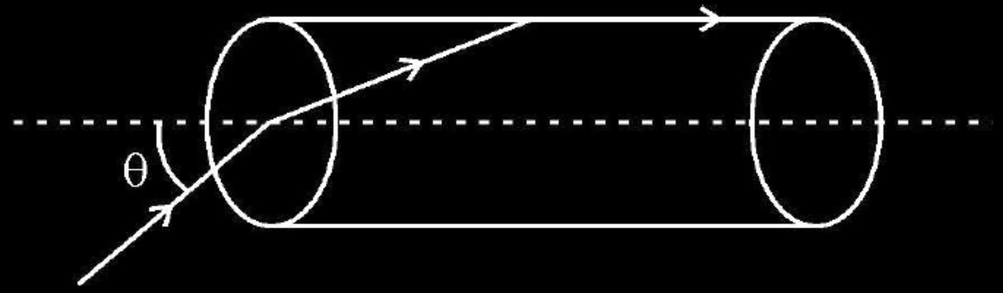
(B) $\frac{3H}{4}$

19. A transparent solid cylindrical rod (refractive index $\frac{2}{\sqrt{3}}$) is kept in air. A ray of light incident on its face travels along the surface of the rod, as shown in figure. Calculate the angle θ. **2**



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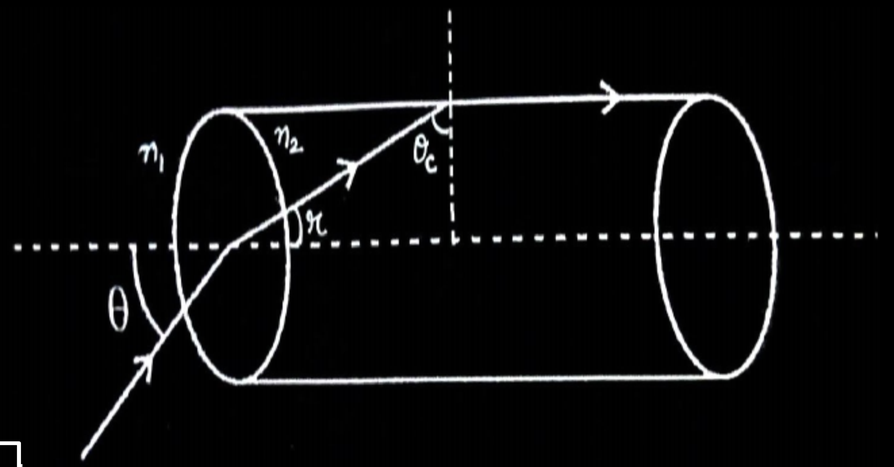

For critical Angle

$$\frac{n_2}{n_1} = \frac{1}{\sin\theta_c} \qquad ½$$

$n_1=1$ $n_2=\frac{2}{\sqrt{3}}$ (given)

$$\frac{2}{\sqrt{3}} = \frac{1}{\sin\theta_c}$$

$$\sin\theta_c = \frac{\sqrt{3}}{2}$$

$$\theta_c = 60°$$

$$r = 90 - \theta_c$$

$$= 30°$$ ½

From Snell's law at air rod interface

$$n_1 \sin i = n_2 \sin r$$ ½

$$n_2 = \frac{\sin\theta}{\sin r}$$

$$\frac{2}{\sqrt{3}} = \frac{\sin\theta}{\sin 30^0}$$

$$\frac{2}{\sqrt{3}} \times \frac{1}{2} = \sin\theta$$

$$\frac{1}{\sqrt{3}} = \sin\theta$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$ ½

6. A point object is placed in air at a distance of 4R on the principal axis of a convex spherical surface of radius of curvature R separating two mediums, air and glass. As the object is moved towards the surface, the image formed is : CBSE 2024 C

(A) always real

(B) always virtual

(C) first virtual and then real

(D) first real and then virtual

6. A point object is placed in air at a distance of 4R on the principal axis of a convex spherical surface of radius of curvature R separating two mediums, air and glass. As the object is moved towards the surface, the image formed is : CBSE 2024 C

(A) always real

(B) always virtual

(C) first virtual and then real

(D) first real and then virtual

(D) first real and then virtual.

13. *Assertion (A) :* Although the surfaces of a goggle lens are curved, it does not have any power.

Reason (R) : In case of goggles, both the curved surfaces are curved on the same side and have equal radii of curvature.

CBSE 2024 C

13. *Assertion (A) :* Although the surfaces of a goggle lens are curved, it does not have any power.

Reason (R) : In case of goggles, both the curved surfaces are curved on the same side and have equal radii of curvature. CBSE 2024 C

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion(A).

20. A point light source rests on the bottom of a bucket filled with a liquid of refractive index $\mu = 1.25$ up to height of 10 cm. Calculate :

(a) the critical angle for liquid-air interface

(b) radius of circular light patch formed on the surface by light emerging from the source.

CBSE 2024 C

2

20. A point light source rests on the bottom of a bucket filled with a liquid of refractive index $\mu = 1.25$ up to height of 10 cm. Calculate :

(a) the critical angle for liquid-air interface

(b) radius of circular light patch formed on the surface by light emerging from the source.

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2

(a) $\sin i_c = \frac{1}{\mu}$ ½

$\sin i_c = \frac{4}{5} \quad (\because \mu = 1.25 = \frac{5}{4})$

$i_c = \sin^{-1}(\frac{4}{5})$ ½

$or\ i_c = 53^o$

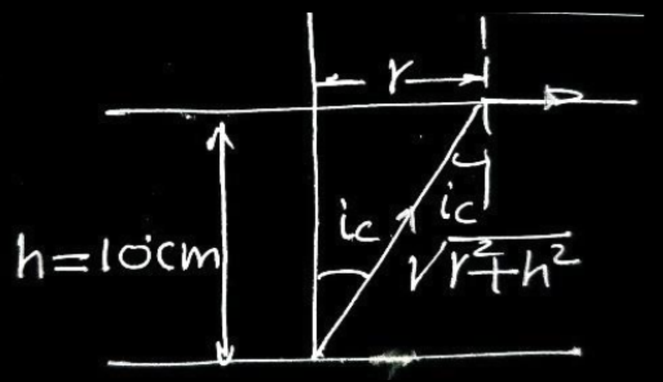
(b) $\sin i_c = \frac{r}{\sqrt{r^2+h^2}}$ ½

$\frac{r^2}{r^2+h^2} = \left(\frac{4}{5}\right)^2$

$25r^2 = 16r^2 + 16h^2$

$9r^2 = 1600$

$r = \frac{40}{3}\text{cm}$ ½



32. (a) (i) What are the two main considerations for designing the objective and eyepiece lenses of an astronomical telescope ? Obtain the expression for magnifying power of the telescope when the final image is formed at infinity.

(ii) A ray of light is incident at an angle of 45° at one face of an equilateral triangular prism and passes symmetrically through the prism. Calculate :

(1) the angle of deviation produced by the prism

(2) the refractive index of the material of the prism CBSE 2024 C *5*

Two main considerations

Objective should have

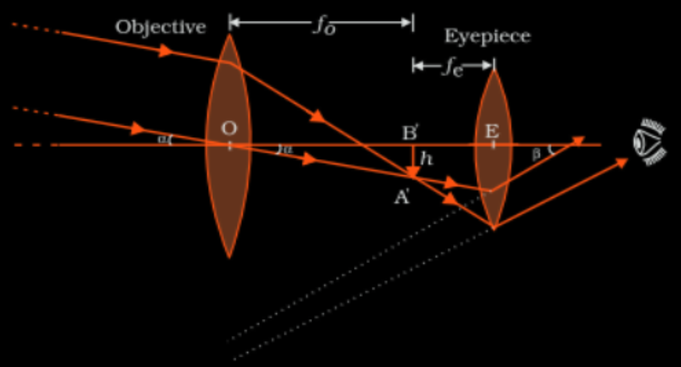
1. Larger diameter
2. Larger focal length

½

Eye piece should have

1. Smaller diameter
2. Smaller focal length

½



1

Magnifying power of telescope

Magnifying power is the ratio of the angle β subtended at the eye by the final images to the angle α which the object subtends at the lens or eye

$$m \approx \frac{\beta}{\alpha} \approx \frac{h}{f_e} \cdot \frac{f_0}{h} = \frac{f_0}{f_e}$$

1

(ii) $i+e = D+A$

$at \quad D = D_m \quad , \quad i = e$

$2i = D_m + A$

$2 \times 45 = D_m + 60^0$

$D_m = 30^0$

1

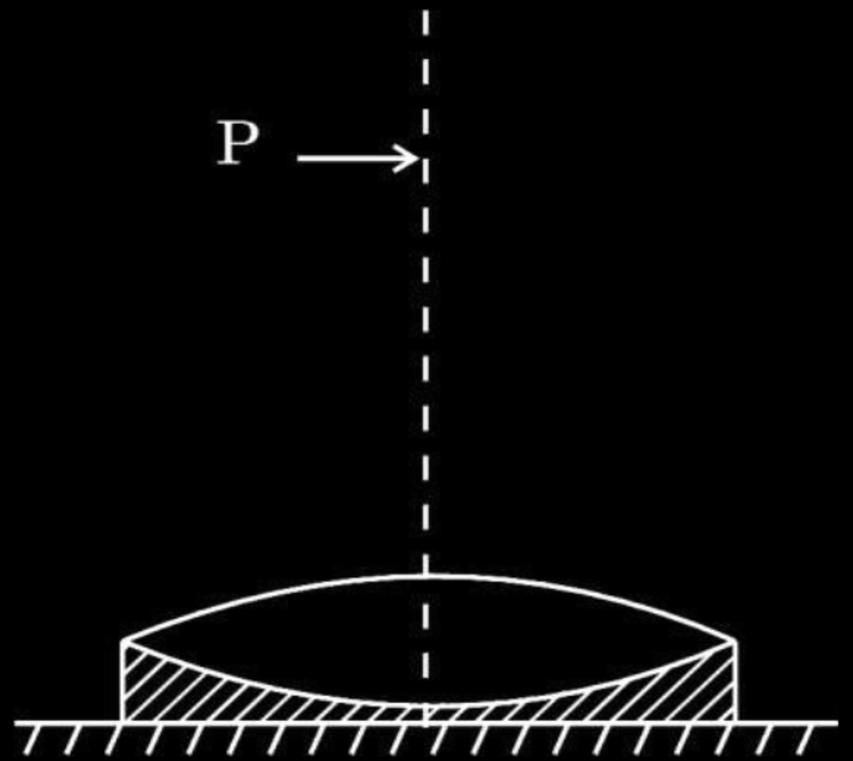
$$\mu = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\mu = \frac{\sin\left(\frac{60^0+30^0}{2}\right)}{\sin\left(\frac{60^0}{2}\right)}$$

1

$= \sqrt{2}$

(ii) The figure below shows an equiconvex lens (of refractive index 1.50) in contact with a liquid layer on top of a plane mirror. A small needle with its tip on the principal axis is moved along the axis until its inverted image is found at the position of the needle. The distance of the needle from the lens is measured to be 45.0 cm. When the liquid is removed and the experiment is repeated, the new distance is 30.0 cm. Find the refractive index of the liquid. CBSE 2024 C



(ii) $\frac{1}{f}=(\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$ ½

Focal length of convex lens, $f_1=30\,\text{cm}$

$\frac{1}{30}=(1.5-1)\left[\frac{1}{R}-\frac{1}{(-R)}\right]$ ½

$R = 30\,\text{cm}$ ½

focal length of combination, f = 45 cm

focal length of plane concave lens of liquid.

$\frac{1}{f_2} = \frac{1}{f}-\frac{1}{f_1}$ ½

$\frac{1}{f_2} = \frac{1}{45}-\frac{1}{30}$

$f_2 = -90\,\text{cm}$

Using lens maker's formula ½

$\frac{1}{-90} = (\mu_l - 1)\left[\frac{1}{-30}-\frac{1}{\infty}\right]$

$\mu_l = \frac{4}{3}$ ½

16. *Assertion (A) :* In optical fibres, the refractive index of cladding must be lesser than that of the core.

Reason (R) : Total internal reflection takes place only when the ray passes from denser to rarer medium. CBSE 2024 C

16. *Assertion (A) :* In optical fibres, the refractive index of cladding must be lesser than that of the core.

Reason (R) : Total internal reflection takes place only when the ray passes from denser to rarer medium. CBSE 2024 C

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

18. (a) Why does a water tank appear shallow ? Explain. Write the relation between real depth and apparent depth of the tank with the refractive index of water. CBSE 2024 C *2*

18. (a) Why does a water tank appear shallow ? Explain. Write the relation between real depth and apparent depth of the tank with the refractive index of water. CBSE 2024 C *2*

It is due to refraction of light. Light coming out of the water surface deviates away from normal and as a result appear to come from a point above the bottom of water tank. Relation between real depth, apparent depth and refractive index of water. $\text{Refractive index} = \frac{\text{Real depth}}{\text{Apparent depth}}$ Alternatively- $\mu_w = \frac{R_d}{A_d}$	1 1

(b) What are optical fibres ? How does a light ray travel through an optical fibre ? Give two applications of optical fibres. CBSE 2024 C *2*

(b) What are optical fibres ? How does a light ray travel through an optical fibre ? Give two applications of optical fibres. CBSE 2024 C *2*

Optical fibres are composite glass/ quartz fibres consisting of a core & a cladding used to transmit audio and video signals through long distances.	½
When a signal in the form of light is directed at one end of the fibre at a suitable angle, it undergoes repeated total internal reflections along the length of the fibre and finally comes out at the other end.	½
Two applications of optical fibres- (1) For communication (2) For decorative purpose (3) For examining internal organs (Any two)	1

26. A glass biconvex lens of focal length 0.6 m and refractive index 1.5 is immersed in a medium of refractive index 2.0. Find the new power of the lens and the nature of lens. CBSE 2024 C *3*

26. A glass biconvex lens of focal length 0.6 m and refractive index 1.5 is immersed in a medium of refractive index 2.0. Find the new power of the lens and the nature of lens. CBSE 2024 C *3*

According to Lens- Makers formula: $\frac{1}{f} = (\mu - 1)(\frac{1}{R_1} - \frac{1}{R_2})$	½
For biconvex lens $\frac{1}{0.6} = (1.5 - 1)(\frac{1}{R_1} - \frac{1}{R_2}) - - - - - - - - - (1)$	½
When immersed in medium:- $\frac{1}{f} = (\frac{1.5}{2} - 1)(\frac{1}{R_1} - \frac{1}{R_2}) - - - - - - - - - (2)$	½
On solving equation (1) and (2):- $\frac{1}{f} = -\frac{5}{6} m$ $P = -\frac{5}{6} D$	1
Nature of lens will be concave or diverging.	½

33. (a) (i) Write the relation between u, v and R for a convex spherical surface when refraction takes place from optically rarer to denser medium. Using it, derive 'lens maker formula' for a double convex lens.

(ii) Two thin lenses of power + 25 D and – 5 D are kept in contact with each other. *5*

(1) Calculate the focal length of the combination.

(2) An object of 4 cm height is placed at a distance of 25 cm from this combination. What is the position and size of the image formed ?

CBSE 2024 C

(i) For a convex spherical surface, for light going from optically rarer to optically denser medium-

$$\frac{n_1}{-u}+\frac{n_2}{v}=\frac{n_2-n_1}{R}$$ ½

For the refraction through the first convex surface (light going from rarer to denser medium) , the relation will be ½

$$\frac{n_1}{-u}+\frac{n_2}{v_1}=\frac{n_2-n_1}{R_1} \quad \text{----------} \quad (1)$$

For the refraction through the second convex surface (light going from denser to rarer medium) , the relation will be

$$\frac{-n_2}{v_1}+\frac{n_1}{v}=\frac{n_1-n_2}{R_2} \quad \text{----------} \quad (2)$$

Adding equation (1) and (2) ½

$$\frac{n_1}{v}-\frac{n_1}{u}=(n_2-n_1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right) \quad \text{---------}(3)$$

From lens formula

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

Using equation (3) and (4)

$$\frac{1}{f}=(n_{21}-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$ ½

(ii)

(1) $P_1 = +25D \Rightarrow f_1 = 4\,cm$

$P_2 = -5D \Rightarrow f_2 = -20\,cm$

$$f = \frac{f_1 f_2}{f_1 + f_2} = \frac{4 \times (-20)}{-16}$$

$f = 5$ cm — 1

(2) From lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$ — ½

$$\frac{1}{v} = \frac{5-1}{25}$$

on solving

$$v = \frac{25}{4}\ cm$$ — 1

Magnification $m = \frac{I}{O} = \frac{v}{u}$

$$I = \frac{v}{u} \times O = \frac{25}{4 \times (-25)} \times 4$$ — ½

$I = -1$ cm

11. The focal lengths of the objective and the eyepiece of a compound microscope are 1 cm and 2 cm respectively. If the tube length of the microscope is 10 cm, the magnification obtained by the microscope for most suitable viewing by relaxed eye is :

(A) 250 (B) 200

(C) 150 (D) 125

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11. The focal lengths of the objective and the eyepiece of a compound microscope are 1 cm and 2 cm respectively. If the tube length of the microscope is 10 cm, the magnification obtained by the microscope for most suitable viewing by relaxed eye is :

(A) 250 (B) 200

(C) 150 (D) 125

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(D) 125

12. A point object is kept 60 cm in front of a spherical convex surface (n = 1.5, radius of curvature 40 cm). The image formed is

(A) real, at a distance 1.8 m from the surface.

(B) virtual, at a distance 1.8 m from the surface.

(C) real, at a distance 3.6 m from the surface.

(D) virtual, at a distance 3.6 m from the surface.

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12. A point object is kept 60 cm in front of a spherical convex surface (n = 1.5, radius of curvature 40 cm). The image formed is

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(B) virtual, at a distance 1.8 m from the surface.

(C) real, at a distance 3.6 m from the surface.

(D) virtual, at a distance 3.6 m from the surface.

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(D) virtual, at a distance of 3.6 m from the surface.

19. A thin converging lens of focal length 10 cm is placed coaxially in contact with a thin diverging lens of focal length 15 cm. How will the combination behave ? Justify your answer.

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It will behave like a conversing lens. Power of converging lens is more than the power of diverging lens. Hence the combination will behave like a conversing lens. **Alternatively** $P = P_1 + P_2$ $= \frac{100}{10} + \frac{100}{-15}$ $P = \frac{10}{3}\,\text{D}$ **Alternatively** $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$ $\frac{1}{f} = \frac{1}{10} - \frac{1}{15}$ $\frac{1}{f} = \frac{1}{30}$ $f = 30\ \text{cm}$	1 1

29. A prism is an optical medium bounded by three refracting plane surfaces. A ray of light suffers successive refractions on passing through its two surfaces and deviates by a certain angle from its original path. The refractive index of the material of the prism is given by $\mu = \sin\left(\frac{A + \delta m}{2}\right) / \sin\frac{A}{2}$. If the angle of incidence on the second surface is greater than an angle called critical angle, the ray will not be refracted from the second surface and is totally internally reflected.

(i) The critical angle for glass is θ_1 and that for water is θ_2. The critical angle for glass-water surface would be (given ${}_a\mu_g = 1.5$, ${}_a\mu_w = 1.33$)

(A) less than θ_2 (B) between θ_1 and θ_2

(C) greater than θ_2 (D) less than θ_1

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(A) less than θ_2 (B) between θ_1 and θ_2

(C) greater than θ_2 (D) less than θ_1

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(C) greater than θ_2

(ii) When a ray of light of wavelength λ and frequency v is refracted into a denser medium

(A) λ and v both increase.

(B) λ increases but v is unchanged.

(C) λ decreases but v is unchanged.

(D) λ and v both decrease.

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(A) λ and v both increase.

(B) λ increases but v is unchanged.

(C) λ decreases but v is unchanged.

(D) λ and v both decrease.

(C) λ decreases but ν is unchanged

(iii) (a) The critical angle for a ray of light passing from glass to water is minimum for

(A) red colour

(B) blue colour

(C) yellow colour

(D) violet colour

(iii) (a) The critical angle for a ray of light passing from glass to water is minimum for

(A) red colour

(B) blue colour

(C) yellow colour

(D) violet colour

(D) violet colour

(iii) (b) Three beams of red, yellow and violet colours are passed through a prism, one by one under the same condition. When the prism is in the position of minimum deviation, the angles of refraction from the second surface are r_R, r_Y and r_V respectively. Then

(A) $r_V < r_Y < r_R$

(B) $r_Y < r_R < r_V$

(C) $r_R < r_Y < r_V$

(D) $r_R = r_Y = r_V$

(iii) (b) Three beams of red, yellow and violet colours are passed through a prism, one by one under the same condition. When the prism is in the position of minimum deviation, the angles of refraction from the second surface are r_R, r_Y and r_V respectively. Then

(A) $r_V < r_Y < r_R$

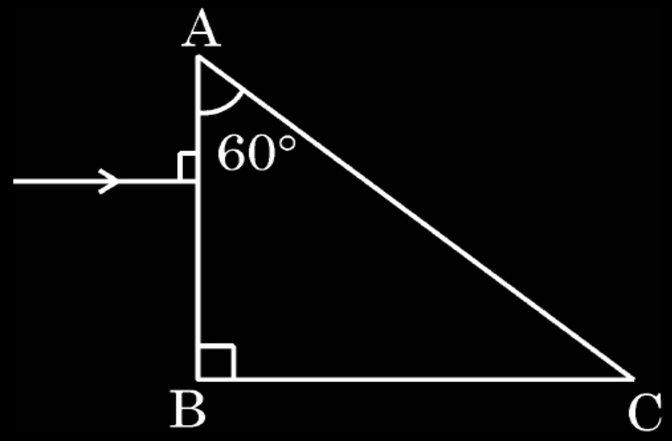
(B) $r_Y < r_R < r_V$

(C) $r_R < r_Y < r_V$

(D) $r_R = r_Y = r_V$

(C) $r_R < r_Y < r_V$

(iv) A ray of light is incident normally on a prism ABC of refractive index $\sqrt{2}$, as shown in figure. After it strikes face AC, it will



(A) go straight undeviated

(B) just graze along the face AC

(C) refract and go out of the prism

(D) undergo total internal reflection

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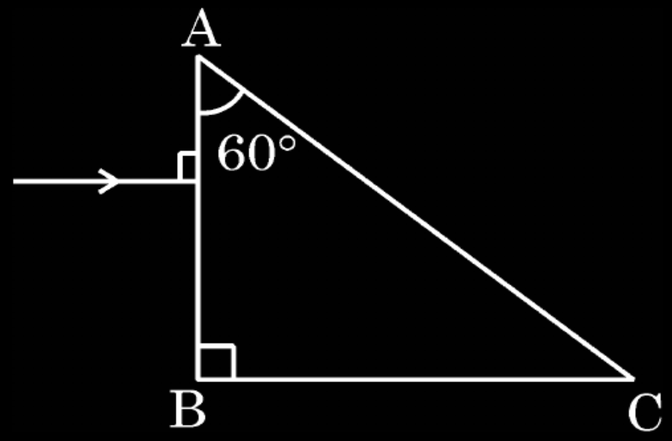

(A) go straight undeviated

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(D) undergo total internal reflection

(ii) An object is placed between the pole and the focus of a concave mirror. Using mirror formula, prove mathematically that it produces a virtual and an enlarged image.

CBSE 2024

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(ii)

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$$

$$v=\frac{uf}{u-f}$$

Following new cartesian sign conversion

$$v=\frac{(-u)(-f)}{-u-(-f)}$$

$$v=\frac{uf}{f-u} \qquad \text{as } f>u$$

1

v is +ve, So image is virtual.

$$m=-\frac{v}{u}=\frac{f}{f-u}>1$$ i.e. Enlarged image

1

19. The radius of curvature of a convex mirror is 30 cm. It forms an image of an object which is half the size of the object. Find the separation between the object and the image.

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19. The radius of curvature of a convex mirror is 30 cm. It forms an image of an object which is half the size of the object. Find the separation between the object and the image.

CBSE 2024

$m = -\frac{v}{u} = \frac{h_I}{h_O} = \frac{1}{2}$	½
$u = -2v$	
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	½
$\frac{1}{15} = \frac{1}{v} - \frac{1}{2v}$	
On solving	
$\lvert v \rvert = 7.5\,\text{cm}$	
$\lvert u \rvert = +15.0\,\text{cm}$	
Separation = 15.0 + 7.5	½
= 22.5 cm	½

19. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate its magnifying power in normal adjustment and the distance of the image formed by the objective. CBSE 2024

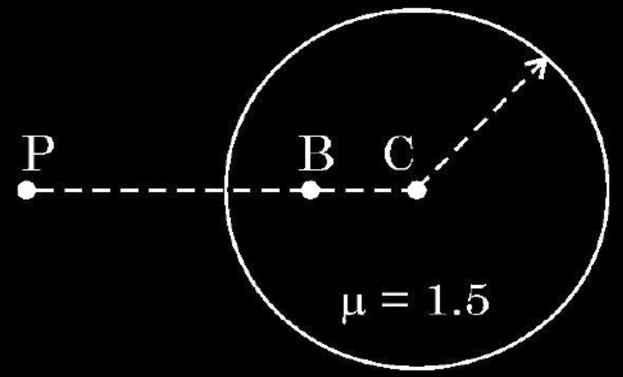
19. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate its magnifying power in normal adjustment and the distance of the image formed by the objective. CBSE 2024

$\lvert m\rvert = \frac{f_0}{f_e}$	½
$= \frac{150}{5} = 30$	½
$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$	½
$\frac{1}{150} = \frac{1}{v} - \frac{1}{\infty}$ $v = 150\,\text{cm}$	½
(Note: Award full credit of this part, if a student writes correct distance of image without calculation i.e. using object position at infinity.)	

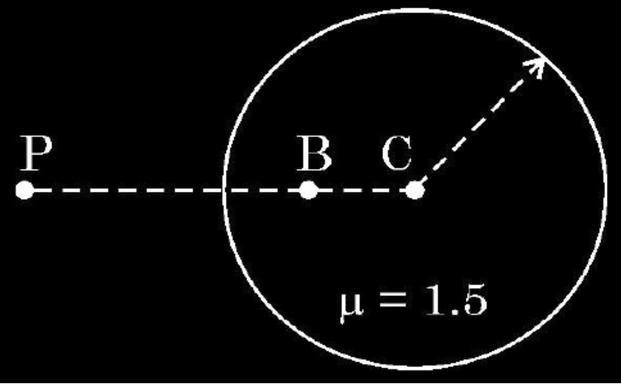
17. (a) An air bubble is trapped at point B (CB = 20 cm) in a glass sphere of radius 40 cm and refractive index 1.5 as shown in figure. Find the nature and position of the image of the bubble as seen by an observer at point P.

CBSE 2024

2



17. (a) An air bubble is trapped at point B (CB = 20 cm) in a glass sphere of radius 40 cm and refractive index 1.5 as shown in figure. Find the nature and position of the image of the bubble as seen by an observer at point P. CBSE 2024 **2**



Using refraction formula at spherical surface from denser to rarer medium n_1 = refractive index of rarer medium n_2 = refractive index of denser medium	½
$\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}$ u = - 20 cm , R= - 40 cm , n_1 = 1 , n_2 = 1.5	½
$\frac{1}{v} - \frac{1.5}{(-20)} = \frac{1-1.5}{(-40)}$	½
v = -16 cm	½
Nature of image is virtual.	

(b) In normal adjustment, for a refracting telescope, the distance between objective and eye piece lens is 1.00 m. If the magnifying power of the telescope is 19, find the focal length of the objective and the eyepiece lens. CBSE 2024 **2**

(b) In normal adjustment, for a refracting telescope, the distance between objective and eye piece lens is 1.00 m. If the magnifying power of the telescope is 19, find the focal length of the objective and the eyepiece lens. CBSE 2024 **2**

Distance between objective and eyepiece $f_o + f_e = 1.00\ m = 100\ cm$	½
Magnifying power $\lvert m \rvert = \frac{f_o}{f_e} = 19$	½
On solving $f_o = 95\ cm = 0.95\ m$	½
$f_e = 5\ cm = 0.05\ m$	½

19. A ray of light is incident normally on one face of an equilateral glass prism of refractive index μ. When the prism is completely immersed in a transparent medium, it is observed that the emergent ray just grazes the adjacent face. Find the refractive index of the medium. CBSE 2024 **2**

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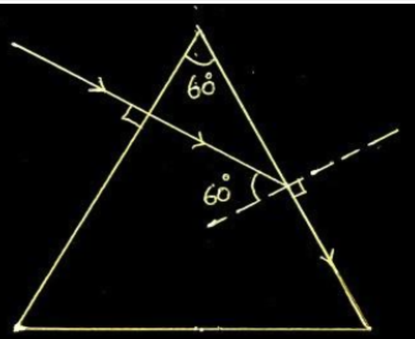

½

From snell's law, $\mu.\sin i = \mu_m.\sin r$ ½

$\mu.\sin 60^0 = \mu_m.\sin 90^0$ ½

$\mu_m = \mu.\frac{\sqrt{3}}{2}$ ½

Alternatively

$\mu_{ga} = \frac{1}{\sin C}$ 1

$\frac{\mu}{\mu_m} = \frac{1}{\sin 60^0}$ ½

$\mu_m = \frac{\sqrt{3}}{2}\mu$ ½

30. A lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical. Applying the formula of image formation by a single spherical surface successively at the two surfaces of a thin lens, a formula known as lens maker's formula and hence the basic lens formula can be obtained. The focal length (or power) of a lens depends on the radii of its surfaces and the refractive index of its material with respect to the surrounding medium. The refractive index of a material depends on the wavelength of light used. Combination of lenses helps us to obtain diverging or converging lenses of desired power and magnification.

CBSE 2024 $4 \times 1 = 4$

(i) A thin converging lens of focal length 20 cm and a thin diverging lens of focal length 15 cm are placed coaxially in contact. The power of the combination is

(A) $\frac{-5}{6}$ D

(B) $\frac{-5}{3}$ D

(C) $\frac{4}{3}$ D

(D) $\frac{3}{2}$ D

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(C) $\frac{4}{3}$ D

(D) $\frac{3}{2}$ D

(B) $\frac{-5}{3} D$

(ii) The radii of curvature of two surfaces of a convex lens are R and 2R. If the focal length of this lens is $\left(\frac{4}{3}\right)$R, the refractive index of the material of the lens is :

(A) $\frac{5}{3}$

(B) $\frac{4}{3}$

(C) $\frac{3}{2}$

(D) $\frac{7}{5}$

(ii) The radii of curvature of two surfaces of a convex lens are R and 2R. If the focal length of this lens is $\left(\frac{4}{3}\right)$R, the refractive index of the material of the lens is :

(A) $\frac{5}{3}$ (B) $\frac{4}{3}$

(C) $\frac{3}{2}$ (D) $\frac{7}{5}$

(C) $\frac{3}{2}$

(iii) The focal length of an equiconvex lens

(A) increases when the lens is dipped in water.

(B) increases when the wavelength of incident light decreases.

(C) increases with decrease in radius of curvature of its surface.

(D) decreases when the lens is cut into two identical parts along its principal axis.

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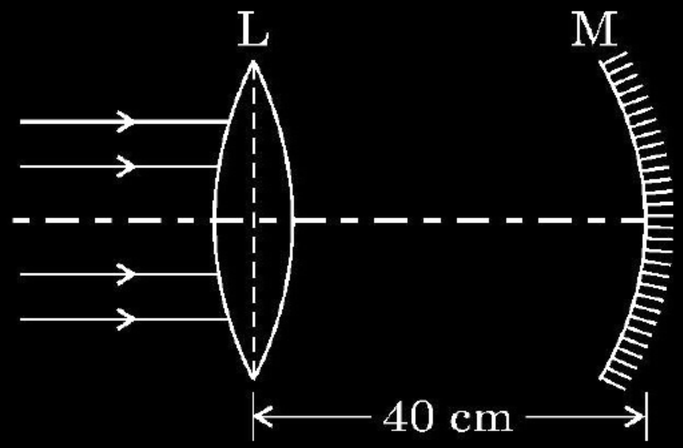
(D) decreases when the lens is cut into two identical parts along its principal axis.

(A) increases when a lens is dipped in water.

(iv) (a) A thin convex lens L of focal length 10 cm and a concave mirror M of focal length 15 cm are placed coaxially 40 cm apart as shown in figure. A beam of light coming parallel to the principal axis is incident on the lens. The final image will be formed at a distance of



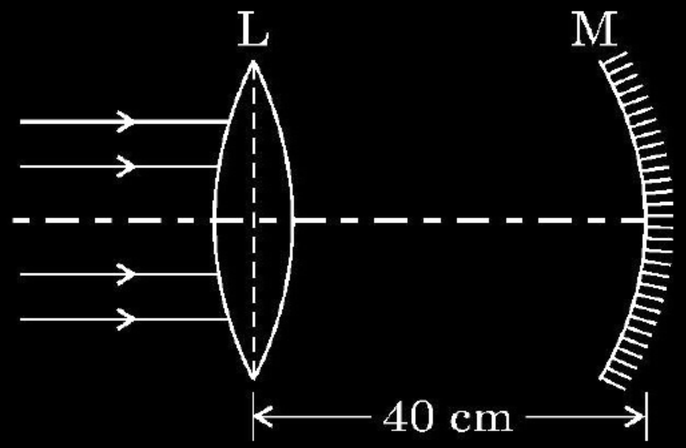
(A) 10 cm, left of lens

(B) 10 cm, right of lens

(C) 20 cm, left of lens

(D) 20 cm, right of lens

(iv) (a) A thin convex lens L of focal length 10 cm and a concave mirror M of focal length 15 cm are placed coaxially 40 cm apart as shown in figure. A beam of light coming parallel to the principal axis is incident on the lens. The final image will be formed at a distance of



(A) 10 cm, left of lens
(B) 10 cm, right of lens
(C) 20 cm, left of lens
(D) 20 cm, right of lens

(B) 10 cm , right from lens.

(iv) (b) A beam of light coming parallel to the principal axis of a convex lens L_1 of focal length 16 cm is incident on it. Another convex lens L_2 of focal length 12 cm is placed coaxially at a distance 40 cm from L_1. The nature and distance of the final image from L_2 will be

(A) real, 24 cm

(B) virtual, 12 cm

(C) real, 32 cm

(D) virtual, 18 cm

(iv) (b) A beam of light coming parallel to the principal axis of a convex lens L_1 of focal length 16 cm is incident on it. Another convex lens L_2 of focal length 12 cm is placed coaxially at a distance 40 cm from L_1. The nature and distance of the final image from L_2 will be

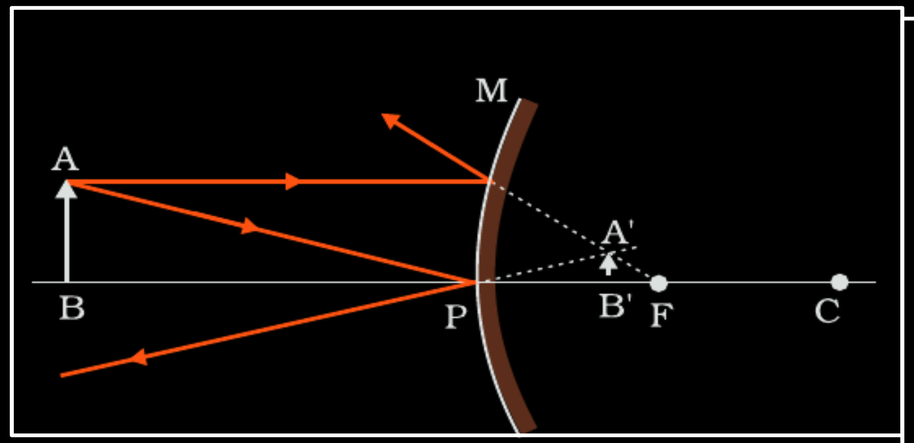
(A) real, 24 cm
(B) virtual, 12 cm
(C) real, 32 cm
(D) virtual, 18 cm

(A) real , 24 cm

32. (a) (i) Draw a ray diagram for the formation of the image of an object by a convex mirror. Hence, obtain the mirror equation.

(ii) Why are multi-component lenses used for both the objective and the eyepiece in optical instruments ?

(iii) The magnification of a small object produced by a compound microscope is 200. The focal length of the eyepiece is 2 cm and the final image is formed at infinity. Find the magnification produced by the objective. CBSE 2024 **5**



For paraxial rays MP can be considered to be a straight line perpendicular to CP,

Therefore right angled triangles $A'B'F$ and MPF are similar

$$\frac{B'A'}{PM} = \frac{B'F}{FP}$$

1

Or $\frac{B'A'}{BA} = \frac{B'F}{FP}$ ($\because$ PM = AB) ----------(1) ½

Since $\angle$ APB = $\angle$ $A'PB'$, the right angled triangles $A'PB'$ and ABP are also similar

Therefore , $\frac{B'A'}{BA} = \frac{B'P}{BP}$ ---------- (2) ½

Comparing eq (1) and (2), we get

$$\frac{B'F}{FP} = \frac{B'P}{BP}$$

$$\frac{PF - PB'}{FP} = \frac{B'P}{BP}$$

Using sign convention

PF = f, PB' = +v , PB = -u ½

on solving $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ ½

ii) To improve image quality by minimizing various optical aberrations in lenses. 1

iii) Magnification produced by compound microscope

$$m = m_o \times m_e$$ ½

$$m_o = \frac{m}{m_e} = \frac{m}{\left|\frac{D}{fe}\right|}$$

16. *Assertion (A) :* A convex lens, when immersed in a liquid, disappears.

Reason (R) : The refractive indices of material of the lens and the liquid are equal.

CBSE 2024

16. *Assertion (A) :* A convex lens, when immersed in a liquid, disappears.

Reason (R) : The refractive indices of material of the lens and the liquid are equal.

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(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

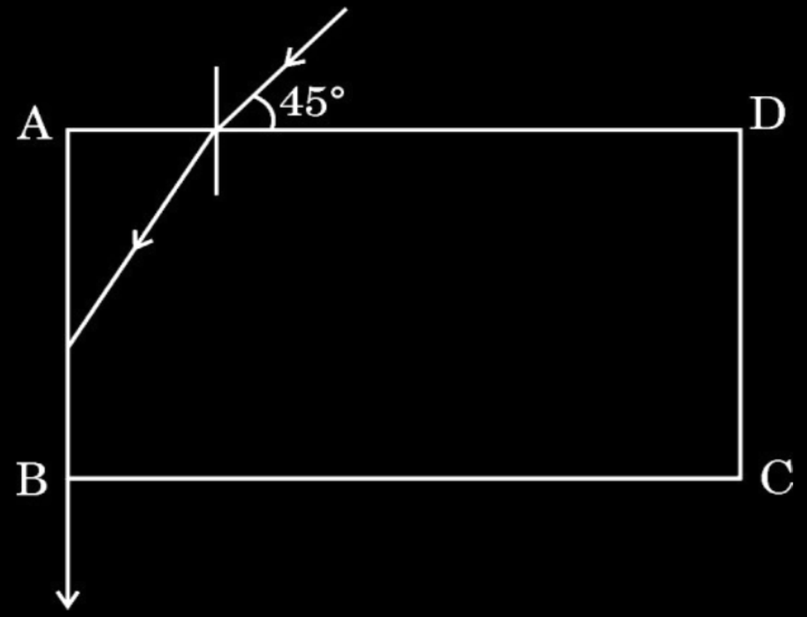
18. The magnifying power of an astronomical telescope is 24. In normal adjustment, distance between its two lenses is 150 cm. Find the focal length of the objective lens. CBSE 2024 *2*

18. The magnifying power of an astronomical telescope is 24. In normal adjustment, distance between its two lenses is 150 cm. Find the focal length of the objective lens. CBSE 2024 *2*

Magnifying power =24 , Distance between lenses =150 cm	
$\frac{f_o}{f_e} = 24$	½
$f_o + f_e = 150\,\text{cm}$	½
$f_e = 6\,\text{cm}$	½
$f_o = 144\,\text{cm}$	½

33. (a) (i) Trace the path of a ray of light showing refraction through a triangular prism and hence obtain an expression for angle of deviation (δ) in terms of A, i and e, where symbols have their usual meanings. Draw a graph showing the variation of angle of deviation with the angle of incidence.

(ii) In the figure, a ray of light is incident on a transparent liquid contained in a thin glass box at an angle of 45° with its one face. The emergent ray passes along the face AB. Find the refractive index of the liquid. *5*



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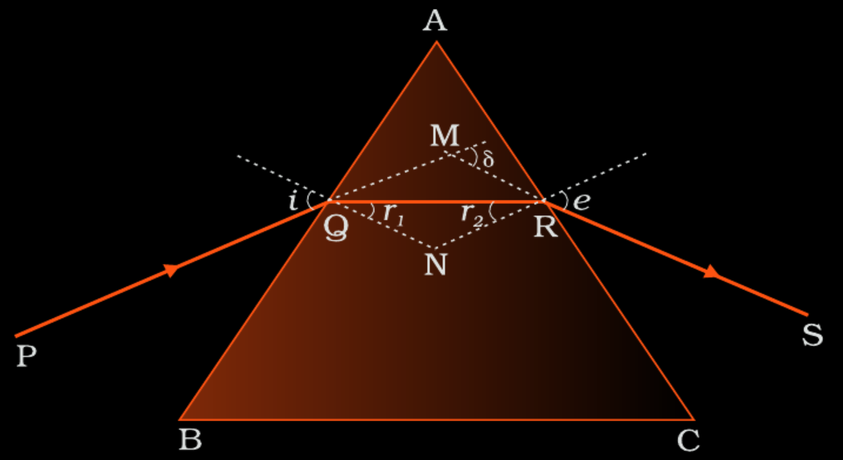
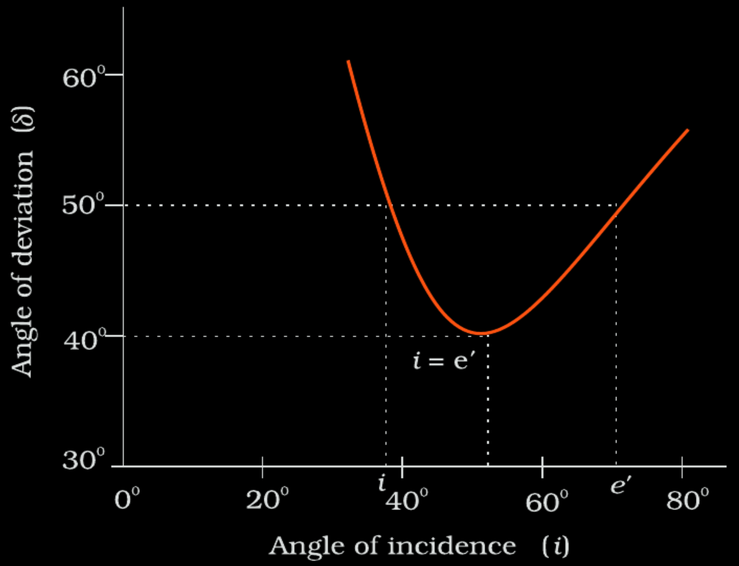




½

1

For quadrilateral AQNR,

$\angle A + \angle QNR = 180^o$ --- (i)

½

For triangle QNR

$r_1 + r_2 + \angle QNR = 180^o$ ---- (ii)

comparing equation (i) and (ii)

$r_1 + r_2 = A$ ------ (iii)

½

The angle of deviation

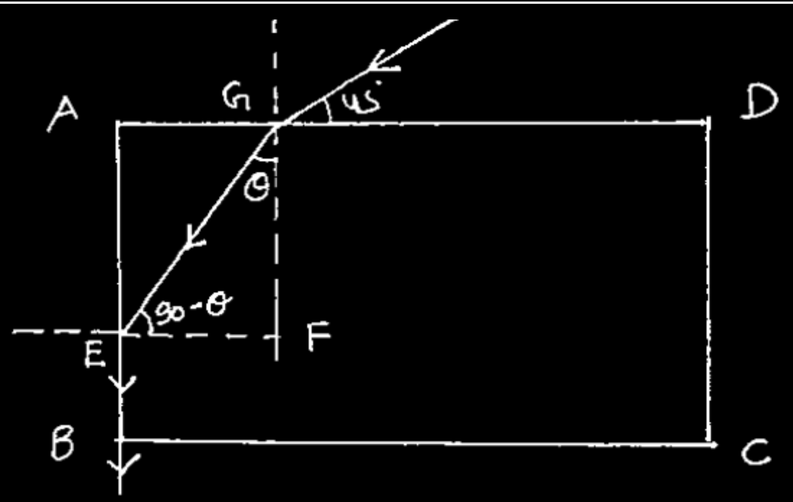
$\delta = (i - r_1) + (e - r_2)$ ------ (iv)

from equation (iii) and (iv)

$\delta = i + e - A$

½

(ii)



$$\frac{\sin 45^0}{\sin \theta} = \mu$$

$$\frac{1}{\sqrt{2}} = \mu \sin \theta$$

½

For second surface,

$$\frac{1}{\sqrt{2}} \frac{\cos \theta}{\sin \theta} = 1$$

½

$$\tan \theta = \frac{1}{\sqrt{2}}$$

½

From the triangle GEF

$$\sin \theta = \frac{1}{\sqrt{3}}$$

$$\mu = \sqrt{\frac{3}{2}}$$

½

19. Explain the following : *2*

(a) For a simple microscope, the angular size of the object equals the angular size of the image. Yet it offers magnification.

(b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ? CBSE 2024

19. Explain the following : *2*

(a) For a simple microscope, the angular size of the object equals the angular size of the image. Yet it offers magnification.

(b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ? CBSE 2024

(a) Yes, it offers magnification. We can keep the small object much closer to the eye than 25 cm and hence have it subtend a large angle. (b) Yes, Rays converging to a point behind a plane or convex mirror are reflected to a point in front of the mirror on a screen	½ ½ ½ ½

14. *Assertion (A) :* The magnifying power of a compound microscope is negative.

Reason (R) : The final image formed is erect with respect to the object.

CBSE 2024

14. *Assertion (A) :* The magnifying power of a compound microscope is negative.

Reason (R) : The final image formed is erect with respect to the object.

CBSE 2024

(C) Assertion(A) is true, but Reason (R) is false

21. A convex lens (n = 1·52) has a focal length of 15·0 cm in air. Find its focal length when it is immersed in liquid of refractive index 1·65. What will be the nature of the lens ?

CBSE 2024

2

21. A convex lens (n = 1·52) has a focal length of 15·0 cm in air. Find its focal length when it is immersed in liquid of refractive index 1·65. What will be the nature of the lens ? CBSE 2024 *2*

For convex lens in air

$$\frac{1}{f_a} = \left(\frac{n_g}{n_a} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For convex lens in liquid.

$$\frac{1}{f_l} = \left(\frac{n_g}{n_l} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ ½

$$\frac{f_l}{f_a} = \frac{\frac{1.52-1}{1}}{\frac{1.52-1.65}{1.65}}$$ ½

$= -6.6$

$f_l = -6.6\, f_a$

$= -99\text{cm}$ ½

Nature of the lens: Diverging/ behaves like a concave lens. ½

29. When a ray of light propagates from a denser medium to a rarer medium, it bends away from the normal. When the incident angle is increased, the refracted ray deviates more from the normal. For a particular angle of incidence in the denser medium, the refracted ray just grazes the interface of the two surfaces. This angle of incidence is called the critical angle for the pair of media involved.

CBSE 2024

(i) For a ray incident at the critical angle, the angle of reflection is : *1*

(A) $0°$ (B) $< 90°$

(C) $> 90°$ (D) $90°$

29. When a ray of light propagates from a denser medium to a rarer medium, it bends away from the normal. When the incident angle is increased, the refracted ray deviates more from the normal. For a particular angle of incidence in the denser medium, the refracted ray just grazes the interface of the two surfaces. This angle of incidence is called the critical angle for the pair of media involved.

CBSE 2024

(i) For a ray incident at the critical angle, the angle of reflection is : *1*

(A) $0°$ (B) $< 90°$

(C) $> 90°$ (D) $90°$

Since no option is correct, award 1 mark to all students.

(ii) A ray of light of wavelength 600 nm is incident in water $\left(n=\frac{4}{3}\right)$ on the water-air interface at an angle less than the critical angle. The wavelength associated with the refracted ray is : *1*

(A) 400 nm (B) 450 nm

(C) 600 nm (D) 800 nm

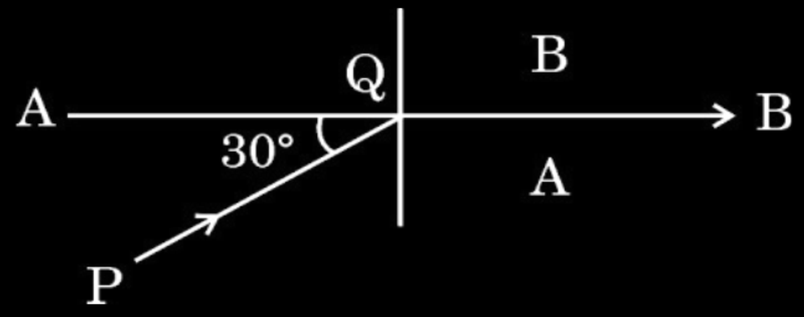
(ii) A ray of light of wavelength 600 nm is incident in water $\left(n=\frac{4}{3}\right)$ on the water-air interface at an angle less than the critical angle. The wavelength associated with the refracted ray is : *1*

(A) 400 nm (B) 450 nm

(C) 600 nm (D) 800 nm

(D) 800 nm

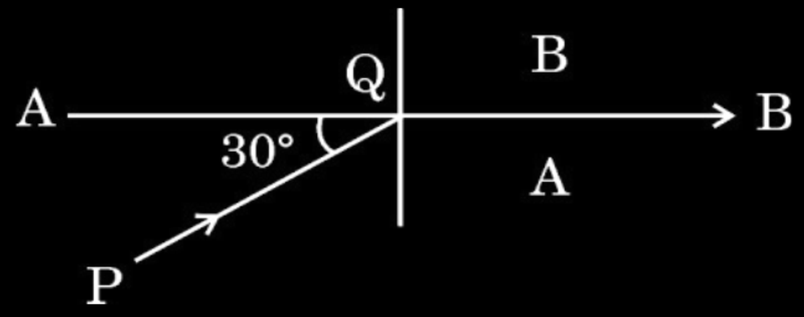
(iii) (a) The interface AB between the two media A and B is shown in the figure. In the denser medium A, the incident ray PQ makes an angle of 30° with the horizontal. The refracted ray is parallel to the interface. The refractive index of medium B w.r.t. medium A is : *1*



(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{5}}{2}$

(C) $\frac{4}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{3}}$

(iii) (a) The interface AB between the two media A and B is shown in the figure. In the denser medium A, the incident ray PQ makes an angle of 30° with the horizontal. The refracted ray is parallel to the interface. The refractive index of medium B w.r.t. medium A is : *1*



(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{5}}{2}$

(C) $\frac{4}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{3}}$

(A) $\frac{\sqrt{3}}{2}$

(b) Two media A and B are separated by a plane boundary. The speed of light in medium A and B is 2×10^8 ms^{-1} and $2{\cdot}5 \times 10^8$ ms^{-1} respectively. The critical angle for a ray of light going from medium A to medium B is : *1*

(A) $\sin^{-1} \frac{1}{2}$ (B) $\sin^{-1} \frac{4}{5}$

(C) $\sin^{-1} \frac{3}{5}$ (D) $\sin^{-1} \frac{2}{5}$

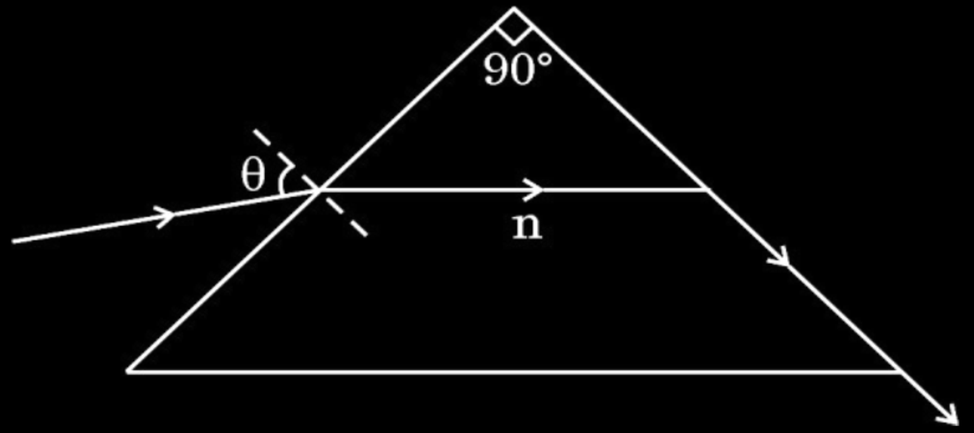
(b) Two media A and B are separated by a plane boundary. The speed of light in medium A and B is 2×10^8 ms^{-1} and $2{\cdot}5 \times 10^8$ ms^{-1} respectively. The critical angle for a ray of light going from medium A to medium B is : *1*

(A) $\sin^{-1} \frac{1}{2}$ (B) $\sin^{-1} \frac{4}{5}$

(C) $\sin^{-1} \frac{3}{5}$ (D) $\sin^{-1} \frac{2}{5}$

(B) $\sin^{-1}\left(\frac{4}{5}\right)$

(iv) The figure shows the path of a light ray through a triangular prism. In this phenomenon, the angle θ is given by : *1*



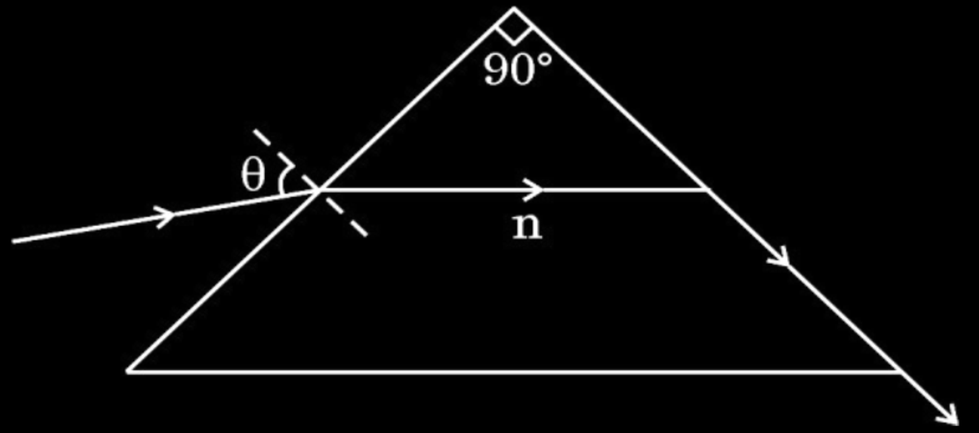
(A) $\sin^{-1}\sqrt{n^2-1}$

(B) $\sin^{-1}(n^2-1)$

(C) $\sin^{-1}\left[\frac{1}{\sqrt{n^2-1}}\right]$

(D) $\sin^{-1}\left[\frac{1}{(n^2-1)}\right]$

(iv) The figure shows the path of a light ray through a triangular prism. In this phenomenon, the angle θ is given by : *1*



(A) $\sin^{-1}\sqrt{n^2-1}$

(B) $\sin^{-1}(n^2-1)$

(C) $\sin^{-1}\left[\frac{1}{\sqrt{n^2-1}}\right]$

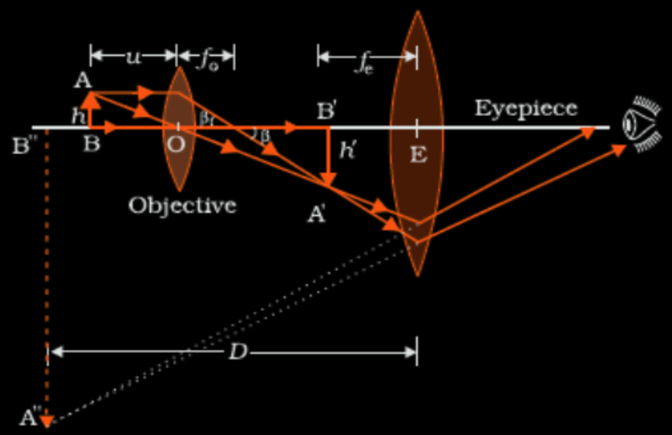
(D) $\sin^{-1}\left[\frac{1}{(n^2-1)}\right]$

(A) $\sin^{-1}\sqrt{n^2-1}$

(b) (i) Draw a labelled ray diagram of a compound microscope showing image formation at least distance of distinct vision. Derive an expression for its magnifying power.

(ii) A telescope consists of two lenses of focal length 100 cm and 5 cm. Find the magnifying power when the final image is formed at infinity. CBSE 2024 *5*

(i)



1

The magnification obtained by eye-piece lens $m_e = \left(1 + \frac{D}{f_e}\right)$ ½

The magnification obtained by objective lens $m_0 = \frac{v_0}{-u_0}$ ½

Hence the total magnifying power is

$m = m_0 \times m_e$ ½

$= \frac{v_0}{-u_0}\left(1 + \frac{D}{f_e}\right)$ ½

$(ii)\ \mathrm{m} = \left|\frac{f_0}{f_e}\right|$ 1

Identification of focal length of objective and eyepiece

$f_0 = 100cm$

$f_e = 5cm$ ½

$m = \left|\frac{100}{5}\right| = 20$ ½

14. *Assertion (A) :* Plane and convex mirrors cannot produce real images under any circumstance.

Reason (R) : A virtual image cannot serve as an object to produce a real image.

CBSE 2024

14. *Assertion (A) :* Plane and convex mirrors cannot produce real images under any circumstance.

Reason (R) : A virtual image cannot serve as an object to produce a real image.

CBSE 2024

(D) Both Assertion (A) and Reason (R) are false.

18. An object is placed 30 cm in front of a concave mirror of radius of curvature 40 cm. Find the (i) position of the image formed and (ii) magnification of the image. CBSE 2024 *2*

18. An object is placed 30 cm in front of a concave mirror of radius of curvature 40 cm. Find the (i) position of the image formed and (ii) magnification of the image. CBSE 2024 *2*

(i) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} + \frac{1}{-30} = \frac{1}{-20}$	½
On solving v = - 60 cm	½
(ii) m = $-\frac{v}{u}$	½
$= -(\frac{-60}{-30}) = -2$	½

(b) A plano-convex lens of focal length 16 cm is made of a material of refractive index 1·4. Calculate the radius of the curved surface of the lens. CBSE 2024 *2*

(b) A plano-convex lens of focal length 16 cm is made of a material of refractive index 1·4. Calculate the radius of the curved surface of the lens. CBSE 2024 *2*

$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ $\frac{1}{16} = (1.4 - 1)\left(\frac{1}{R} - \frac{1}{\infty}\right)$ $\frac{1}{16} = 0.4 \times \frac{1}{R}$ $R = 16 \times 0.4$ $R = 6.4$ cm	1 1

29. A lens is a transparent medium bounded by two surfaces, with one or both surfaces being spherical. The focal length of a lens is determined by the radii of curvature of its two surfaces and the refractive index of its medium with respect to that of the surrounding medium. The power of a lens is reciprocal of its focal length. If a number of lenses are kept in contact, the power of the combination is the algebraic sum of the powers of the individual lenses.

CBSE 2024

(i) A double-convex lens, with each face having same radius of curvature R, is made of glass of refractive index n. Its power is : *1*

(A) $\frac{2\,(n-1)}{R}$

(B) $\frac{(2n-1)}{R}$

(C) $\frac{(n-1)}{2R}$

(D) $\frac{(2n-1)}{2R}$

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CBSE 2024

(i) A double-convex lens, with each face having same radius of curvature R, is made of glass of refractive index n. Its power is : *1*

(A) $\frac{2\,(n-1)}{R}$ (B) $\frac{(2n-1)}{R}$

(C) $\frac{(n-1)}{2R}$ (D) $\frac{(2n-1)}{2R}$

(A) $\frac{2(n-1)}{R}$

(ii) A double-convex lens of power P, with each face having same radius of curvature, is cut into two equal parts perpendicular to its principal axis. The power of one part of the lens will be : *1*

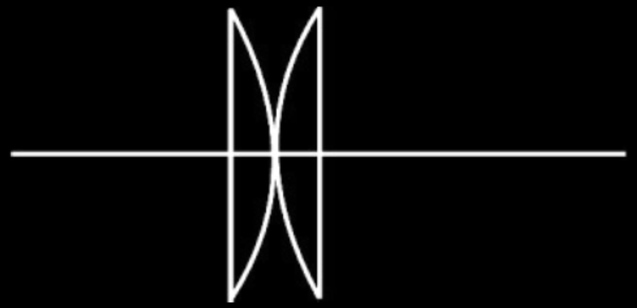
(A) 2P (B) P (C) 4P (D) $\frac{P}{2}$

(ii) A double-convex lens of power P, with each face having same radius of curvature, is cut into two equal parts perpendicular to its principal axis. The power of one part of the lens will be : *1*

(A) 2P (B) P (C) 4P (D) $\frac{P}{2}$

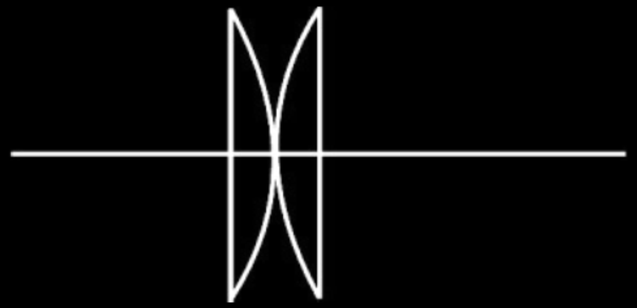
(D) P/2

(iii) The above two parts are kept in contact with each other as shown in the figure. The power of the combination will be : *1*



(A) $\frac{P}{2}$ (B) P (C) 2P (D) $\frac{P}{4}$

(iii) The above two parts are kept in contact with each other as shown in the figure. The power of the combination will be : *1*



(A) $\frac{P}{2}$ (B) P (C) 2P (D) $\frac{P}{4}$

(B) P

(iv) (a) A double-convex lens of power P, with each face having same radius of curvature, is cut along its principal axis. The two parts are arranged as shown in the figure. The power of the combination will be : *1*

(A) Zero

(B) P

(C) 2P

(D) $\frac{P}{2}$

(iv) (a) A double-convex lens of power P, with each face having same radius of curvature, is cut along its principal axis. The two parts are arranged as shown in the figure. The power of the combination will be : *1*

(A) Zero (B) P

(C) 2P (D) $\frac{P}{2}$

(C) 2P

(b) Two convex lenses of focal lengths 60 cm and 20 cm are held coaxially in contact with each other. The power of the combination is : *1*

(A) 6·6 D

(B) 15 D

(C) $\frac{1}{15}$ D

(D) $\frac{1}{80}$ D

(b) Two convex lenses of focal lengths 60 cm and 20 cm are held coaxially in contact with each other. The power of the combination is : *1*

(A) 6·6 D (B) 15 D

(C) $\frac{1}{15}$ D (D) $\frac{1}{80}$ D

(A) 6.6 D

33. (a) (i) A ray of light passes through a triangular prism. Show graphically, how the angle of deviation varies with the angle of incidence ? Hence define the angle of minimum deviation.

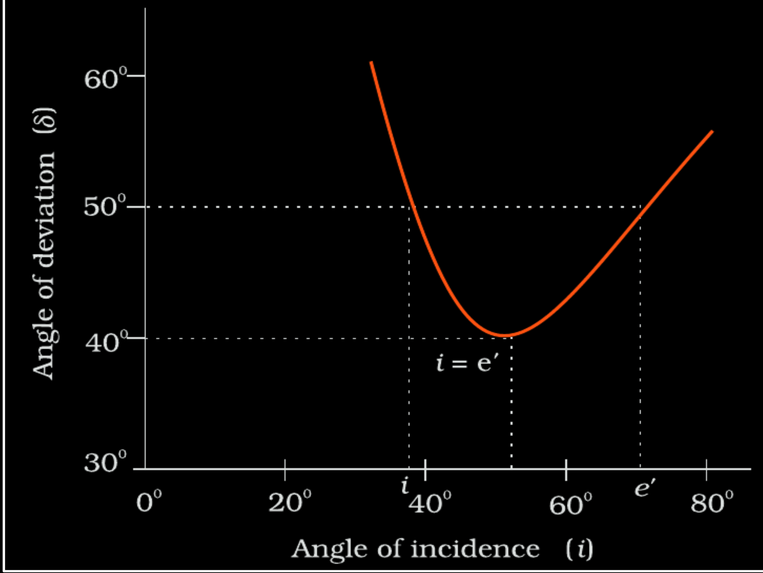
(ii) A ray of light is incident normally on a refracting face of a prism of prism angle A and suffers a deviation of angle δ. Prove that the refractive index n of the material of the prism is given by $n = \frac{\sin (A + \delta)}{\sin A}$.

(iii) The refractive index of the material of a prism is $\sqrt{2}$. If the refracting angle of the prism is 60°, find the

(1) Angle of minimum deviation, and

(2) Angle of incidence.

CBSE 2024



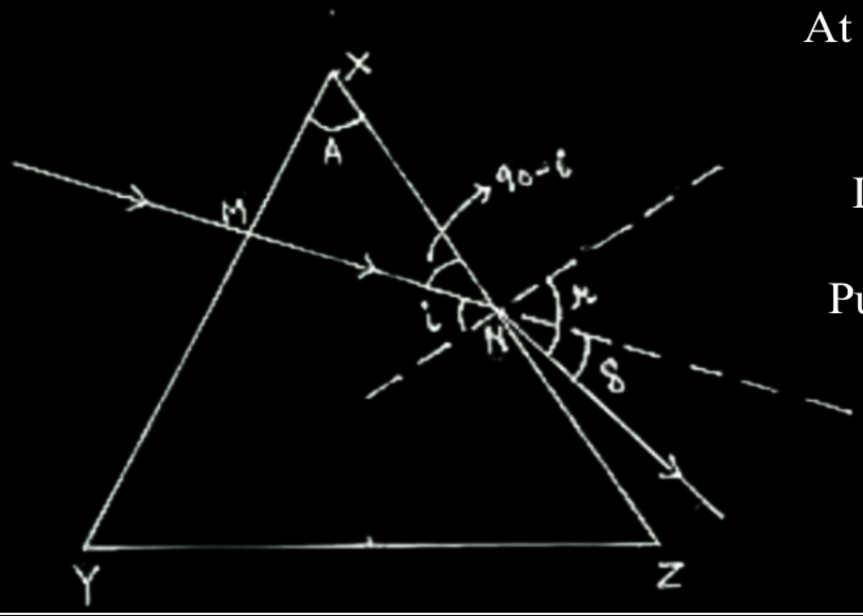
Minimum deviation angle is defined as the angle at which angle of incidence is equal to the angle of emergence.

Alternatively

At minimum deviation refracted ray inside the prism becomes parallel to the base of the prism.

1

1



At the face XZ :-

$\mu \sin i = 1 \times \sin r$ ----- (1)

$r = i + \delta$ [from diagram] ----- (2)

In ΔXMN ; $A + (90 - i) + 90 = 180$

$\Rightarrow A = i$ ----- (3)

Putting eq. (3) & (2) in eq. (1)

$\mu \sin A = \sin (A + \delta)$

$$\mu = \frac{\sin (A + \delta)}{\sin A}$$

½

½

(iii)

(1) $$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$$

$$\sqrt{2} = \frac{\sin\left(\frac{60+\delta_m}{2}\right)}{\sin 30^\circ}$$ ½

$$\Rightarrow \sin\left(\frac{60+\delta_m}{2}\right) = \frac{1}{\sqrt{2}} = \sin 45^\circ$$

$$\frac{60+\delta_m}{2} = 45^\circ \Rightarrow \delta_m = 30^\circ$$ ½

(2) $$i = \frac{A+\delta_m}{2}$$ ½

$$\Rightarrow i = \frac{60+30}{2}$$

$$i = 45^\circ$$ ½

16. *Assertion (A) :* Signals through an optical fibre can be transmitted without any appreciable loss of energy.

Reason (R) : An optical fibre works on the principle of total internal reflection.

CBSE 2024

16. *Assertion (A) :* Signals through an optical fibre can be transmitted without any appreciable loss of energy.

Reason (R) : An optical fibre works on the principle of total internal reflection.

CBSE 2024

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

18. (a) Name the factors on which focal length of a lens depends. How will a thin biconvex glass lens submerged in water behave ? Give reason for your answer. CBSE 2024 *2*

18. (a) Name the factors on which focal length of a lens depends. How will a thin biconvex glass lens submerged in water behave ? Give reason for your answer. CBSE 2024 *2*

Two factors: Radius of curvature Refractive index of the material of the lens w.r.t. the surroundings	½+ ½
Nature of the lens will not change but the focal length will increase	½
Reason: Focal length of a lens is inversely proportional to the refractive index of the material of lens w.r.t. the surroundings.	½

19. The power of the combination of two thin convex lenses L_1 and L_2, kept coaxially in contact, is +10 D. If the focal length of L_1 is four times that of L_2, find their focal lengths. CBSE 2024 *2*

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$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$	½
$f_1 = 4f_2$ (given)	
$P = \frac{1}{f} = \frac{1}{4f_2} + \frac{1}{f_2}$	½
$10 = \frac{5}{4f_2}$	
$f_2 = \frac{50}{4}\,cm$	½
$\therefore f_1 = 50\,cm$	½

29. In a homogeneous medium, light travels along a straight line path. But when a ray of light travels from one medium to another, it deviates from its initial path. This bending of light is called refraction of light. The extent to which a ray deviates in medium 2 from its initial path in medium 1, is determined by the refractive index of medium 2 with respect to medium 1. The refractive index is also related to the ratio of speeds of light in the two media. CBSE 2024

The phenomenon of total internal reflection occurs when light passes from a denser medium into a rarer medium and the angle of incidence exceeds the critical angle for the given pair of media.

(i) A ray of monochromatic light of wavelength λ and speed v travels from air into a glass slab. Inside the glass slab, *1*

(A) both λ and v decrease

(B) both λ and v increase

(C) λ decreases and v increases

(D) λ increases and v decreases

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CBSE 2024

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(A) both λ and v decrease

(B) both λ and v increase

(C) λ decreases and v increases

(D) λ increases and v decreases

(A) both λ & v decrease

(ii) (a) A ray of light travelling from air into a medium, is incident on the interface at an angle of 45°. If the ray deviates by 15° from its initial path, the refractive index of the medium is : *1*

(A) 2 (B) $\sqrt{2}$

(C) $2\sqrt{2}$ (D) $\sqrt{3}$

(ii) (a) A ray of light travelling from air into a medium, is incident on the interface at an angle of 45°. If the ray deviates by 15° from its initial path, the refractive index of the medium is : *1*

(A) 2 (B) $\sqrt{2}$

(C) $2\sqrt{2}$ (D) $\sqrt{3}$

(B) $\sqrt{2}$

(ii) (b) A point source of light is kept at the bottom of a tank filled with a transparent liquid up to a height H. If the radius of the circular patch on the surface from which light can come out is $\frac{H}{\sqrt{3}}$, the refractive index of the liquid is close to : *1*

(A) 1·3 (B) 1·5

(C) 1·7 (D) 2·0

(ii) (b) A point source of light is kept at the bottom of a tank filled with a transparent liquid up to a height H. If the radius of the circular patch on the surface from which light can come out is $\frac{H}{\sqrt{3}}$, the refractive index of the liquid is close to : *1*

(A) 1·3 (B) 1·5

(C) 1·7 (D) 2·0

(D) 2

(iii) In an optical fiber, the refractive indices of the core and cladding are μ_1 and μ_2 respectively. Then : *1*

(A) $\mu_1 = \mu_2$

(B) $\mu_1 < \mu_2$

(C) μ_1 is slightly greater than μ_2

(D) μ_1 is large compared to μ_2

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(A) $\mu_1 = \mu_2$

(B) $\mu_1 < \mu_2$

(C) μ_1 is slightly greater than μ_2

(D) μ_1 is large compared to μ_2

(C) μ_1 is slightly greater than μ_2

(iv) Light of wavelength λ travelling with speed c in air, enters a medium of refractive index μ. The frequency of light in the medium is : *1*

(A) $\frac{c}{\mu\lambda}$

(B) $\frac{\mu c}{\lambda}$

(C) $\frac{\mu\lambda}{c}$

(D) $\frac{c}{\lambda}$

(iv) Light of wavelength λ travelling with speed c in air, enters a medium of refractive index μ. The frequency of light in the medium is : *1*

(A) $\frac{c}{\mu\lambda}$

(B) $\frac{\mu c}{\lambda}$

(C) $\frac{\mu\lambda}{c}$

(D) $\frac{c}{\lambda}$

(D) $\frac{c}{\lambda}$

(ii) An object is kept 20 cm from a convex mirror of radius of curvature 60 cm. Find the position of the image formed. Will the image be real or virtual ?

CBSE 2024

(ii) An object is kept 20 cm from a convex mirror of radius of curvature 60 cm. Find the position of the image formed. Will the image be real or virtual ?

CBSE 2024

(ii) u= -20cm

$$f = \frac{R}{2} = 30\text{cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \left(-\frac{1}{20}\right) = \frac{1}{30}$$ ½

$$\frac{1}{v} = \frac{1}{30} + \frac{1}{20} = \frac{3+2}{6} = \frac{5}{60}$$ ½

$$v = 12\,\text{cm}$$ ½

Virtual Image. ½

(i) Answer the following giving reasons : CBSE 2024

(1) In modern microscopes, multicomponent lenses are used for both the objective and the eyepiece.

(2) In a compound microscope, both the objective and the eyepiece have small focal lengths.

(3) When viewing through a compound microscope, our eyes should be positioned not on the eyepiece, but a short distance away from it for best viewing.

(i) (1)To improve image quality by minimizing various optical aberrations in lenses. (2) To achieve large magnification Alternatively: Magnifying Power $m = \left(\frac{L}{f_0}\right)\left(\frac{D}{f_e}\right)$ (3) Because of this position our eye will collect all the light refracted by the objective which results in relaxation of the eye. Alternatively: To increase the field of view & for relaxation of eye.	1 1 1

22. An equiconvex lens of focal length f is cut into two equal halves perpendicular to its principal axis. Find the focal length of each half of the lens. CBSE 2023 C *2*

22. An equiconvex lens of focal length f is cut into two equal halves perpendicular to its principal axis. Find the focal length of each half of the lens. CBSE 2023 C *2*

$\frac{1}{f} = (n-1)(\frac{1}{R_1} - \frac{1}{R_2})$ ½

For equiconvex lens $R_1 = R$, $R_2 = -R$

$\frac{1}{f} = (n-1)(\frac{1}{R} + \frac{1}{R})$

$\frac{1}{f} = (n-1)(\frac{2}{R})$ _____(i) ½

After cutting in two parts

$R_1 = R \qquad R_2 = \infty$

$\frac{1}{f'} = (n-1)(\frac{1}{R_1} + \frac{1}{\infty})$

$\frac{1}{f'} = (n-1)(\frac{1}{R})$ _____(ii) ½

From (i) and (ii)

$f' = 2f$ ½

(b) (i) Name the factors on which the focal length of a lens depends.

(ii) A converging lens is immersed in a transparent medium. Explain the condition under which the lens (1) becomes invisible and (2) behaves as a diverging lens.

(iii) Is it possible for two lenses in contact to have zero power ? Explain.

CBSE 2023 C

5

(i) Radius of curvature of two faces of lens Refractive index of lens with respect to surrounding	1
(ii) (1) If the refractive index of material of surrounding medium is equal to refractive index of lens	1
(2)If the refractive of surrounding medium is more than the refractive index of converging lens	1
(iii) yes	½
If one lens is converging and other is diverging, both having same power.	½

35. When a ray of light passes from a denser to a rarer medium, it bends away from the normal. If we go on increasing the angle of incidence in denser medium, then for a particular value of angle of incidence, the incident ray is refracted along the interface of the two media. This angle is called critical angle (i_c). If incident angle (i) is greater than critical angle (i_c), then the ray of light is reflected back in the same denser medium. This phenomenon is called total internal reflection of light.
Based on the above facts, answer the following questions :

(i) If the refractive index of the denser medium increases, how will the corresponding value of critical angle for the pair of mediums change ? *1*

(ii) Explain how this phenomenon can be used in a prism to deviate a ray of light by 90°. *1*

CBSE 2023 C

(iii) What is an optical fibre ? Write its two uses. *2*

OR

(iii) A ray of light travels from an optically denser medium to a rarer medium and is incident at critical angle (i_c). Find the angle of deviation for this ray. *2*

CBSE 2023 C

(i) $\sin i_c = \frac{1}{\mu}$. If μ of denser medium increases, critical angle (i_c) decreases.	1
(ii) For this we take isosceles right angled prism. We allow incident light to fall normally on one of the two mutually perpendicular faces. It will deviated by 90° after total internal reflection	1
(iii) Optical fibres are very fine fibres made of glass. Through these fibres optical signals can be transmitted without appreciable loss of energy, using the phenomena of total internal reflection. Uses: Endoscopy Communication	1 ½ + ½
OR	
(iii) Angle of deviation = $90^o - i_c$	2

33. (a) (i) Draw a ray diagram to show the formation of image by a concave mirror when the object is placed between f and 2f. Using this diagram, derive the mirror equation.

(ii) An object is kept 30 cm in front of a convex mirror of focal length 12 cm. Find the nature and position of the image formed. *5*

CBSE 2023 C

From $\triangle A'B'C' \sim \triangle DPF$

$$\frac{B'A'}{BA} = \frac{B'F}{FP} (\because PD = AB) \quad \text{---------(1)}$$

Form $\triangle A'B'P \sim \triangle ABP$ ½

$$\frac{B'A'}{BA} = \frac{B'P}{BP} \quad \text{---------(2)}$$

½

Comparing equation (1) and (2)

$$\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{BP}$$

Here B'P = - v

FP = -f

BP = - u

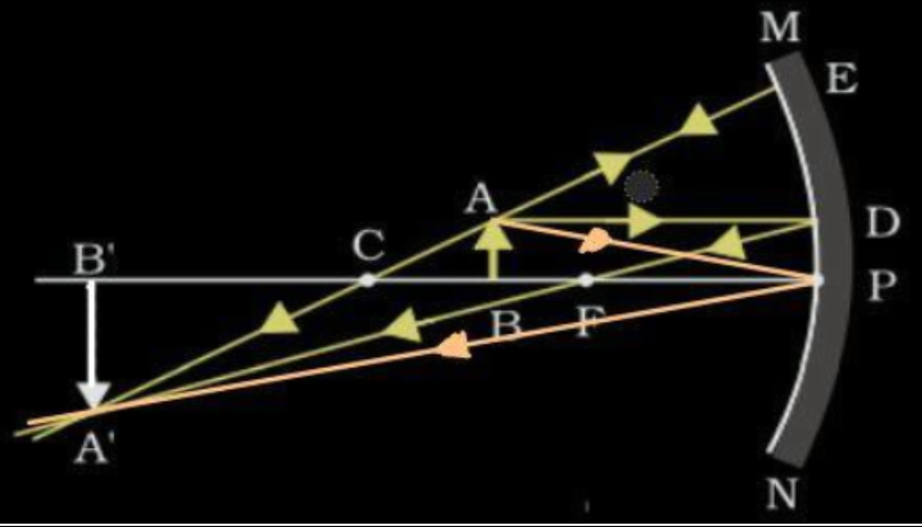
$$\frac{-v + f}{-f} = \frac{-v}{-u}$$ ½

$$\frac{v - f}{f} = \frac{v}{u}$$

$$\frac{v}{f} = 1 + \frac{v}{u}$$

On solving we get

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$ ½



(ii) u= -30 cm

f= + 12 cm

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{12} + \frac{1}{30} = \frac{7}{60}$$ 1

v= $\frac{60}{7}$ cm ½

Since v is positive image is virtual and erect.

(b) (i) Draw a ray diagram showing the image formation by a compound microscope. Hence obtain the expression for total magnification when the image is formed at least distance of distinct vision.

(ii) A compound microscope consists of an objective lens of focal length 2·0 cm and an eyepiece of focal length 6·0 cm. If they are separated by a distance of 24 cm, find the total magnification when the image is formed at infinity. *5*

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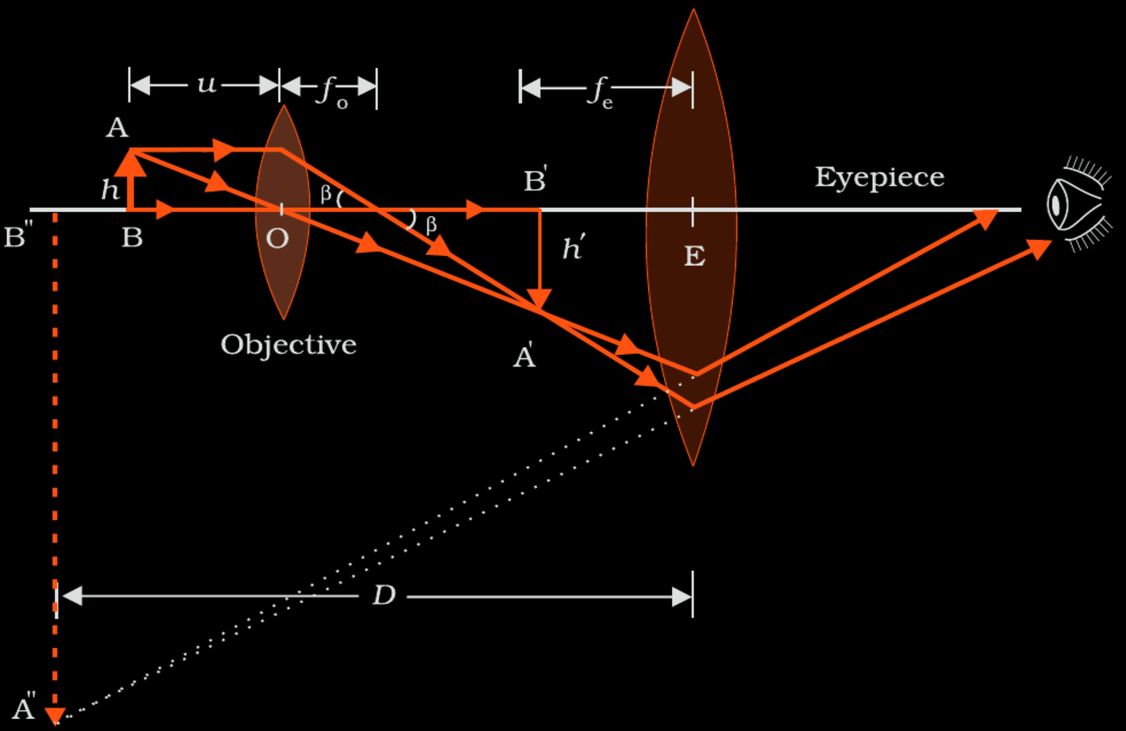
Magnification due to objective

$$m_0 = \frac{h'}{h} = \frac{L}{f_0} \qquad \left(\because \tan\beta = \frac{h}{f_0} = \frac{h'}{L} \right)$$

Magnification due to eye piece when final image is formed at the near point

$$m_e = 1 + \frac{D}{f_e}$$



Total magnification

$$m = m_0\, m_e$$

$$m = \frac{L}{f_0}\left(1 + \frac{D}{f_e}\right)$$

$$m = \frac{L}{f_0}\left(1 + \frac{D}{f_e}\right)$$

(ii) $$m = \frac{LD}{f_o f_e}$$

$$m = \frac{24 \times 25}{2 \times 6} = \frac{600}{12} = 50$$

½

½

½

½

1

1

35. A prism is a solid transparent medium bounded by three rectangular faces with triangular base and top. A ray of light incident at angle i on one face of a prism suffers two refractions on passing through a prism. Hence it deviates through a certain angle δ from its original path. The angle of deviation becomes minimum ($\delta = \delta_m$) for a certain value of angle i. In such a condition, the refracted ray inside the prism becomes parallel to its base. An expression for refractive index μ of the material of the prism can be obtained in terms of angle A and angle δ_m.

(a) Show in a figure the variation of angle δ with angle of incidence i. *1*

(b) Show that for a prism of small angle A, the refractive index μ of its material can be written as $\mu = 1 + \frac{\delta_m}{A}$. *1*

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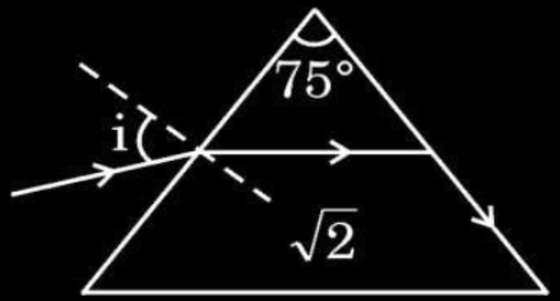
(c) A ray of light passes through an equilateral prism such that both the angle of incidence and the angle of emergence are equal to the angle of prism A. Find the refractive index of the material of the prism, in terms of A. *2*

OR

(c) A ray of light passes through a prism of angle 75°, as shown in the figure. The refractive index of the material of the prism, with respect to its surrounding is $\sqrt{2}$. Find the angle of incidence i. *2*



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(b)

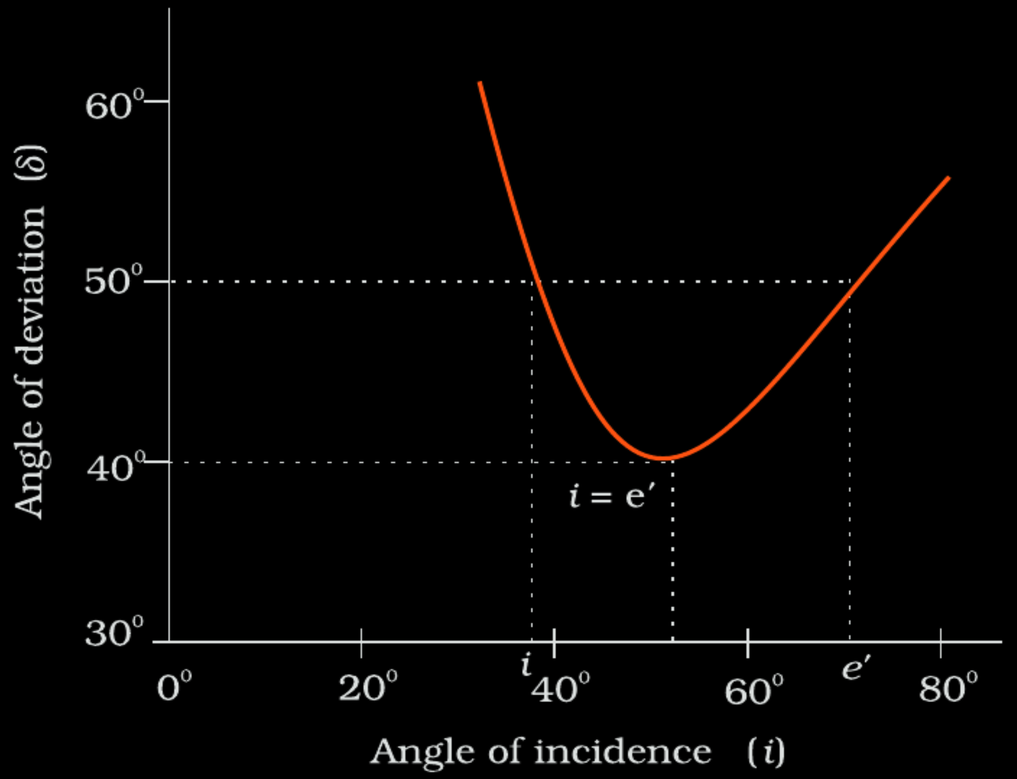
$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin A/2}$$

For small angle

$$\mu = \frac{\frac{A+\delta_m}{2}}{A/2}$$

$$\mu = \frac{A+\delta_m}{A}$$

$$\mu = 1+\frac{\delta_m}{A}$$



½

1

½

(c) $i+e=A$

$$r = \frac{A}{2}$$

$$\mu = \frac{\sin i}{\sin r}$$

$$\mu = \frac{\sin A}{\sin A/2}$$

$$\mu = \frac{2\sin(A/2)\cos(A/2)}{\sin(A/2)} = 2\cos(A/2)$$

½

½

½

½

$$\frac{\sin i}{\sin r} = \sqrt{2}$$

$$\frac{\sin r_2}{\sin 90^0} = \frac{1}{\sqrt{2}}$$ ½

$$\sin r_2 = \frac{1}{\sqrt{2}}$$

$$r_2 = 45^0$$ ½

$$r_1 + r_2 = A$$

$$r_1 + 45^0 = 75^0$$

$$r_1 = 30^0$$ ½

$$\frac{\sin i}{\sin r_1} = \sqrt{2}$$

$$\frac{\sin i}{\sin 30^0} = \sqrt{2}$$

$$\sin i = \sqrt{2} \times \frac{1}{2}$$ ½

$$\sin i = \frac{1}{\sqrt{2}}$$

$$i = 45^0$$

19. A bright point source is placed at the bottom of a tank containing a transparent liquid (refractive index μ) to a height H. Let r be the radius of the circular area of the liquid surface through which light can emerge out. Find the ratio (r/H).

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2

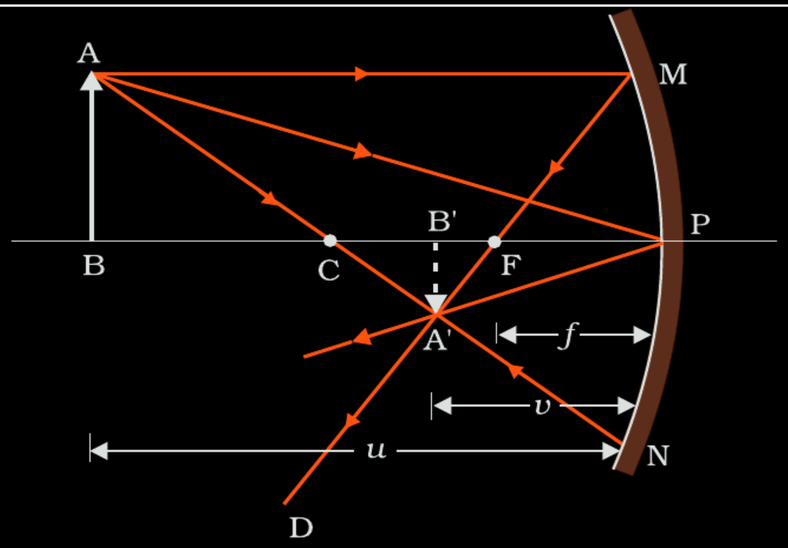
19. A bright point source is placed at the bottom of a tank containing a transparent liquid (refractive index μ) to a height H. Let r be the radius of the circular area of the liquid surface through which light can emerge out. Find the ratio (r/H). CBSE 2023 C *2*

$\text{Sin } i_c = \frac{1}{\mu}$	½
$\cos i_c = \sqrt{1-\sin^2 i} = \frac{\sqrt{\mu^2-1}}{\mu}$	½
$\tan i_c = \frac{1}{\sqrt{\mu^2-1}}$	½
$\frac{r}{H} = \frac{1}{\sqrt{\mu^2-1}}$	½

31. (a) Draw a ray diagram for formation of a real and diminished image of an object kept in front of a concave mirror. Hence derive the mirror equation.

(b) A concave mirror of focal length 10 cm produces a real image which is 3 times the size of the object. Find the distance of the object from the mirror. *5*

CBSE 2023 C



1

$\Delta A'B'F \sim \Delta MPF$

$\frac{B'A'}{BA} = \frac{B'F}{FP}$ $(\because PM = AB)$ _____(1) ½

Also, $\Delta A'B'P \sim \Delta ABP$

$\frac{B'A'}{BA} = \frac{B'P}{BP}$ _____(2) ½

Comparing eq. (1) and (2)

$\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{FP}$ ½

$B'P = -v$

$FP = -f$

$BP = -u$

On solving we get , $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ ½

(b)

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ ½

$\frac{1}{u} = \frac{1}{f} - \frac{1}{v}$ ½

$-\frac{v}{u} = -3$

$v = 3u$ ½

$\frac{1}{u} = \frac{1}{f} - \frac{1}{3u}$

$\frac{1}{u} + \frac{1}{3u} = \frac{1}{f}$

$\frac{4}{3u} = \frac{1}{f}$ ½

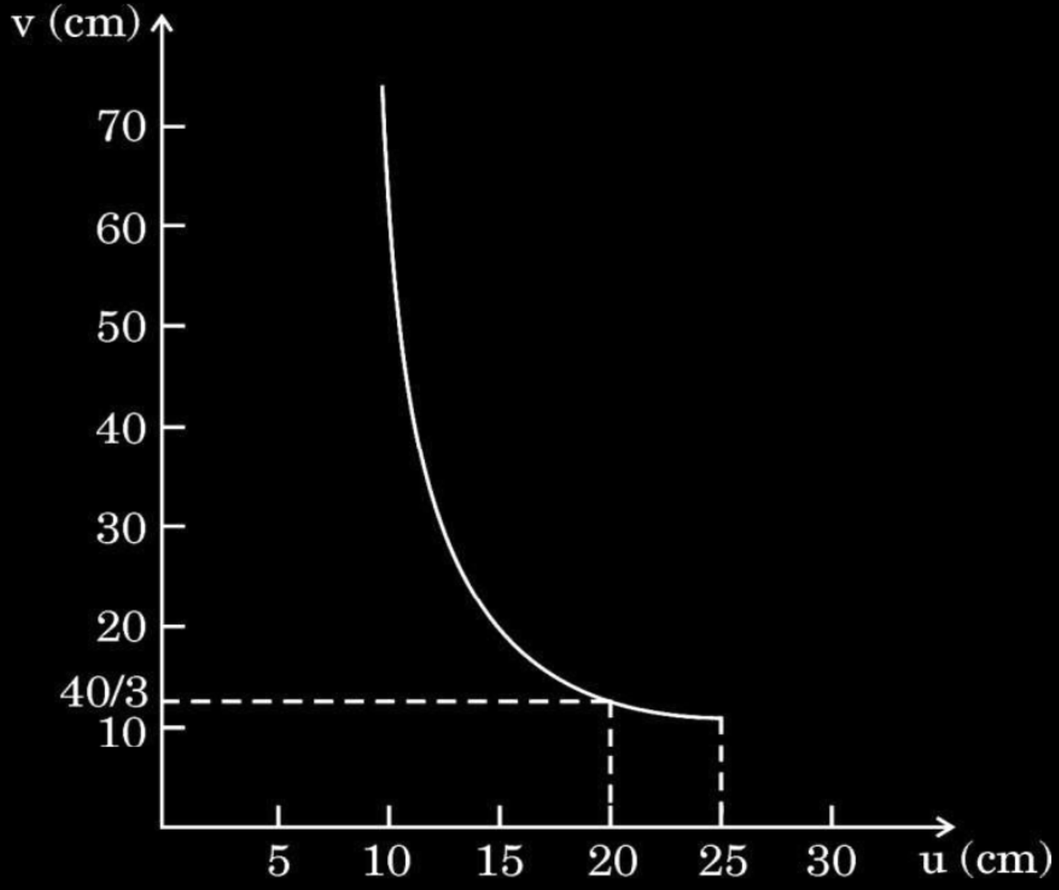
$u = \frac{4f}{3} = -\frac{40}{3} cm$ ½

(a) Two converging lenses of focal lengths f_1 and f_2 are placed coaxially in contact. Derive the expression for the focal length of the combination using the ray diagram for the formation of the image of an object by the combination.

(b) The figure shows the variation of image distance v with object distance u for a lens.

(i) What is the nature of the lens ?

(ii) Using the graph, find the focal length of the lens. *5*



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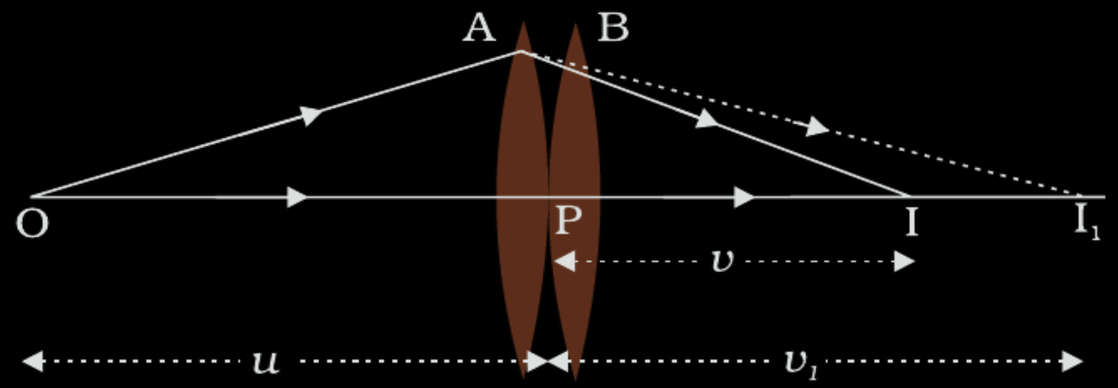

Image formed by the first lens A, ½

$$\frac{1}{v_1}-\frac{1}{u}=\frac{1}{f_1} \quad \text{_____(1)}$$

Image formed by second lens B, ½

$$\frac{1}{v}-\frac{1}{v_1}=\frac{1}{f_2} \quad \text{_____(2)}$$

Adding eq (1) and (2), we get,

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f_1}+\frac{1}{f_2} \quad \text{_____(3)}$$

If the two lens system is regarded as equivalent to a single lens of focal length f, we have

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \quad \text{_____(4)}$$ ½

From equation (3) and (4) ,we get

$$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$$ ½

(b) (i) From graph we see that as the object moves away from the lens the image comes nearer. Therefore it is a convex lens ½+½

(ii) $u=-20cm$

$$v=\frac{40}{3}cm$$

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$ ½

$$=\frac{3}{40}+\frac{1}{20}$$ ½

$$f=8cm$$

1. A small object lies at the bottom of a vessel filled with water (refractive index 4/3) up to a height H. When viewed from a point above the surface of water, the object appears raised by n percent of H. The value of n is :

(a) 15 (b) 20

(c) 25 (d) 33

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1. A small object lies at the bottom of a vessel filled with water (refractive index 4/3) up to a height H. When viewed from a point above the surface of water, the object appears raised by n percent of H. The value of n is :

(a) 15 (b) 20

(c) 25 (d) 33

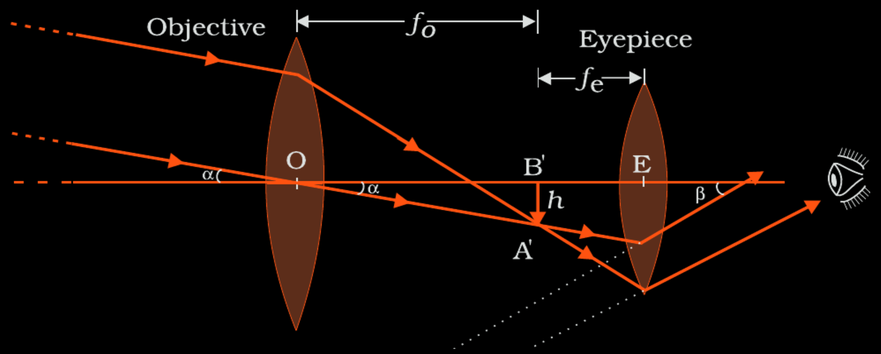
CBSE 2023 C

(c) 25

31. (a) (i) Draw a labelled ray diagram of an astronomical telescope to show the image formation of a distant object by it in normal adjustment. What are the main considerations required in selecting the objective and eyepiece lenses so that the telescope has large magnifying power and high resolution ?

(ii) A biconvex lens of focal length 20 cm is immersed in water, whose refractive index is $\frac{4}{3}$. Find the change, if any, in the nature and the focal length of the lens. Refractive index of the material of convex lens is $\frac{3}{2}$.

CBSE 2023 C *5*



For large magnifying power
$f_o > f_e$ ½
For high resolution
Aperture of objective should be large ½

(ii) $\frac{1}{f_a} = (n_g - 1)[\frac{1}{R_1} - \frac{1}{R_2}]$ ½

$\frac{1}{f_a} = (\frac{n_g}{n_w} - 1)[\frac{1}{R_1} - \frac{1}{R_2}]$

$\frac{f_w}{f_a} = \frac{(n_g - 1)}{(\frac{n_g}{n_w} - 1)}$

$\frac{f_w}{20} = \frac{(\frac{3}{2} - 1)}{(\frac{3}{2} \times \frac{3}{4} - 1)} = \frac{1/2}{1/8} = 4$ ½

f_w= 80 cm ½

Changes in focal length $\Delta f = 80 - 60$ ½
$= 20$ cm

Nature of the lense is converging because f_w is positive

(b) (i) Draw a ray diagram showing refraction of light through a prism of angle A and obtain the relation between μ, A and the angle of minimum deviation δ_m.

(ii) An equiconvex lens of radius of curvature R and made of glass of refractive index μ is cut into two identical plano-convex lenses. Find the focal length of the plano-convex lenses. *5*

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From the figure

$\angle A+\angle QNR= 180^o$

From $\triangle QNR$

$r_1+ r_2+\angle QNR=180^o$

comparing the above equations

$r_1+ r_2=A$ ------- (1) ½

$\delta = (i-r_1)+(e - r_2)$

$\delta = i + e - A$ ------- (2) ½

At the minimum derivation D_m

$\delta = D_m$, i=e and $r_1 = r_2$

From equation (1) ½

$2r= A$

$r= \frac{A}{2}$

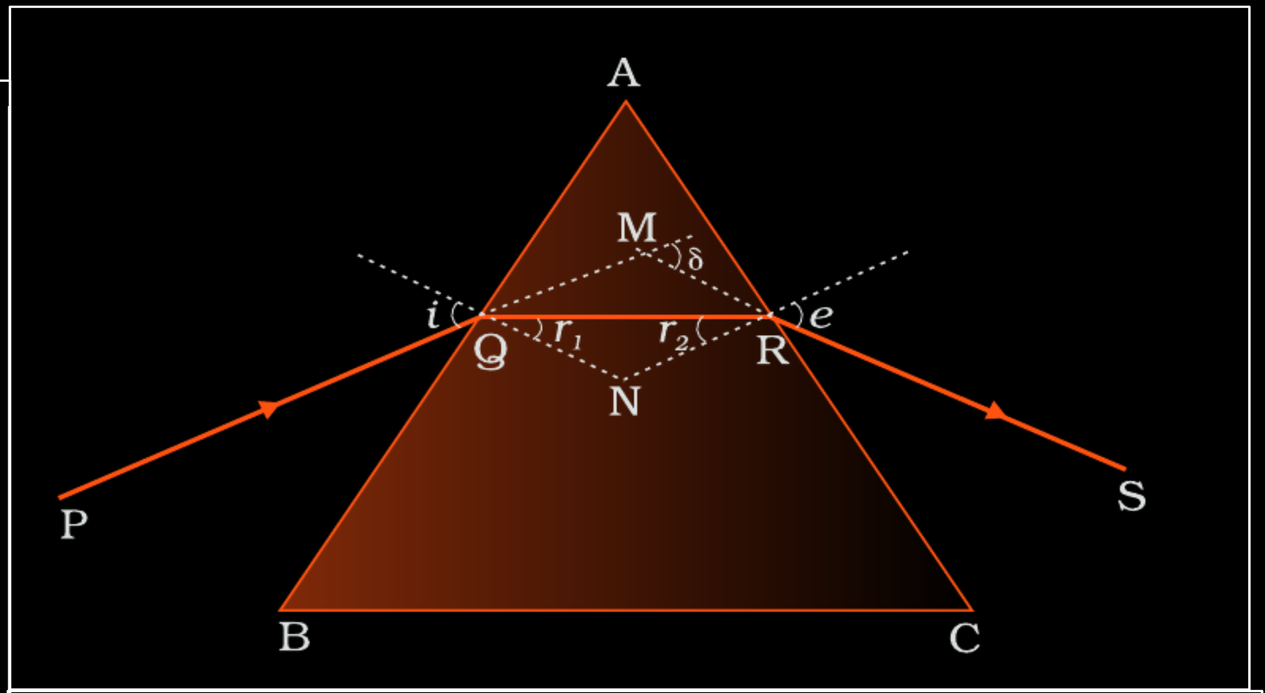
From equation (2)

$D_m =2i-A$

$i= \frac{A+D_m}{2}$

The refractive index of the prism is

$n_{21} = \frac{\sin i}{\sin r} = \frac{\sin[\frac{(A+D_m)}{2}]}{\sin[A/2]}$ ½



(ii)

$\frac{1}{f} = (n-1)[\frac{1}{R_1} - \frac{1}{R_2}]$ ½

$\frac{1}{f_1} = (n-1)[\frac{2}{R}]$ ½

$\frac{1}{f_2} = (n-1)[\frac{1}{R}]$ ½

$f_2 = 2 f_1$ ½

10. An object is placed at a distance of 10 cm in front of a concave mirror of radius of curvature 15 cm. The nature and position the image formed is :

(a) real and inverted, 30 cm in front of the mirror

(b) real and inverted, 15 cm in front of the mirror

(c) virtual and erect, 30 cm behind the mirror

(d) virtual and erect, 15 cm behind the mirror

CBSE 2023

10. An object is placed at a distance of 10 cm in front of a concave mirror of radius of curvature 15 cm. The nature and position the image formed is :

(a) real and inverted, 30 cm in front of the mirror

(b) real and inverted, 15 cm in front of the mirror

(c) virtual and erect, 30 cm behind the mirror

(d) virtual and erect, 15 cm behind the mirror

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(a) real and inverted, 30 cm in front of the mirror

16. *Assertion (A) :* The speed of light decreases when it passes from air into a denser medium.

Reason (R) : The speed of light in a denser medium is given by $v = \frac{c}{\mu}$, where c is the speed of light in air and μ is the refractive index of the denser medium.

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16. *Assertion (A) :* The speed of light decreases when it passes from air into a denser medium.

Reason (R) : The speed of light in a denser medium is given by $v = \frac{c}{\mu}$, where c is the speed of light in air and μ is the refractive index of the denser medium.

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(a) [A]- True [R]- True and correct explanation

21. A glass jug is filled with water to a height of 9·6 cm. The depth of a coin lying on its bottom as measured by a microscope is 7·2 cm. Find the refractive index of the water and the speed of light in water. CBSE 2023 *2*

21. A glass jug is filled with water to a height of 9·6 cm. The depth of a coin lying on its bottom as measured by a microscope is 7·2 cm. Find the refractive index of the water and the speed of light in water. CBSE 2023 *2*

$\mu = \frac{real\ depth}{\text{apparent } depth}$	½
$\mu = \frac{9.6}{7.2} = \frac{4}{3}$	½
$v = \frac{c}{\mu}$	½
$= \frac{3\times10^8}{4}\times3 = 2.25\times10^8\ \text{ms}^{-1}$	½

22. A convex lens is made of the material of refractive index n_1. What will happen if it were immersed in a medium of refractive index n_2 in the following cases ? *2*

(a) $n_2 > n_1$

(b) $n_2 = n_1$

22. A convex lens is made of the material of refractive index n_1. What will happen if it were immersed in a medium of refractive index n_2 in the following cases ? CBSE 2023 *2*

(a) $n_2 > n_1$

(b) $n_2 = n_1$

Focal length of a lens in a medium

$$\frac{1}{f} = \left(\frac{n_1}{n_2} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

n_1 = refractive index od lens

n_2 = refractive index of medium

(a) $n_2 > n_1$

f becomes negative

The convex lens will behave as a concave lens/diverging lens — 1

(b) $n_2 = n_1$

$$\frac{1}{f} = 0$$

$$f = \infty$$

Lens will be invisible in the medium / behave as a glass plate. — 1

33. (a) (i) What is meant by total internal reflection of light ? Write the two conditions necessary for this phenomenon to occur. Briefly explain one of its technological applications.

(ii) A thin converging lens of focal length 10 cm is placed coaxially in contact with a thin diverging lens of focal length 15 cm. Find the nature and focal length of the combined lens. CBSE 2023 *5*

(i) Total internal reflection is the phenomenon in which light waves reflected back in the same medium while travelling from denser to rarer medium without undergoing any refraction.	1
Two conditions:- (i) The incident ray must propagate from denser to rarer medium. (ii) The incident angle must be greater than critical angle.	½+ ½
Application: Telecommunication, medical application, data transmission any one or any other relevant technological application. Through optical fibre a signal goes from its one end to another by multiple total internal reflections without any loss in signal.	1
(ii) Focal length of combination of two lenses	
$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$	½
$f_1 = +10\,\text{cm}$	
$f_2 = -15\,\text{cm}$	
$\frac{1}{f} = \frac{1}{10} + \frac{1}{-15} = \frac{3-2}{30} = \frac{1}{30}$	½
$f = +30\,\text{cm}$	½
Nature- Converging	½

(i) Answer the following giving reasons :

(1) The angular size of the image equals the angular size of the object in a simple microscope, yet it offers magnification.

(2) Both the objective and the eyepiece of a compound microscope have short focal lengths.

(3) A microscope and a telescope play different roles with respect to resolution and magnification of the image.

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(i) 1. Because the image is at minimum distance of distinct vision(D) and object is very close to the eye at distance d (D>d)	1
Angular magnification $\theta = \frac{h_2}{h_1} = \frac{D}{d} \Rightarrow h_2 > h_1$ h_1 -size of object h_2 -size of image	
2. Magnification of a compound microscope $m = (\frac{L}{f_0})(\frac{D}{f_e})$	½
To achieve large magnification (m), f_0 and f_e must be short.	½
3. Microscope - magnifies Telescope - Resolves	½ ½

33. (a) (i) Draw a ray diagram to show how the final image is formed at infinity in an astronomical refracting telescope. Obtain an expression for its magnifying power.

(ii) Two thin lenses L_1 and L_2, L_1 being a convex lens of focal length 24 cm and L_2 a concave lens of focal length 18 cm are placed coaxially at a separation of 45 cm. A 1 cm tall object is placed in front of the lens L_1 at a distance of 36 cm. Find the location and height of the image formed by the combination. *5*

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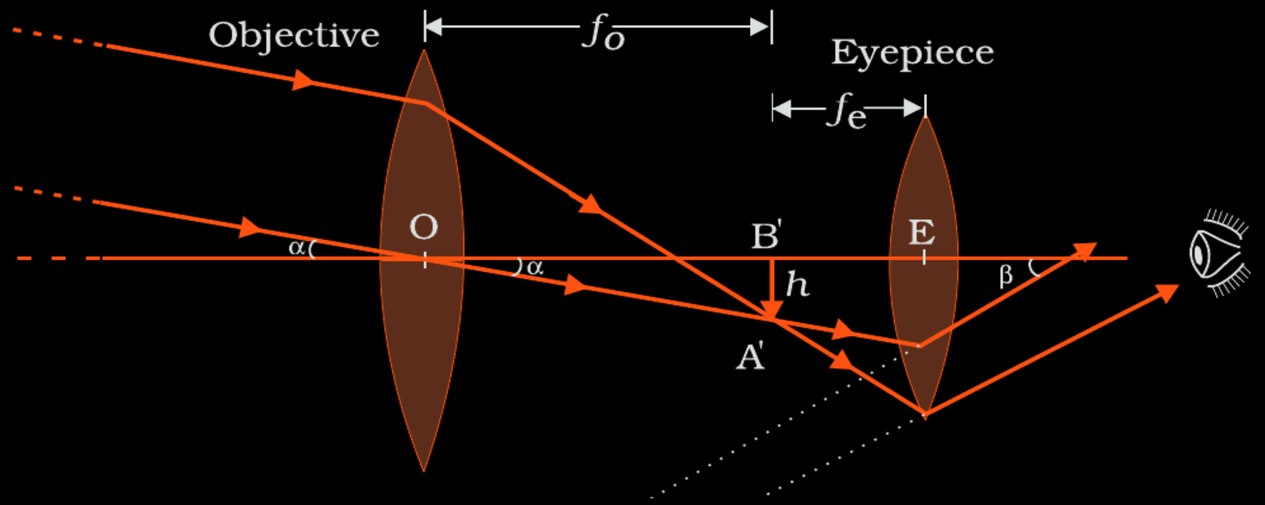


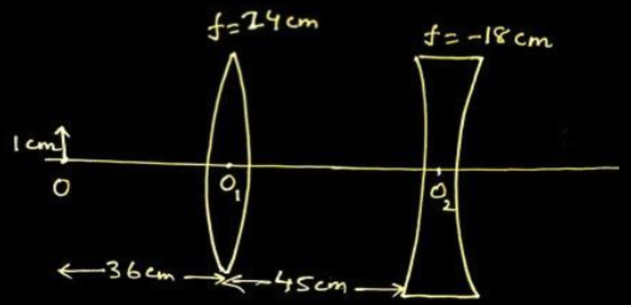
1 ½

From the diagram $\beta = \frac{h}{f_e}$ ½

and $\alpha = \frac{h}{f_o}$ ½

Magnifying Power $= \frac{f_o}{f_e}$ ½

(ii)



For lens L_1,

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ ½

$\frac{1}{v} - \frac{1}{-36} = \frac{1}{24}$

$\frac{1}{v} = \frac{1}{24} - \frac{1}{36}$

$\frac{1}{v} = \frac{3-2}{72} = \frac{1}{72}$

$v = 72cm$ ½

For lens L_2 :

$\frac{1}{v'} - \frac{1}{u'} = \frac{1}{f'}$

$\frac{1}{v'} - \frac{1}{(72-45)} = \frac{1}{-18}$

$\frac{1}{v'} = \frac{1}{-18} + \frac{1}{27}$

$\frac{1}{v'} = \frac{-3+2}{54} = \frac{-1}{54}$

$v' = -54cm$ ½

Final distance $v_1' = -54 - (-45)$

$v_1' = -9cm$ (to the left of convex lens)

Magnification $\frac{h_i}{h_o} = \frac{v_1'}{u}$

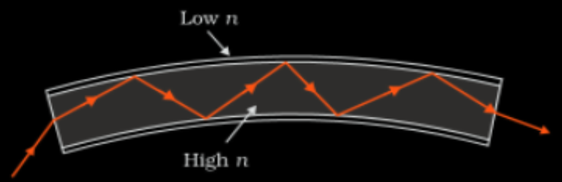
$\frac{h_i}{1} = \frac{-9}{-36} \Rightarrow h_i = +\frac{1}{4}cm$ ½

(b) (i) Explain the working principle of an optical fibre with the help of a diagram. Mention one use of a light pipe.

(ii) A ray of light is incident at an angle of 60° on one face of a prism with the prism angle $A = 60°$. The ray passes symmetrically through the prism. Find the angle of minimum deviation (δ_m) and refractive index of the material of the prism. If the prism is immersed in water, how will δ_m be affected ? Justify your answer. CBSE 2023 *5*

(i) Working Principle:

Optical fibre uses the optical principle of total internal reflection to capture the light transmitted in an optical fibre and confine the light to the core of the fibre. **1**



½

Uses : Transmission of audio and video signal / Examination of internal organs / Endoscopy ½

(ii) $\delta_m = i + e - A$ ½

$\delta_m = 2i - A$

$\delta_m = 60^\circ$ ½

Refractive Index

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin A/2}$$ ½

$$\mu = \frac{\sin\frac{120^o}{2}}{\sin\frac{60^0}{2}}$$

$$\mu = \frac{\sin 60^0}{\sin 30^0} = \frac{\sqrt{3}/2}{1/2}$$

$\mu = \sqrt{3}$ ½

If the prism is immersed in water μ decreases and consequently angle of minimum deviation decreases. Since δ_m depends on μ through equation given above. 1

35. The lens maker's formula is useful to design lenses of desired focal lengths using surfaces of suitable radii of curvature. The focal length also depends on the refractive index of the material of the lens and the surrounding medium. The refractive index depends on the wavelength of the light used. The power of a lens is related to its focal length.

Answer the following questions based on the above :

(a) How will the power of a lens be affected with an increase of wavelength of light ? *1*

(b) The radius of curvature of two surfaces of a convex lens is R each. For what value of μ of its material will its focal length become equal to R ? *1*

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(a) $P = \frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ $\mu \propto \frac{1}{\lambda} \Rightarrow P \propto \frac{1}{\lambda}$ So P decreases as λ increases.	1
(b) $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$	½
$\frac{1}{R} = (\mu - 1)\frac{2}{R} \qquad (R_1 = R,\ R_2 = -R)$	
$\Rightarrow \mu = 3/2$	½

(c) The focal length of a concave lens of $\mu = 1{\cdot}5$ is 20 cm in air. It is completely immersed in water of $\mu = \frac{4}{3}$. Calculate its focal length in water. *2*

OR

(c) An object is placed in front of a lens which forms its erect image of magnification 3. The power of the lens is 5 D. Calculate the distance of the object and the image from the lens. *2*

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(c) $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ½

$\frac{1}{20} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ½

When the lens is immersed in water

$\frac{1}{f_{unknown}} = \left(\frac{9}{8} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ½

$\frac{\frac{1}{20}}{\frac{1}{f}} = \frac{1/2}{1/8} = \Rightarrow f = 80 \text{ cm}$ ½

OR

$m = 3 = \frac{v}{u} \Rightarrow v = 3u$ ½

$f = \frac{100}{+5} = +20 \text{ cm}$ ½

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{3u} - \frac{1}{u} = \frac{1}{20}$

$\frac{1-3}{3u} = \frac{1}{20}$

$u = -(40/3) \text{ cm}$ ½

$v = -40 \text{ cm}$ ½

24. The refractive indices of two media A and B are 2 and $\sqrt{2}$ respectively. What is the critical angle for their interface ? CBSE 2023 *2*

24. The refractive indices of two media A and B are 2 and $\sqrt{2}$ respectively. What is the critical angle for their interface ? CBSE 2023 *2*

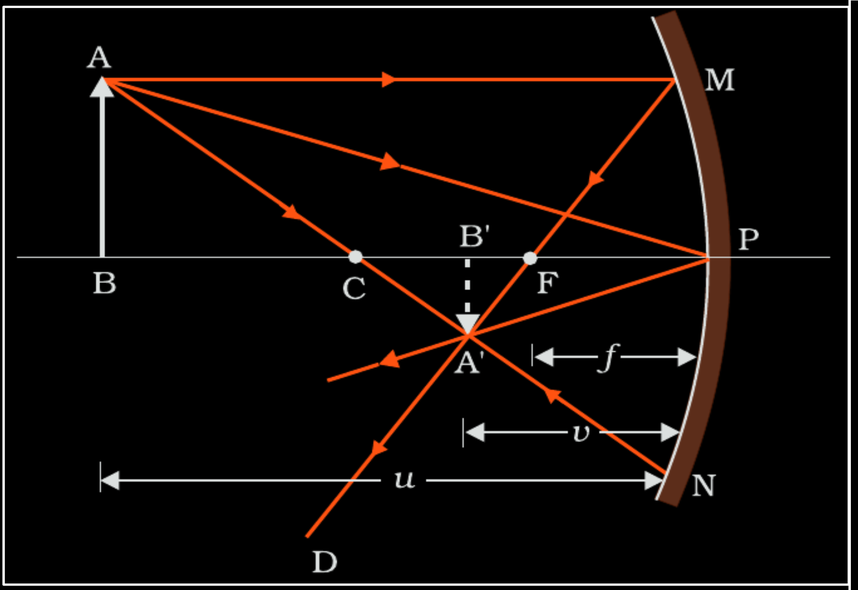
From Snell 's law:-	
$\mu_A \sin i_c = \mu_B \text{Sin}\, 90°$	½
$2 \times \sin i_c = \sqrt{2} \times 1$	½
$\sin i_c = \frac{1}{\sqrt{2}}$	½
$i_c = 45°$	½

31. (a) (i) Draw a ray diagram showing the formation of a real image of an object placed at a distance 'u' in front of a concave mirror of radius of curvature 'R'. Hence, obtain the relation for the image distance 'v' in terms of u and R.

(ii) A 1·8 m tall person stands in front of a convex lens of focal length 1 m, at a distance of 5 m. Find the position and height of the image formed.

CBSE 2023

5



From Fig. the two right-angled triangles A′B′F and MPF are similar. (For paraxial rays, MP can be considered to be a straight line perpendicular to CP.) Therefore,

$$\frac{B'A'}{PM} = \frac{B'F}{FP}$$

$$\text{or } \frac{B'A'}{BA} = \frac{B'F}{FP} \quad (\because PM = AB)$$

-----(i) ½

Since ∠ APB = ∠ A′PB′, the right angled triangles A′B′P and ABP are also similar. Therefore,

$$\frac{B'A'}{BA} = \frac{B'P}{BP}$$

----(ii) ½

Comparing equations (i) and (ii)

$$\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{BP}$$

-----(iii) ½

B′ P = – v, FP = – f, BP = – u;

Using these in Eq.(iii) we get $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$ ½

Alternatively:- If the result derived by any other method, full credit to be given.

(ii) For lens: $\frac{1}{\mathrm{v}} - \frac{1}{u} = \frac{1}{f}$ ½

$u = -5m; \quad f = +1m$

$\frac{1}{\mathrm{v}} - \frac{1}{-5} = \frac{1}{(+1)}$

$\Rightarrow \mathrm{v} = \frac{5}{4}m = 1.25m$ ½

$m = \frac{I}{O} = \frac{\mathrm{v}}{u} = \frac{\left(+5/4\right)}{(-5)}$ ½

I = (- 0.25) × (1.8)

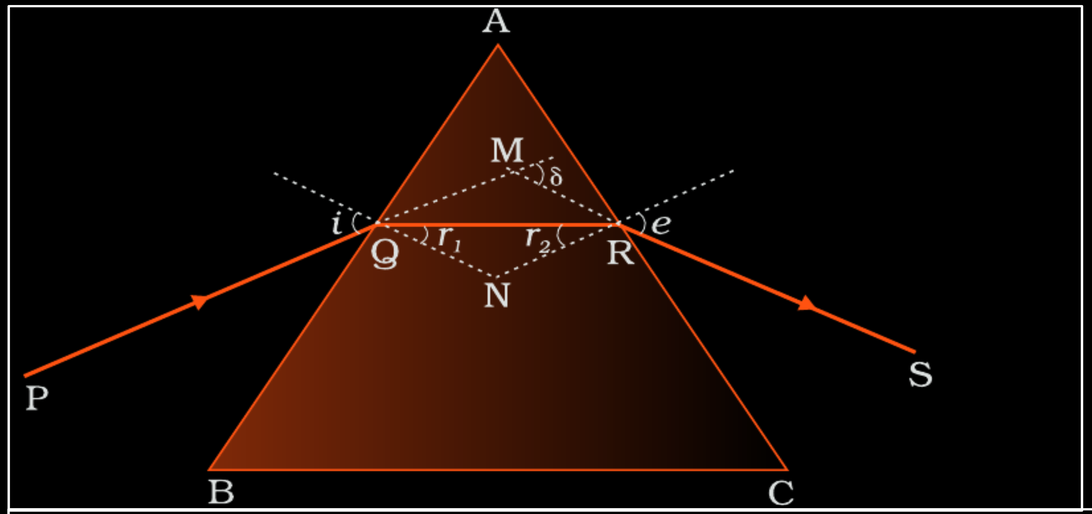
I = - 0.45 m ½

(b) (i) Draw a ray diagram showing refraction of a ray of light through a triangular glass prism. Hence, obtain the relation for the refractive index (μ) in terms of angle of prism (A) and angle of minimum deviation (δ_m).

(ii) The radii of curvature of the two surfaces of a concave lens are 20 cm each. Find the refractive index of the material of the lens if its power is – 5·0 D.

CBSE 2023

5



In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral is 180º.

$\angle A + \angle QNR = 180^\circ$

From the triangle QNR, $r_1 + r_2 + \angle QNR = 180^\circ$

Comparing these two equations, we get

$r_1 + r_2 = A$ -------(i) ½

The total deviation δ is the sum of deviations at the two faces, ½

$\delta = (i - r_1) + (e - r_2)$ that is, $\delta = i + e - A$ -------(ii) ½

When $\delta = \delta_m$; $i = e$ & $r_1 = r_2$

From (i); $2r = A$ or $r = A/2$

From (ii); $\delta_m = 2i - A$ or $i = \frac{A+\delta_m}{2}$

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin(\frac{A+\delta_m}{2})}{\sin A/2}$$ ½

(ii) Given; P = - 5D

$$f \text{ (in cm)} = \frac{100}{(-5)} = -20 \text{ cm}$$ ½

Using Lens Maker's formula ; $\frac{1}{f} = (\mu - 1)[\frac{1}{R_1} - \frac{1}{R_2}]$ ½

$$\frac{1}{(-20)} = (\mu - 1)[\frac{1}{(-20)} - \frac{1}{(+20)}]$$

$$\frac{1}{(-20)} = (\mu - 1)[-\frac{1}{10}]; \quad \mu - 1 = \frac{1}{2}$$ ½

$$\Rightarrow \mu = \frac{3}{2} = 1.5$$ ½

13. Which of the following pairs of media has the least value of critical angle ?

(a) Glass to air

(b) Glass to water

(c) Diamond to water

(d) Diamond to air

CBSE 2023

13. Which of the following pairs of media has the least value of critical angle ?

(a) Glass to air

(b) Glass to water

(c) Diamond to water

(d) Diamond to air

CBSE 2023

(d) Diamond to air

19. The radii of curvature of the faces of a double-convex lens are 20 cm and 30 cm. Its power is $\frac{25}{6}$ D. What is the refractive index of the glass of the lens ?

CBSE 2023

2

19. The radii of curvature of the faces of a double-convex lens are 20 cm and 30 cm. Its power is $\frac{25}{6}$ D. What is the refractive index of the glass of the lens ?

CBSE 2023

2

R_1 =20 cm, R_2 =-30 cm P = $\frac{25}{6}D = \frac{25}{600}cm^{-1}$	
P= (n-1) ($\frac{1}{R_1} - \frac{1}{R_2}$)	½
$\frac{25}{600} = (n-1)(\frac{1}{20} + \frac{1}{30})$	½
$\frac{25}{600} = (n-1)(\frac{50}{20 \times 30})$	
25= (n-1)50	½
n-1= $\frac{1}{2}$	
n= $\frac{3}{2}$	½

(b) Why is a reflecting telescope preferred over a refracting telescope ? Justify your answer giving two reasons. CBSE 2023 *2*

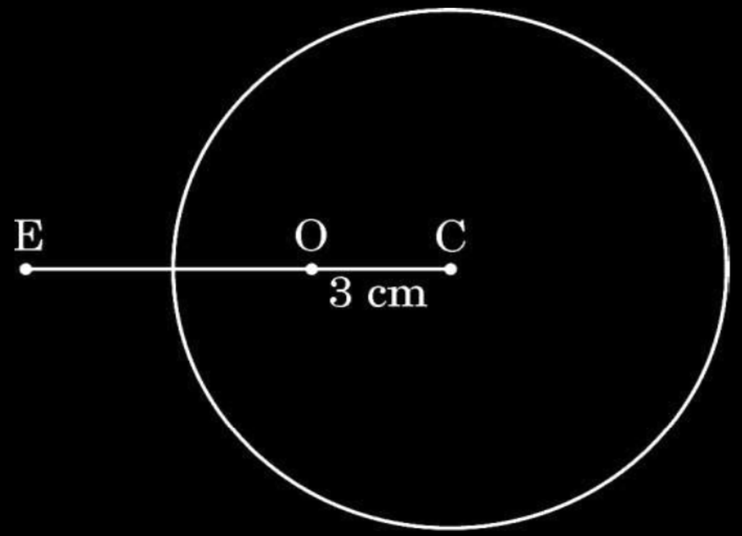
(b) Why is a reflecting telescope preferred over a refracting telescope ?

Justify your answer giving two reasons. CBSE 2023 *2*

1 No Chromatic Aberration - No refraction in mirrors 2 No Spherical Aberration - Due to use of parabolic reflector 3 Easy mechanical support required - Mirrors weigh less and can be supported over entire back surface. 4 High resolving power – Due to Mirror with large diameter are better. 5 Brighter image – Large mirrors gather more light waves. **(Any two)**	1+1

(b) (i) A spherical surface of radius of curvature R separates two media of refractive indices n_1 and n_2. A point object is placed in front of the surface at distance u in medium of refractive index n_1 and its image is formed by the surface at distance v, in the medium of refractive index n_2. Derive a relation between u and v.

(ii) A solid glass sphere of radius 6·0 cm has a small air bubble trapped at a distance 3·0 cm from its centre C as shown in the figure. The refractive index of the material of the sphere is 1·5. Find the apparent position of this bubble when seen through the surface of the sphere from an outside point E in air.



CBSE 2023

5

Assume that the aperture of the surface is small as compared to other distance involved, so that small angle approximation can be made.
For small angles

for ΔNOC, I is the exterior angle

$\therefore$ I = $\angle NOM + \angle NCM$

$$i = \frac{MN}{OM} + \frac{MN}{MC} \qquad \text{(i)}$$

Similarly r = $\angle NCM - \angle NIM$

$$= \frac{MN}{MC} - \frac{MN}{MI} \qquad \text{(ii)}$$

By Snell's law

$n_1 \text{Sin I} = n_2 \text{ Sin r}$ ½

for small angles

$n_1 \text{ I} = n_2 \text{ r}$

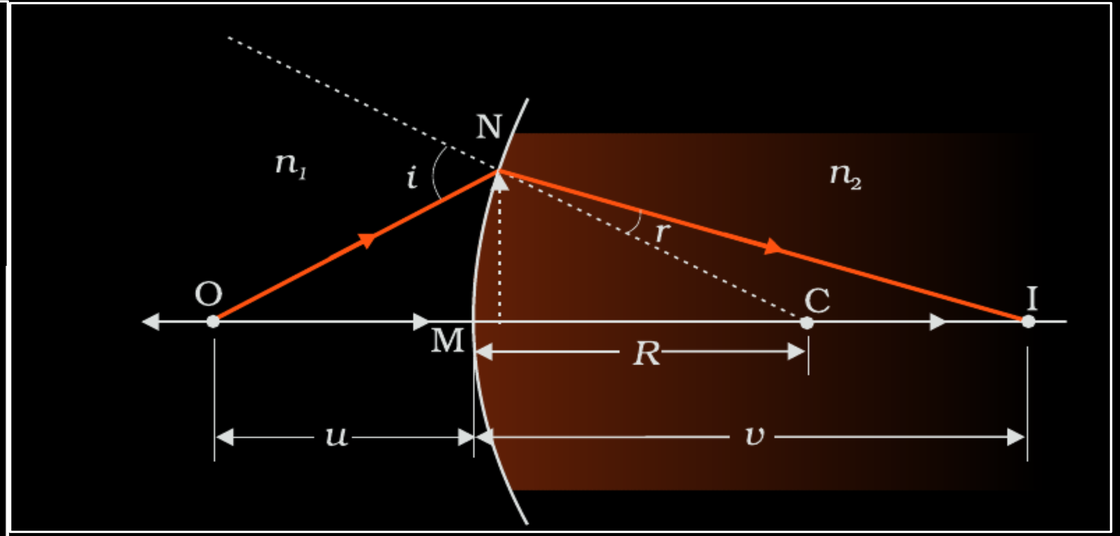
substituting I and r from (i) and (ii) we get ½

$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}$$

Applying Cartesian coordinates

OM= -u, MI= + v , MC= +R ½

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$



(ii) $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ ½

R= - 6 cm, u = - 3cm , n_1 =1.5 n_2 = 1

$\frac{1}{v} + \frac{1.5}{3} = \frac{1-1.5}{-6}$ ½

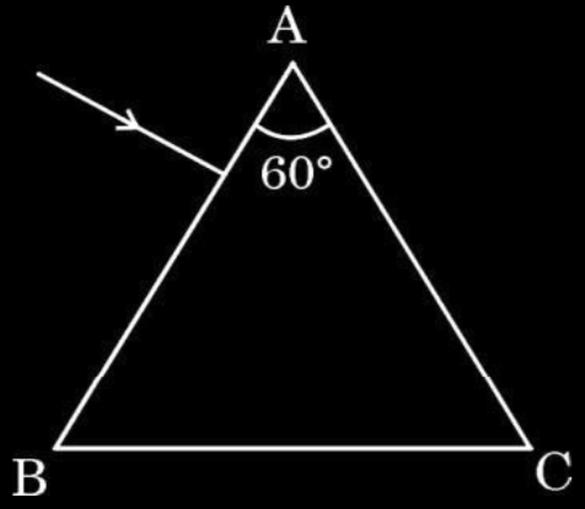
$\frac{1}{v} = \frac{0.5}{6} - \frac{1.5}{3}$ ½

$\frac{1}{v} = \frac{0.5-3}{6}$

$\frac{1}{v} = \frac{-2.5}{6}$ ½

v = - 2.4 cm

from the left surface inside the sphere ½



35. Strontium titanate is a rare oxide — a natural mineral found in Siberia. It is used as a substitute for diamond because its refractive index and critical angle are 2·41 and 24·5°, respectively, which are approximately equal to the refractive index and critical angle of diamond. It has all the properties of diamond. Even an expert jeweller is unable to differentiate between diamond and strontium titanate. A ray of light is incident normally on one face of an equilateral triangular prism ABC made of strontium titanate.

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Answer the following questions based on the above :

(a) Trace the path of the ray showing its passage through the prism. *1*

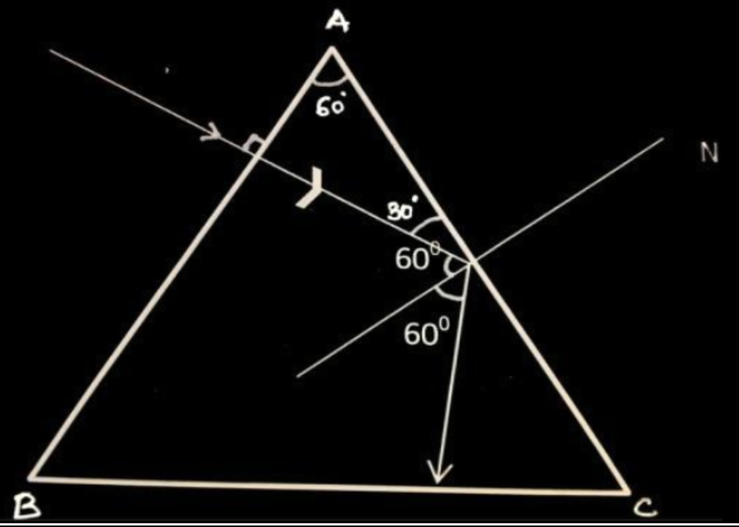
(b) Find the velocity of light through the prism. *1*

(c) Briefly explain two applications of total internal reflection. *2*

OR

(c) Define total internal reflection of light. Give two conditions for it. *2*

a)



From fig. angle of incidence on second face $\angle i = 60^0$
critical angle $\angle i_c = 24.5^o$
$(\angle i) > (\angle i_c)$
$\therefore$ TIR takes place — 1

b) $n = \frac{c}{v}$

$v = \frac{c}{n} = \frac{3\times10^8}{2.41} = 1.24 \times 10^8$ m/s — 1

c) Optical Fibre / Brilliance of diamond / mirage (any two) — 1+1
Note: Give full credit if students mention the names of applications only.

OR

c) When light travels from optically denser medium to rarer medium at an interface and gets reflected back into the same medium the phenomenon is called as total internal reflection. — 1

Conditions for TIR
1. Light must travel from optically denser medium to rarer medium.
2. Angle of incidence at the interface must be greater than the critical angle for the pair of media. — ½ + ½

3. The minimum distance between an object and its real image formed by a convex lens of focal length f is :

(a) f (b) 2f

(c) 3f (d) 4f

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3. The minimum distance between an object and its real image formed by a convex lens of focal length f is :

(a) f (b) 2f

(c) 3f (d) 4f

CBSE 2023

(d) 4f

24. A double-convex lens of power 4 D is manufactured from a glass of refractive index 1·5. What is the radius of curvature of each face of this lens if both faces have the same radius of curvature ? CBSE 2023 *2*

24. A double-convex lens of power 4 D is manufactured from a glass of refractive index 1·5. What is the radius of curvature of each face of this lens if both faces have the same radius of curvature ? CBSE 2023 *2*

P= 4D, n=1.5	
$P=\frac{1}{f}=(n-1)(\frac{1}{R_1}-\frac{1}{R_2})$	½
$R_1 = R$, $R_2 = -R$	
$P=\frac{1}{f}=(n-1)(\frac{2}{R})$	½
$4=(1.5-1)\ (\frac{2}{R})$	½
$4=\frac{0.5\times 2}{R}$	
$R=\frac{1}{4}=0.25\text{m}=25\text{ cm}$	½

9. For a concave mirror of focal length 'f', the minimum distance between the object and its real image is :

(a) zero

(b) f

(c) 2f

(d) 4f

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9. For a concave mirror of focal length 'f', the minimum distance between the object and its real image is :

(a) zero (b) f

(c) 2f (d) 4f

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(a) Zero

12. A ray of light travels a distance of 12.0 m in a transparent sheet in 60 ns. The refractive index of the sheet is **1**

(a) 1.33 (b) 1.50

(c) 1.65 (d) 1.75

CBSE 2023

12. A ray of light travels a distance of 12.0 m in a transparent sheet in 60 ns. The refractive index of the sheet is **1**

(a) 1.33 (b) 1.50

(c) 1.65 (d) 1.75

CBSE 2023

(b) 1.50

14. A ray of monochromatic light propagating in air, is incident on the surface of water. Which of the following will be the same for the reflected and refracted rays ? **1**

(a) Energy carried
(b) Speed
(c) Frequency
(d) Wavelength

CBSE 2023

14. A ray of monochromatic light propagating in air, is incident on the surface of water. Which of the following will be the same for the reflected and refracted rays ? **1**

(a) Energy carried (b) Speed

(c) Frequency (d) Wavelength

CBSE 2023

(c) Frequency

23. The power of a thin lens is +5 D. When it is immersed in a liquid, it behaves like a concave lens of focal length 100 cm. Calculate the refractive index of the liquid. Given refractive index of glass = 1.5. **2**

CBSE 2023

23. The power of a thin lens is +5 D. When it is immersed in a liquid, it behaves like a concave lens of focal length 100 cm. Calculate the refractive index of the liquid. Given refractive index of glass = 1.5. **2**

CBSE 2023

$P = +5\ D, \quad f_e = -100\ cm \quad , \quad \mu_g = 1.5 , \quad \mu_l = ?$

$$f_a = \frac{1}{P} = \frac{1}{5} = 0.2\,m = 20\,cm$$ ½

$$\frac{1}{f_a} = (\mu_{ga} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{20} = (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{----------(1)}$$ ½

$$\frac{1}{f_l} = \left(\frac{\mu_g}{\mu_l} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{(-100)} = \left(\frac{1.5}{\mu_l} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{----------(2)}$$ ½

From (1) and (2), on solving

$$\mu_l = \frac{5}{3} = 1.67$$ ½

26. A ray of light is refracted by a glass prism. Obtain an expression for the refractive index of the glass in terms of the angle of prism A and the angle of minimum deviation δm.

CBSE 2023 **3**

26. A ray of light is refracted by a glass prism. Obtain an expression for the refractive index of the glass in terms of the angle of prism A and the angle of minimum deviation δm.

CBSE 2023 **3**

At minimum deviation , a ray of light passes through a prism symmetrically

i.e $\angle i = \angle e$, $\angle r_1 = \angle r_2 = \angle r$ ½

We know that $i + e = A + \delta$ and $r_1 + r_2 = A$ ½

$\therefore \quad i + i = A + \delta m$

$$i = \frac{A+\delta m}{2}$$

And also

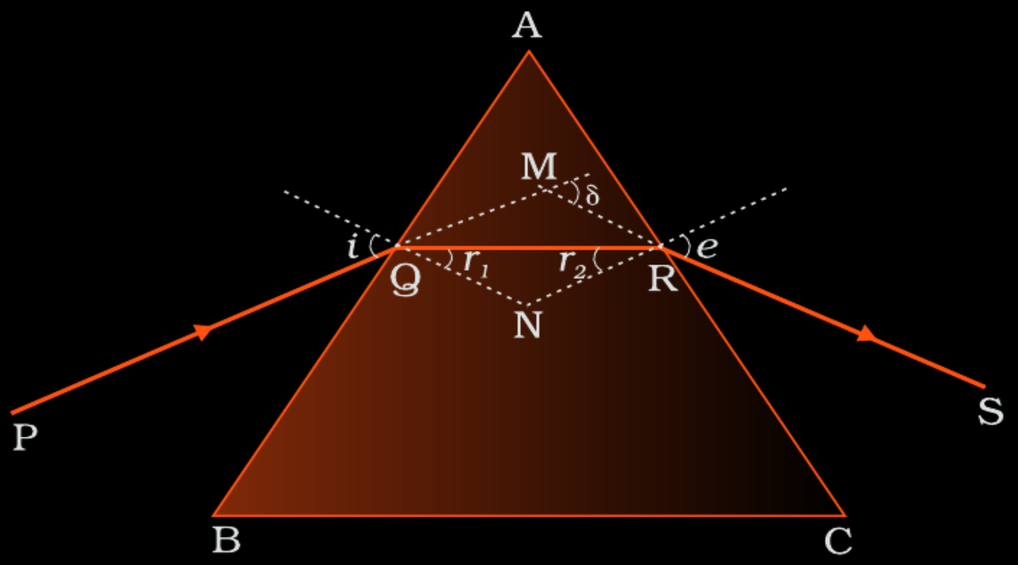
$r + r = A$

$$\Rightarrow r = \frac{A}{2}$$ ½

Using Snell's law

$$\mu = \frac{\sin i}{\sin r}$$ ½

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$ ½



(ii) A concave mirror of focal length 12 cm forms a three times magnified virtual image of an object. Find the distance of the object from the mirror.

CBSE 2023

(ii) A concave mirror of focal length 12 cm forms a three times magnified virtual image of an object. Find the distance of the object from the mirror.

CBSE 2023

(ii) $m = +3$, f = -12 cm , $u = ?$	
$m = -\frac{v}{u} = 3 \Rightarrow v = -3u$	½
using mirror formula	
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	½
$\frac{1}{-3u} + \frac{1}{u} = \frac{1}{-12}$	½
u = - 8 cm	½

(b) (i) Draw a labelled ray diagram showing the image formation by a refracting telescope. Define its magnifying power. Write two limitations of a refracting telescope over a reflecting telescope.

(ii) The focal lengths of the objective and the eye-piece of a compound microscope are 1.0 cm and 2.5 cm respectively. Find the tube length of the microscope for obtaining a magnification of 300.

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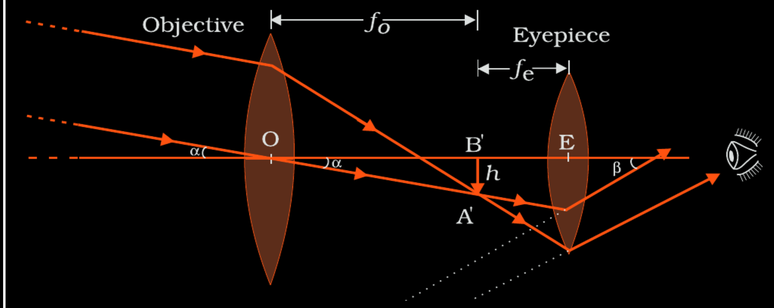

(Note deduct ½ mark if a student does not show the direction of propagation of the light.)

Alternatively

Give full credit for ray diagram if a candidate draws ray diagram for final image at the near point.

Magnifying power of a telescope – It is defined as the ratio the angle subtended at the eye by the final image to the angle subtended by the object at the lens or the eye. 1

Two limitations of a refracting telescope over a reflecting telescope.

(i) Less resolving power.

(ii) Difficult mechanical support.

(iii) Less bright image.

(iv) Suffers chromatic aberration.

(v) Image suffers with spherical aberration. ½ + ½

(Any two of the above)

f_o = 1.0 cm , f_e = 2.5 cm , m = 300 , D=25 cm , L=?

$$|m| = \frac{L}{f_0} . \frac{D}{f_e} \qquad ½$$

$$300 = \frac{L}{1.0} . \frac{25}{2.5} \qquad ½$$

$$L = 30 \text{ cm} \qquad ½$$

14. A ray of light of wavelength 600 nm propagates from air into a medium. If its wavelength in the medium becomes 400 nm, the refractive index of the medium is **1**

(a) 1.4 (b) 1.5

(c) 1.6 (d) 1.8

CBSE 2023

14. A ray of light of wavelength 600 nm propagates from air into a medium. If its wavelength in the medium becomes 400 nm, the refractive index of the medium is **1**

(a) 1.4 (b) 1.5

(c) 1.6 (d) 1.8

CBSE 2023

(b) 1.5

25. A point object in air is placed symmetrically at a distance of 60 cm in front of a concave spherical surface of refractive index 1.5. If the radius of curvature of the surface is 20 cm, find the position of the image formed. **2**

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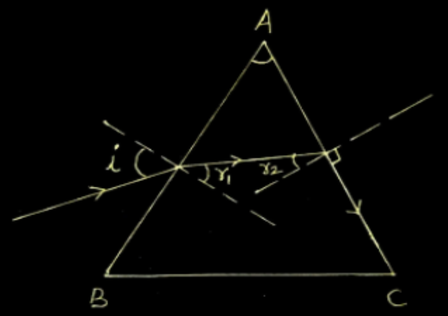
25. A point object in air is placed symmetrically at a distance of 60 cm in front of a concave spherical surface of refractive index 1.5. If the radius of curvature of the surface is 20 cm, find the position of the image formed. **2**

CBSE 2023

μ = 1.5 , u= - 60 cm , R = -20 cm , v=?	
$\frac{\mu}{v}-\frac{1}{u}=\frac{\mu-1}{R}$	½
$\frac{1.5}{v}-\frac{1}{(-60)}=\frac{1.5-1}{(-20)}$	1
v = -36 cm	½

27. A ray of light is incident on a glass prism of refractive index μ and refracting angle A. If it just suffers total internal reflection at the other face, obtain a relation between the angle of incidence, angle of prism and critical angle. CBSE 2023 **3**

27. A ray of light is incident on a glass prism of refractive index μ and refracting angle A. If it just suffers total internal reflection at the other face, obtain a relation between the angle of incidence, angle of prism and critical angle. CBSE 2023 **3**



1

∵ the light ray just suffers total internal reflection

∴ $\angle r_2 = \angle i_c$ ½

$$A = r_1 + r_2$$

$$A = r_1 + i_c$$ ½

From Snell's law at surface AB

$$\sin i = \mu . \sin r_1$$ ½

$$\sin i = \mu . \sin(A - i_c)$$ ½

9. A beam of light travels from air into a medium. Its speed and wavelength in the medium are 1.5×10^8 ms^{-1} and 230 nm respectively. The wavelength of light in air will be **1**

(a) 230 nm (b) 345 nm

(c) 460 nm (d) 690 nm

CBSE 2023

9. A beam of light travels from air into a medium. Its speed and wavelength in the medium are 1.5×10^8 ms^{-1} and 230 nm respectively. The wavelength of light in air will be **1**

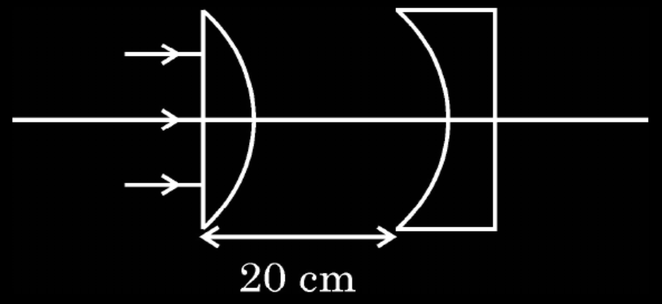
(a) 230 nm (b) 345 nm

(c) 460 nm (d) 690 nm

CBSE 2023

(c)
460 nm

21. In the given figure the radius of curvature of curved face in the plano-convex and the plano-concave lens is 15 cm each. The refractive index of the material of the lenses is 1.5. Find the final position of the image formed. **2**



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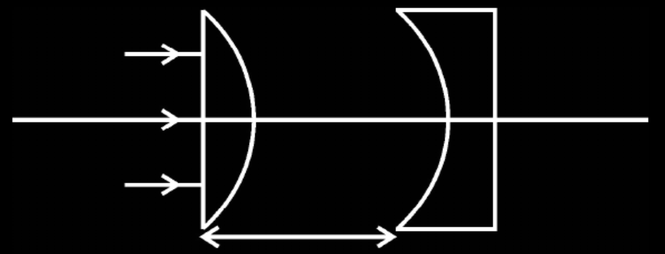
21. In the given figure the radius of curvature of curved face in the plano-convex and the plano-concave lens is 15 cm each. The refractive index of the material of the lenses is 1.5. Find the final position of the image formed. **2**



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Using the formula $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$	½
• Focal length of plano-convex lens = +30 cm Focal length of plano-concave lens = -30 cm	½
• For plano-convex lens As object is at ∞, its real image will be formed at its focus i.e +30 cm v_1 = +30 cm	
• For plano-concave lens u = +(30-20) cm = +10 cm	
$\frac{1}{f_2}=\left(\frac{1}{v_2}-\frac{1}{u_2}\right)$	½
$\frac{1}{-30}=\left(\frac{1}{v_2}-\frac{1}{10}\right)$ $\therefore v_2$ = 15 cm	½

29. Define critical angle for a given pair of media and total internal reflection. Obtain the relation between the critical angle and refractive index of the medium. **3**

CBSE 2023

29. Define critical angle for a given pair of media and total internal reflection. Obtain the relation between the critical angle and refractive index of the medium. **3**

CBSE 2023

Critical angle - When a ray of light passes from a denser to a rarer medium, the value of angle of incidence for which the angle of refraction becomes 90^0 is called critical angle for that pair of media. 1

Total internal Reflection – When a ray of light passes from a denser to rarer medium and the angle of incidence exceeds the critical angle for pair of media, the ray under goes reflection. This is called total internal reflection. 1

From Snell's law $\dfrac{\sin i}{\sin r} = \mu_{rd}$ ½

When angle of incidence is equal to critical angle ($\angle i = \angle i_c$) , $\angle r = 90^0$

$$\therefore \quad \frac{\sin i_c}{\sin 90^0} = \frac{1}{\mu_{dr}}$$

$$\therefore \quad \mu_{dr} = \frac{1}{\sin i_c}$$ ½

32. (a) (i) Draw a ray diagram to show the working of a compound microscope. Obtain the expression for the total magnification for the final image to be formed at the near point. **3**

(ii) In a compound microscope an object is placed at a distance of 1.5 cm from the objective of focal length 1.25 cm. If the eye-piece has a focal length of 5 cm and the final image is formed at the near point, find the magnifying power of the microscope. **2**

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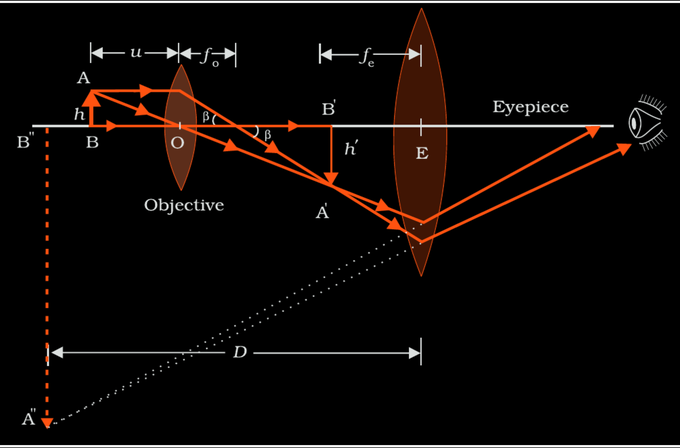

1½

Linear magnification due to the objective is

$$\tan\beta = \left(\frac{h}{f_0}\right) = \left(\frac{h'}{L}\right)$$

$$m_0 = \frac{h'}{h} = \frac{L}{f_0}$$ ½

Here L is the distance between the second focal point of the objective and the first focal point of the eyepiece.

Linear magnification due to eyepiece is

$$m_e = \left(1 + \frac{D}{f_e}\right)$$ ½

Thus, the total magnification is, $m = m_0 \times m_e$

$$m = \frac{L}{f_0}\left(1 + \frac{D}{f_e}\right)$$ ½

Note: full credit of the derivation should be given if a student derives

$$m = \frac{v_0}{u_0}\left(1 + \frac{D}{f_e}\right)$$

$u_0 = -1.5\,\text{cm}$

$f_0 = 1.25\,\text{cm}$

$f_e = 5\,\text{cm}$

$D = 25\,\text{cm}$

$$\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$$

$$\frac{1}{v_0} = \frac{1}{1.25} - \frac{1}{1.5} = \frac{2}{15}$$ ½

$$v_0 = \frac{15}{2}\,\text{cm}$$

$$|m_0| = \frac{v_0}{u_0} = \frac{15}{2} \times \frac{1}{1.5} = 5$$ ½

$$|m_e| = \left(1 + \frac{D}{f_e}\right)$$ ½

$$|m_e| = 1 + \frac{25}{5} = 6$$

$$m = m_0 \times m_e = 5 \times 6 = 30$$ ½

(b) (i) Draw a ray diagram for the formation of image of an object by an astronomical telescope, in normal adjustment. Obtain the expression for its magnifying power.

(ii) The magnifying power of an astronomical telescope in normal adjustment is 2.9 and the objective and the eyepiece are separated by a distance of 150 cm. Find the focal lengths of the two lenses.

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5

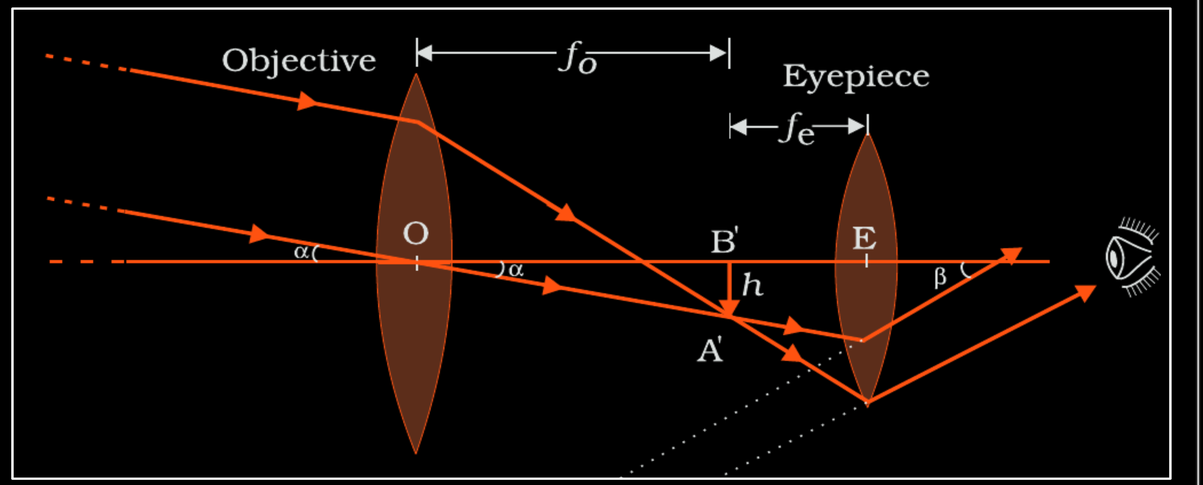
Magnifying power (m)

$$m \approx \frac{\beta}{\alpha}$$

$$\beta = \frac{h}{f_e}$$

$$\alpha = \frac{h}{f_0}$$

$$m = \frac{f_0}{f_e}$$



1½

½

½

½

(ii) $m = 2.9$, $d = 150\,cm$ (Given)

$$m = \frac{f_o}{f_e} = \frac{29}{10}$$

½

$$f_0 + f_e = 150$$

½

$$f_e = 38.5\,cm$$

½

$$f_o = 150 - 38.5 = 111.5\,cm$$

½

34. A lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical. Considering image formation by a single spherical surface successively at the two surfaces of a lens, lens maker's formula is obtained. It is useful to design lenses of desired focal length using surfaces of suitable radii of curvature. This formula helps us obtain a relation between u, v and f for a lens. Lenses form images of objects and they are used in a number of optical devices, for example microscopes and telescopes.

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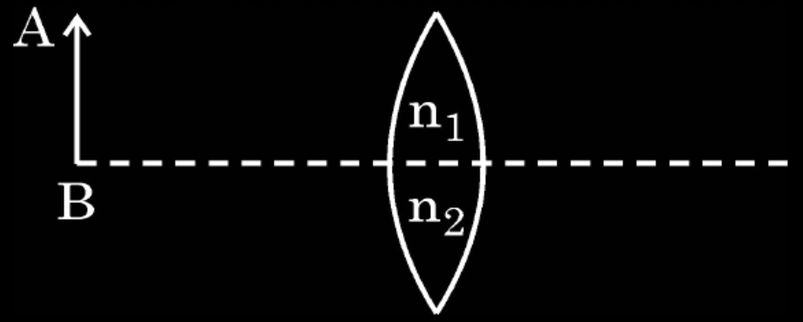
4

(i) An object AB is kept in front of a composite convex lens, as shown in figure. Will the lens produce one image ? If not, explain.



(ii) A real image of an object formed by a convex lens is observed on a screen. If the screen is removed, will the image still be formed ? Explain.

(iii) A double convex lens is made of glass of refractive index 1.55 with both faces of the same radius of curvature. Find the radius of curvature required if focal length is 20 cm.

OR

(iii) Two convex lenses A and B of focal lengths 15 cm and 10 cm respectively are placed coaxially 'd' distance apart. A point object is kept at a distance of 30 cm in front of lens A. Find the value of 'd' so that the rays emerging from lens B are parallel to its principal axis.

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(i) No, ½

The lens is made up of two materials of different refractive indices. It has two focal lengths. ½

(ii) Yes ½

Rays are still intersecting/ converging at the location of image. ½

(iii) $\frac{1}{f} = (n-1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$ ½

$R_1 = +R\ ,\ R_2 = -R$ ½

$\frac{1}{f} = (n-1)\left[\frac{2}{R}\right]$ ½

$R = 2(n-1)f$

R= 2 (1.55-1) x 20=22 cm ½

For lens A (f =15 cm)

u = -30 cm

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ ½

$\frac{1}{v_1} = \frac{1}{15} - \frac{1}{30} = \frac{1}{30}$

v_1 = 30 *cm* ½

For lens B

For rays to go parallel to principal axis out of lens B the image formed by lens A must lie at the focus of B

So d= 30+10=40 cm 1

Alternatively: (Object is kept at 2f so image will also be formed at 2f on the other side of the lens i.e. at 30 cm. Now the final image is to be formed at infinity so the image formed must lie at the focus of the second lens (B). Thus separation is 30 +10 =40 cm)

7. (a) Define the term 'Power of a lens' and mention its SI unit.

(b) A diverging lens is made of a material of refractive index 1·6. The radii of curvature of its two surfaces are 10 cm and 30 cm. Calculate its power. CBSE 2022 C *3*

7. (a) Define the term 'Power of a lens' and mention its SI unit.

(b) A diverging lens is made of a material of refractive index 1·6. The radii of curvature of its two surfaces are 10 cm and 30 cm. Calculate its power. CBSE 2022 C *3*

(a) Power of a lens is defined as the tangent of the angle by which it converges and diverges a beam of light falling at unit distance from the optical centre. Alternatively Power :- Ability to converge or diverge the beam of light incident on a lens. Alternatively Power of the lens is given by the reciprocal of its focal length.	½
S.I unit is dioptre or m^{-1}.	½
(b) $P = \frac{1}{f} = (\mu-1)(\frac{1}{R_1} - \frac{1}{R_2})$	½
$= (1.6 - 1)(\frac{1}{-0.1} + \frac{1}{0.3})$	½
$= (0.6)(\frac{-3+1}{0.3})$	½

8. (a) (i) Both the objective and the eyepiece of a compound microscope have short focal lengths. Explain.

(ii) Two convex lenses of focal lengths 1·25 cm and 5·0 cm are used to design a compound microscope with an angular magnification of 30. Find the required tube length of the microscope for the final image to form at infinity. CBSE 2022 C *3*

8. (a) (i) Both the objective and the eyepiece of a compound microscope have short focal lengths. Explain.

(ii) Two convex lenses of focal lengths 1·25 cm and 5·0 cm are used to design a compound microscope with an angular magnification of 30. Find the required tube length of the microscope for the final image to form at infinity.

CBSE 2022 C *3*

(a) (i) The magnification produced by a compound microscope is inversely proportional to the focal length of the objective and the eyepiece. Hence short focal length increases magnification.

Alternatively

1

$m = \left(\frac{L}{f_0}\right) \quad \left(\frac{D}{f_e}\right)$, where f_0 and f_e are small then magnification m will be large.

1

(ii) $m = m_0 \times m_e = \frac{L \times D}{f_0 \times f_e}$

$30 = \frac{L \times (25)}{1.25 \times 5}$ ½

$\therefore$ L = 7.5 cm ½

10. (a) For an equilateral triangular prism, the angle of minimum deviation is 30°. Find the refractive index of the material of the prism.

CBSE 2022 C

(b) The angle of minimum deviation for a prism is found to be equal to the angle of the prism (A). Show that the refractive index of the material of the prism, $\mu = 2 \cos\left(\frac{A}{2}\right)$. *3*

10. (a) For an equilateral triangular prism, the angle of minimum deviation is 30°. Find the refractive index of the material of the prism.

CBSE 2022 C

(b) The angle of minimum deviation for a prism is found to be equal to the angle of the prism (A). Show that the refractive index of the material of the prism, $\mu = 2 \cos\left(\frac{A}{2}\right)$. *3*

(a) $\mu = \dfrac{\sin\left(\dfrac{A+D_m}{2}\right)}{\sin A/2}$	½
$= \dfrac{\sin\left(\dfrac{60^o+30^o}{2}\right)}{\sin 60/2}$	½
$= \dfrac{\sin 45^o}{\sin 30^o} = \sqrt{2}$	½
(b) $\mu = \dfrac{\sin\left(\dfrac{A+D_m}{2}\right)}{\sin A/2}$	½
$= \dfrac{\sin\left(\dfrac{A+A}{2}\right)}{\sin A/2} = \dfrac{\sin A}{\sin A/2}$	½
$\mu = \dfrac{2\sin A/2 \cos A/2}{\sin A/2} = 2\cos A/2$	½

6. (a) Draw a ray diagram to show the formation of an image at the least distance of distinct vision, by a compound microscope. Hence, obtain an expression for its angular magnification. CBSE 2022 C *3*

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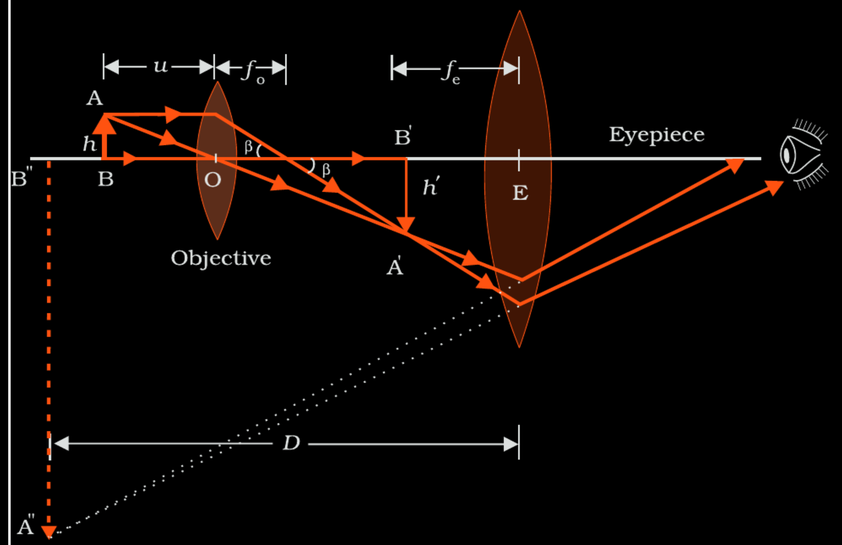


Linear magnification due to objective (m_o)

$$m_o = \frac{h'}{h} = \frac{L}{f_o} \text{ ---------- (1)}$$ ½

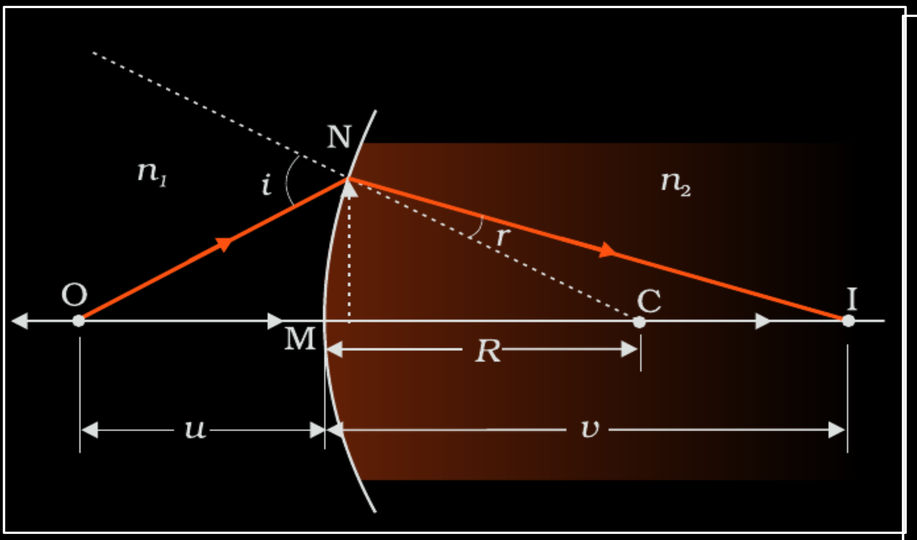
Linear magnification due to eye piece (m_e) when final image is formed at near point (D)

$$m_e = 1 + \frac{D}{f_e} \text{ ----------(2)}$$ ½

Total magnification $m = m_o \times m_e$ ½

$$m = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$$ ½

(b) Consider a convex spherical surface of radius of curvature R, separating two media of refractive indices n_1 and n_2 ($> n_1$). A point object is placed at a distance u in front of the surface in medium of refractive index n_1. Its real image is formed at a distance v. Obtain a relation between u and v in terms of n_1, n_2 and R. CBSE 2022 C *3*



for small angles,

$\tan \angle NOM = \frac{MN}{OM}$

$\tan \angle NCM = \frac{MN}{MC}$

$\tan \angle NIM = \frac{MN}{MI}$

Now, for ΔNOC,∠ i is the exterior angle. Therefore, ∠i = ∠NOM + ∠NCM

$i = \frac{MN}{OM} + \frac{MN}{MC}$ ½

Similarly,

i.e. $r = \frac{MN}{MC} - \frac{MN}{MI}$ ½

now, by Snell's law

$n_1 \sin i = n_2 \sin r$

or for small angles , $\sin i \approx i$ and $\sin r \approx r$

$n_1\ i = n_2\ r$

Substituting i and r , we get

$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}$ ½

Here , OM , MI and MC represent magnitude of distances. Applying the Cartesian sign convention

OM = -u , MI = +v , MC = +R

Substituting these in equation , we get

$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ ½

1

(b) Light travels from a rarer to a denser medium. How will

(i) the energy carried, and

(ii) the wavelength of the light wave,

be affected ?

CBSE 2022 C

1

(b) Light travels from a rarer to a denser medium. How will

(i) the energy carried, and

(ii) the wavelength of the light wave,

be affected ?

CBSE 2022 C

1

(b) (i) No effect
(c) (ii) Wavelength decreases

12. A lens is a portion of a transparent medium bounded by two surfaces and one of these surfaces is essentially spherical. It is used to converge or diverge the light incident on it. Power of a lens is the measure of its ability to converge or diverge the light incident on it. Power of a lens depends on the refractive index of the material of lens with respect to the surrounding medium and the radii of curvature of its two surfaces. CBSE 2022 C

(a) The power of a convex lens

(i) increases when the lens is dipped in water.

(ii) increases when the wavelength of incident light decreases.

(iii) decreases when another convex lens is placed in contact with it.

(iv) increases when the lens is cut into two identical plano-convex lenses.

12. A lens is a portion of a transparent medium bounded by two surfaces and one of these surfaces is essentially spherical. It is used to converge or diverge the light incident on it. Power of a lens is the measure of its ability to converge or diverge the light incident on it. Power of a lens depends on the refractive index of the material of lens with respect to the surrounding medium and the radii of curvature of its two surfaces. CBSE 2022 C

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(i) increases when the lens is dipped in water.

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(iii) decreases when another convex lens is placed in contact with it.

(iv) increases when the lens is cut into two identical plano-convex lenses.

(ii)

(b) The focal length of a concave lens is 40 cm. The power of the lens is :

(i) 0·025 D

(ii) 2·5 D

(iii) – 0·025 D

(iv) – 2·5 D

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(i) 0·025 D

(ii) 2·5 D

(iii) – 0·025 D

(iv) – 2·5 D

(iv)

(c) The focal length of a concave lens (μ_g = 1·5) in air is 20 cm. What should be the refractive index of the surrounding medium so that the lens behaves as a converging lens of focal length 60 cm ?

(i) 1·4

(ii) 1·8

(iii) 1·7

(iv) 1·2

(c) The focal length of a concave lens (μ_g = 1·5) in air is 20 cm. What should be the refractive index of the surrounding medium so that the lens behaves as a converging lens of focal length 60 cm ?

(i) 1·4

(ii) 1·8

(iii) 1·7

(iv) 1·2

(ii)

(d) Beams of red light, blue light, yellow light and violet light are incident on a convex lens, one-by-one. Which one of them converges nearest to the lens ?

(i) Blue light

(ii) Violet light

(iii) Red light

(iv) Yellow light

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(i) Blue light

(ii) Violet light

(iii) Red light

(iv) Yellow light

(ii)

(e) A beam of light coming parallel to the principal axis of a convex lens L_1 of focal length 15 cm is incident on it. Another convex lens L_2 of focal length 25 cm is placed coaxially at a distance d from L_1. For the final image to be formed at infinity, the value of d should be :

(i) 10 cm

(ii) 15 cm

(iii) 25 cm

(iv) 40 cm

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(i) 10 cm

(ii) 15 cm

(iii) 25 cm

(iv) 40 cm

(iv)

4. (a) (i) Define SI unit of power of a lens.

(ii) A plano convex lens is made of glass of refractive index 1.5. The radius of curvature of the convex surface is 25 cm.

(ii.i) Calculate the focal length of the lens.

(ii.ii) If an object is placed 50 cm in front of the lens, find the nature and position of the image formed.

(i)

Dioptre is the reciprocal of focal length of lens in metre. — 1

Alternatively:

$$\text{Dioptre} = \frac{1}{\text{focal length (m)}}$$

Alternatively:

One dioptre is the power of a lens of focal length of one metre

ii.i)

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ — ½

$R_1 = \infty,\ R_2 = -25\ \text{cm},\ \mu = 1{\cdot}5$

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{\infty} + \frac{1}{25}\right)$$

$$or\ \frac{1}{f} = 0{\cdot}5 \times \frac{1}{25}\ or\ f = 50\ \text{cm}$$ — ½

ii.ii) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\frac{1}{50} = \frac{1}{v} - \frac{1}{-50}$$

$$\frac{1}{v} = 0$$ — ½

$\therefore\ v = \infty$, — ½

Thus the image will be real and inverted

(Note: Award only ½ mark if a student draws ray diagram showing correct position of the image instead of doing calculations)

6. With the help of a ray diagram, show how a compound microscope forms a magnified image of a tiny object, at least distance of distinct vision. Hence derive an expression for the magnification produced by it. 3

CBSE 2022

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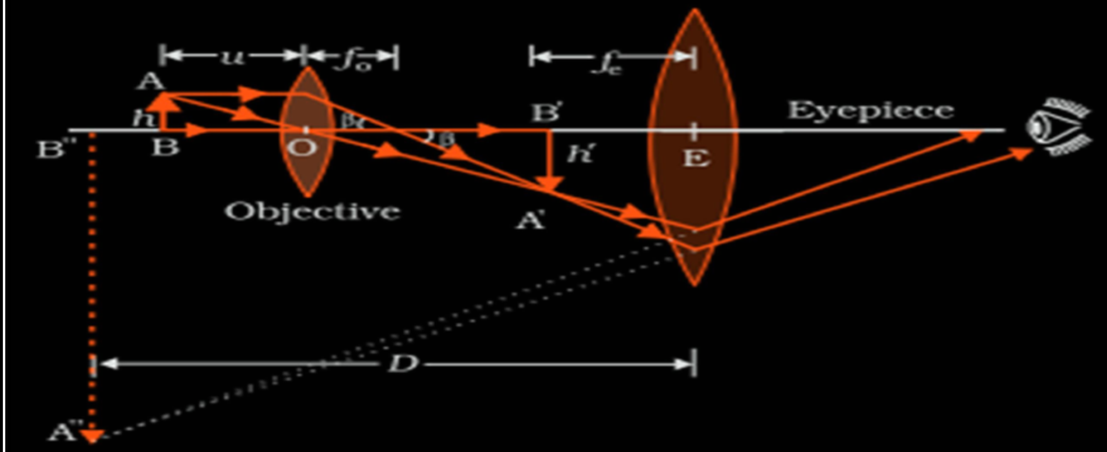

1½

(**Note:** Deduct ½ mark if the arrows are not drawn in the ray diagram)

The linear magnification due to objective lens

$m_0 = \frac{h'}{h} = \frac{L}{f_0}$ ----------(i) ½

h= size of object

h'= size of first image

As $\tan\beta = \frac{h}{f_0} = \frac{h'}{L}$

magnification due to eye piece

$m_e = 1 + \frac{D}{f_e}$ ----------(ii) ½

Total Magnification $m = m_o \times m_e$

$m_. = \frac{L}{f_0}\left(1 + \frac{D}{f_e}\right)$ ½

(b) With the help of a ray diagram explain the working of a reflecting telescope. Mention two advantages of a reflecting telescope over a refracting telescope. 3

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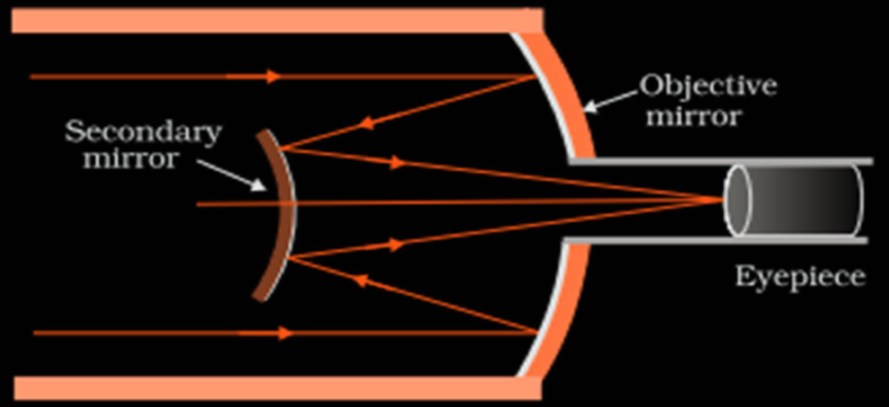

1 ½

Working: Parallel beam of light gathered by objective mirror is reflected to the secondary mirror, which further forms the image in front of the eyepiece. ½

(**Note:** Deduct ½ marks for not showing the direction of propagation of rays and give full credit for the ray diagram of Newtonian telescope)

Two Advantages (Any Two) ½ + ½

- High resolving power
- No chromatic aberration
- Reduced spherical aberration
- Brighter image is formed
- Easy mechanical support
- Large magnifying power

10. A ray of light passes through a prism of refractive index $\sqrt{2}$ as shown in the figure. Find 3

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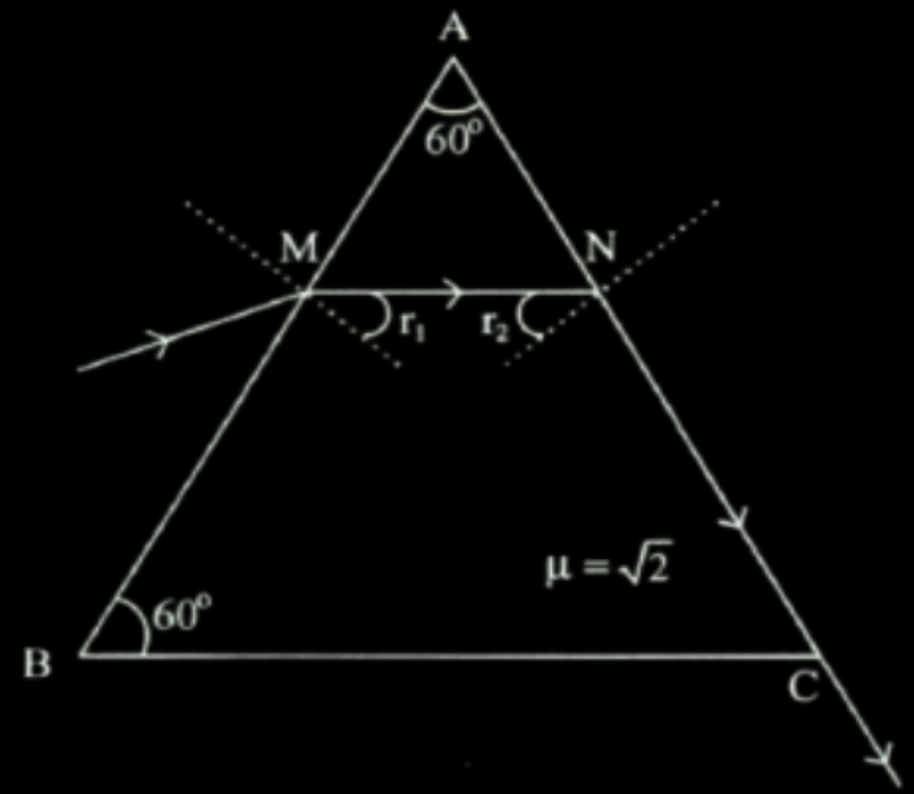

(i) The angle of incidence ($\angle r_2$) at face AC.

(ii) The angle of minimum deviation for this prism.

(i) As the emergent ray grazes along the side AC, therefore	
$\frac{1}{\sqrt{2}} = \frac{\sin r_2}{\sin 90^0}$	1
$\therefore \; r_2 = 45°$	½
(ii) $\mu = \frac{\sin(\frac{A+\delta m}{2})}{\sin\frac{A}{2}}$	½
$\sqrt{2} = \frac{\sin(\frac{60^0+\delta m}{2})}{\sin 30^0}$	½
$\therefore \delta_m = 30^0$	½

12. A ray of light travels from a denser to a rarer medium. After refraction, it bends away from the normal. When we keep increasing the angle of incidence, the angle of refraction also increases till the refracted ray grazes along the interface of two media. The angle of incidence for which it happens is called critical angle. If the angle of incidence is increased further the ray will not emerge and it will be reflected back in the denser medium. This phenomenon is called total internal reflection of light. 5

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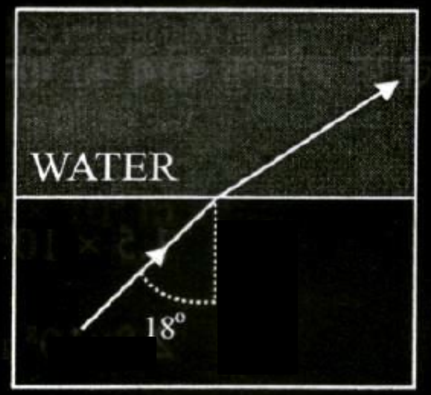
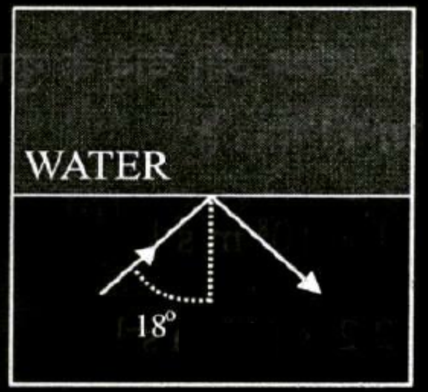
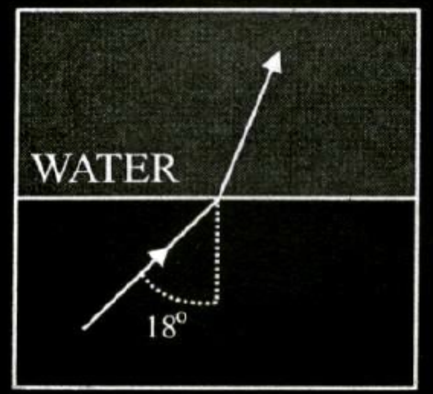
(i) A ray of light travels from a medium into water at an angle of incidence of 18°. The refractive index of the medium is more than that of water and the critical angle for the interface between the two media is 20°. Which one of the following figures best represents the correct path of the ray of light ?

(A)

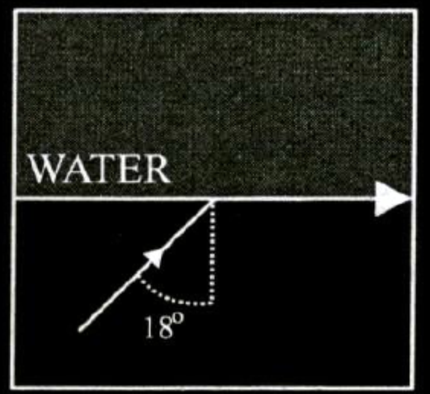


(B)





(C)



(D)

(i) A ray of light travels from a medium into water at an angle of incidence of 18°. The refractive index of the medium is more than that of water and the critical angle for the interface between the two media is 20°. Which one of the following figures best represents the correct path of the ray of light ?

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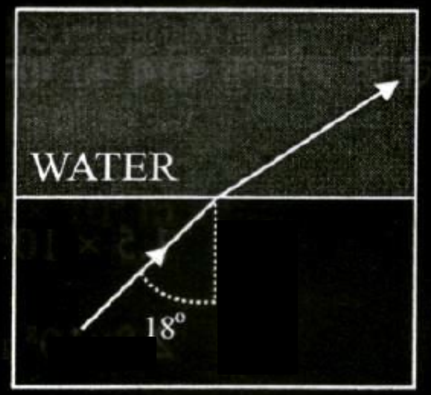
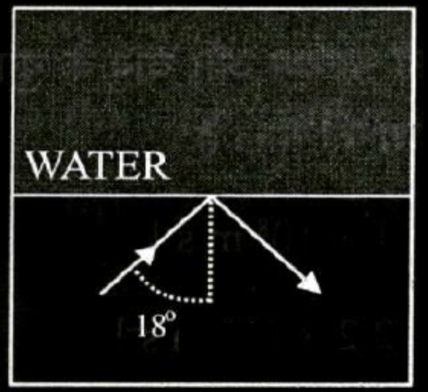
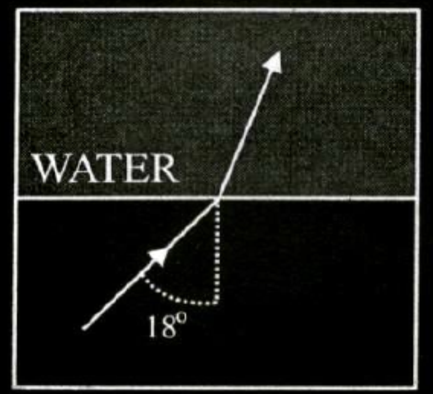
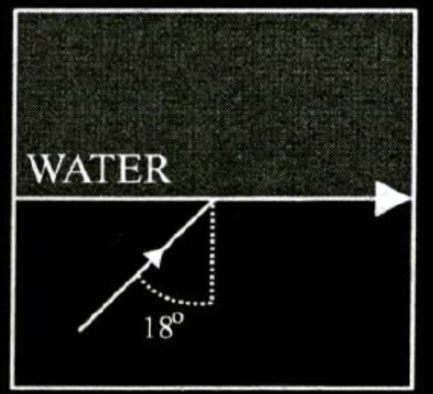


(B)





(C)



(D)

(i)—a

(ii) A point source of light is placed at the bottom of a tank filled with water, of refractive index μ, to a depth d. The area of the surface of water through which light from the source can emerge, is :

(a) $\frac{\pi d^2}{2(\mu^2-1)}$ (b) $\frac{\pi d^2}{(\mu^2-1)}$ (c) $\frac{\pi d^2}{\sqrt{2}\sqrt{\mu^2-1}}$ (d) $\frac{2\pi d^2}{(\mu^2-1)}$

(ii) A point source of light is placed at the bottom of a tank filled with water, of refractive index μ, to a depth d. The area of the surface of water through which light from the source can emerge, is :

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(ii)—b

(iii) For which of the following media, with respect to air, the value of critical angle is maximum ?

(a) Crown glass (b) Flint glass (c) Water (d) Diamond

(iii) For which of the following media, with respect to air, the value of critical angle is maximum ?

(a) Crown glass (b) Flint glass (c) Water (d) Diamond

(iii)—c

(iv) The critical angle for a pair of two media A and B of refractive indices 2.0 and 1.0 respectively is :

(a) 0^o (b) 30^o (c) 45^o (d) 60^o

(iv) The critical angle for a pair of two media A and B of refractive indices 2.0 and 1.0 respectively is :

(a) 0^o (b) 30^o (c) 45^o (d) 60^o

(iv)—b

(v) The critical angle of pair of a medium and air is 30^{o}. The speed of light in the medium is :

(a) $1 \times 10^{8}\ m\ s^{-1}$ (b) $1.5 \times 10^{8}\ m\ s^{-1}$

(c) $2.2 \times 10^{8}\ m\ s^{-1}$ (d) $2.8 \times 10^{8}\ m\ s^{-1}$

(v) The critical angle of pair of a medium and air is 30°. The speed of light in the medium is :

(a) 1×10^8 m s^{-1}

(b) 1.5×10^8 m s^{-1}

(c) 2.2×10^8 m s^{-1}

(d) 2.8×10^8 m s^{-1}

(v) —b

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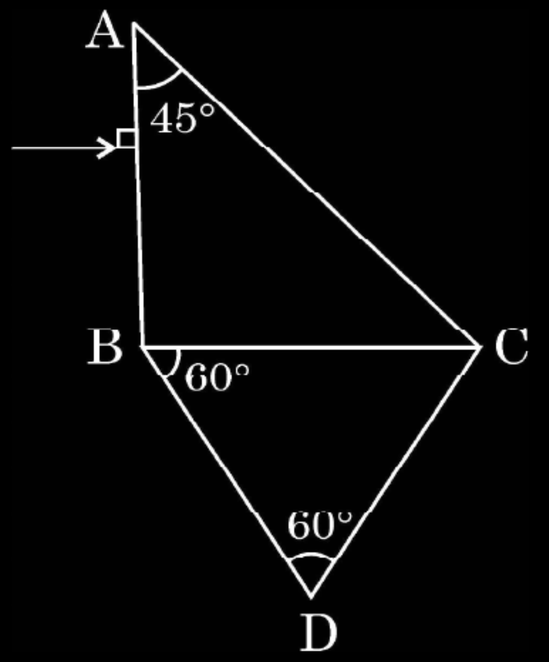
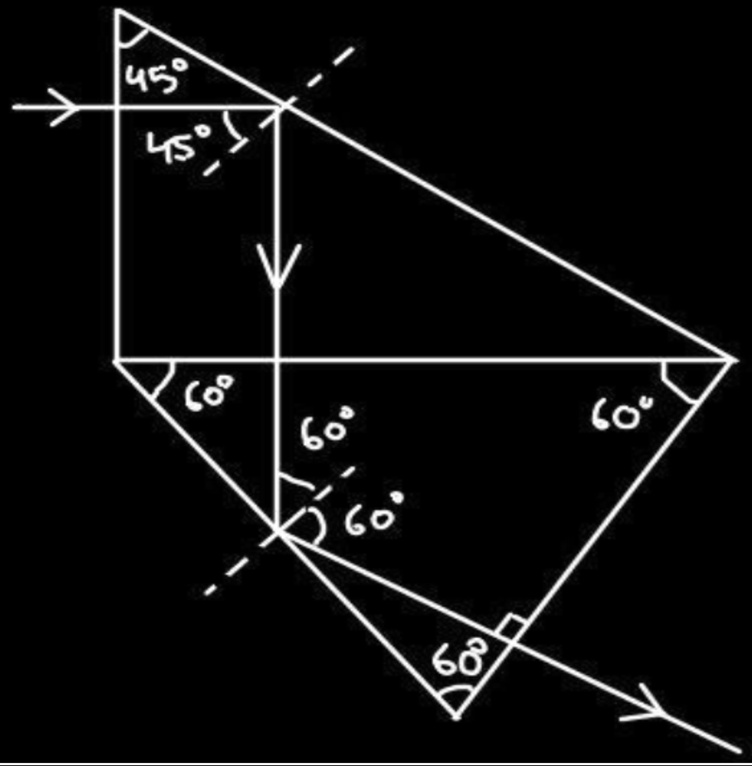
4. (a) Write two necessary conditions for total internal reflection.

(b) Two prisms ABC and DBC are arranged as shown in figure.

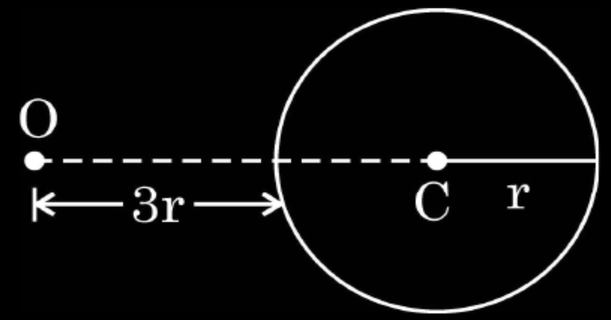


The critical angles for the two prisms with respect to air are 41.1° and 45° respectively. Trace the path of the ray through the combination. **3**

a) Two conditions	
(i) The light must travel from an optically denser medium to a rarer medium.	1
(ii) Angle of incidence should be greater than the critical angle.	1
(b) 	1

(a) An object is placed in front of a converging lens. Obtain the conditions under which the magnification produced by the lens is (i) negative and (ii) positive.

(b) A point object is placed at O in front of a glass sphere as shown in figure.



Show the formation of image by the sphere.

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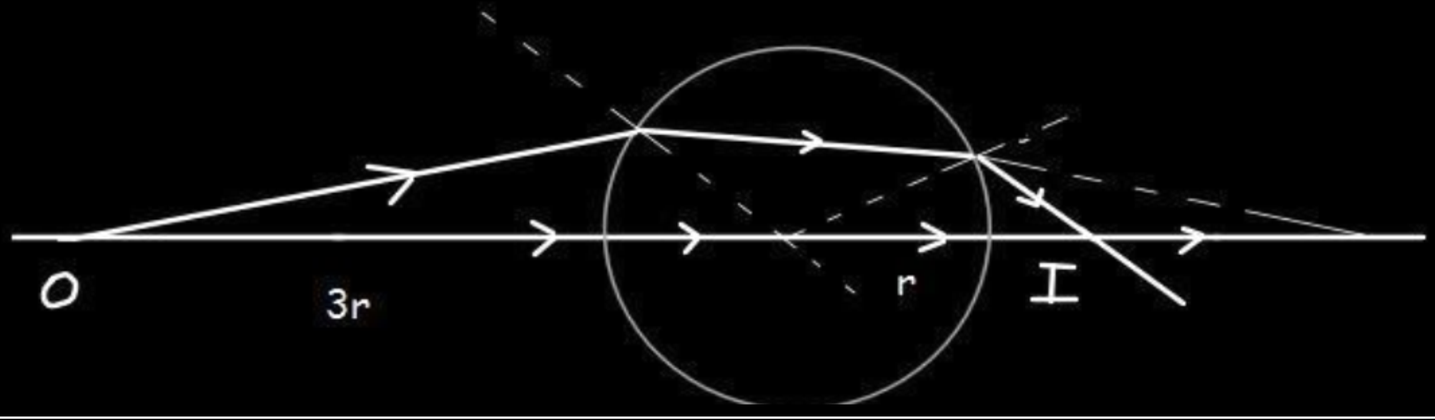
(a) Two conditions $m = \frac{h'}{h} = \frac{v}{u}$

(i) For real images / when object is placed beyond F, As u is negative and v is positive.. ½ + ½

(ii) for virtual image / when object is kept between F and the optical centre of the lens. As u and v both are negative. ½ + ½



1

12. A compound microscope consists of two converging lenses. One of them, of smaller aperture and smaller focal length is called objective and the other of slightly larger aperture and slightly larger focal length is called eye-piece. Both the lenses are fitted in a tube with an arrangement to vary the distance between them. A tiny object is placed in front of the objective at a distance slightly greater than its focal length. The objective produces the image of the object which acts as an object for the eye-piece. The eye piece, in turn produces the final magnified image. **1 × 5 = 5**

I. In a compound microscope the images formed by the objective and the eye-piece are respectively

(A) virtual, real
(B) real, virtual
(C) virtual, virtual
(D) real, real

CBSE 2022

12. A compound microscope consists of two converging lenses. One of them, of smaller aperture and smaller focal length is called objective and the other of slightly larger aperture and slightly larger focal length is called eye-piece. Both the lenses are fitted in a tube with an arrangement to vary the distance between them. A tiny object is placed in front of the objective at a distance slightly greater than its focal length. The objective produces the image of the object which acts as an object for the eye-piece. The eye piece, in turn produces the final magnified image. **1 × 5 = 5**

I. In a compound microscope the images formed by the objective and the eye-piece are respectively

(A) virtual, real
(B) real, virtual
(C) virtual, virtual
(D) real, real

CBSE 2022

(B) real , virtual

II. The magnification due to a compound microscope *does not* depend upon

(A) the aperture of the objective and the eye-piece

(B) the focal length of the objective and the eye-piece

(C) the length of the tube

(D) the colour of the light used

II. The magnification due to a compound microscope *does not* depend upon

(A) the aperture of the objective and the eye-piece

(B) the focal length of the objective and the eye-piece

(C) the length of the tube

(D) the colour of the light used

(A) The aperture of the objective and the eye piece

III. Which of the following is *not correct* in the context of a compound microscope ?

(A) Both the lenses are of short focal lengths.

(B) The magnifying power increases by decreasing the focal lengths of the two lenses.

(C) The distance between the two lenses is more than (fo + fe).

(D) The microscope can be used as a telescope by interchanging the two lenses.

III. Which of the following is *not correct* in the context of a compound microscope ?

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IV. A compound microscope consists of an objective of 10X and an eye-piece of 20X. The magnification due to the microscope would be

(A) 2
(B) 10
(C) 30
(D) 200

IV. A compound microscope consists of an objective of 10X and an eye-piece of 20X. The magnification due to the microscope would be

(A) 2 (B) 10

(C) 30 (D) 200

(D) 200

V. The focal lengths of objective and eye-piece of a compound microscope are 1.2 cm and 3.0 cm respectively. The object is placed at a distance of 1.25 cm from the objective. If the final image is formed at infinity, the magnifying power of the microscope would be

(A) 100 (B) 150

(C) 200 (D) 250

V. The focal lengths of objective and eye-piece of a compound microscope are 1.2 cm and 3.0 cm respectively. The object is placed at a distance of 1.25 cm from the objective. If the final image is formed at infinity, the magnifying power of the microscope would be

(A) 100 (B) 150

(C) 200 (D) 250

(C) 200

5. Monochromatic light of wavelength 600 nm is incident from air on a water surface. The refractive index of water is 1.33. Find the (i) wavelength, (ii) frequency and (iii) speed, of reflected and refracted light. CBSE 2022 **3**

5. Monochromatic light of wavelength 600 nm is incident from air on a water surface. The refractive index of water is 1.33. Find the (i) wavelength, (ii) frequency and (iii) speed, of reflected and refracted light. CBSE 2022 **3**

(a) Reflected high	
$\lambda' = \lambda = 600$ nm.	
No change in wavelength	½
c = 3 x 10^8 m/s in air	
	½
$\upsilon = \frac{c}{\lambda} = \frac{3 \times 10^8}{5 \times 10^{14}}$ Hz	½
Refracted light	
$\lambda' = \frac{\lambda}{\mu} = \frac{600}{1.33} = 451.1$ nm	½
$\simeq 450$nm	
$\upsilon = \frac{c}{\mu} = \frac{3\ X10^8}{1.33}$	½
$= 2.26 \times 10^8$ m/s	
$= 5 \times 10^{14}$ Hz	½

(b) A thin equiconvex lens of radius of curvature R made of material of refractive index $\mu 1$ is kept coaxially, in contact with an equiconcave lens of the same radius of curvature and refractive index $\mu 2$ ($> \mu 1$). Find :

(i) the ratio of their powers, and

(ii) the power of the combination and its nature.

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(i) From $P = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ½

$$P_1 = P_{convex} = (\mu - 1)\left(\frac{1}{R_1} - (-\frac{1}{R_2})\right)$$

$$= (\mu - 1)(\frac{2}{R}) \quad \text{½}$$

$$P_2 = P_{concave} = (\mu - 1)\left(-\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$= -(\mu - 1)(\frac{2}{R})$$

$$\therefore \frac{P_1}{P_2} = \frac{(\mu_1 - 1)}{-(\mu_2 - 1)} = \frac{(\mu_1 - 1)}{(1 - \mu_2)} \quad \text{½}$$

(ii) $P = P_1 + P_2$ ½

$$= (\mu_1 - 1)\left(\frac{2}{R}\right) + (-(\mu_2 - 1))\left(\frac{2}{R}\right)$$

$$P = \frac{2(\mu_1 - \mu_2)}{R} \quad \text{½}$$

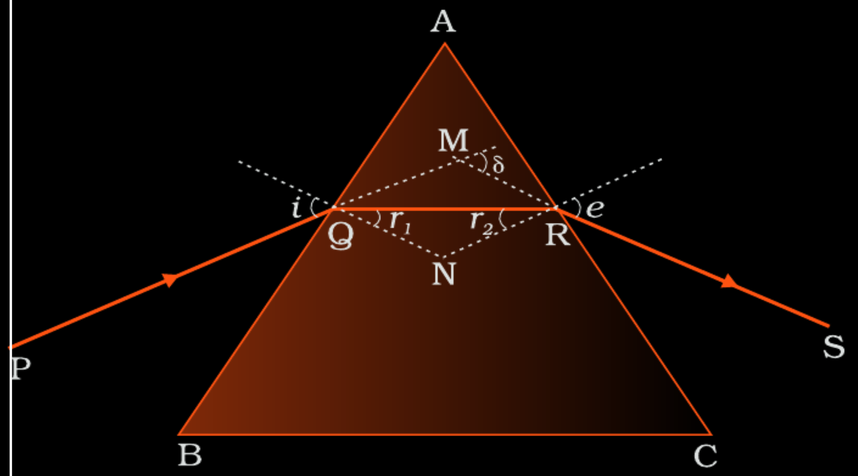
As $\mu_2 > \mu_1$, P is negative

$\therefore$ Nature is diverging. ½

(b) (i) Trace the path of a ray of light PQ which is incident at an angle i on one face of a glass prism of angle A. It then emerges out from the other face at an angle e. Use the ray diagram to prove that the angle through which the ray is deviated is given by $\angle\delta = \angle i + \angle e - \angle A$.

(ii) What will be the minimum value of δ if the ray passes symmetrically through the prism ?

CBSE 2022



$\delta = (i - r_1) + (e - r_2)$ ½

$\delta = (i + e) - (r_1 + r_2)$

In Quadrilateral AQOR

$\angle Q = \angle R = 90^\circ$ $\quad \therefore \angle A + \angle O = 180^\circ$ ---------(1)

In ΔQOR

$O + r_1 + r_2 = 180^\circ$ ---------(2) ½

Comparing (1) and (2)

$\therefore A = r_1 + r_2$ ½

$\therefore \delta = (i + e) - A$

i) If a ray passes symmetrically through a prism (parallel to base of prism), the value of angle of deviation is minimum. ½
At this angle $\angle i = \angle e$ and $\angle r_1 = \angle r_2$

11. For a glass prism of refractive index $\sqrt{3}$, the value of angle of minimum deviation δ is equal to the angle of prism A. Find the angle of prism A. How will the angle δ be affected if the prism were immersed in water ? **3**

CBSE 2022

11. For a glass prism of refractive index $\sqrt{3}$, the value of angle of minimum deviation δ is equal to the angle of prism A. Find the angle of prism A. How will the angle δ be affected if the prism were immersed in water ? **3**

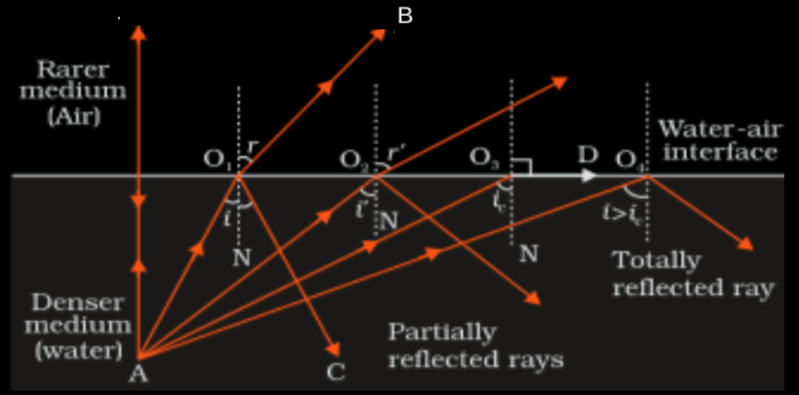
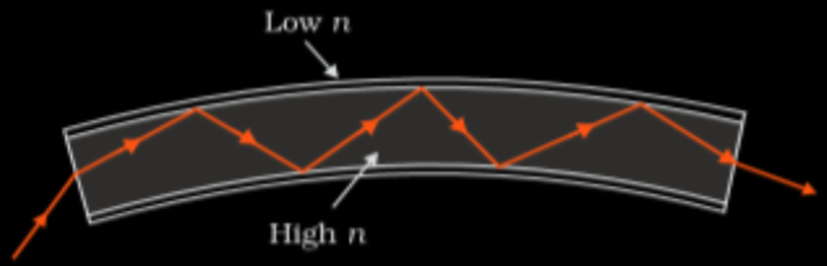
CBSE 2022

$\mu = \sqrt{3}$, $\delta = A$	
$\mu = \dfrac{\sin\dfrac{(A+\delta_m)}{2}}{\sin(A/2)}$	½
$\sqrt{3} = \dfrac{\sin\dfrac{(A+A)}{2}}{\sin(A/2)} = \dfrac{\sin A}{\sin\dfrac{A}{2}}$	½
$\sqrt{3} = \dfrac{2\sin\dfrac{A}{2}\cos\dfrac{A}{2}}{\sin\dfrac{A}{2}} = 2\cos\dfrac{A}{2}$	½
$\dfrac{\sqrt{3}}{2} = \cos\dfrac{A}{2}$	
$\dfrac{A}{2} = 30$	
$\therefore A = 60^\circ$	½
With increase in refractive index of surrounding medium, angle of deviation (δ_m) will decrease	1

4. Explain with the help of a suitable diagram, the phenomenon on which an optical fibre works. Mention any two uses of optical fibres. CBSE 2022 **3**

4. Explain with the help of a suitable diagram, the phenomenon on which an optical fibre works. Mention any two uses of optical fibres. **3**

CBSE 2022

An optical fibre works on the principle of Total internal reflection.	1
	1
Alternatively 	
Uses of optical fibres(any two) i) Medical and optical examination (endoscopy). ii) Transmission and reception of signals iii) Photometric sensors.	½+½

10. Find the two possible positions of an object kept in front of a lens of + 5.0 D, so that the image formed in both cases is four times magnified. CBSE 2022 **3**

10. Find the two possible positions of an object kept in front of a lens of + 5.0 D, so that the image formed in both cases is four times magnified. CBSE 2022 **3**

Power of lens = +5 D Focal length of lens = $\frac{100}{5}$ = 20 cm Magnification ,m = ± 4	½
Case I ,m = 4 = $\frac{v}{u}$ $\quad \therefore v = 4u$ $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ $\frac{1}{20} = \frac{1}{4u} - \frac{1}{u}$ $u = -15cm$	½ 1
Case II, m = - 4 = $\frac{v}{u}$ $\quad \therefore v = -4u$ $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ $\frac{1}{20} = \frac{1}{-4u} - \frac{1}{u}$ $u = -25cm$	½ ½

5. A point source in air is kept 24 cm in front of a concave spherical glass surface (${}_a\mu_g = 1.5$) and radius of curvature 60 cm. Find the nature of the image formed and its distance from the point source. CBSE 2022 **3**

5. A point source in air is kept 24 cm in front of a concave spherical glass surface ($_a\mu_g = 1.5$) and radius of curvature 60 cm. Find the nature of the image formed and its distance from the point source. CBSE 2022 **3**

Relation for concave spherical surface $\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$	½
$\frac{1}{-u} + \frac{\mu}{v} = \frac{\mu - 1}{R}$	½
$\frac{1}{-(-24)} + \frac{1{\cdot}5}{v} = \frac{1{\cdot}5 - 1}{-60}$	½
$\therefore v = -30$ cm	
Distance of image from point source = $-24 - (-30) = 6$ cm	½
Nature of image = Virtual image	1

7. A ray of light passes through an equilateral triangular prism and suffers minimum deviation of 30°. Calculate :

(a) the refractive index of the material of the prism.

(b) the angle at which the ray is incident on the prism. *3*

CBSE 2022

7. A ray of light passes through an equilateral triangular prism and suffers minimum deviation of 30°. Calculate :

(a) the refractive index of the material of the prism.

(b) the angle at which the ray is incident on the prism. *3*

CBSE 2022

(b) $\angle i = \dfrac{\angle A + \angle \delta m}{2} = \dfrac{60^\circ + 30^\circ}{2} = 45^\circ$ — 1½

(a) $\mu = \dfrac{\sin\dfrac{(A+\delta m)}{2}}{\sin(\dfrac{A}{2})} = \dfrac{\sin(\dfrac{60^\circ+30^\circ}{2})}{\sin\dfrac{60^\circ}{2}} = \dfrac{\sin 45^\circ}{\sin 30^\circ}$ — 1

$= \dfrac{\dfrac{1}{\sqrt{2}}}{\dfrac{1}{2}} = \dfrac{2}{\sqrt{2}} = \sqrt{2}$ — ½

(b) (i) What is meant by total internal reflection of light ? Write the two necessary conditions for total internal reflection to occur.

(ii) Calculate the speed of light in a medium for which the critical angle with respect to air is 45°. CBSE 2022 *3*

(b) (i) What is meant by total internal reflection of light ? Write the two necessary conditions for total internal reflection to occur.

(ii) Calculate the speed of light in a medium for which the critical angle with respect to air is 45°. CBSE 2022 *3*

I.	(i) The phenomenon due to which the light entering from a optically denser to a rarer medium gets reflected to the denser medium.	½
	(ii) Two conditions:	
	• Light should travel from a denser medium to rarer medium	½
	• Angle of incidence should be more than the critical angle.	½
II.	$n = \frac{1}{\sin C} = \frac{c_o}{c}$	½
	$c = c_o \sin C$ where c_o- speed of light in vacuum (3×10^8 m/s)	
	$= 3\times10^8 \text{ m/s} \times \frac{1}{\sqrt{2}} = \frac{3}{\sqrt{2}}\times10^8 \text{ m/s}$	½ + ½

10. Answer the following : *3*

(a) A convex lens produces a real image of an object. If the upper half of the lens is covered with a black paper, then how will the image be affected ?

(b) The time difference between the apparent sunrise and the actual sunrise is about 2 minutes. Explain.

(c) Modern telescopes use a concave mirror rather than a lens for the objective. Why ?

CBSE 2022

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(a) A convex lens produces a real image of an object. If the upper half of the lens is covered with a black paper, then how will the image be affected ?

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(c) Modern telescopes use a concave mirror rather than a lens for the objective. Why ?

CBSE 2022

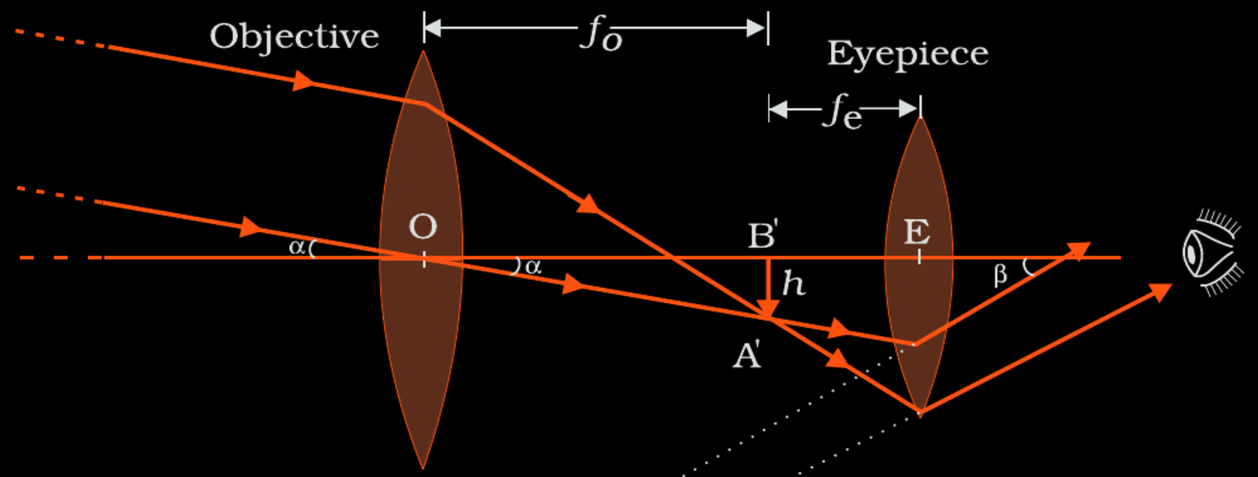
(a) No change in the shape / size and position of the image formed but the intensity of the image decreases	½ + ½
(b) Due to atmospheric refraction the light coming from the sun keeps bending towards the normal, hence reaches earlier	1
(c) Use of mirror makes the image free form chromatic aberration / improves the quality of image / image is comparatively brighter (any other)	1

7. (a) (i) Draw a labelled ray diagram showing the formation of the image at infinity by an astronomical telescope.

(ii) A telescope consists of an objective of focal length 150 cm and an eyepiece of focal length 6·0 cm. If the final image is formed at infinity, then calculate :

(I) the length of the tube in this adjustment, and

(II) the magnification produced. CBSE 2022 *3*



1

(ii) Given $f_o = 150$ cm, $f_e = 6\ cm$

(I) Length of the tube $L = f_o + f_e$ ½

$= 150 + 6$

$L = 156\ cm$ ½

(II) $m = \dfrac{f_o}{f_e}$ ½

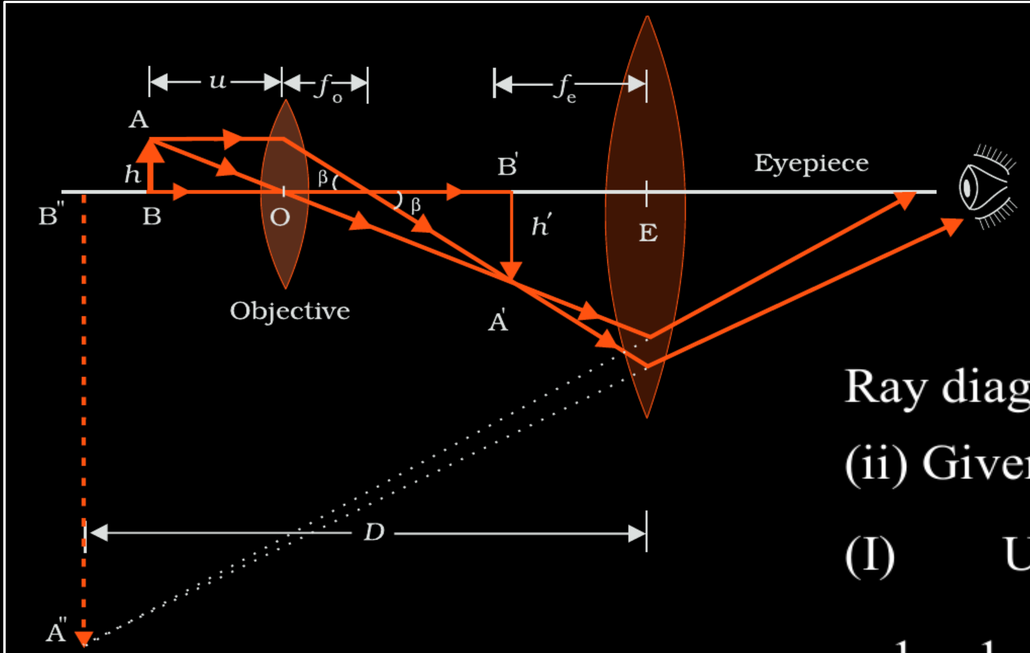
$\dfrac{150}{6} = 25$ ½

(b) (i) Draw a labelled ray diagram showing the formation of the image at least distance of distinct vision by a compound microscope.

(ii) A small object is placed at a distance of 3·0 cm from a magnifier of focal length 4·0 cm. Find :

(I) the position of the image formed, and

(II) the linear magnification produced. CBSE 2022 *3*



Ray diagram of image formation by a compound microscope

(ii) Given u= -3 cm f = 4 cm

(I) Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{1}{-3{\cdot}0} + \frac{1}{4{\cdot}0}$$

$$\frac{1}{v} = \frac{-4+3}{12} = -\frac{1}{12} \qquad v = -12 \text{ cm}$$

(II) Linear magnification $m = \frac{v}{u}$

$$m = \frac{-12}{-3} = 4$$

1

½

½

½

½

11. A diver looking up through water ($\mu = \frac{4}{3}$) sees the outside world contained in a circular area on the surface of water. If the diver's eyes are $\sqrt{7}$ m below the surface of water, then calculate the area of the circle. *3*

CBSE 2022

11. A diver looking up through water ($\mu = \frac{4}{3}$) sees the outside world contained in a circular area on the surface of water. If the diver's eyes are $\sqrt{7}$ m below the surface of water, then calculate the area of the circle. *3*

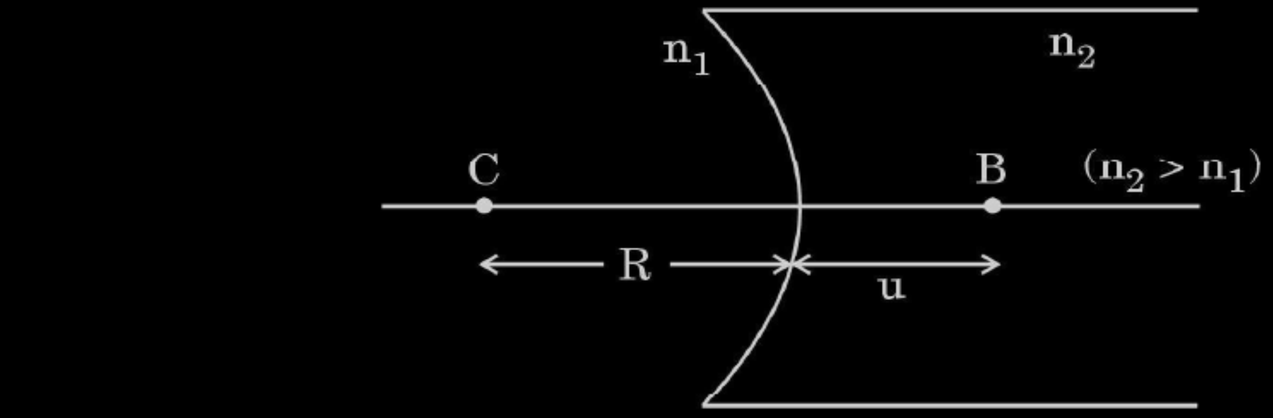
CBSE 2022

$\text{Sin C} = \frac{1}{\mu}$	½
$\text{Sin C} = \frac{3}{4}$	½
$\text{Sin C} = \frac{R}{\sqrt{R^2+7}}$	½
$\frac{3}{4} = \frac{R}{\sqrt{R^2+7}}$	½
$R = 3\text{m}$	½
$\text{Area} = \pi R^2$	
$= 3.14 \times (3)^2$	
$= 28.26\ \text{m}^2$	½

12. Two transparent media of refractive indices n_1 and n_2 are separated by a spherical transparent surface. The rays of light incident on the surface get refracted into the medium on the other side. The laws of refraction are valid at each point of the spherical surface. A lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. The focal length of a lens is determined by the radii of curvature (R_1 and R_2) of its two surfaces and the refractive index (n) of the medium of the lens with respect to the surrounding medium. Depending on R_1 and R_2, a lens behaves as a diverging or a converging lens. The ability of a lens to diverge or converge a beam of light incident on it defines its power.

CBSE 2022

(a) An object is placed at the point B as shown in the figure. The object distance (u) and the image distance (v) are related as



(i) $\frac{1}{v} - \frac{1}{u} = \left(\frac{n_2 - n_1}{n_1}\right)\frac{1}{R}$

(ii) $\frac{1}{v} - \frac{1}{u} = \left(\frac{n_1 - n_2}{n_2}\right)\frac{1}{R}$

(iii) $\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R}$

(iv) $\frac{n_1}{v} - \frac{n_2}{u} = \frac{(n_1 - n_2)}{R}$

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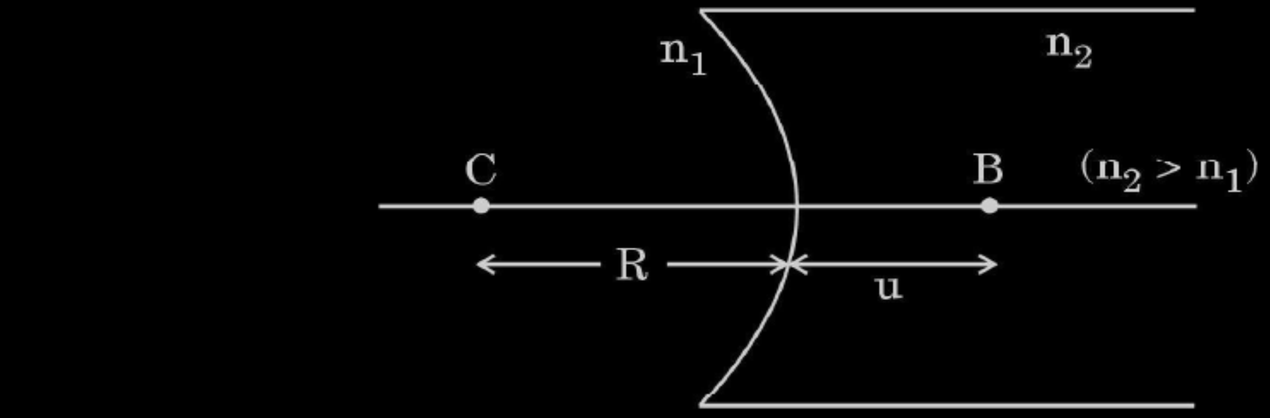

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(ii) $\frac{1}{v} - \frac{1}{u} = \left(\frac{n_1 - n_2}{n_2}\right)\frac{1}{R}$

(iii) $\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R}$

(iv) $\frac{n_1}{v} - \frac{n_2}{u} = \frac{(n_1 - n_2)}{R}$

(iv) $\frac{n_1}{v} - \frac{n_2}{u} = \frac{(n_1 - n_2)}{R}$

(b) A point object is placed in air at a distance 'R' in front of a convex spherical refracting surface of radius of curvature R. If the medium on the other side of the surface is glass, then the image is :

(i) real and formed in glass.

(ii) real and formed in air.

(iii) virtual and formed in glass.

(iv) virtual and formed in air.

(b) A point object is placed in air at a distance 'R' in front of a convex spherical refracting surface of radius of curvature R. If the medium on the other side of the surface is glass, then the image is :

(i) real and formed in glass.

(ii) real and formed in air.

(iii) virtual and formed in glass.

(iv) virtual and formed in air.

(iv) virtual and formed in air

(c) An object is kept at 2F in front of an equiconvex lens. The image formed is :

(i) real and of the size of the object.

(ii) virtual and of the size of the object.

(iii) real and enlarged.

(iv) virtual and diminished.

(c) An object is kept at 2F in front of an equiconvex lens. The image formed is :

(i) real and of the size of the object.

(ii) virtual and of the size of the object.

(iii) real and enlarged.

(iv) virtual and diminished.

(i) real and of the size of the object

(d) A thin converging lens of focal length 10 cm and a thin diverging lens of focal length 20 cm are placed coaxially in contact. The power of the combination is :

(i) – 5 D

(ii) + 5 D

(iii) + 15 D

(iv) – 15 D

(d) A thin converging lens of focal length 10 cm and a thin diverging lens of focal length 20 cm are placed coaxially in contact. The power of the combination is :

(i) – 5 D

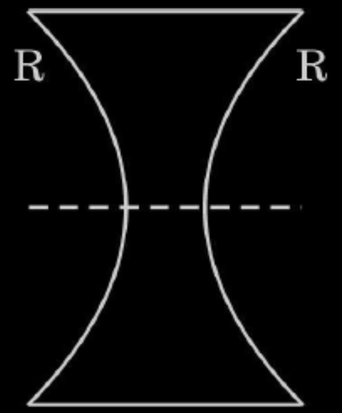
(ii) + 5 D

(iii) + 15 D

(iv) – 15 D

(ii) + 5D

(e) An equiconcave lens of focal length 'f' is cut into two identical parts along the dotted line as shown in the figure. The focal length of each part will be :



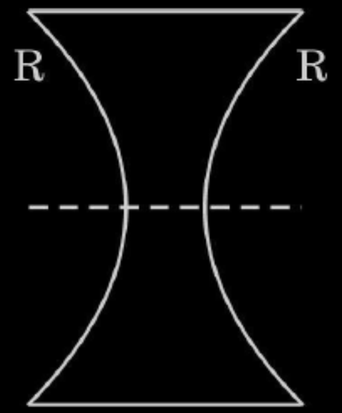
(i) $\frac{f}{4}$

(ii) $\frac{f}{2}$

(iii) f

(iv) 2f

(e) An equiconcave lens of focal length 'f' is cut into two identical parts along the dotted line as shown in the figure. The focal length of each part will be :



(i) $\frac{f}{4}$

(ii) $\frac{f}{2}$

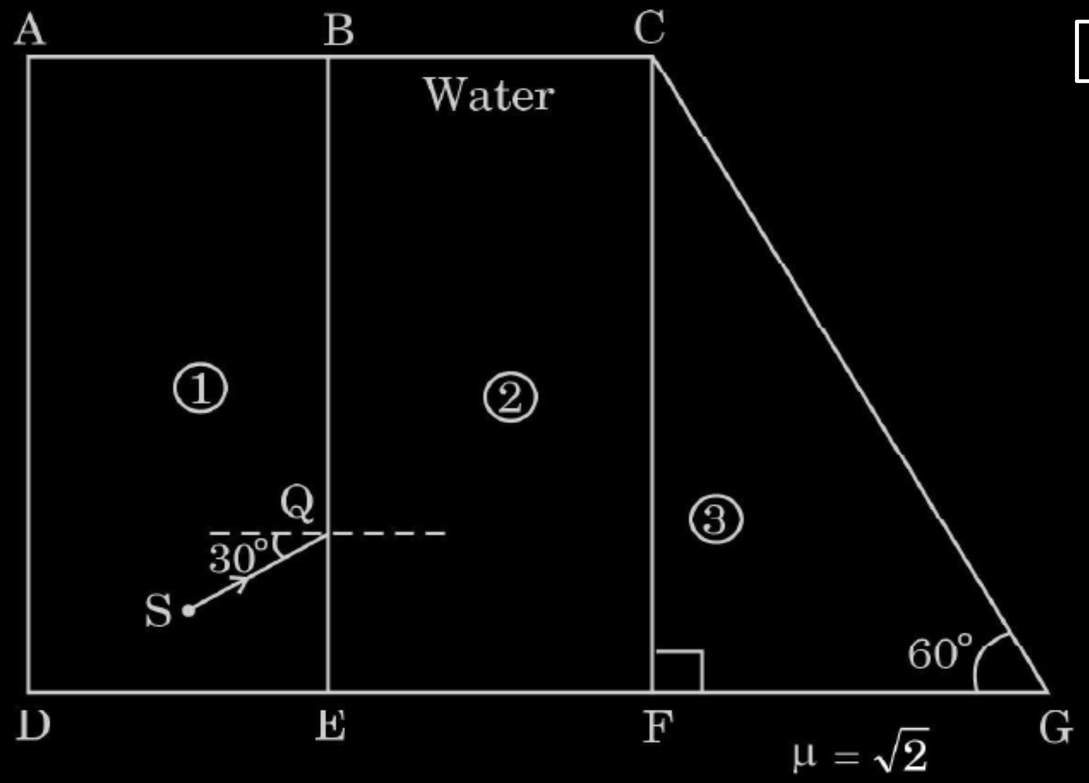
(iii) f

(iv) 2f

(iii) f

11. The figure shows a water column BCFE surrounded by two media ① and ③ of the same refractive index $\sqrt{2}$. A ray of light from a point source S is incident on surface BE at an angle of 30°.

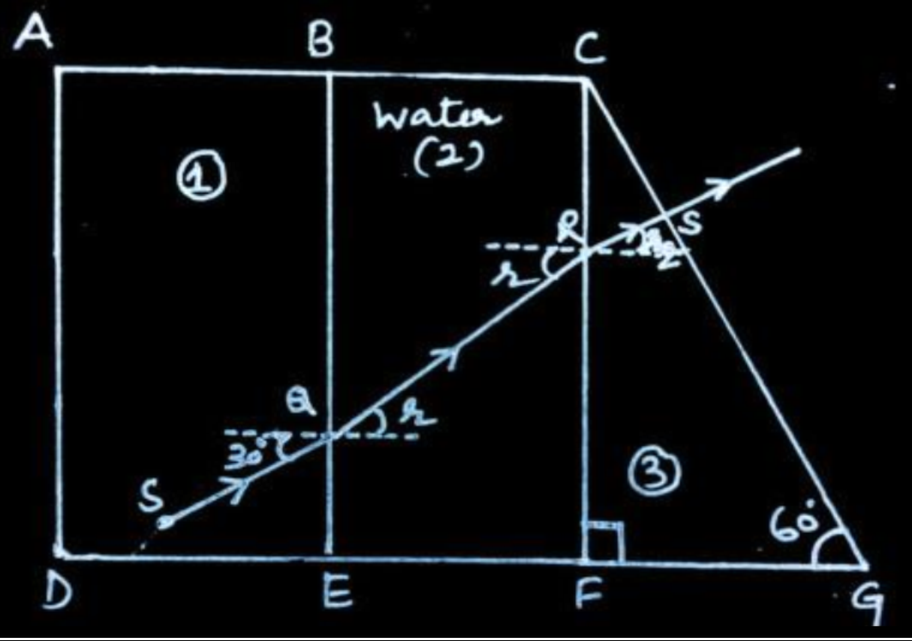
CBSE 2022



(a) Trace the path of ray through media ② and ③ as it emerges out of face CG.

(b) Find the angle of emergence at face CG. *3*

a)



b) Refraction at Q , $\sqrt{2}\ \text{Sin}\ 30^o = \frac{4}{3}\ \text{Sin r}$

$\text{Sin r} = \frac{3}{4\sqrt{2}}$ ½

Refraction from face CF

$\frac{4}{3}\ \text{Sin r} = \sqrt{2}\ \text{Sin r}_2$ ½

$\frac{4}{3} \times \frac{3}{4\sqrt{2}} = \sqrt{2}\ \text{Sin r}_2$ ½

$r_2 = 30^0$

$\angle C + \angle CRS + \angle CSR = 180^0$

$30^o + 60^o + \angle CSR = 180^0$ ½

$\angle CSR = 90^0$

Ray goes perpendicularly out from face CG

$e = 0$

∴ Ray RS incidents perpendicular to CS

∴ angle of emergence of the ray is zero.

Alternatively-

As medium is same on both side of the water column.

Angle of emergence

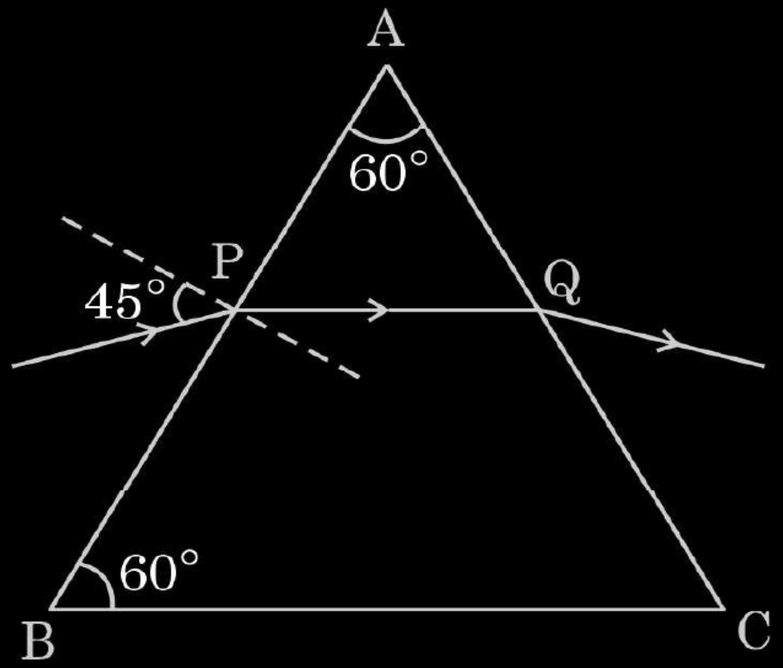
Now $r_1 + r_2 = A$, ½

$r_2 = 30^0$ ½

$r_1 + 30 = 30$ ½

$r_1 = 0$, Hence $e = 0$ ½

10. A ray of light is incident on a prism at an angle of 45° and passes symmetrically as shown in the figure. Calculate : CBSE 2022 *3*



(a) the angle of minimum deviation,

(b) the refractive index of the material of the prism, and

(c) the angle of refraction at the point P.

(a)	As the ray passes symmetrically	
	$\angle i = \angle e$ and $\delta = \delta_m$	
	$\therefore \quad i + e = A + \delta$	½
	$2i = A + \delta_m$	
	$\delta_m = 2i - A = 90° - 60°$	
	$= 30°$	½
(b)		
	$\mu = \dfrac{\sin \dfrac{(A + \delta_m)}{2}}{\sin A/2}$	½
	$= \dfrac{\sin 45°}{\sin 30°} = \sqrt{2}$	½
(c)		
	As $\delta = \delta_m$	
	$r_1 = r_2$ (Alternatively)	
	Hence, r_1 (Angle of refraction at P)	
	$1 \times \sin 45 = \mu \sin r_1$	
	$1 \times \dfrac{1}{\sqrt{2}} = \sqrt{2} \sin r_1$	½
	$\therefore \angle r_1 = 30°$	½

6. An equiconvex lens forms a two times enlarged real image when an object is kept 16 cm from it. The lens is cut into two identical plano-convex lenses. If the object is again kept 16 cm in front of one of these lenses, then find the nature and position of the image formed. CBSE 2022 *3*

6. An equiconvex lens forms a two times enlarged real image when an object is kept 16 cm from it. The lens is cut into two identical plano-convex lenses. If the object is again kept 16 cm in front of one of these lenses, then find the nature and position of the image formed. CBSE 2022 *3*

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$ ½

$$\frac{1}{f} = \frac{1}{32} + \frac{1}{16} = \frac{1+2}{32} = \frac{3}{32}$$ ½

$$f = \frac{32}{3} cm$$ ½

When lens is cut into two equal halves the new focal length = $f' = 2f = \frac{64}{3}$ cm

$$\frac{1}{f'} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{3}{64} = \frac{1}{v} + \frac{1}{16}$$ ½

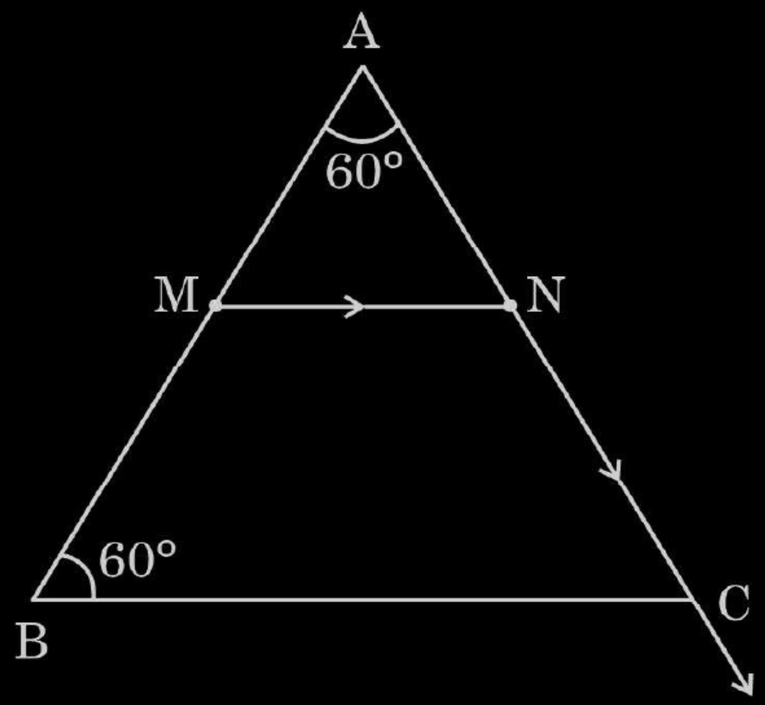
$$\frac{1}{v} = \frac{3}{64} - \frac{1}{16} = \frac{3-4}{64} = \frac{-1}{64}$$

v = - 64 cm ½

Image formed is virtual & erect. ½

7. A ray is incident on a prism of material of refractive index $\sqrt{2}$ at point M such that it grazes along NC after emerging from the prism, as shown in the figure.

CBSE 2022



Find : *3*

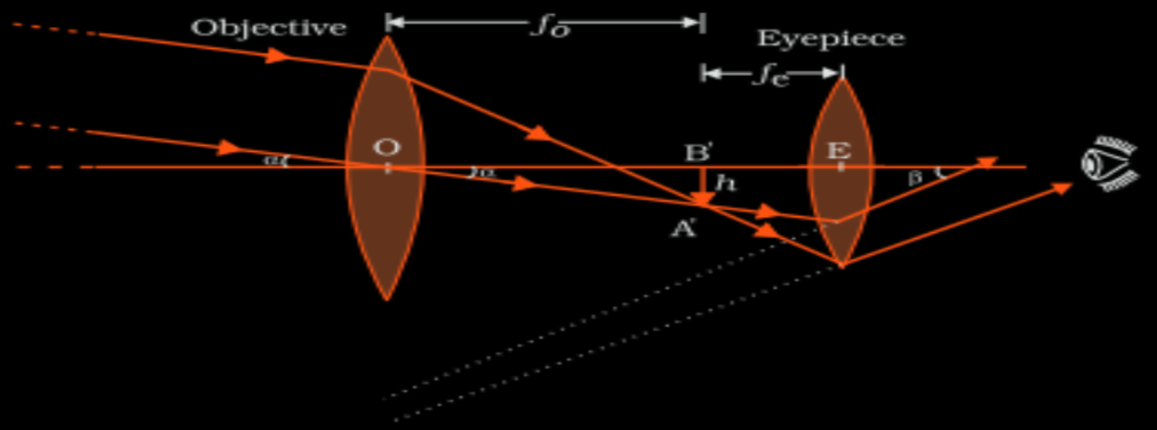
(a) the critical angle for the prism.

(b) the angle of refraction at face AB.

a)	
$\mu = \frac{1}{\sin i_c} \Rightarrow \sqrt{2} = \frac{1}{\sin i_c}$	½
$i_c = 45°$	½
b)	
$r_2 = 45° = i_c$	½
$A = r_1 + r_2$	½
$60 = r_1 + 45°$	½
$r_1 = 15°$	½

9. (a) Draw a labelled ray diagram showing the formation of an image by an astronomical refracting telescope in normal adjustment. Hence, obtain the expression for its magnifying power. CBSE 2022 *3*

9. (a) Draw a labelled ray diagram showing the formation of an image by an astronomical refracting telescope in normal adjustment. Hence, obtain the expression for its magnifying power. CBSE 2022 *3*



1½

$$m = \frac{\tan\beta}{\tan\alpha} \approx \frac{\beta}{\alpha} \text{ (as } \alpha, \beta \text{ are small angles)}$$ ½

$$\approx \frac{h}{f_e} x \frac{f_0}{h}$$ ½

$$= \frac{f_0}{f_e}$$ ½

(Note: ½ mark to be deducted if arrows not shown or labelling is not done)

10. (a) Is the speed of light in glass independent of the colour of light ? Give reason.

(b) A small bulb is placed at the bottom of a tank containing water to a depth of 70 cm. Find the area of the surface of water through which light from the bulb can emerge out. Given refractive index of water is $\frac{4}{3}$.

CBSE 2022

3

a) No

$$\mu \alpha \frac{1}{v_m} \alpha \frac{1}{\lambda}$$

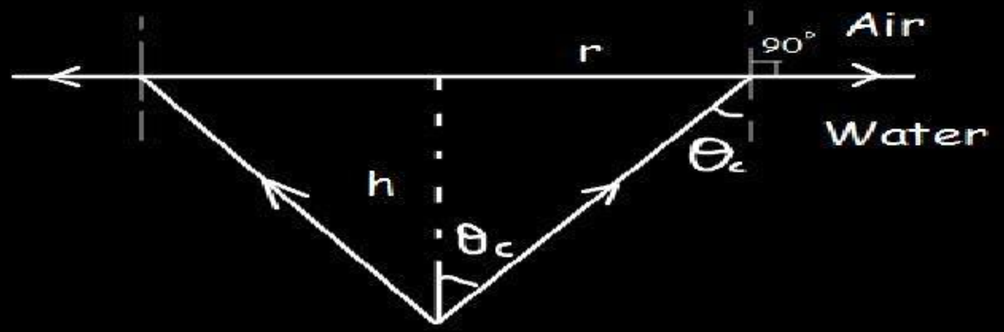
½

Where v_m velocity of light in medium

∴ velocity depends on wavelength

½

b)



½

$$\mu \sin\theta_c = 1 \Rightarrow \sin\theta_c = \frac{1}{\mu} = \frac{3}{4}$$

½

$$\tan\theta_c = \frac{r}{h} = \frac{1}{\mu} \times \frac{1}{\sqrt{1-\frac{1}{\mu^2}}} = \frac{1}{\sqrt{\mu^2-1}}$$

Area $A = \pi r^2 = \pi(h\tan\theta_c)^2$

½

$$A = \frac{\pi h^2}{\mu^2 - 1}$$

$$A = \frac{\pi \times 70 \times 70}{\left(\frac{4}{3}\right)^2 - 1} = 19800 \text{ cm}^2$$

½

11. The refractive index of an equilateral triangular prism kept in air is $\sqrt{2}$.

Calculate : *3*

(a) the angle of minimum deviation.

(b) the critical angle for the prism.

CBSE 2022

11. The refractive index of an equilateral triangular prism kept in air is $\sqrt{2}$.

Calculate : *3*

(a) the angle of minimum deviation.

(b) the critical angle for the prism.

CBSE 2022

a)

$$\mu = \frac{\sin(\frac{\delta_m + A)}{2}}{\sin(A/2)}$$

½

$$\sqrt{2} = \frac{\sin\left(\frac{\delta_m + 60°}{2}\right)}{\sin 30°} \Rightarrow \sin\left(\frac{\delta_m + 60°}{2}\right) = \sqrt{2} \times \frac{1}{2}$$

½

$$\sin\left(\frac{\delta_m + 60°}{2}\right) = \frac{1}{\sqrt{2}} \Rightarrow \left(\frac{\delta_m + 60°}{2}\right) = 45°$$

½

$$\delta_m = 90° - 60° \Rightarrow \delta_m = 30°$$

½

b) $\sin\theta_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}}$

½

$\theta_c = 45°$

½

5. (a) State the conditions for total internal reflection to take place.

(b) A tank is filled with a transparent liquid to height 'H'. A coin suspended by a thread in the liquid is gradually lowered till it touches the bottom. The apparent depth is determined corresponding to different positions of the coin.

(i) Plot a graph showing variation of the apparent depth with the real depth of the coin.

(ii) What is the physical significance of the slope of the graph ? *3*

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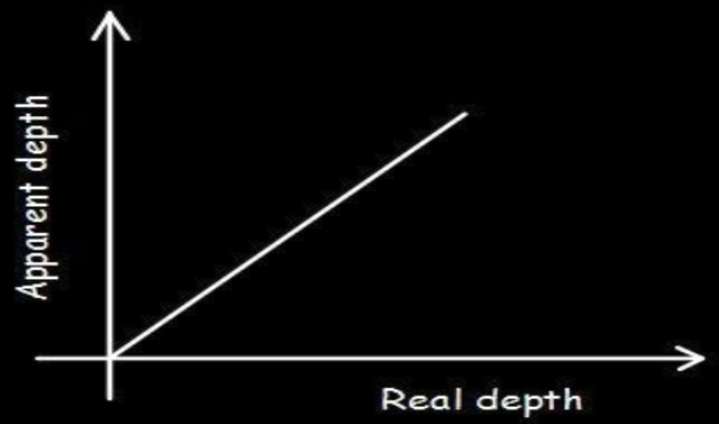
(a) The conditions for total internal reflection to take place are:	
(i) The light should travel from denser to rarer medium.	½
(ii) The angle of incidence should be greater than the critical angle for a given pair of media.	½
(b) (i) 	1
(ii)The slope will give 1/μ where μ is refractive index as per the graph above.	1

7. A converging lens made of glass ($\mu = 1{\cdot}5$) has its spherical faces of radii of curvature 10 cm and 20 cm. Find its focal length

(a) in air, and

(b) when it is immersed in a liquid of refractive index 1·25. *3*

CBSE 2022

7. A converging lens made of glass ($\mu = 1{\cdot}5$) has its spherical faces of radii of curvature 10 cm and 20 cm. Find its focal length

(a) in air, and

(b) when it is immersed in a liquid of refractive index 1·25. *3*

CBSE 2022

(a)

$$\frac{1}{f_a} = ({}_a\mu^g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ ½

$R_1 = 10$ cm & $R_2 = -20$ cm , ${}_a\mu^g = 1.5$ ½

$$\frac{1}{f_a} = (1.5-1)\left(\frac{1}{10} - \frac{1}{-20}\right)$$

$$f_a = \frac{40}{3} = 13.3 \text{ cm}$$ ½

b) $$\frac{1}{f_l} = ({}_l\mu^g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ ½

$$= \left(\frac{1.5}{1.25} - 1\right)\left(\frac{1}{10} - \frac{1}{-20}\right)$$ ½

$$f_l = \frac{100}{3} = 33.3 \text{ cm}$$ ½

10. A biconcave lens of power P vertically splits into two identical plano concave parts. The power of each part will be

CBSE 2020

(a) 2P (b) P/2

(c) P (d) $P/\sqrt{2}$ **1**

10. A biconcave lens of power P vertically splits into two identical plano concave parts. The power of each part will be

CBSE 2020

(a) 2P (b) P/2

(c) P (d) $P/\sqrt{2}$ 1

(b) P/2

32. An optical instrument uses a lens of power 100 D for objective lens and 50 D for its eyepiece. When the tube length is kept at 25 cm. the final image is formed at infinity.

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(a) Identify the optical instrument

(b) Calculate the magnification produced by the instrument. **3**

Objective lens with a power of 100 D, has a focal length of 1cm (very short focal length) Eye piece, with a power of 50 D, has a focal length of 2cm (short focal length)	½
The optical instrument is therefore a compound microscope. (**Note:** Award this one mark if a student writes directly compound microscope without justifying.)	½
When the final image is formed at infinity, the magnification of a compound microscope equals $m = \left(\frac{L}{f_0}\right)\left(\frac{D}{f_e}\right)$ $L = 25\,cm, D = 25\,cm$ $f_0 = 1\,cm, f_e = 2\,cm$	1
$m = \left(\frac{25}{1} \times \frac{25}{2}\right)$	½
$= 312.5$	½

35. (a) Define the term 'focal length of a mirror'. With the help of a ray diagram, obtain the relation between its focal length and radius of curvature.

(b) Calculate the angle of emergence (e) of the ray of light incident normally on the face AC of a glass prism ABC of refractive index $\sqrt{3}$. How will the angle of emergence change qualitatively, if the ray of light emerges from the prism into a liquid of refractive index 1.3 instead of air ?

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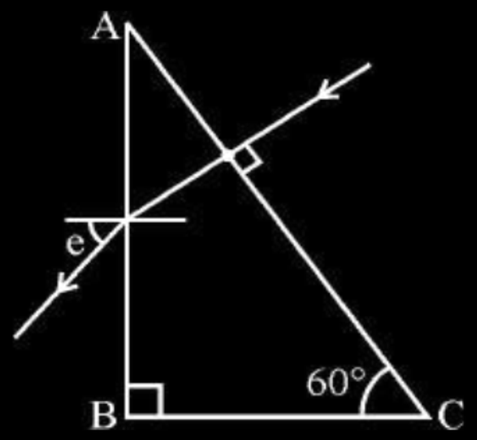
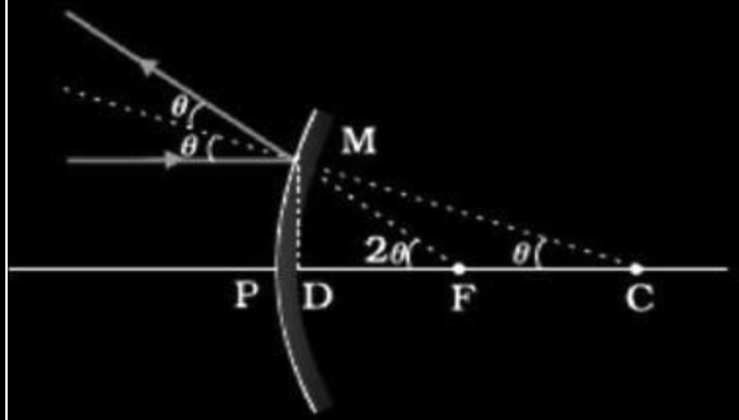


(a) **Focal length of mirror**: It is the distance of the point from the pole of mirror through which ray of light moving parallel to its principle axis passes (or appear to come from). 1

Alternatively: It is half of the distance of its centre of curvature from the pole of a mirror.



Let C be the centre of curvature of mirror, MD be the perpendicular from M to the principal axis.

$\angle MCP = \theta$ *and* $\angle MFP = 2\theta$

$$\tan\theta = \frac{MD}{CD} \quad , \quad \tan 2\theta = \frac{MD}{FD} \qquad (1)$$ ½

For small angles, $\tan\theta \approx \theta$ and $\tan 2\theta \approx 2\theta$ ½

From equation 1, $\dfrac{MD}{FD} = 2\dfrac{MD}{CD}$

$$FD = \frac{CD}{2} \qquad ----equation\ (2)$$

For small θ, the point D is very close to the point P

$$\therefore \ \text{FD} \simeq \text{FP} = f \text{ and } \text{CD} \simeq \text{CP} = R$$

$\therefore$ from equation 2, we get $\text{f} = \dfrac{R}{2}$ ½

(b) Applying Snell's law at face AB, we get $\sqrt{3}\sin 30 = 1 \cdot \sin e$ $\sqrt{3} \times \frac{1}{2} = \sin e$ $\frac{\sqrt{3}}{2} = \sin e$ $\sin 60 = \sin e$ $e = 60^0$ When the medium (the air) in which the prism is kept is replaced with a liquid of refractive index 1.3 the angle of emergence would decrease. It is because bending in the ray of light will be lesser.	½ ½ ½ ½ ½

(b) A screen is placed 80 cm from an object. The image of the object on the screen is formed by a convex lens placed between them at two different locations separated by a distance 20 cm. Determine the focal length of the lens.

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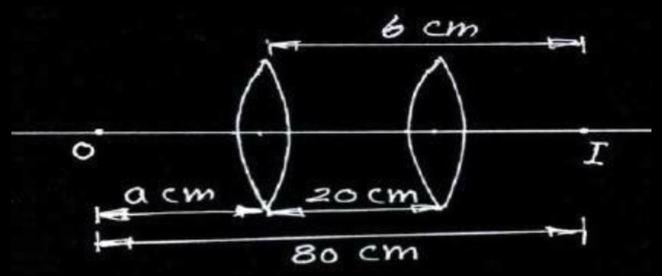
The two positions of the lens are as shown



for position 1

$u = -a\ cm$

$v = b = +(80 - a)\ cm$ ½

for position 2

$u = -(a + 20)\ cm$

$v = +(b - 20) = +(60 - a)\ cm$ ½

By lens formula ½

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\therefore \frac{1}{(80-a)} + \frac{1}{a} = \frac{1}{(60-a)} + \frac{1}{(20+a)}$$

This gives

$a(80-a)=(a+20)(60-a)$

or $80a - a^2 = 40a + 1200 - a^2$

or $a = 30cm$

$$\therefore \frac{1}{f} = \frac{1}{50} + \frac{1}{30} = \frac{3+5}{150} = \frac{8}{150}$$

$$\therefore f = \frac{150}{8}\,cm = 18.75cm$$ ½

Alternatively

$$F = \left(\frac{D^2 - x^2}{4D}\right) = \frac{80^2 - 20^2}{4 \times 80} = 18.75\,cm$$ 1+1

Alternatively

The values of

$a = |u|$ *and* $b = |v|$ simply get interchanged in the two positions. ½

$b+a = 80cm$

$b-a = 20cm$ ½

This gives $b = 50cm$ *and* $a = 30cm$

$$\therefore \frac{1}{f} = \frac{1}{50} - \left(\frac{1}{-30}\right) = \frac{8}{150}$$ ½

$f = 18.75\ cm$ ½

19. An astronomical telescope may be a refracting type or a reflecting type. Which of the two produces image of better quality ? Justify your answer. **1**

CBSE 2020

19. An astronomical telescope may be a refracting type or a reflecting type. Which of the two produces image of better quality ? Justify your answer. **1**

CBSE 2020

Reflecting type telescope Reason/Justification :- Mirror have large aperture/high resolving power/ free from chromatic aberration /free from spherical aberration. (Any one)	½ ½

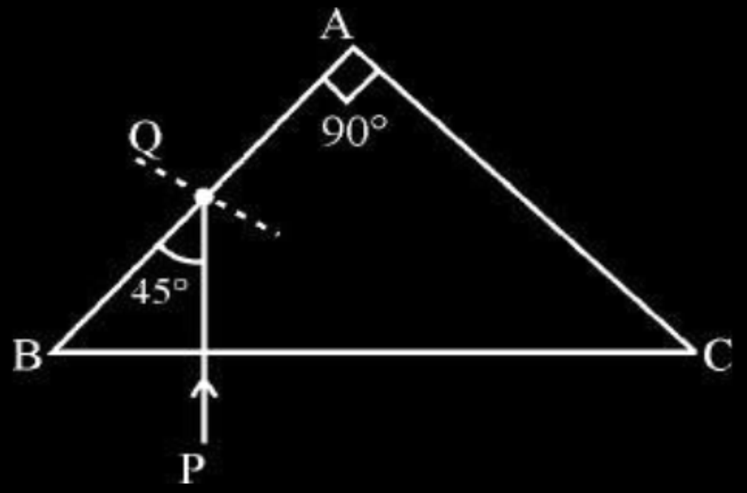
34. Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm, form real images of equal size. Size of object P is three times size of object Q. If the distance of P is 50 cm from the mirror, find the distance of Q from the mirror. CBSE 2020 **3**

34. Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm, form real images of equal size. Size of object P is three times size of object Q. If the distance of P is 50 cm from the mirror, find the distance of Q from the mirror. CBSE 2020 **3**

For object P $$m = \frac{h_2}{h_1} = \frac{f}{f - u_1}$$ For Object Q	½
$$m' = \frac{h_2'}{h_1'} = \frac{f}{f - u_2}$$	½
Now $h_1 = 3h_1'$; $h_2 = h_2'$; $f = -20\ cm$; $u_1 = -50\ cm$ $$\frac{m}{m'} = \frac{h_2}{h_1} \times \frac{h_1'}{h_2'} = \frac{f - u_2}{f - u_1}$$	½
$$\frac{m}{m'} = \frac{1}{3} = \frac{-20 - u_2}{-20 + 50}$$	1
$10 = -20 - u_2 \Rightarrow\!\!\Rightarrow\ u_2 = -30\ cm$	½

36. (a) Draw the ray diagram showing refraction of ray of light through a glass prism. Derive the expression for the refractive index μ of the material of prism in terms of the angle of prism A and angle of minimum deviation δ_m.

(b) A ray of light PQ enters an isosceles right angled prism ABC of refractive index 1.5 as shown in figure.



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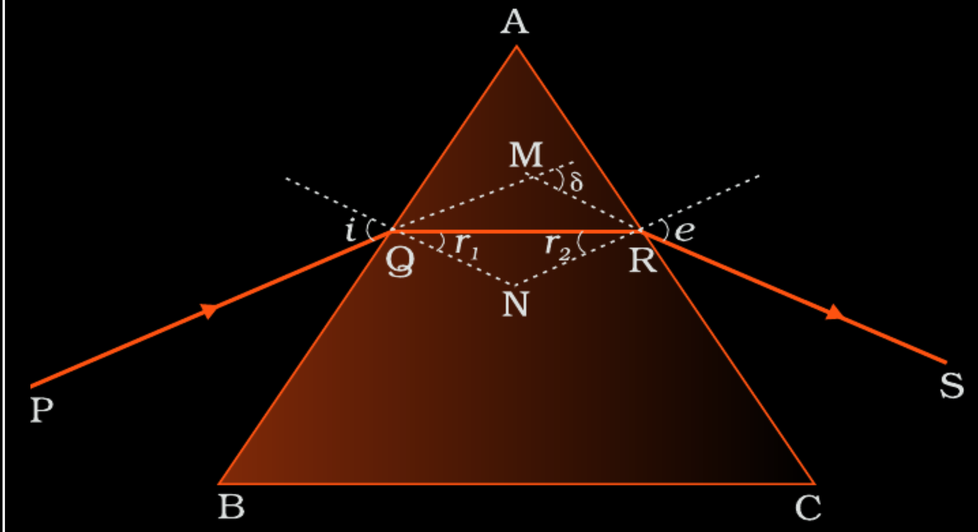
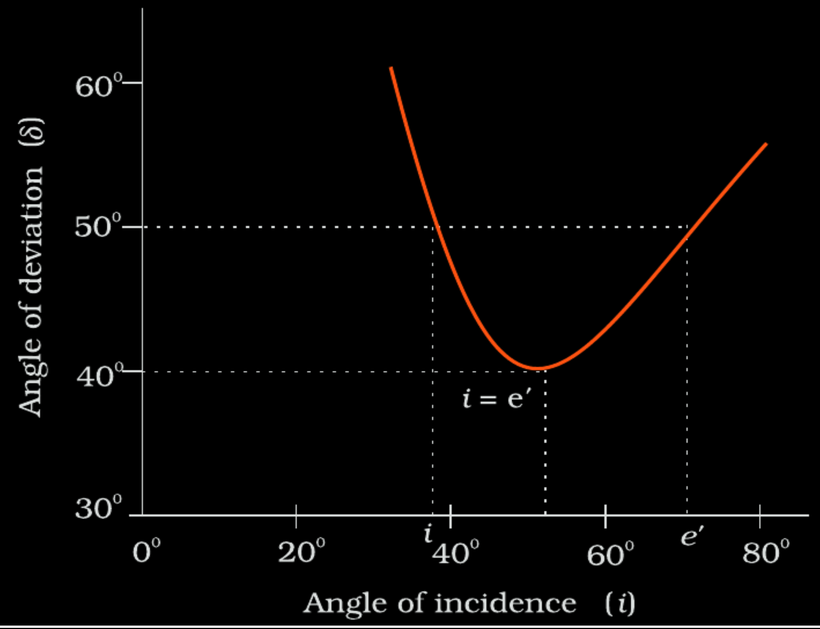
(i) Trace the path of the ray through the prism.

(ii) What will be the effect on the path of the ray if the refractive index of the prism is 1·4 ?

5





(ii) Derivation

From the figure

$\angle A + \angle QNR = 180^0$ (i) ½

In $\Delta QNR \quad r_1 + r_2 + \angle QNR = 180^0$ (ii)

Comparing equation (i) and (ii) we get

$r_1 + r_2 = A$ (iii)

Total deviation produced $\delta = (i - r_1) + (e - r_2)$

$\delta = i + e - (r_1 + r_2) = i + e - A$ (iv) ½

From the graph δ *vs* i we find that when δ becomes minimum i.e. δ_m

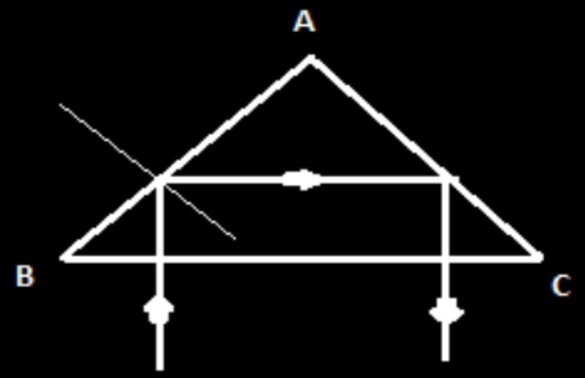
i = e and $r_1 = r_2$ ½

From (iv) $i = \frac{(A + \delta_m)}{2}$

and from (iii) $r = \frac{A}{2}$

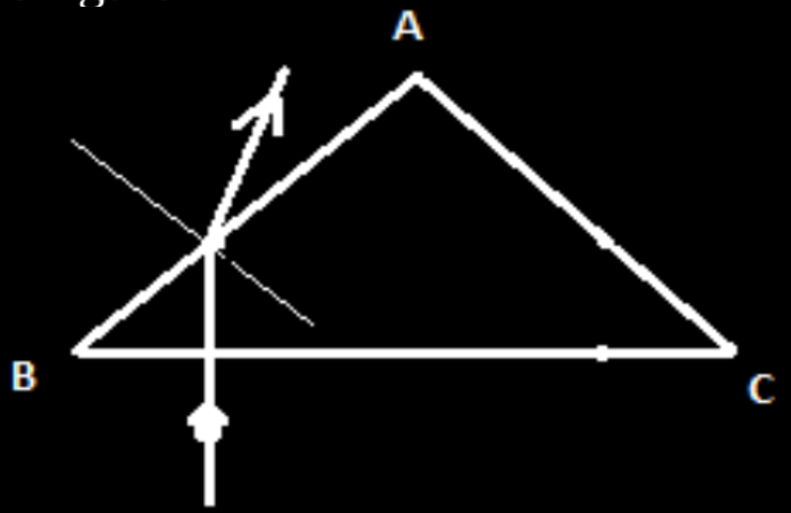
$\mu = \frac{\sin i}{\sin r} = \frac{\sin {}^{(A+\delta_m)}/_2}{\sin \frac{A}{2}}$ ½

(a) (i)



1

(ii) If $\mu = 1.4$ Total Internal Reflection will not occur as shown in the figure



1

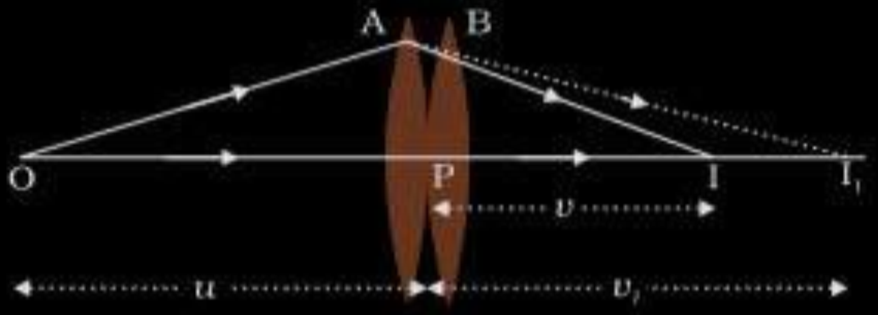
(Note: Award this last one mark if student does not draw the diagram and conclude correctly.)

(a) Two thin lenses are placed coaxially in contact. Obtain the expression for the focal length of this combination in terms of the focal lengths of the two lenses.

(b) A converging lens of refractive index 1.5 has a power of 10 D. When it is completely immersed in a liquid, it behaves as a diverging lens of focal length 50 cm. Find the refractive index of the liquid. **5**

CBSE 2020

(a)



½

For lens A

$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$ (i)

½

For lens B :

virtual image I_1 formed by A acting as object

$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$ (ii)

½

Adding equations (i) and (ii)

$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u}$

½

Hence $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Therefore $\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{F}$

½

(b)

In air $P_1 = \frac{1}{f_1} = (a^{\mu_g} - 1)(\frac{1}{R_1} - \frac{1}{R_2})$ (i) ½

For Liquid

$P_2 = \frac{1}{f_2} = (l^{\mu_g} - 1)(\frac{1}{R_1} - \frac{1}{R_2})$ (ii) ½

From (i) and (ii) ½

$\frac{P_1}{P_2} = \frac{(a^{\mu_g} - 1)}{(l^{\mu_g} - 1)}$

$\frac{10}{-2} = \frac{(1.5 - 1)}{(\frac{1.5}{\mu_l} - 1)}$ ½

$\mu_l = \frac{5}{3}$ ½

12. The refractive index of the material of a converging lens is 1·5. If air is replaced by a medium of refractive index 1·6, then the lens will now behave as a _____ lens. *1*

CBSE 2020

12. The refractive index of the material of a converging lens is 1·5. If air is replaced by a medium of refractive index 1·6, then the lens will now behave as a _____ lens. *1*

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Divergent lens/ Concave lens

25. A converging lens of focal length f_1 is placed coaxially in contact with a diverging lens of focal length f_2 ($f_1 > f_2$). Determine the power and nature of the combination in terms of f_1 and f_2. CBSE 2020 *2*

25. A converging lens of focal length f_1 is placed coaxially in contact with a diverging lens of focal length f_2 ($f_1 > f_2$). Determine the power and nature of the combination in terms of f_1 and f_2. CBSE 2020 *2*

$$\frac{1}{f} = \frac{1}{f_1} - \frac{1}{f_2}$$	½
$$\frac{1}{f} = \frac{f_2 - f_1}{f_1 f_2}$$	½
$$\therefore P = \frac{f_2 - f_1}{f_1 f_2}$$	½
Because $f_2 < f_1$ ∴ P is negative	½
∴ nature is diverging lens	

30. (a) Using the necessary ray diagram, derive the mirror formula for a concave mirror.

(b) In the magnified image of a measuring scale (with equidistant markings) lying along the principal axis of a concave mirror, the markings are not equidistant. Explain. *3*

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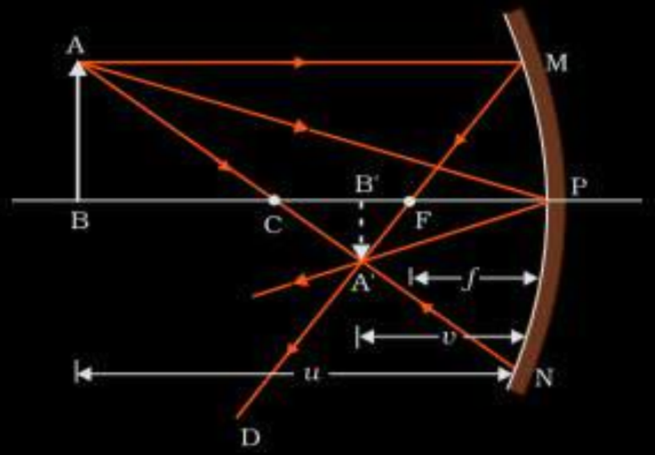
a) Ray diagrams for concave mirror



½

Derivation of Mirror Formula

From the diagram,

$\Delta A'B'F$ & ΔMPF are similar

$$\therefore \frac{B'A'}{PM} = \frac{B'F}{FP}$$

$$\frac{B'A'}{BA} = \frac{B'F}{FP} \ (\because PM = AB) - - - - - - - - eq1$$

½

Since

$$\angle APB = \angle A'PB'$$

$\Delta A'B'P$ & ΔABP are also similar

$$\frac{B'A'}{BA} = \frac{B'P}{BP} \quad - - - - - - - - - - eq\ 2$$

½

Comparing eq. 1 and eq. 2

$$\frac{B'P}{BP} = \frac{B'P - FP}{FP}$$

As per the sign convention

$$B'P = -v, \qquad FP = -f, \quad \text{BP = -u}$$

$$\frac{-v + f}{-f} = \frac{-v}{-u} = \frac{v}{u}$$

$$-vu + uf = -vf$$

Dividing by uvf

$$\Rightarrow \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

1

b) Magnification is different for different object distances

½

25. An object is kept 20 cm in front of a concave mirror of radius of curvature 60 cm. Find the nature and position of the image formed. CBSE 2020 *2*

25. An object is kept 20 cm in front of a concave mirror of radius of curvature 60 cm. Find the nature and position of the image formed. CBSE 2020 *2*

The focal length $f = \frac{-R}{2} = -30\ cm\ u = -20\ cm$	½
$\frac{1}{V} + \frac{1}{u} = \frac{1}{f}$	½
$\therefore \frac{1}{V} - \frac{1}{20} = -\frac{1}{30} \Rightarrow \frac{1}{V} = -\frac{1}{30} + \frac{1}{20}$	½
$\therefore V = +60\ cm$	
Nature of image: virtual, erect and magnified	½

31. What is the difference in the construction of an astronomical telescope and a compound microscope ? The focal lengths of the objective and eyepiece of a compound microscope are 1·25 cm and 5·0 cm, respectively. Find the position of the object relative to the objective in order to obtain an angular magnification of 30 when the final image is formed at the near point. *3*

CBSE 2020

Aperture of telescope objective lens is large whereas aperture of microscope objective is small — ½

$f_o > f_e$ in telescope
$f_o < f_e$ in microscope — ½

[Alternatively, focal length of telescope objective is large whereas focal length of microscope objective is very small]
[Award full 1 mark even if a student writes only one difference]

$$m = m_o \times m_e \quad \text{½}$$

$$m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{5} = 6$$

$$\therefore m_o = \frac{30}{6} = -5 \quad \text{½}$$

$$m_o = \frac{v}{u} = -5$$

$$v = -5u$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad \text{½}$$

$$\frac{1}{1.25} = \frac{1}{5u_o} + \frac{1}{u_0} \qquad ; u = -u_0$$

$$u_0 = \frac{6}{5} \times 1.25 = 1.5\text{cm} \quad \text{½}$$

[Alternatively, $m_o = \frac{v}{u} = \frac{f_o}{f_o + u_o}$

$$-5 = \frac{1.25}{1.25 + u_o}$$

$$-7.5 = 5u_o$$

$u_o = -1.5\text{cm}$ (for last ½ mark)]

37. (a) Derive lens maker's formula for a biconvex lens.

(b) A point object is placed at a distance of 12 cm on the principal axis of a convex lens of focal length 10 cm. A convex mirror is placed coaxially on the other side of the lens at a distance of 10 cm. If the final image coincides with the object, sketch the ray diagram and find the focal length of the convex mirror. *5*

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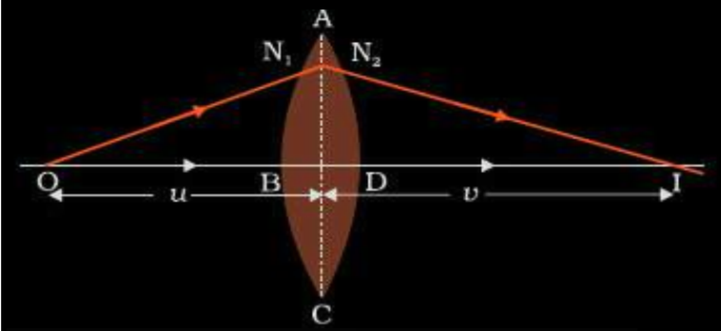
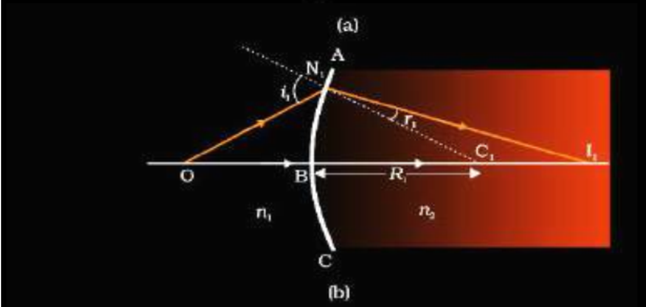




½

½

For first refracting surface

$$\frac{\mu_2}{v_1} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1} \text{----------1}$$

For second refracting surface ADC

$$\frac{\mu_1}{v} - \frac{\mu_2}{v_1} = \frac{\mu_1 - \mu_2}{R_2} \text{----------2}$$

Adding equations 1 and 2, we get

$$\frac{\mu_1}{v} - \frac{\mu_1}{u} = (\mu_2 - \mu_1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

½

$$\Rightarrow \frac{1}{v} - \frac{1}{u} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

$$\because \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

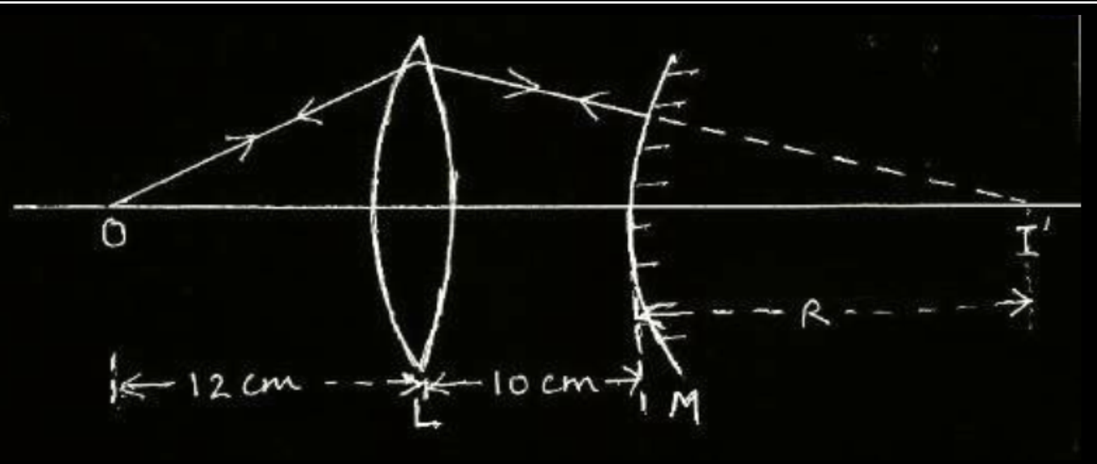
$$\therefore \frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

½

also

$$\frac{\mu_2}{\mu_1} = \mu$$

$$\therefore \frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$



1

$f_1 = 10\,cm$

$u = -12\,cm$

Aplying lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$ ½

$$\frac{1}{10} = \frac{1}{v} - \frac{1}{-12}$$

$\Rightarrow v = 60\,cm$ ½

$\therefore$ *radius of curvature of the mirror*

$R = 60\,cm - 10\,cm = 50\,cm$

hence focal length of the morror $f_m = \frac{R}{2} = 25\,cm$ ½

7. A biconvex lens of glass having refractive index 1·47 is immersed in a liquid. It becomes invisible and behaves as a plane glass plate. The refractive index of the liquid is *1*

CBSE 2020

(A) 1·47

(B) 1·62

(C) 1·33

(D) 1·51

7. A biconvex lens of glass having refractive index 1·47 is immersed in a liquid. It becomes invisible and behaves as a plane glass plate. The refractive index of the liquid is *1*

CBSE 2020

(A) 1·47

(B) 1·62

(C) 1·33

(D) 1·51

(A)

1.47

8. For a glass prism, the angle of minimum deviation will be smallest for the light of *1*

(A) red colour.

(B) blue colour.

(C) yellow colour.

(D) green colour.

CBSE 2020

8. For a glass prism, the angle of minimum deviation will be smallest for the light of *1*

(A) red colour.

(B) blue colour.

(C) yellow colour.

(D) green colour.

CBSE 2020

(A)

red colour

15. A ray of light on passing through an equilateral glass prism, suffers a minimum deviation equal to the angle of the prism. The value of refractive index of the material of the prism is _____ . CBSE 2020 *1*

15. A ray of light on passing through an equilateral glass prism, suffers a minimum deviation equal to the angle of the prism. The value of refractive index of the material of the prism is $\sqrt{3}$. CBSE 2020 *1*

Using lens maker's formula, derive the thin lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ for a biconvex lens. *2*

CBSE 2020

Using lens maker's formula, derive the thin lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ for a biconvex lens. 2

CBSE 2020

$\because \frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) - - - - - - - 1$	1
When u=∞ and $v = f$ $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) - - - - - - - - - - - 2$	½
$\left[n = \frac{n_2}{n_1}\right]$ From Eq 1 and 2 $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ *then lens formula* [Even if the student derives $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ for biconvex lens, award 1 ½ marks]	½

37. (a) Draw the ray diagram of an astronomical telescope when the final image is formed at infinity. Write the expression for the resolving power of the telescope.

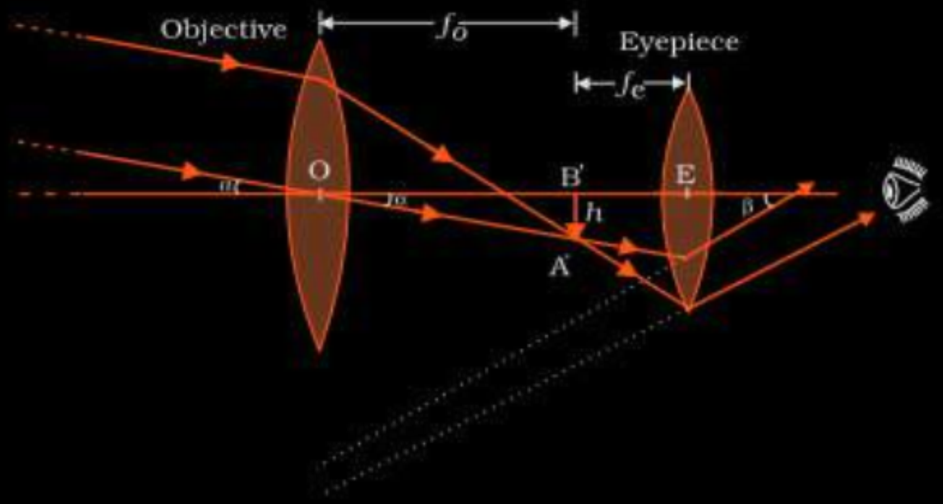
(b) An astronomical telescope has an objective lens of focal length 20 m and eyepiece of focal length 1 cm.

(i) Find the angular magnification of the telescope.

(ii) If this telescope is used to view the Moon, find the diameter of the image formed by the objective lens. Given the diameter of the Moon is $3{\cdot}5 \times 10^6$ m and radius of lunar orbit is $3{\cdot}8 \times 10^8$ m.

CBSE 2020 *5*

a)



2

Resolving power of telescope = $\frac{D}{1.22\lambda}$

1

b) (i) Angular magnification $m = \frac{\beta}{\alpha} = \frac{f_o}{f_e} = \frac{20m}{10^{-2}m} = 2000$

1

(ii)

$$\frac{D}{d} = \frac{x}{f_o}$$

½

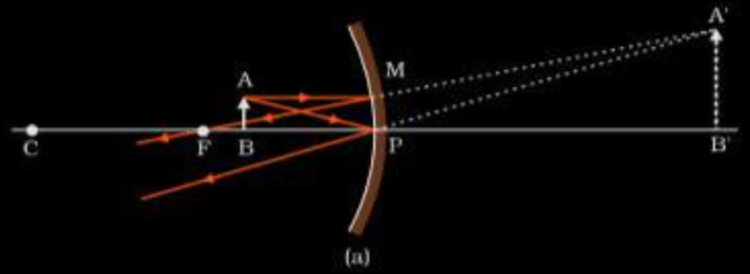
$$d = \frac{Df_0}{x} = \frac{3.5 \times 10^6 \times 20}{3.8 \times 10^8} = .18\text{m}$$

½

(a) An object is placed in front of a concave mirror. It is observed that a virtual image is formed. Draw the ray diagram to show the image formation and hence derive the mirror equation $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$.

(b) An object is placed 30 cm in front of a plano-convex lens with its spherical surface of radius of curvature 20 cm. If the refractive index of the material of the lens is 1·5, find the position and nature of the image formed.

CBSE 2020 *5*



1

$\Delta ABP \sim A'B'P$

$$\frac{A'B'}{AB} = \frac{PB'}{PB} - - - - - - - - - - 1$$

½

Also $\Delta A'B'F \sim MNP$ (for small curvature)

$$\therefore \frac{A'B'}{MP} = \frac{B'F}{PF}$$

$$\frac{A'B'}{AB} = \frac{B'F}{PF} - - - - - - - - - - - - 2$$

From 1 and 2

$$\frac{PB'}{PB} = \frac{B'F}{PF} - - - - - - - - - - - - 3$$

½

$$\frac{PB'}{PB} = \frac{B'P + PF}{PF} - - - - - - - - -4$$

½

$PB = -u \qquad PB' = v \qquad PF = -f$

$$\frac{v}{-u} = \frac{v - f}{-f}$$

$$-vf = -vu + uf$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

½

(b) According to lens maker's formula

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ ½

for plano convex lens $R_1 \rightarrow$ R and $R_2 \rightarrow \infty$

$$\frac{1}{f} = \frac{(\mu - 1)}{R} = \frac{1.5 - 1}{20}$$ ½

$\therefore$f=40 cm

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{40} = \frac{1}{v} - \frac{1}{-30}$$

$$v = -12\ cm$$ ½

Nature: virtual ½

A screen is placed 90 cm from an object. The image of the object on the screen is formed by a convex lens at two different positions separated by 20 cm. Calculate the focal length of the lens.

CBSE 2019

A screen is placed 90 cm from an object. The image of the object on the screen is formed by a convex lens at two different positions separated by 20 cm. Calculate the focal length of the lens.

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v+u = 90	½
v-u = 20	½
v = 55 cm	½
u = 35 cm	½
$f = \frac{55x35}{55+35}$ = 21.4 cm	1

A convex lens of focal length 20 cm and a concave lens of focal length 15 cm are kept 30 cm apart with their principal axes coincident. When an object is placed 30 cm in front of the convex lens, calculate the position of the final image formed by the combination. Would this result change if the object were placed 30 cm in front of the concave lens ? Give reason.

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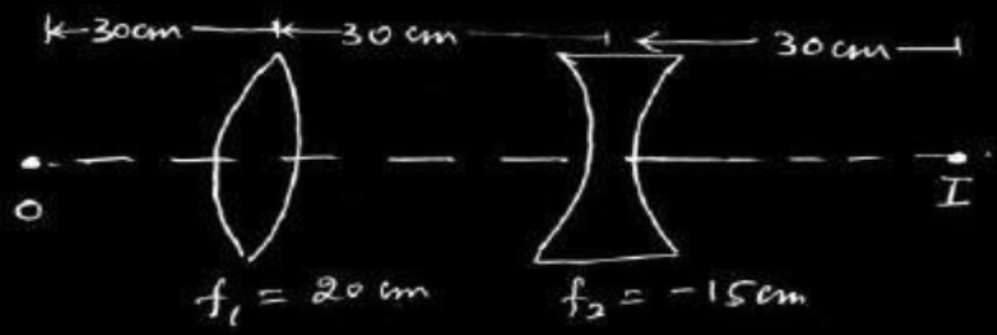
For Image formed by convex lens:

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$



½

$$\frac{1}{20}=\frac{1}{v}+\frac{1}{30}$$

$$v=\frac{20X30}{30-20}=60cm$$

1

u for concave lens = + 30cm

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$$\frac{1}{-15}=\frac{1}{v}-\frac{1}{30}$$

$$v=\frac{15X30}{15-30}=-30cm$$

1

No, the result will not change from principle of reversibility

½

(a) Draw a labelled ray diagram of compound microscope, when final image forms at the least distance of distinct vision.

(b) Why is its objective of short focal length and of short aperture, compared to its eyepiece ? Explain.

(c) The focal length of the objective is 4 cm while that of eyepiece is 10 cm. The object is placed at a distance of 6 cm from the objective lens.

(i) Calculate the magnifying power of the compound microscope, if its final image is formed at the near point.

(ii) Also calculate length of the compound microscope.

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(b) The magnifying power of compound microscope

$$m = m_0 \times m_e = \frac{L}{f_0} \times (1 + \frac{D}{f_e})$$

To have high magnifying power and high resolution, the focal length of the objective and its aperture should be short.
Focal length of eyepiece is comparatively greater than the objective so that image formed by objective lens may form within the focal length of eyepiece and the final magnified image may be formed.
Aperture in short for higher resolution. 1

(c)

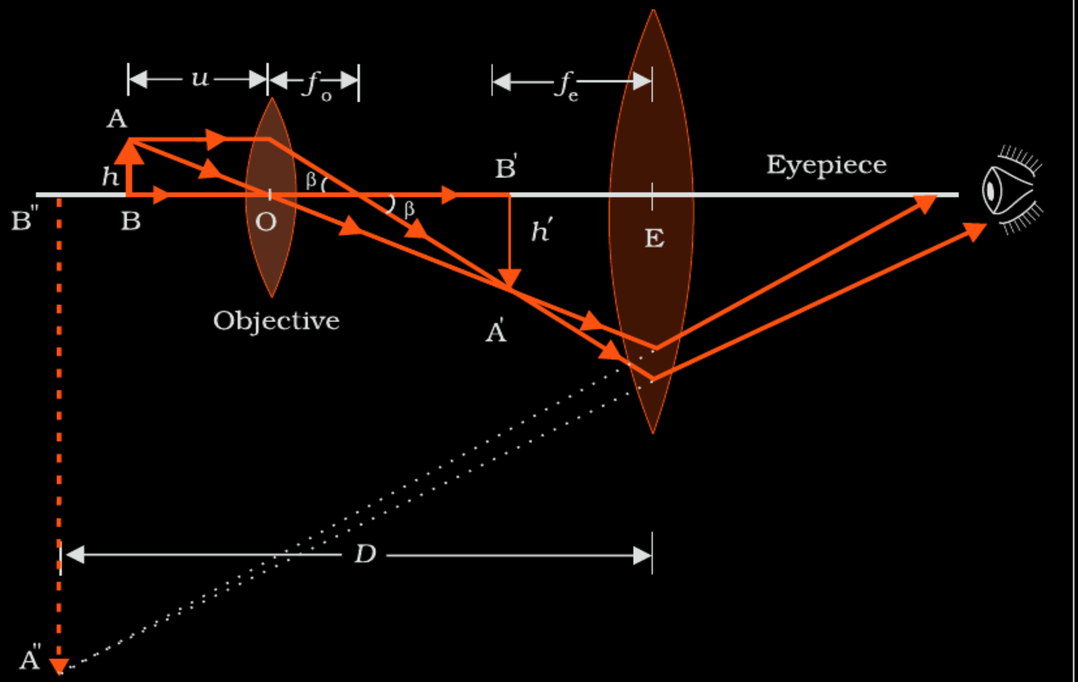
$u_0 = -6cm$

$for \quad objective\, lens$

$$\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$$

$$\frac{1}{v_0} = \frac{1}{f_0} + \frac{1}{u_0} \quad = \frac{1}{4} + \frac{1}{-6}$$

$v_0 = 12cm$



1

Since for eyepiece $v_e = -D = -25cm$

$$\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{-25} - \frac{1}{10}$$

$$u_e = -7.14\,cm$$

(i) For magnifying power of compound microscope

1

$$m = \frac{v_o}{u_o}\left(1 + \frac{D}{f_e}\right)$$

$$m = \frac{12}{6}\left(1 + \frac{25}{10}\right) = 7$$

(ii) Length of Microscope

1

$$L = v_0 + |u_e| = 12 + 7.14$$

$$L = 19.14\,cm$$

(a) With the help of a labelled ray diagram, explain the construction and working of a Cassegrain reflecting telescope.

(b) An amateur astronomer wishes to estimate roughly the size of the Sun using his crude telescope consisting of an objective lens of focal length 200 cm and an eyepiece of focal length 10 cm. By adjusting the distance of the eyepiece from the objective, he obtains an image of the Sun on a screen 40 cm behind the eyepiece. The diameter of the Sun's image is measured to be 6·0 cm. Estimate the Sun's size, given that the average Earth-Sun distance is $1{\cdot}5 \times 10^{11}$ m.

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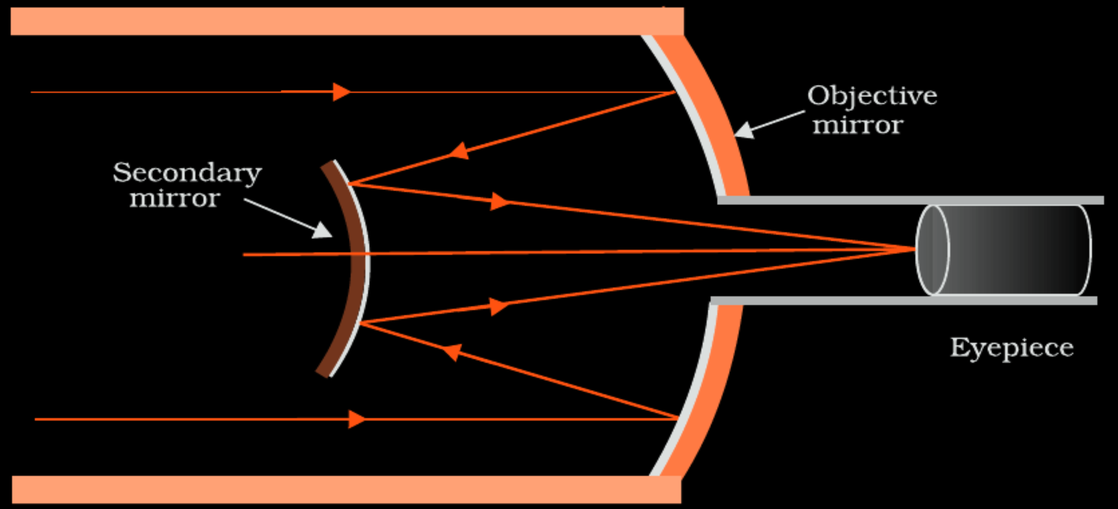
It consists for large concave (primary) paraboidal mirror having in its central part a hole. There is a small convex (secondary) mirror near the focus of concave mirror. Eye pieces if placed near the hole of the concave mirror .

1

The parallel rays from distance object are reflected by the large concave mirror . These rays fall on the convex mirror which reflects these rays outside the hole. The final magnified image in formed.



(b) For eyepiece.

$$\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$$

$$or \quad \frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{40} - \frac{1}{10}$$

½

$$u_e = \frac{40}{3} cm$$

½

Magnification produced by eye pieces is

$$m_e = \frac{v_e}{|u_e|} = \frac{40}{40/3} = 3$$

Diameter of the image formed by the objective is

½

d = 6/3 = 2cm

If D be the diameter of the SUN then the angle subtended by it on the objective will be

$$\alpha = \frac{D}{1.5\times10^{11}} rad$$

Angle subtended by the image at the objective

= angle subtended by the SUN

$$\therefore \alpha = \frac{size\,of\ image}{f_o} = \frac{2}{200} = \frac{1}{100} rad.$$

$$\therefore \frac{D}{1.5\times10^{11}} = \frac{1}{100}$$

½

or

$$D = 1.5\times10^{9} m$$

A beam of light converges at a point P. Now a convex lens is placed in the path of the convergent beam at 15 cm from P. At what point does a beam converge if the convex lens has a focal length 10 cm ?

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A beam of light converges at a point P. Now a convex lens is placed in the path of the convergent beam at 15 cm from P. At what point does a beam converge if the convex lens has a focal length 10 cm ?

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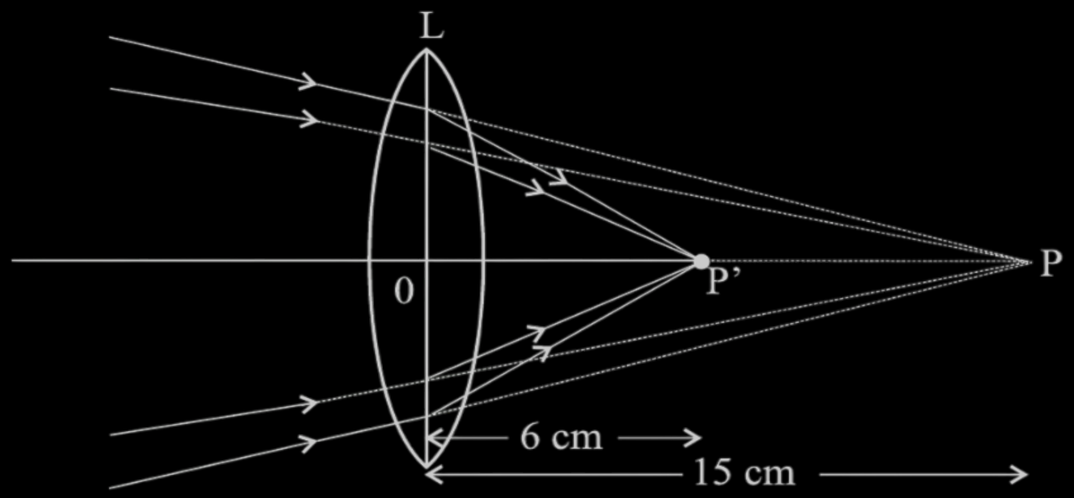

½

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ (lens formula) ½

Here $u = +15\text{cm}$; $f = +10$ cm ½

$\therefore \quad \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{10} + \frac{1}{15}$

$\Rightarrow \quad v = 6$ cm 1/

An object is kept in front of a concave mirror of focal length 15 cm. The image formed is real and three times the size of the object. Calculate the distance of the object from the mirror.

CBSE 2019

An object is kept in front of a concave mirror of focal length 15 cm. The image formed is real and three times the size of the object. Calculate the distance of the object from the mirror.

CBSE 2019

Here, $m = +3$ and $f = -15$ cm	
$m = \frac{v}{u} = 3 \quad \therefore \quad v = 3u$	½
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	½
$\frac{1}{-15} = \frac{1}{3u} + \frac{1}{u}$	½
$\Rightarrow \quad u = -20$ cm	½

Draw a ray diagram to show the image formation of a distant object by a refracting telescope. Write the expression for its angular magnification in terms of the focal lengths of the lenses used. State the important considerations required to achieve large resolution and their consequent limitations.

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Draw a ray diagram to show the image formation of a distant object by a refracting telescope. Write the expression for its angular magnification in terms of the focal lengths of the lenses used. State the important considerations required to achieve large resolution and their consequent limitations.

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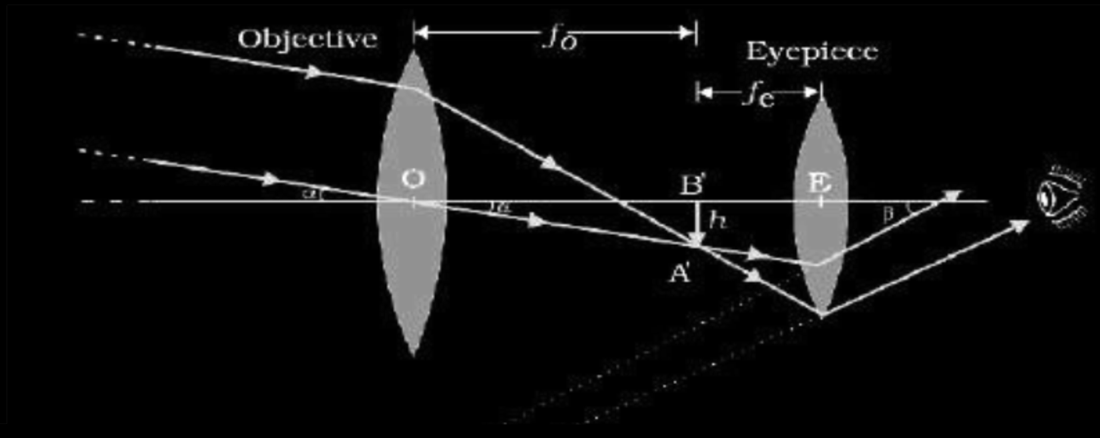


1½

Angular magnification $m = \frac{-f_0}{f_e}$ or $\frac{f_0}{f_e}$ ½

Important considerations :

For achieving large resolution, the objective of large aperture is required. ½

Consequent Limitation : Heavy, hence difficult to make and support by their edge / suffers with chromatic aberrations (any one of above) ½

(a) Plot a graph for angle of deviation as a function of angle of incidence for a triangular prism.

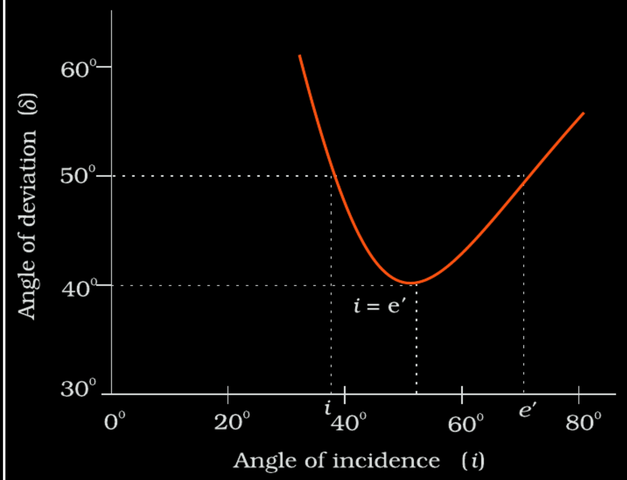
(b) Derive the relation for the refractive index of the prism in terms of the angle of minimum deviation and angle of prism.

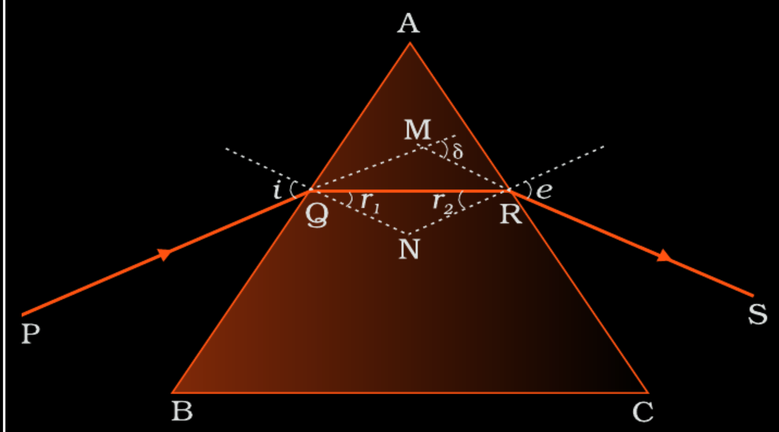
CBSE 2019

(a) Plot a graph for angle of deviation as a function of angle of incidence for a triangular prism.

(b) Derive the relation for the refractive index of the prism in terms of the angle of minimum deviation and angle of prism.

CBSE 2019





Since $n_{21} = \dfrac{\sin i}{\sin r}$ ½

From the figure and calculations

$r_1 + r_2 = A$

At minimum deviation i.e. $\delta = \delta m$, $i = e$ and $r_1 = r_2 = r$ 1

$\therefore \quad r = A/2$ ..... (eqⁿ 1)

From the figure

$\delta = (i - r_1) + (e - r_2)$

$\therefore \quad \delta_m = (i + e) + (r_1 + r_2)$ ½

$i = \dfrac{A + \delta_m}{2}$ ..... (eqⁿ 2)

$\therefore \quad n_{21} = \dfrac{\sin i}{\sin r} = \dfrac{\text{Sin}\,\dfrac{A + \delta_m}{2}}{\sin A/2}$ ½

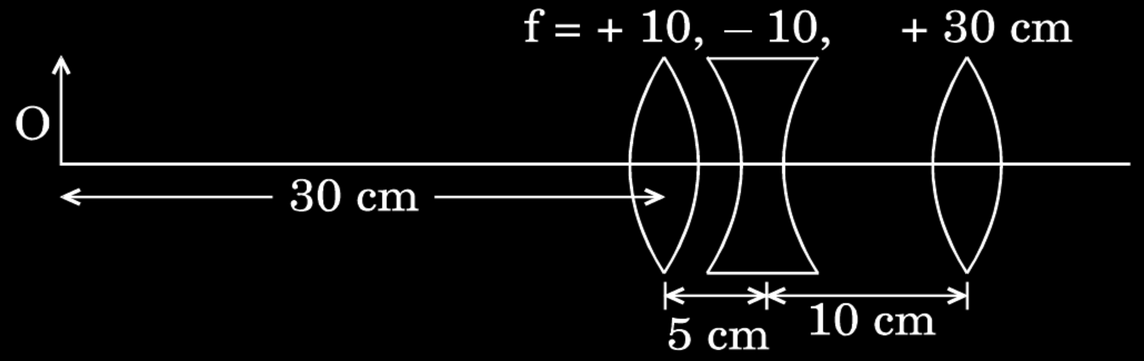
½

(a) Using the ray diagram for a system of two lenses of focal lengths f_1 and f_2 in contact with each other, show that the two lens system can be regarded as equivalent to a single lens of focal length f, where

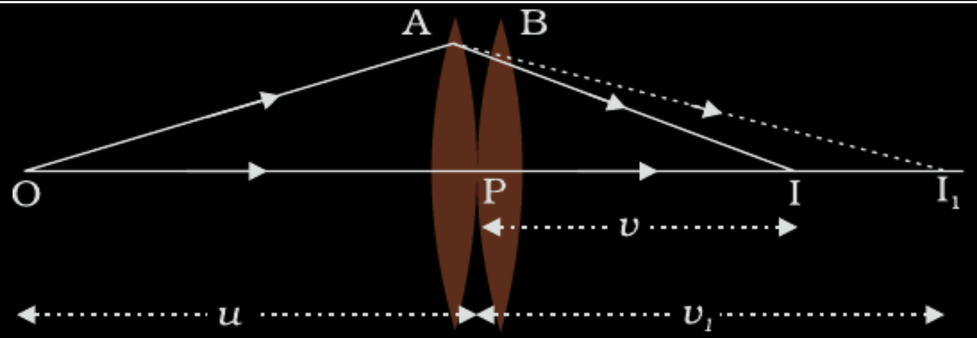
$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}.$$

Also write the relation for the equivalent power of the lens combination.

(b) Determine the position of the image formed by the lens combination given in the figure.



CBSE 2019



For less A

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u}$$ eg. (I) ½

For less B

$$\frac{1}{f_2} = \frac{1}{v} - \frac{1}{v_1}$$ eg. (ii) ½

Adding eqn. (i) & eqn. (ii) ½

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v_1} - \frac{1}{u} + \frac{1}{v} - \frac{1}{v_1}$$

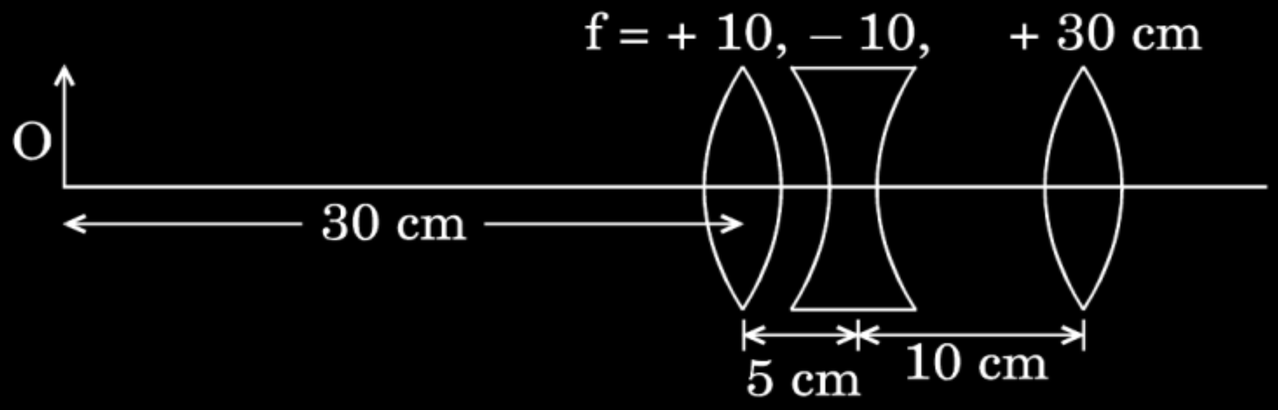
$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{F}$$ ½

$\therefore$ equivalent power

$P = P_2 + P_2$ ½

b) Image formed by lense of f = +10 cm



$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

$$\frac{1}{v_1} - \frac{1}{30} = \frac{1}{10}$$

$\therefore\ v_1 = 15$ cm

This image formed by the lens act as object from concave lens

$\therefore\ u_2 = 15\text{-}5 = 10$ cm

$$\frac{1}{f_2} + \frac{1}{v_2} = \frac{1}{u_2}$$

$$\frac{1}{-10} = \frac{1}{v} - \frac{1}{10}$$

$v = \infty$

Therefore virtual image forms at right of concave lens at $v = \infty$ and act as convex lens. (f = +30 cm)

$\therefore\ u_2 = 15 - 5 = 10$ cm ½

$$\frac{1}{v_3} = \frac{1}{4} - \frac{1}{f_3}$$

$$\frac{1}{v_3} = \frac{1}{\infty} = \frac{1}{30}$$

$v_3 = 30$ cm ½

(If student calculate upto $v = \infty$, give full marks)

State the condition under which a large magnification can be achieved in an astronomical telescope.

OR

How does the angle of minimum deviation of a glass prism vary if the incident violet light is replaced by red light ?

CBSE 2019

State the condition under which a large magnification can be achieved in an astronomical telescope.

OR

How does the angle of minimum deviation of a glass prism vary if the incident violet light is replaced by red light ?

CBSE 2019

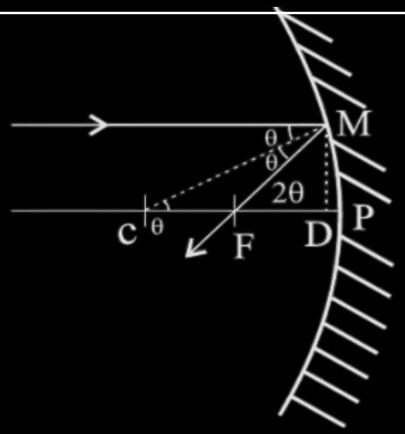
$f_0 >> f_e$ Focal length of objective must be much greater than focal length of eyepiece OR Angle of minimum deviation decreases.	1 1

For paraxial rays, show that the focal length of a spherical mirror is one-half of its radius of curvature.

CBSE 2019

For paraxial rays, show that the focal length of a spherical mirror is one-half of its radius of curvature.

CBSE 2019



1

From ray diagram

$$\tan\theta = \frac{MD}{CD} \quad \text{and} \quad \tan 2\theta = \frac{MD}{FD}$$

½

for small θ, $\tan\theta \cong \theta$, $\tan 2\theta \cong 2\theta$

$$\therefore \frac{MD}{FD} = 2\frac{MD}{CD}$$

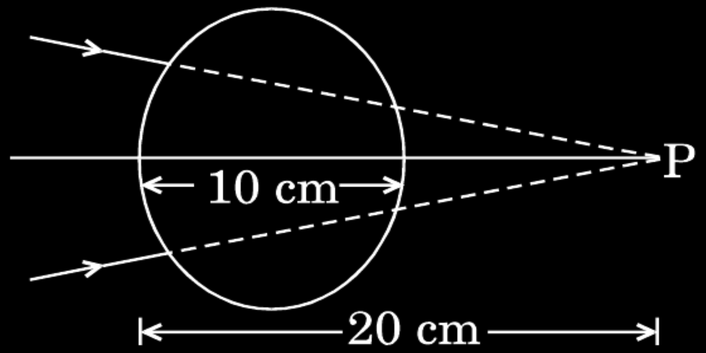
$$FD = \frac{CD}{2} \cong \frac{CP}{2}$$

½

$$\therefore \quad f = \frac{R}{2}$$

A converging beam of light travelling in air converges at a point P as shown in the figure. When a glass sphere of refractive index 1·5 is introduced in between the path of the beam, calculate the new position of the image. Also draw the ray diagram for the image formed.

CBSE 2019

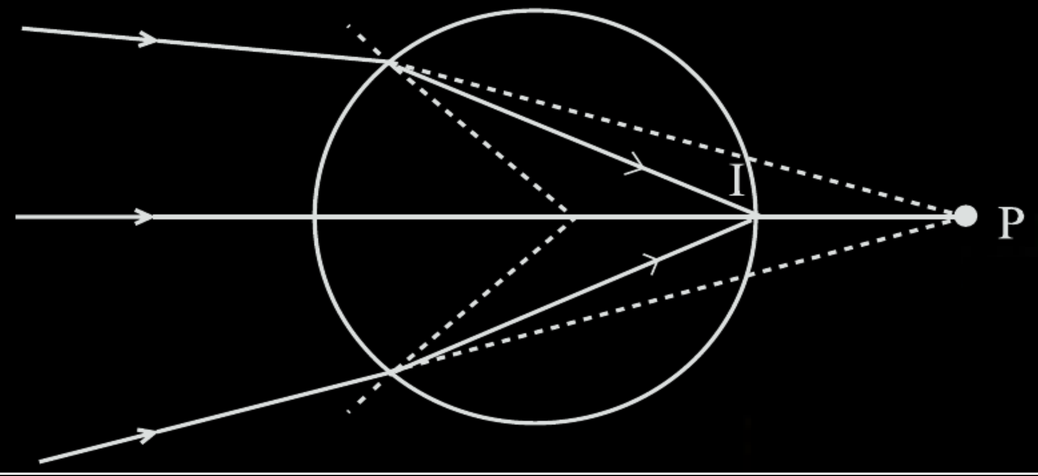


a) $u = 20cm$, $n_2 = 1.5$, $n_1 = 1$, $R = 5cm$

Using $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ ½

$\frac{1.5}{v} - \frac{1}{20} = \frac{1.5-1}{5}$ 1

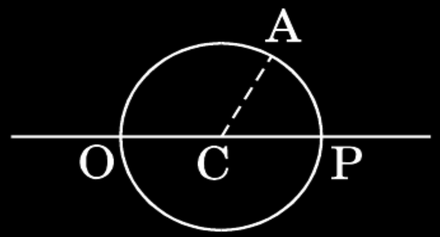
$v = 10$ cm ½



1

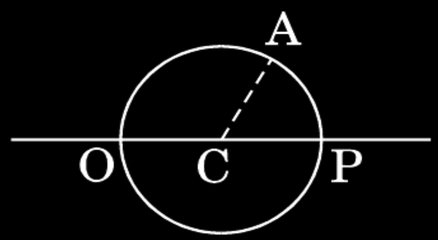
A point ‘O’ marked on the surface of a glass sphere of diameter 20 cm is viewed through glass from the position directly opposite to the point O. If the refractive index of the glass is 1·5, find the position of the image formed. Also, draw the ray diagram for the formation of the image.

CBSE 2019



A point ‘O’ marked on the surface of a glass sphere of diameter 20 cm is viewed through glass from the position directly opposite to the point O. If the refractive index of the glass is 1·5, find the position of the image formed. Also, draw the ray diagram for the formation of the image.

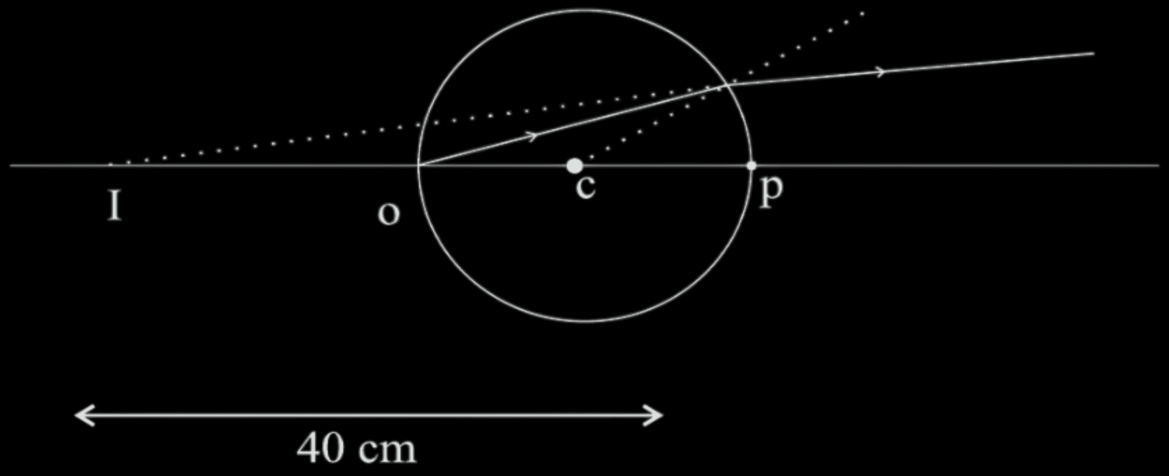
CBSE 2019



$$\frac{n_1}{-u}+\frac{n_2}{v}=\frac{n_2-n_1}{R}$$

$$\frac{1.5}{20}+\frac{1}{v}=\frac{1-1.5}{-10}$$

$v = -40\text{cm}$



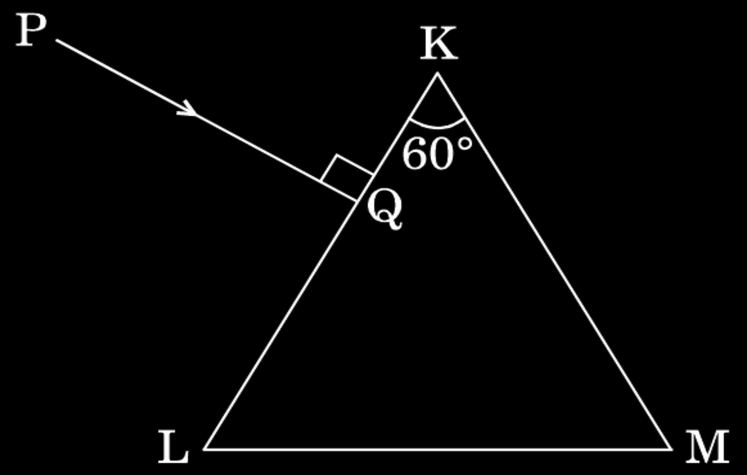
1

½

½

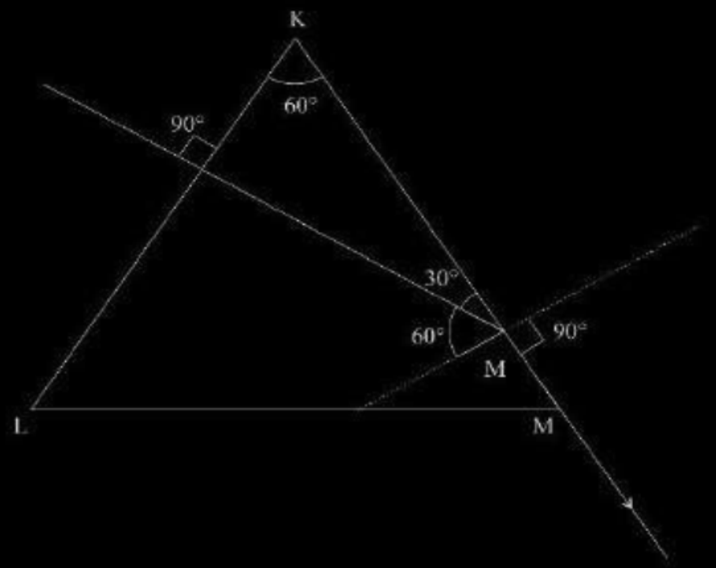
1

A triangular prism of refracting angle $60°$ is made of a transparent material of refractive index $2/\sqrt{3}$. A ray of light is incident normally on the face KL as shown in the figure. Trace the path of the ray as it passes through the prism and calculate the angle of emergence and angle of deviation.



CBSE 2019

Ray diagram:



1

- A=60°
- $\frac{2}{\sqrt{3}}\sin 60^{o} = \sin r$ 1½
- $\sin r = \frac{2\sqrt{3}}{2\sqrt{3}} = 1$
- $r = 90^{o}$ ½

Angle of deviation is equal to 30°

Calculate the radius of curvature of an equi-concave lens of refractive index 1.5, when it is kept in a medium of refractive index 1.4, to have a power of –5D ?

CBSE 2019

Calculate the radius of curvature of an equi-concave lens of refractive index 1.5, when it is kept in a medium of refractive index 1.4, to have a power of –5D ?

CBSE 2019

$f = \frac{1}{P} = \frac{1}{-5}\,m = -\frac{100}{5}\,cm = -20\,cm$ ½

$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R1} - \frac{1}{R2}\right)$ ½

$\mu_2 = 1.5, \quad \mu_1 = 1.4, \quad R_1 = -R, \quad R_2 = R$

$\frac{1}{-20} = \left(\frac{1.5}{1.4} - 1\right)\left(-\frac{1}{R} - \frac{1}{R}\right)$ ½

$\frac{1}{-20} = \left(\frac{0.1}{1.4}\right)\left(-\frac{2}{R}\right)$

$R = \frac{20}{7}\,cm \;(= 2.86\,cm)$ ½

An equilateral glass prism has a refractive index 1.6 in air. Calculate the angle of minimum deviation of the prism, when kept in a medium of refractive index $4\sqrt{2}/5$.

CBSE 2019

An equilateral glass prism has a refractive index 1.6 in air. Calculate the angle of minimum deviation of the prism, when kept in a medium of refractive index $4\sqrt{2}/5$.

CBSE 2019

$$\mu = \frac{\sin\frac{(A + D_m)}{2}}{\sin A/2}$$ ½

$$\mu = \frac{\mu_2}{\mu_1} = \frac{1.6}{\frac{4}{5}\sqrt{2}} = \frac{8}{4\sqrt{2}} = \sqrt{2}$$ ½

$$\sqrt{2} = \frac{\sin(\frac{60+D_m}{2})}{\sin 60/2} = \frac{\sin\frac{(60+D_m)}{2}}{\sin 30}$$

$$\therefore \quad \sin(\frac{60+D_m}{2}) = \sqrt{2} \cdot \frac{1}{2} = \frac{1}{\sqrt{2}} = \text{Sin } 45^0$$ ½

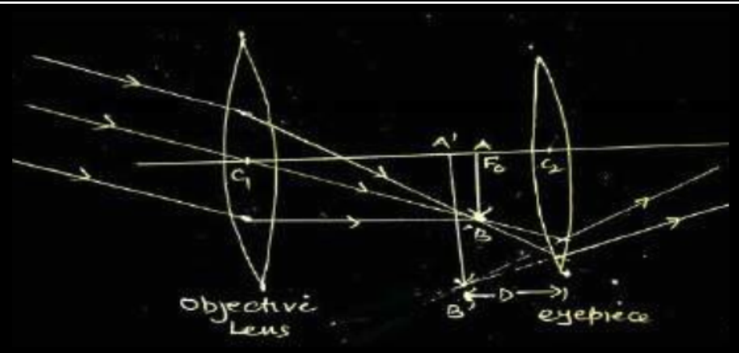
$$\therefore \frac{60+D_m}{2} = 45^o$$ ½

$$\therefore \quad D_m = 30^0$$

Draw a labelled ray diagram of an astronomical telescope in the near point adjustment position.

A giant refracting telescope at an observatory has an objective lens of focal length 15 m and an eyepiece of focal length 1.0 cm. If this telescope is used to view the Moon, find the diameter of the image of the Moon formed by the objective lens. The diameter of the Moon is 3.48×10^6 m, and the radius of lunar orbit is 3.8×10^8 m.

CBSE 2019



1½

Angular magnification of the telescope = $\frac{f_o}{f_e} = \frac{15}{0.01} = 1500$ ½

For objective lens, $\tan\alpha = \frac{3.48X\ 10^6}{3.8X10^8}$

For eyepiece $\tan\beta = \frac{h_i}{f_e} = \frac{h_i}{10^{-2}}$ ½

∴ Magnifying power $= \frac{\beta}{\alpha} = \frac{\frac{h_i}{10^{-2}}}{\frac{3.48\ X10^6}{3.8\ X\ 10^8}}$

$= \frac{h_i X3.8X10^8}{3.48X10^6X10^{-2}} = 1500$

h_i = 13.73 cm ½

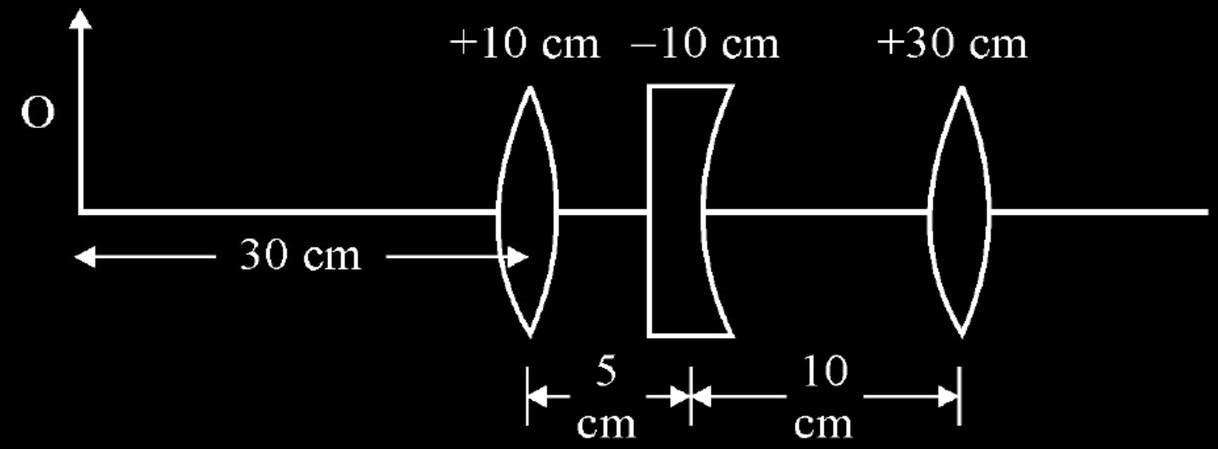
Also accept angular magnification of the telescope

$= \frac{f_o}{f_e}\left(1 + \frac{f_e}{d}\right) = \frac{15}{0.01}\left(1 + \frac{0.01}{0.25}\right) = 1560$

So, h_i = 14.29 cm

(a) Under what conditions is the phenomenon of total internal reflection of light observed ? Obtain the relation between the critical angle of incidence and the refractive index of the medium.

(b) Three lenses of focal lengths +10 cm, –10 cm and +30 cm are arranged coaxially as in the figure given below. Find the position of the final image formed by the combination.



CBSE 2019

(a) (i) Light travels from denser to rarer medium. 1

(ii) Angle of incidence is more than the critical angle 1

For the Grazing incidence

μ sin i_c = 1 sin 90° ½

μ = $\frac{1}{sini_c}$ ½

(b) For convex lens of focal Length 10 cm

$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u_1}$ ½

$\frac{1}{10} = \frac{1}{v_1} - \frac{1}{-30}$ => v_1 = 15 cm ½

Object distance for concave lens u_2= 15-5 =10 cm

$\frac{1}{f_2} = \frac{1}{v_2} - \frac{1}{u_2}$

$\frac{1}{-10} = \frac{1}{v_2} - \frac{1}{10}$ ½

$v_2 = \infty$

For third lens

$\frac{1}{f_3} = \frac{1}{v_3} - \frac{1}{u_3}$ ½

$\frac{1}{3_\circ} = \frac{1}{v_3} - \frac{1}{\infty} => v_3 = 30\ cm$

(a) With the help of a ray diagram, show how a concave mirror is used to obtain an erect and magnified image of an object.

(b) Using the above ray diagram, obtain the mirror formula and the expression for linear magnification.

CBSE 2018

Define the magnifying power of a compound microscope when the final image is formed at infinity. Why must both the objective and the eyepiece of a compound microscope has short focal lengths ? Explain.

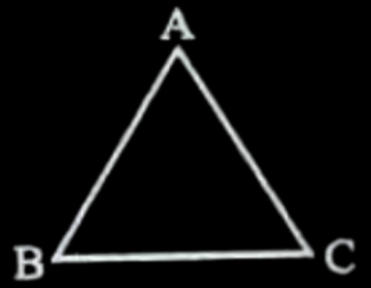
CBSE 2017

Define the magnifying power of a compound microscope when the final image is formed at infinity. Why must both the objective and the eyepiece of a compound microscope has short focal lengths ? Explain.

CBSE 2017

Magnifying power is defined as the angle subtended at the eye by the image to the angle subtended (at the unaided eye) by the object. (Alternatively: Also accept this definition in the form of formula) — — To increase the magnifying power both the objective and eyepiece must have short focal lengths (as — —	1 ½ +½

(i) A ray of light incident on face AB of an equilateral glass prism, shows minimum deviation of 30°. Calculate the speed of light through the prism.



(ii) Find the angle of incidence at face AB so that the emergent ray grazes along the face AC.

CBSE 2017

(b) A double convex lens is made of a glass of refractive index 1·55, with both faces of the same radius of curvature. Find the radius of curvature required, if the focal length is 20 cm. CBSE 2017

(b) A double convex lens is made of a glass of refractive index 1·55, with both faces of the same radius of curvature. Find the radius of curvature required, if the focal length is 20 cm. CBSE 2017

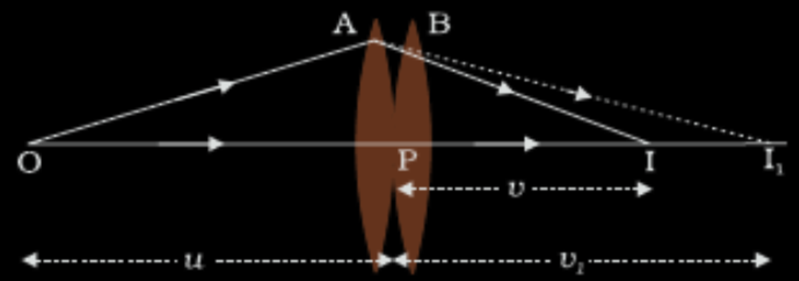
b) $\frac{1}{f} = \left[\frac{\mu_2}{\mu_1} - 1\right]\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$	½
$\therefore \frac{1}{20} = \left[\frac{1.55}{1} - 1\right]\frac{2}{R}$	½
$\therefore R = (20 \times 1.10)\text{cm} = 22\text{ cm}$	½

(a) Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact. Obtain the expression for the power of this combination in terms of the focal lengths of the lenses.

(b) A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is $\frac{3}{4}^{\text{th}}$ of the angle of prism. Calculate the speed of light in the prism.

CBSE 2017

a)



1

Two thin lenses, of focal length f_1 and f_2 are kept in contact. Let O be the position of object and let u be the object distance. The distance of the image (which is at I_1), for the first lens is v_1.

This image serves as object for the second lens. ½

Let the final image be at I. We then have

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u}$$

$$\frac{1}{f_2} = \frac{1}{v} - \frac{1}{v_1}$$ ½

Adding , we get

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$ ½

$$\therefore \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\therefore P = P_1 + P_2$$ ½

b) At minimum deviation

$$r = {}^{A}/_{2} = 30^0$$ ½

We are given that

$$i = \frac{3}{4}A = 45^0$$ ½

$$\therefore \mu = \frac{\sin 45^0}{\sin 30^0} = \sqrt{2}$$ ½

$\therefore$ Speed of light in the prism $= \frac{c}{\sqrt{2}}$

$(\cong 2.1 \times 10^8\ \text{ms}^{-1})$

[Award ½ mark if the student writes the formula:

$$\mu = \frac{\sin(A + D_m)/2}{\sin({}^{A}/_{2})}$$

but does not do any calculations.]

(a) Calculate the distance of an object of height h from a concave mirror of radius of curvature 20 cm, so as to obtain a real image of magnification 2. Find the location of image also.

(b) Using mirror formula, explain why does a convex mirror always produce a virtual image.

CBSE 2016

(a) Calculate the distance of an object of height h from a concave mirror of radius of curvature 20 cm, so as to obtain a real image of magnification 2. Find the location of image also.

(b) Using mirror formula, explain why does a convex mirror always produce a virtual image.

CBSE 2016

a) Given R = -20 cm, and magnification m= -2

Focal length of the mirror $f = \frac{R}{2} = -10\ cm$ ½

Magnification (m) = $-\frac{v}{u}$ ½

$-2 = -\frac{v}{u}$

$\Rightarrow v = 2u$

Using mirror formula

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ ½

$\Rightarrow -\frac{1}{10} = \frac{1}{2u} + \frac{1}{u}$

$\Rightarrow u$ =-15 cm

$\therefore v = 2 \times -15$ cm= -30 cm ½

b)

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

Using sign convention, for convex mirror, we have

f>0, u<0 ½

From the formula

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$\because f$ is positive and u is negative,

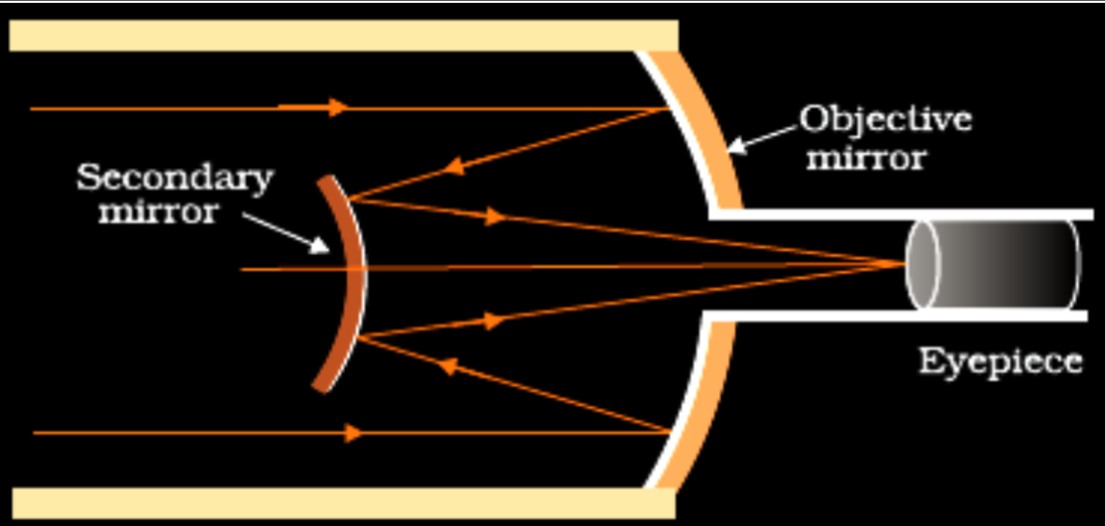
$\Rightarrow$ v is always positive, hence image is always virtual. ½

Draw a schematic ray diagram of reflecting telescope showing how rays coming from a distant object are received at the eye-piece. Write its two important advantages over a refracting telescope.

CBSE 2016

Draw a schematic ray diagram of reflecting telescope showing how rays coming from a distant object are received at the eye-piece. Write its two important advantages over a refracting telescope.

CBSE 2016



2

(i) Large gathering power
(ii) Large magnifying power
(iii) No chromatic aberration
(iv) Spherical aberration is also removed
(v) Easy mechanical support
(vi) Large resolving power

(Any Two)

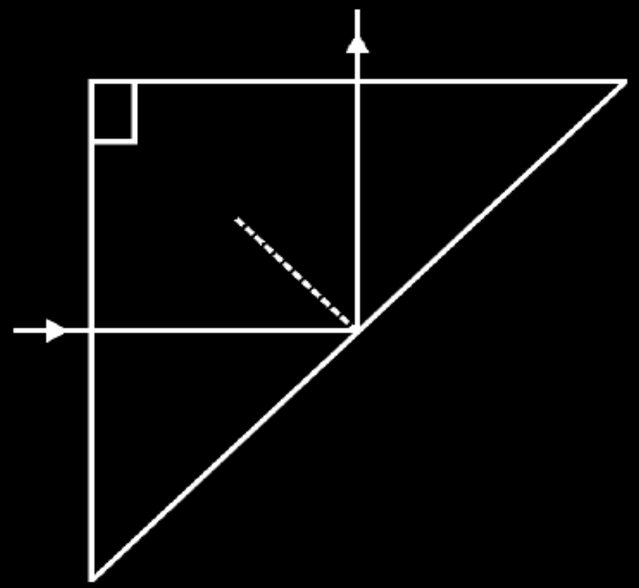
½ +½

(i) Plot a graph to show variation of the angle of deviation as a function of angle of incidence for light passing through a prism. Derive an expression for refractive index of the prism in terms of angle of minimum deviation and angle of prism.

(ii) What is dispersion of light ? What is its cause ?

(iii) A ray of light incident normally on one face of a right isosceles prism is totally reflected as shown in fig. What must be the minimum value of refractive index of glass ? Give relevant calculations.

CBSE 2016



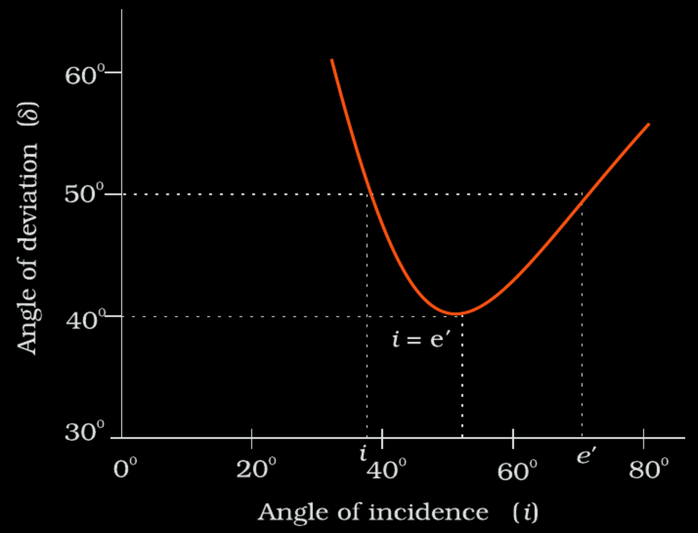
From figure $\delta = D_m, i = e$ which implies $r_1 = r_2$

$2r$ = A, or $r = {}^{A}/_{2}$ ½

Using $\delta = i + e - A$
$D_m = 2i - A$

$$i = \frac{A + D_m}{2}$$ ½

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin(\frac{A + D_m}{2})}{\sin {}^{A}/_{2}}$$ ½



(ii) The phenomenon of splitting of white light into its constituent colours. ½

Cause: Refractive index of the material is different for different colours
According to the equation, $\delta \cong (\mu - 1)A$, where A is the angle of prism, different colours will deviate through different amount. ½

For total internal reflection,
$\angle i \geq \angle i_c$ (critical angle)

$\Rightarrow 45^0 \geq \angle i_c$, i.e. , $\angle i_c \leq 45^0$
$\sin i_c \leq \sin 45^0$

$$\leq \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sin i_c} \geq \sqrt{2}$$

$\Rightarrow \mu \geq \sqrt{2}$

Hence, the minimum value of refractive index must be $\sqrt{2}$

(i) For a glass prism $\left(\mu = \sqrt{3}\right)$ the angle of minimum deviation is equal to the angle of the prism. Calculate the angle of the prism.

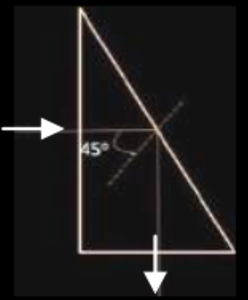
(ii) Draw ray diagram when incident ray falls normally on one of the two equal sides of a right angled isosceles prism having refractive index $\mu = \sqrt{3}$.

CBSE 2016

(i) For a glass prism $\left(\mu = \sqrt{3}\right)$ the angle of minimum deviation is equal to the angle of the prism. Calculate the angle of the prism.

(ii) Draw ray diagram when incident ray falls normally on one of the two equal sides of a right angled isosceles prism having refractive index $\mu = \sqrt{3}$.

CBSE 2016

(i) $\mu = \frac{sin\left(\frac{A+D}{2}\right)}{sin\frac{A}{2}}$	½
$= \frac{sin\left(\frac{2A}{2}\right)}{Sin\frac{A}{2}} = 2\cos A/2 = \sqrt{3}$	½
$\therefore A = 60°$	½
(ii) $\mu = \sqrt{3} = \frac{1}{Sini_c}$ 	
$\therefore Sini_c = \frac{1}{\sqrt{3}} \cong 0.58$	½
Lies between 30^0 and 45^0	
Hence, TIR takes place.	
Alternatively,	
$sinc = \frac{1}{\sqrt{3}}$ which is less than $\frac{1}{\sqrt{2}}$	
$\therefore$ angle of incidence > i_c	
$\therefore$TIR	1

(i) Draw a ray diagram showing the geometry of formation of image of a point object situated on the principal axis and on the convex side of a spherical surface of radius of curvature R. Taking the rays as incident from a rarer medium of refractive index n_1 to a denser medium of refractive index n_2, derive the relation.

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$, where symbols have their usual meaning.

(ii) Explain briefly how the focal length of a convex lens changes with increase in wavelength of incident light.

(iii) What happens to the focal length of convex lens when it is immersed in water ? Refractive index of the material of lens is greater than that of water.

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$$\tan\alpha = \frac{AN}{ON} \approx \alpha$$

$$\tan\beta = \frac{AN}{ON} \approx \beta$$

$$\tan\gamma = \frac{AN}{ON} \approx \gamma$$

$$\alpha + \gamma = i; r = \gamma - \beta$$

$$\frac{AN}{ON} + \frac{AN}{CN} = i; r = \frac{AN}{CN} - \frac{AN}{NI}$$

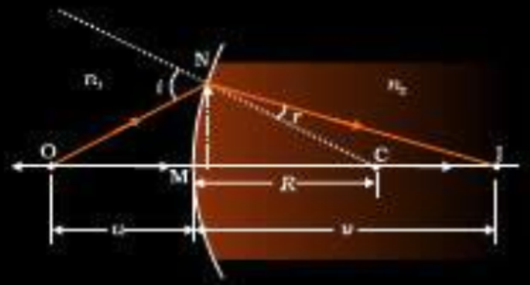
$$n_{21} = \frac{\sin i}{\sin r} \approx \frac{i}{r}$$

$$\frac{n_2}{n_1} = \frac{\frac{AN}{ON} + \frac{AN}{CN}}{\frac{AN}{CN} - \frac{AN}{NI}}$$

$$n_2\left(\frac{AN}{CN} - \frac{AN}{NI}\right) = n_1\left(\frac{AN}{ON} + \frac{AN}{CN}\right)$$

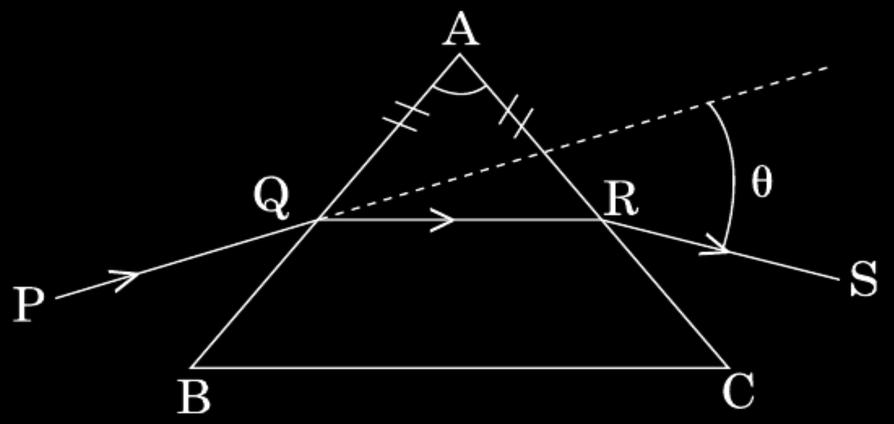
$$CN = R; NI = V; ON = -u$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$



(ii) focal length increases with increase of wavelength	½
$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\frac{2}{R}$ as wavelength increases μ_2/μ_1 decreases hence focal length increases	½
(iii)As μ_1 increases focal length increases	½
$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\frac{2}{R}$	½

A ray PQ incident on the refracting face BA is refracted in the prism BAC as shown in the figure and emerges from the other refracting face AC as RS such that AQ = AR. If the angle of prism A = 60° and refractive index of material of prism is $\sqrt{3}$, calculate angle θ.



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Since AQ = AR, we have QR|| BC $\therefore$ θ is the angle of minimum deviation. (Alternatively: Since AQ=AR, we get $\angle r_1 = \angle r_2$ $\therefore$ θ is the angle of minimum deviation.)	½
$\mu = \dfrac{\sin\left(\dfrac{A+\delta m}{2}\right)}{\sin(A/2)}$	½
$\therefore \sqrt{3} = \dfrac{\sin\left(\dfrac{60+\delta m}{2}\right)}{\sin 30^0}$	½
$\therefore \dfrac{\sqrt{3}}{2} = \sin\left(\dfrac{60+\delta m}{2}\right)$ $\therefore \dfrac{60+\delta m}{2} = 60$ or $\delta m = 60^0$	½

(i) A screen is placed at a distance of 100 cm from an object. The image of the object is formed on the screen by a convex lens for two different locations of the lens separated by 20 cm. Calculate the focal length of the lens used.

(ii) A converging lens is kept coaxially in contact with a diverging lens - both the lenses being of equal focal length. What is the focal length of the combination ?

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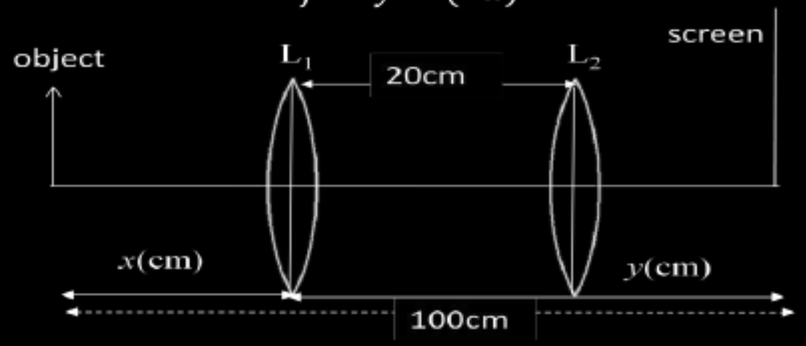
(i) For first position of the lens , we have

$$\frac{1}{f} = \frac{1}{y} - \frac{1}{(-x)}$$



½

For second position of the lens , we have

$$\frac{1}{f} = \frac{1}{y-20} - \frac{1}{(-(x+20))}$$

½

$$\frac{1}{y} + \frac{1}{x} = \frac{1}{(y-20)} + \frac{1}{(x+20)}$$

$$\frac{x+y}{xy} = \frac{(x+20)+(y-20)}{(y-20)(x+20)}$$

$$\therefore xy = (y-20)(x+20)$$

$$= xy - 20x + 20y - 400$$

$$\therefore x - y = -20$$

½

Also , x + y =100

∴ x = 40 cm

and y = 60 cm

$$\therefore \frac{1}{f} = \frac{1}{60} - \frac{1}{-40} = \frac{2+3}{120} = \frac{5}{120}$$

½

$$\therefore f = 24\ cm$$

Alternatively ,

We have

$$f = \frac{D^2 - d^2}{4D}$$ 1

$$= \frac{100^2 - 20^2}{4 \times 100}$$ ½

$$= \frac{120 \times 80}{400}$$ ½

= 24 cm

Alternatively,

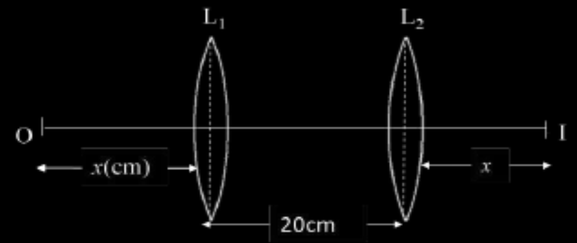
For the two positions of the lens , the values of the magnitudes of u and v , get interchanged. ½

Hence , $|u+v| = 100$ ½

$|u-v| = 20$, This gives $|u| = 60$ $|v| = 40$ ½

$\therefore f = 24\ cm$ ½

Alternatively ,



$2x + 20 = 100$

$\therefore x = 40$cm ½

For lens at position L_1 ; u = - x = -40cm ½

v = 20 + 40 = 60cm ½

This gives f = 24 cm

(i) For combination of two lenses in contact .	
Net Power of combination ,	
$P = P_1 + P_2$	½
$P_{1 = +} P$, $P_2 = -P$	
So P= 0 and F= infinite	½
Alternatively , $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$	½
$= \frac{1}{f} + \left(\frac{-1}{f}\right) = 0$	
F = infinite	½

(i) What is total internal reflection ? Under what conditions does it occur ?

(ii) Find a relation between critical angle and refractive index.

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(iii) Name one phenomenon which is based on total internal reflection.

(i) What is total internal reflection ? Under what conditions does it occur ?

(ii) Find a relation between critical angle and refractive index.

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(iii) Name one phenomenon which is based on total internal reflection.

(i) When a ray of light travels from a denser medium into a rarer medium at an angle greater than the critical angle, it reflects back into the denser medium. This phenomenon is called total internal reflection. Conditions for total internal reflection	½
(a) Light should travel from denser medium to rarer medium.	½
(b) Angle of incidence should be greater than critical angle.	½
(ii) $\frac{1}{\mu} = \frac{\sin i}{\sin r}$, for total internal reflection to occur $i \geq i_c$ at critical angle, angle of refraction $r = 90^\circ$, hence $\frac{1}{\mu} = \frac{\sin i_c}{\sin 90^\circ}$	½
$\Rightarrow \mu = \frac{1}{\sin i_c}$	½
(iii) Mirage/ sparkling of diamond/ optical fiber/ totally reflecting Prism/ shinning of air bubbles in water.(any one)	½

(a) Draw a schematic diagram of a reflecting telescope.

(b) State the advantages of reflecting telescope over refracting telescope.

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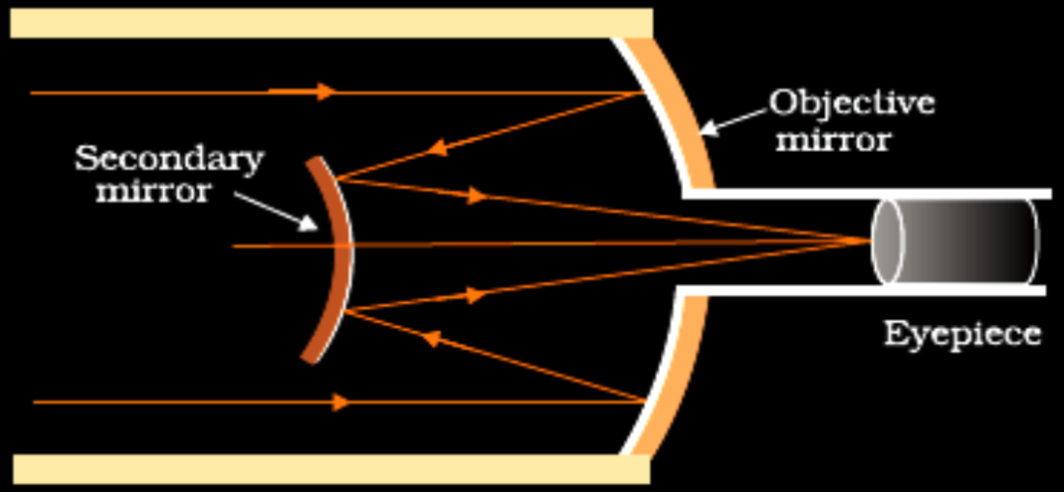
(a)



2

(b) Advantages (any two)

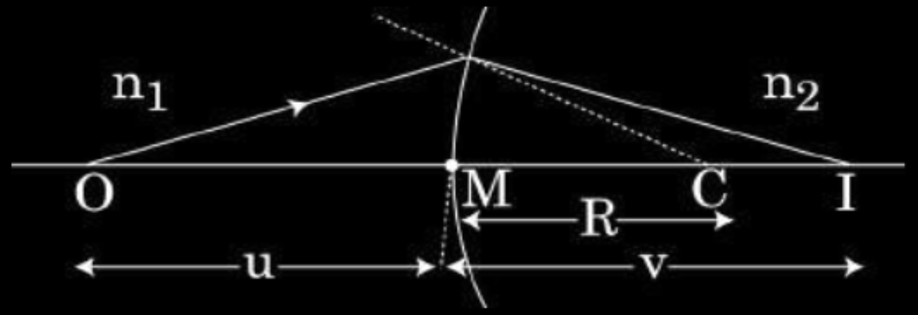
(i) There is no chromatic aberration in a mirror.
(ii) Brighter image
(iii) High resolving Power
(iv) Large light gathering power
(v) Large magnifying power

½+½

(a) A point object O on the principle axis of a spherical surface of radius of curvature R separating two media of refractive indices n_1 and n_2 forms an image 'I' as shown in the figure.

Prove that

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$



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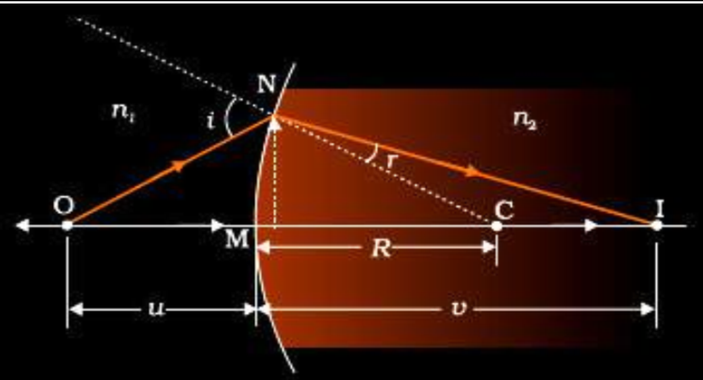
(b) Use this expression to derive lens maker's formula. Draw the necessary diagram.

(c) A convex lens is placed over a plane mirror. A pin is now positioned so that there is no parallax between the pin and its image formed by this lens-mirror combination. How will you use this observation to find focal length of the lens ? Explain briefly.

(a)



½

$$i = \frac{MN}{OM} + \frac{MN}{MC}$$ ½

$$r = \frac{MN}{MC} - \frac{MN}{MI}$$ ½

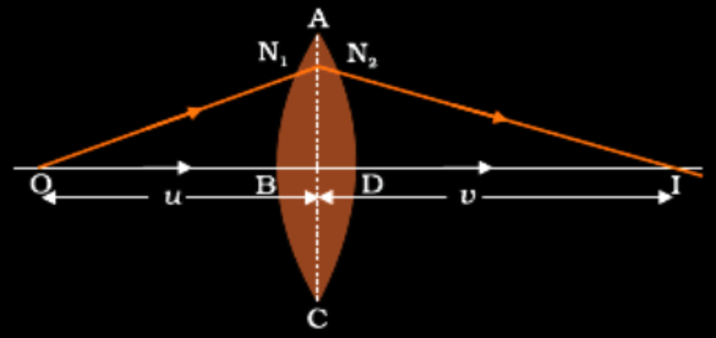
$$n_1 i = n_2 r$$

$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}$$

$$OM = -u, MI = +v, MC = +R$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$ ½

(b) Applying above relation to refraction of light through a convex lens ABCD



½

For interface ABC $\frac{n_2}{v_1} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1}$ ½

For interface ADC $\frac{n_1}{v} - \frac{n_1}{v_1} = \frac{n_1 - n_2}{R_2}$ ½

$$\therefore \frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ ½

or $\frac{1}{f} = (n_{21} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ½

(c) Focal length = distance of the pin from the mirror.

The rays from the object after refraction from lens should fall normally on the Plane mirror. So they retrace their path. Hence, rays must be originating from focus and thus distance of the pin from the plane mirror gives focal lenth of the lens. ½

A ray PQ incident normally on the refracting face BA is refracted in the prism BAC made of material of refractive index 1.5. Complete the path of ray through the prism. From which face will the ray emerge ? Justify your answer.

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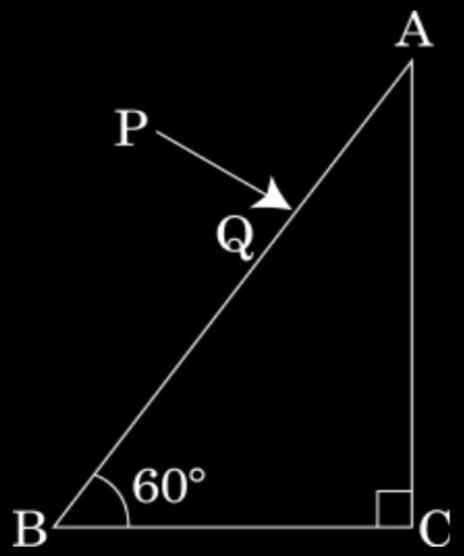

A ray PQ incident normally on the refracting face BA is refracted in the prism BAC made of material of refractive index 1.5. Complete the path of ray through the prism. From which face will the ray emerge ? Justify your answer.

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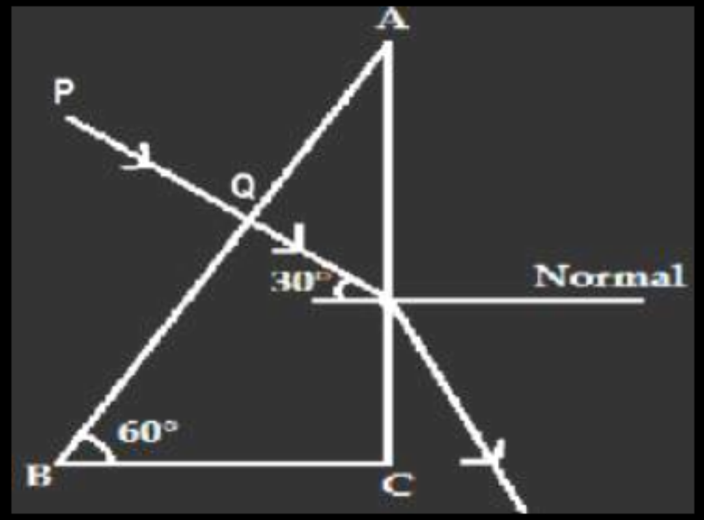

1

Face-AC

Here $i_c = \sin^{-1}(\frac{2}{3})$ ½

$= \sin^{-1}(0.6)$

$\angle i$ on face AC is 30° which is less than $\angle i_c$.Hence the ray get replaced here. ½

(i) Derive the mathematical relation between refractive indices n_1 and n_2 of two radii and radius of curvature R for refraction at a convex spherical surface. Consider the object to be a point since lying on the principle axis in rarer medium of refractive index n_1 and a real image formed in the denser medium of refractive index n_2. Hence, derive lens maker's formula.

(ii) Light from a point source in air falls on a convex spherical glass surface of refractive index 1.5 and radius of curvature 20 cm. The distance of light source from the glass surface is 100 cm. At what position is the image formed ?

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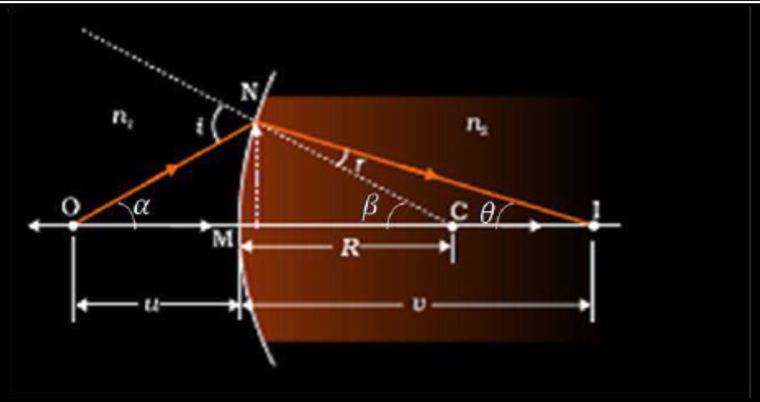


½

Ray diagram showing real image formation as per prescription

$\theta_1 = \alpha + \beta$

$\theta_2 = \beta - \gamma$ $\quad \therefore \gamma = \beta - \theta$

½

For paraxial rays θ_1 and θ_2 are small

Therefore, $n_2 \sin\theta_2 = n_1 \sin\theta_2$ (Snells law)

Reduces to

At N $\frac{Sini}{Sinr} \sim \frac{i}{r} = \frac{n_2}{n_1}$

$\therefore\ n_1 = rXn_2$

$$(\alpha+\beta)n_1 = (\beta-\theta)n_2$$

$$n_1\left(\frac{NM}{OM}+\frac{NM}{MC}\right) = \left(\frac{NM}{MC}-\frac{NM}{MI}\right)n_2$$

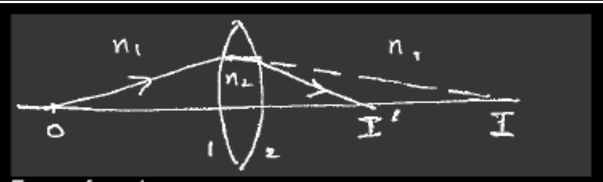
$$n_1\left(\frac{1}{-u}+\frac{1}{+R}\right) = \left(\frac{1}{+R}-\frac{1}{u}\right)n_2$$

½

$$\frac{n_2}{v'}-\frac{n_1}{u} = \frac{(n_2-n_1)}{R_1}$$

1

Appying above relations to refraction through a lens



½

For surface 1

$$\frac{n_2-n_1}{R_1} = \frac{n_2}{v'}-\frac{n_1}{u} \qquad ...(i)$$

For surface 2

$$\frac{n_1-n_2}{R_2} = \frac{n_1}{v}-\frac{n_2}{v'} \qquad ...(ii)$$

½

Adding eqn. (i) and (ii)

$$(n_2-n_1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right) = n_1\left(\frac{1}{v}-\frac{1}{u}\right)$$

For u=∝ v=*f*

$$\therefore \frac{n_1}{f} = (n_2-n_1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$$\Rightarrow \frac{1}{f} = \left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

½

(iii) R = 20 cm n_2 = 1.5 n_1 = 1 u = -100 cm

$$\frac{n_2}{v} = \frac{(n_2-n_1)}{R}+\frac{n_1}{u}$$

½

$$= \frac{0.5}{20cm}-\frac{1}{100cm}$$

$$= \frac{1.5}{100}\text{ cm}$$

$\Rightarrow V = 100cm$ a real image on the other side, 100 cm away from the surface.

½

(a) Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power.

(b) You are given three lenses of power 0.5 D, 4 D and 10 D to design a telescope.

(i) Which lenses should he used as objective and eyepiece ? Justify your answer.

(ii) Why is the aperture of the objective preferred to be large ?

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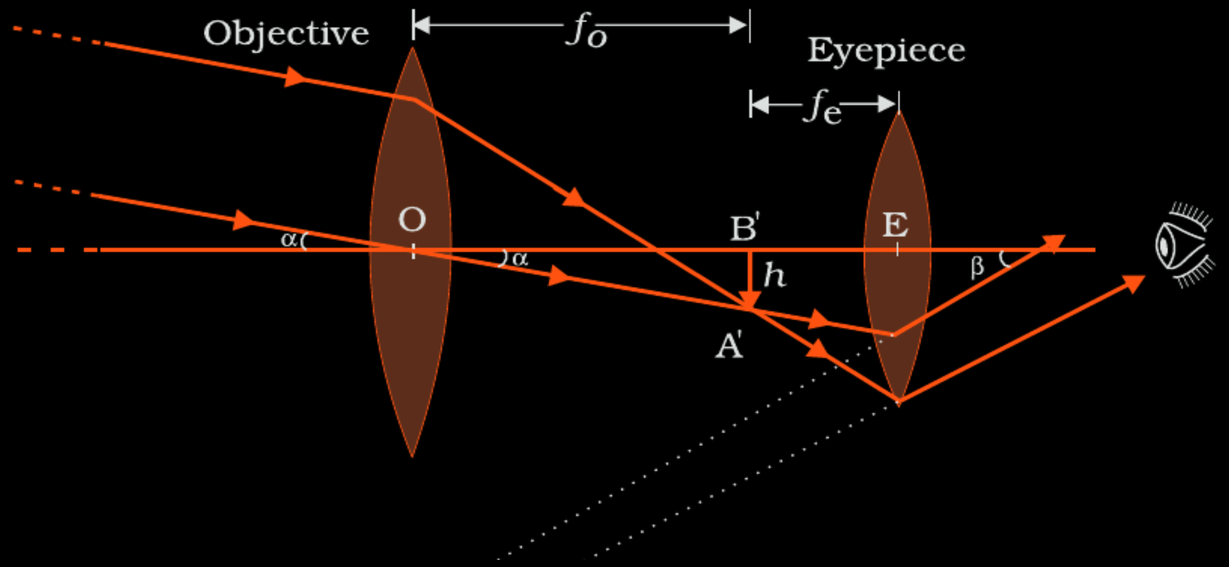

1½

Definition-It is the ratio of the angle subtended at the eye, by the final image, to the angle which the object subtends at the lens, or the eye. 1

b.

i.Objective=.5D ½

½

Eye lens = 10D

This choice would give higher magnification as

M= $\frac{f_o}{f_e} = \frac{P_e}{P_o}$ ½ +½

ii. High resolving power/ Brighter image / lower limit of resolution **(any one)** ½